SECTION 1

Introduction



Section 1: Exam Blueprint

Exam Details



Format
Multiple choice, multiple answer



Type
Foundational



Delivery Method
Testing center or online proctored exam



Time
90 minutes to complete the exam



Cost
100 USD (Practice Exam: 20 USD)



Language
Available in English, Japanese, Korean, and Simplified Chinese

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Section 1: Exam Blueprint

Recommended knowledge:

- At least 6 months experience with AWS Cloud in any role including technical, managerial, sales, purchasing or financial – not necessary for this course!

AWS Whitepapers:

- Overview of Amazon Web Services whitepaper
- Architecting for the Cloud: AWS Best Practices whitepaper
- How AWS Pricing Works whitepaper
- Cost Management in the AWS Cloud whitepaper

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Section 1: Exam Blueprint

Time and Length:

- 90 minutes
- 65 questions – don't worry, most questions are very straightfoward!

Scoring:

- Scaled score between 100 – 1000
- Minimum passing score of 700

Question format:

- Multiple-choice: Has one correct response and three incorrect responses
- Multiple-response: Has two or more correct responses out of five or more options

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Section 1: Exam Blueprint

Domain 1: Cloud Concepts

- 1.1 Define the AWS Cloud and its value proposition
- 1.2 Identify aspects of AWS Cloud economics
- 1.3 List the different cloud architecture design principles

Domain 2: Security

- 2.1 Define the AWS Shared Responsibility model
- 2.2 Define AWS Cloud security and compliance concepts
- 2.3 Identify AWS access management capabilities
- 2.4 Identify resources for security support

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Section 1: Exam Blueprint

Domain 3: Technology

- 3.1 Define methods of deploying and operating in the AWS Cloud
- 3.2 Define the AWS global infrastructure
- 3.3 Identify the core AWS services
- 3.4 Identify resources for technology support

Domain 4: Billing and Pricing

- 4.1 Compare and contrast the various pricing models for AWS
- 4.2 Recognize the various account structures in relation to AWS billing and pricing
- 4.3 Identify resources available for billing support

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Section 1: Exam Blueprint

Domain	% of Examination
Domain 1: Cloud Concepts	28%
Domain 2: Security	24%
Domain 3: Technology	36%
Domain 4: Billing and Pricing	12%
TOTAL:	100%

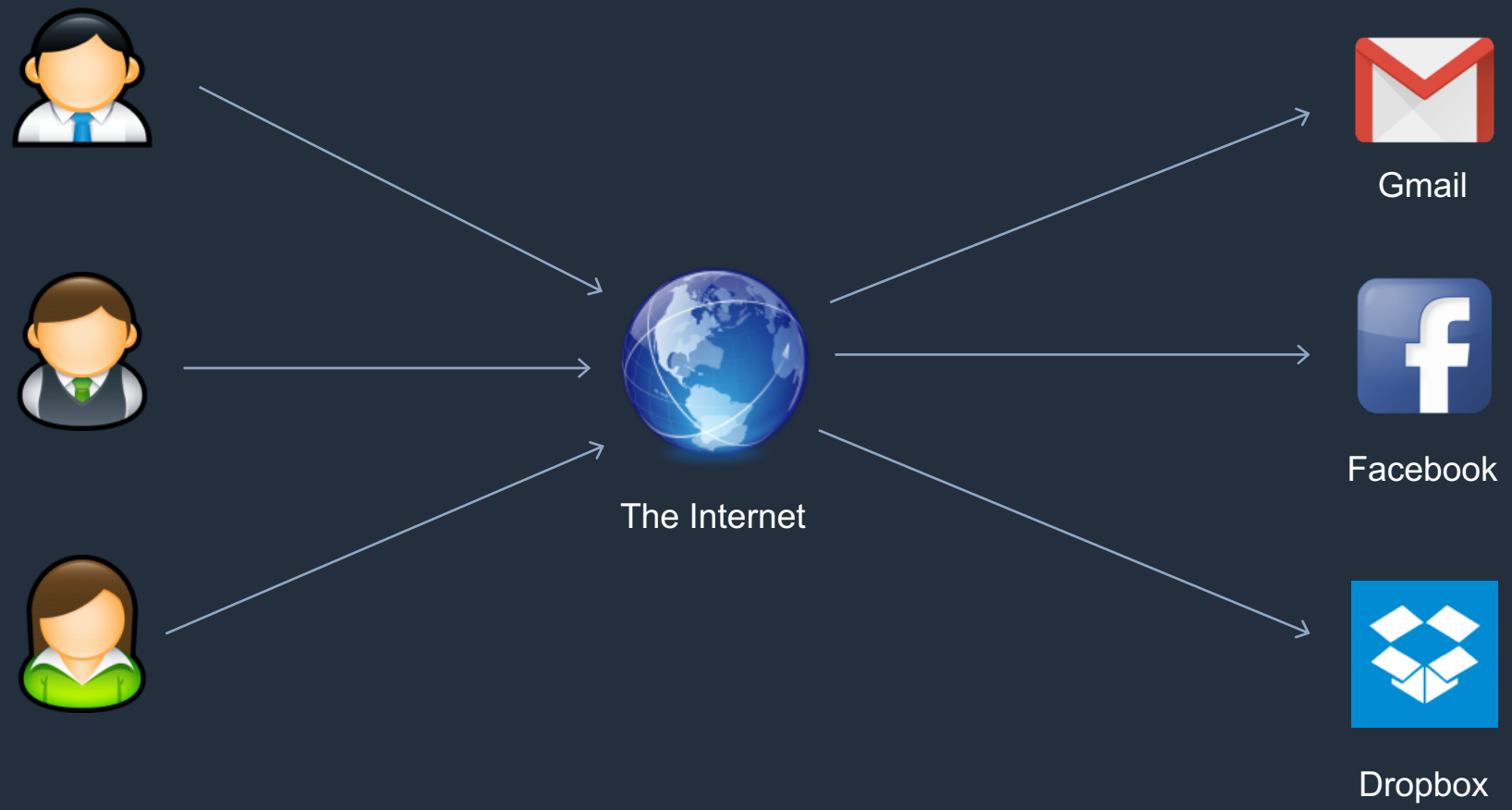
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SECTION 2

Overview of Cloud Computing



Section 2: What is Cloud? Consumer Examples



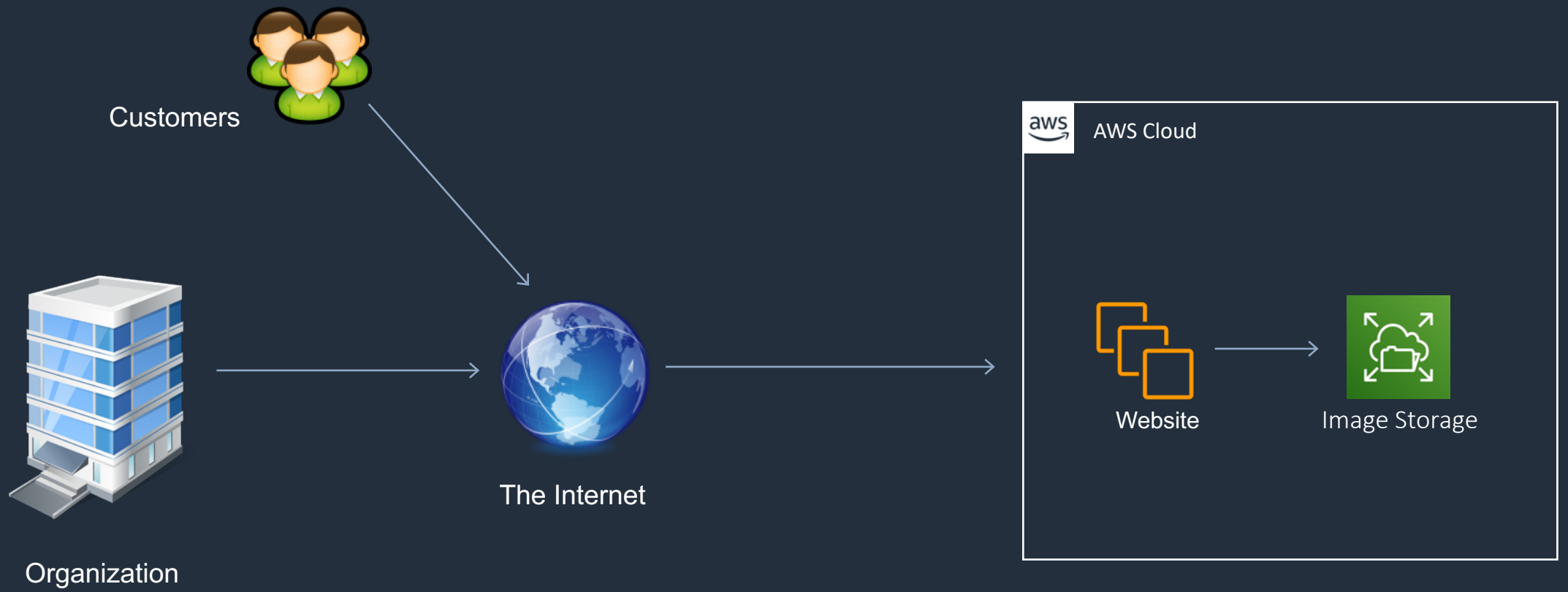
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Section 2: Cloud Terminology

Term	Description
Cloud Computing	Cloud computing is the on-demand delivery of IT services from a third-party provider over the Internet
Cloud Service	The IT capability that is being provided by the cloud provider
Cloud Provider / Cloud Service Provider	A company that provides a cloud service to organizations and/or individuals
Consumer	The organization or individual who uses the cloud service
”Pay as you go” or “pay per use”	You are charged only for what you use. Analogous to a utility bill
Multi-tenant	Multiple customers consume services delivered using shared infrastructure
“x” as a service	Some cloud capability is delivered to consumers as a service

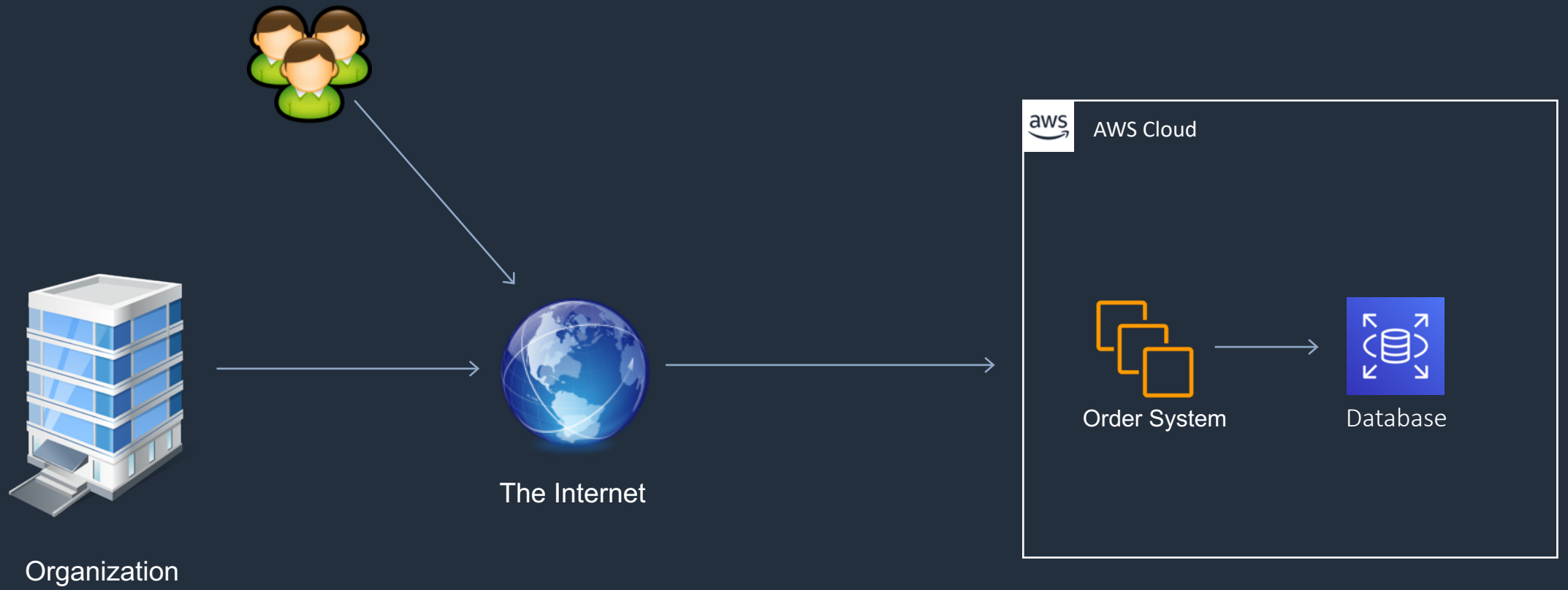
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Section 2: Enterprise Example - Website



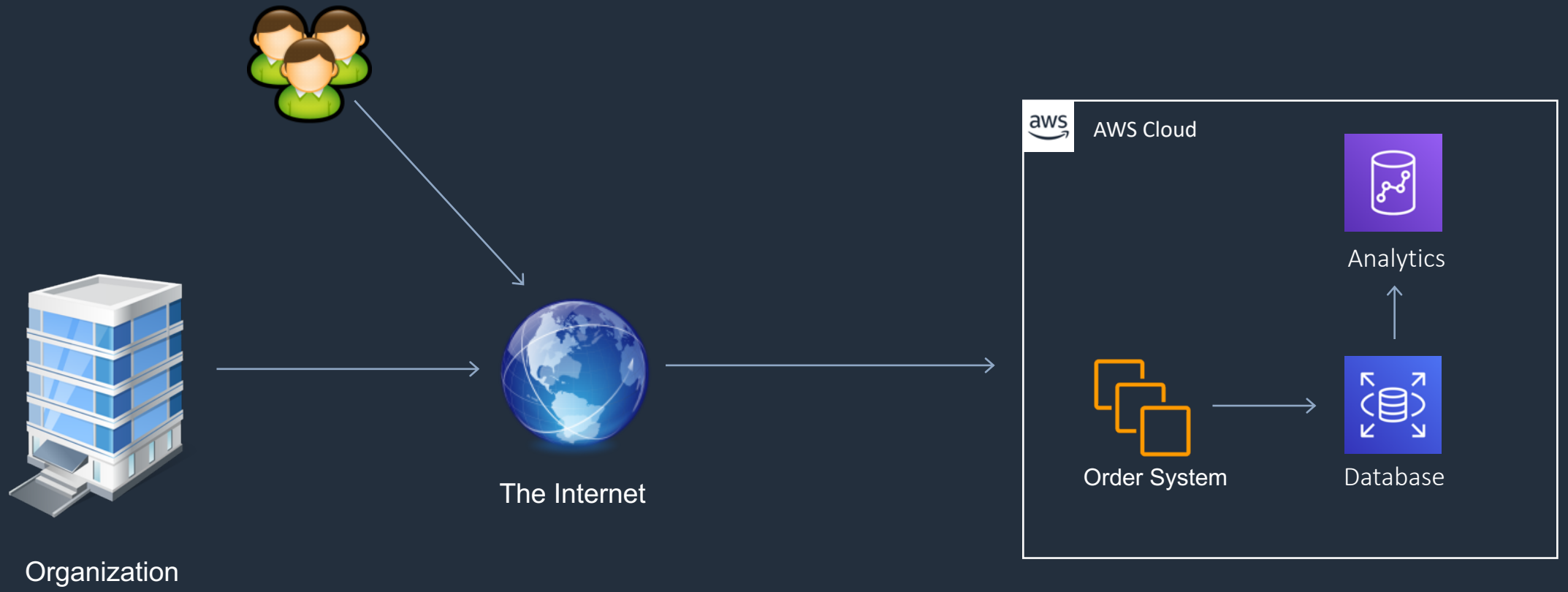
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Section 2: Enterprise Example – Order System



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Section 2: Enterprise Example – Analytics



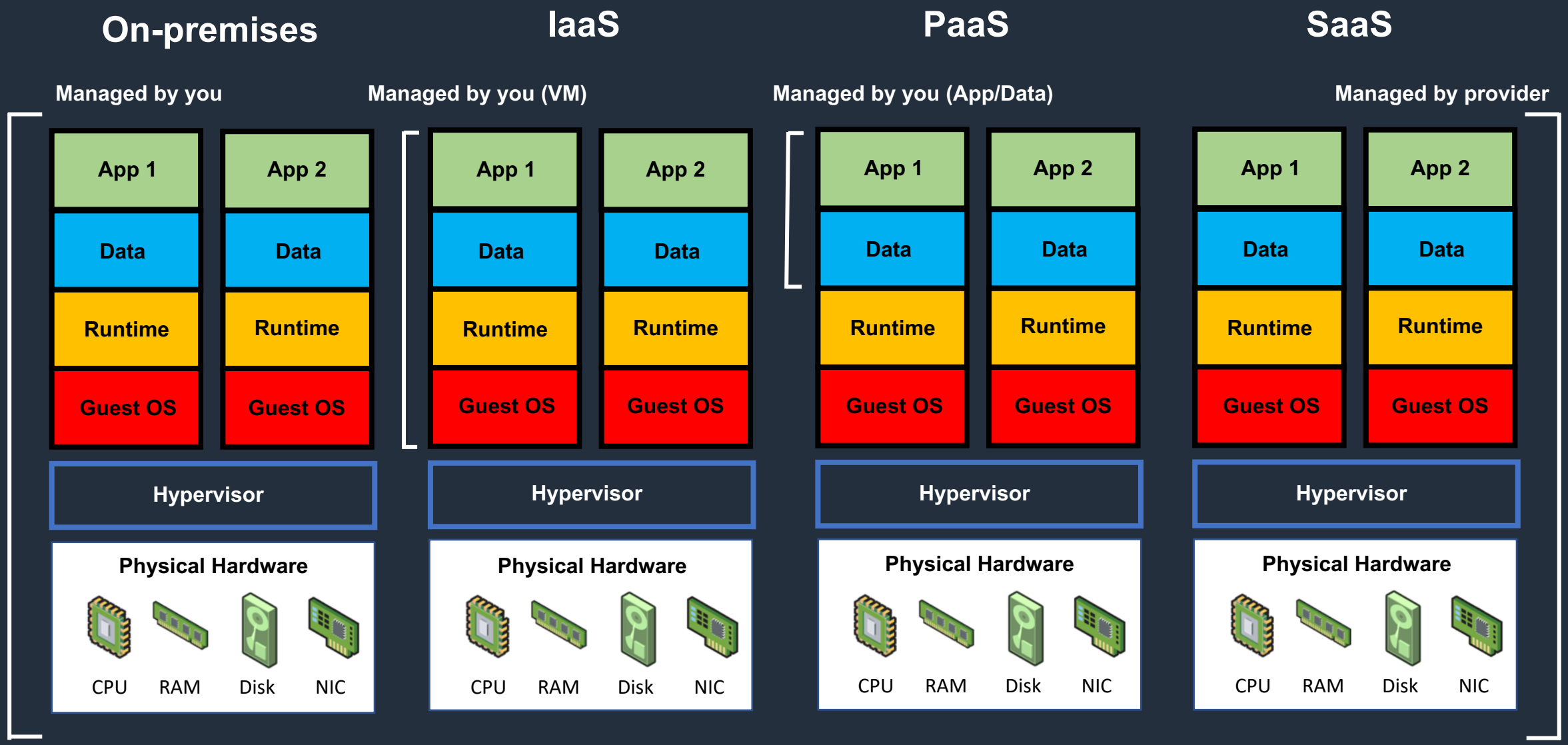
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Section 2: Key Characteristics of Cloud Computing

Name	Description
On-demand, self-service	A user can consume cloud resources, as needed, automatically, and without human interaction
Broad network access	Capabilities are available over the network using standard mechanisms. Can be the Internet or a Wide Area Network (WAN)
Resource pooling	The providers resources are pooled and serve multiple consumers using a multi-tenant model
Rapid elasticity	Capabilities can scale “elastically” based on demand
Measured service	Resource usage is monitored and metered

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Section 2: Cloud Computing Service Models



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Section 2: IaaS, PaaS, and SaaS Examples



System Admin

IaaS

Request a virtual server



Amazon EC2



Developer

PaaS

Upload code



AWS Elastic Beanstalk



Sales Team

SaaS

Log sales data



Salesforce.com

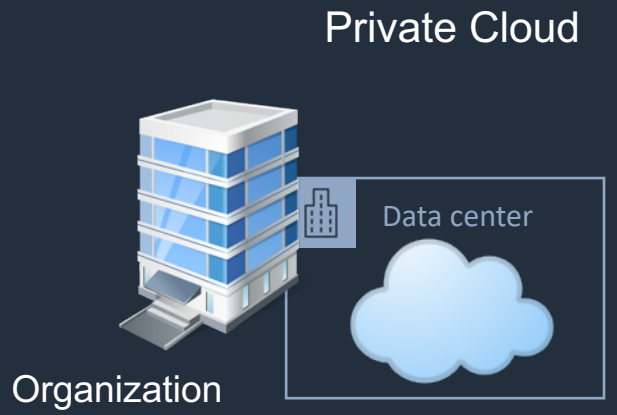
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Section 2: Cloud Computing Deployment Models

Name	Description	Examples
Private Cloud	An enterprise deploys their own infrastructure and applications into their own data center	VMware, Microsoft, RedHat, OpenStack
Public Cloud	The IT services that you consume are hosted and delivered from a third-party and accessed over the Internet	AWS, Microsoft Azure, Google Cloud Platform
Hybrid Cloud	A combination of on-premises, private cloud, and public cloud services are consumed	
Multicloud	Usage of two or more public clouds at a time, and possibly multiple private clouds	

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Section 2: Deployment Models – Private Cloud



Benefits:

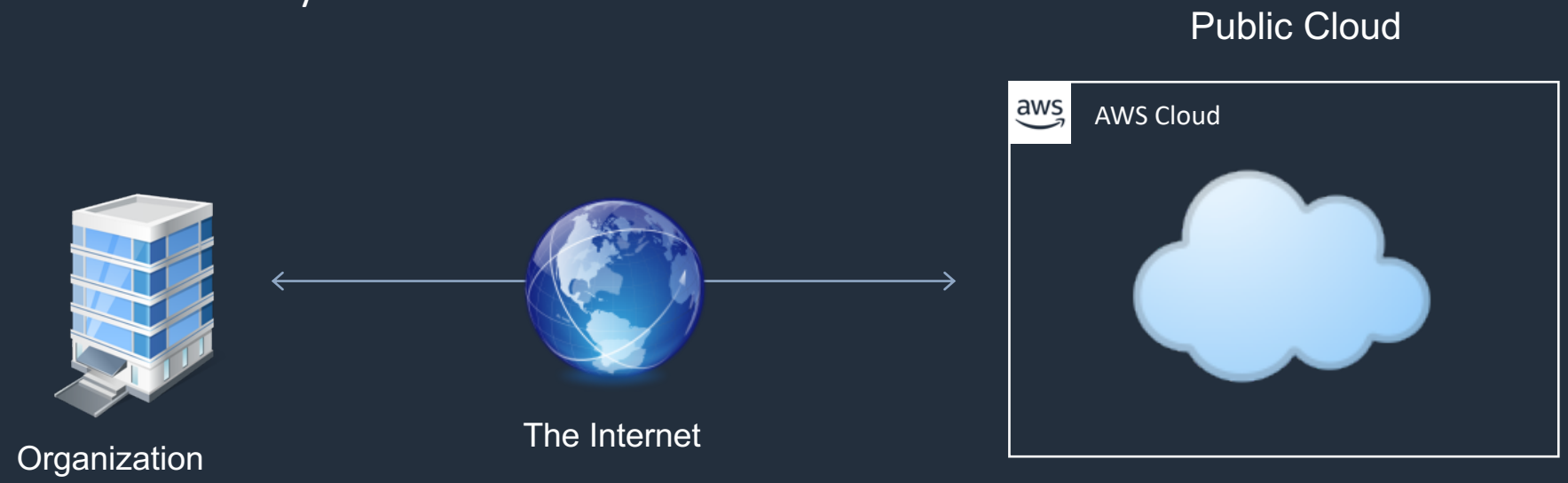
- Complete control of the entire stack
- Security – in a few cases, organizations may need to keep all or some of their applications and data in house

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Section 2: Deployment Models – Public Cloud

Benefits:

- Variable expense, instead of capital expense
- Economies of scale
- Massive elasticity



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Section 2: Deployment Models – Hybrid Cloud

Benefits:

- Allows companies to keep the critical applications and sensitive data in a traditional data center environment or private cloud
- Take advantage of public cloud resources like SaaS, for the latest applications, and IaaS, for elastic virtual resources
- Facilitates portability of data, apps and services and more choices for deployment models



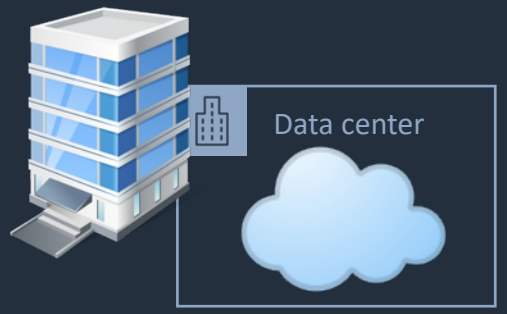
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Section 2: Legacy IT



Costs:

- Data center
- Power
- Computing Equipment
- Software licenses
- Maintenance contracts
- Staff wages
- Etc.



Limitations:

- Very capital heavy (CAPEX) – need to purchase everything
- Operational overhead is high
- Scalability is very limited



Operations

- Data center operations and security
- Equipment installation and maintenance
- System monitoring and operations activities
- Backup/recovery
- Etc.

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Section 2: The 6 Advantages of Cloud

	Name	Description
1	Trade capital expense for variable expense	Instead of investing in data centers before you know how you're going to use them, pay only when, and for how much, you consume
2	Benefit from massive economies of scale	Achieve a lower variable cost due to AWS' scale
3	Stop guessing about capacity	Eliminate guessing, scale as demand dictates
4	Increase speed and agility	Easily and quickly scale your usage
5	Stop spending money running and maintaining data centers	Focus on business growth and innovation instead!
6	Go global in minutes	Easily deploy applications in multiple regions around the world

https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html

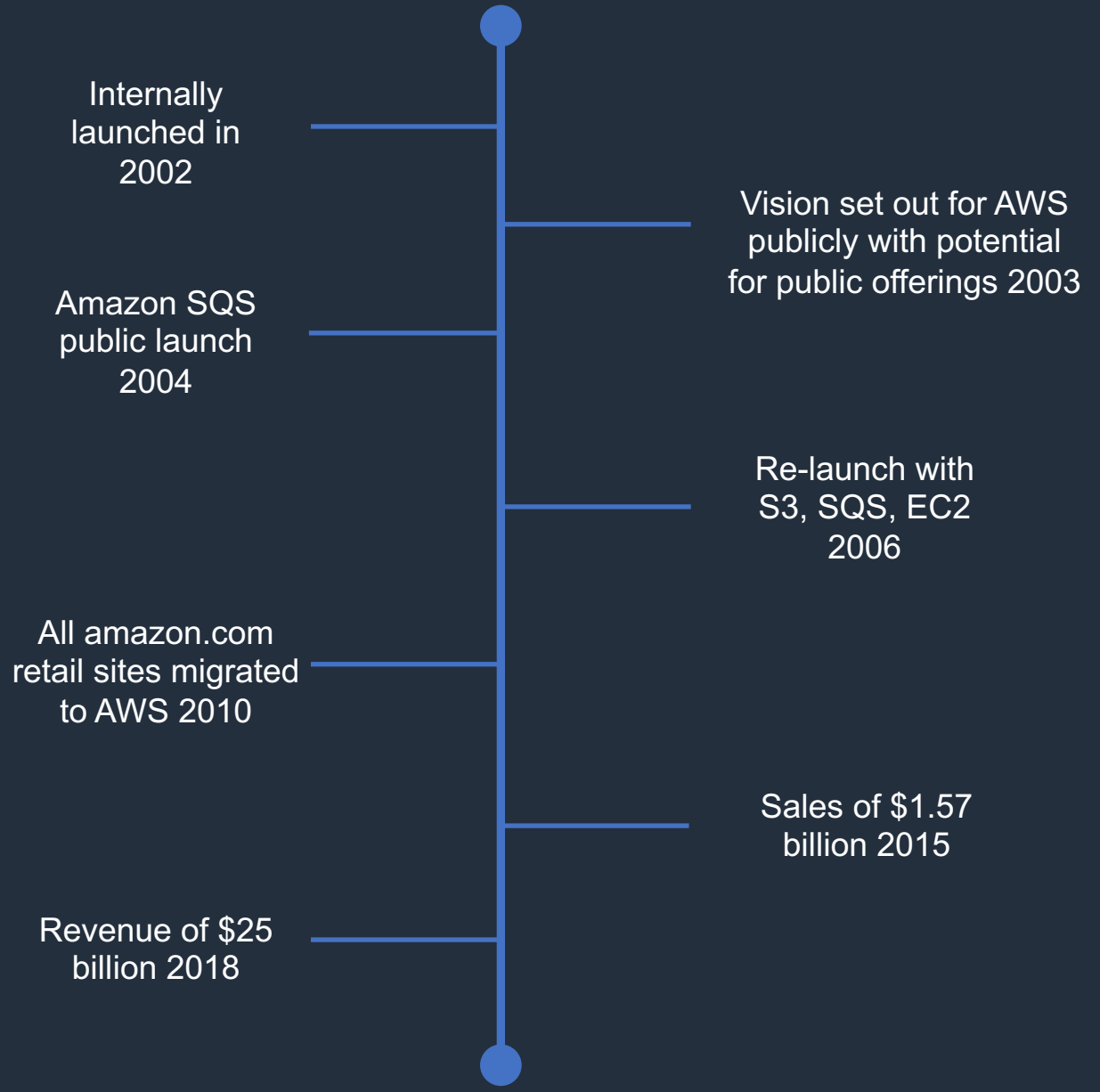
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SECTION 3

AWS Cloud Overview

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Section 3: Amazon Web Services History



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Section 3: Amazon Web Services (AWS) Today

- Over 165 services including computing, storage, networking, database, analytics, media services, machine learning, management, mobile, and IoT
- 22 geographical regions of presence
- $25 billion in revenue in 2018

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Section 3: Gartner Magic Quadrant 2018

- According to Gartner in 2018 AWS was the leader in IaaS with 47.8% share
- …way ahead of Microsoft Azure 15.5%, Alibaba 7.7%, Google 4%, IBM 1.8%
- AWS has been the leader for 9 years in a row!



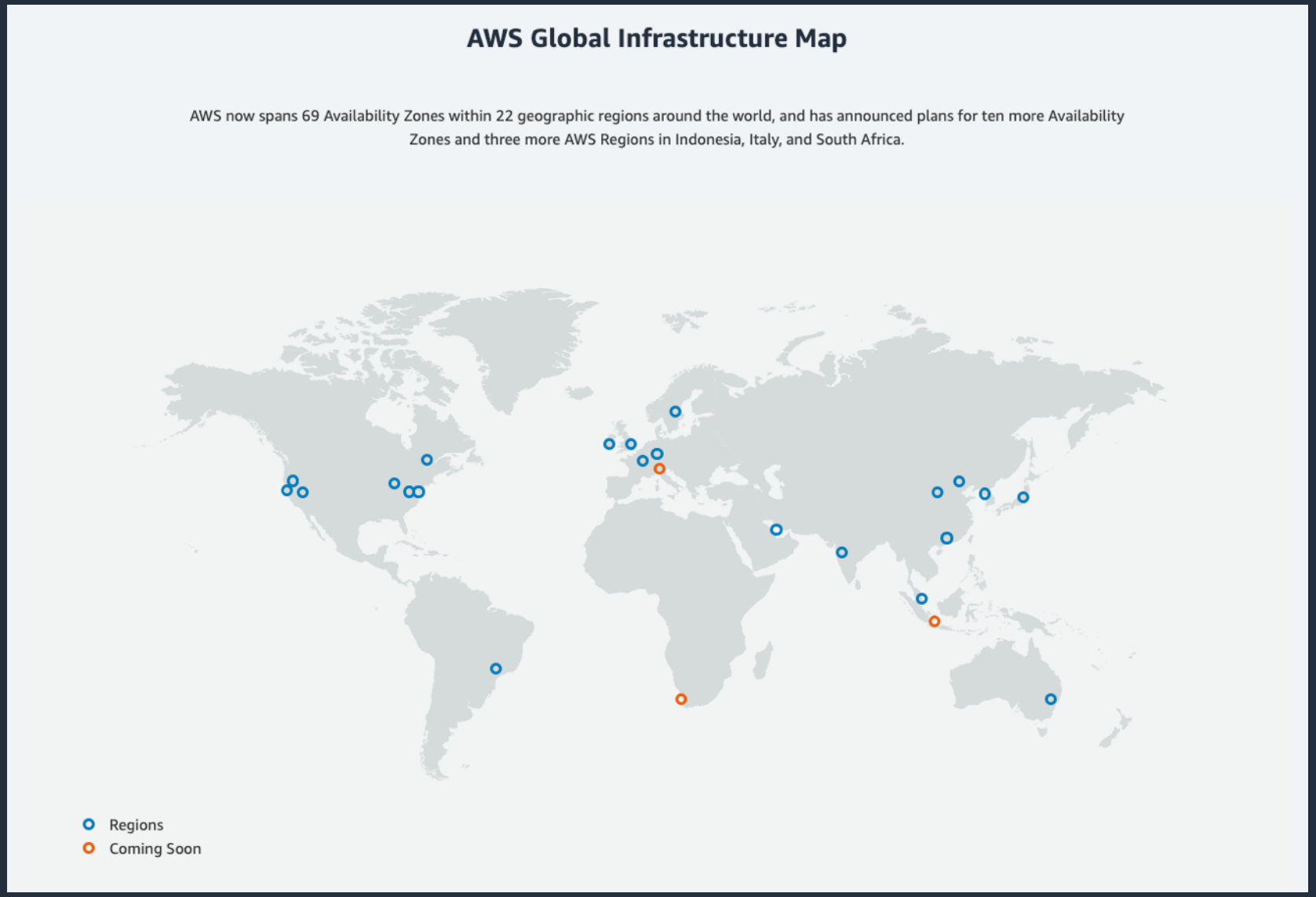
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Section 3: AWS Global Infrastructure

Name	Description
Region	A geographical area with 2 or more AZs, isolated from other AWS regions
Availability Zone (AZ)	One or more data centers that are physically separate and isolated from other AZs
Edge Location	A location with a cache of content that can be delivered at low latency to users – used by CloudFront
Regional Edge Cache	Also part of the CloudFront network. These are larger caches that sit between AWS services and Edge Locations
Global Network	Highly available, low-latency private global network interconnecting every data center, AZ, and AWS region

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Section 3: AWS Global Infrastructure Map



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Section 3: AWS Regions

- An AWS region is a geographical area
- Each region consists of 2 or more availability zones
- Each Amazon Region is designed to be completely isolated from the other Amazon Regions

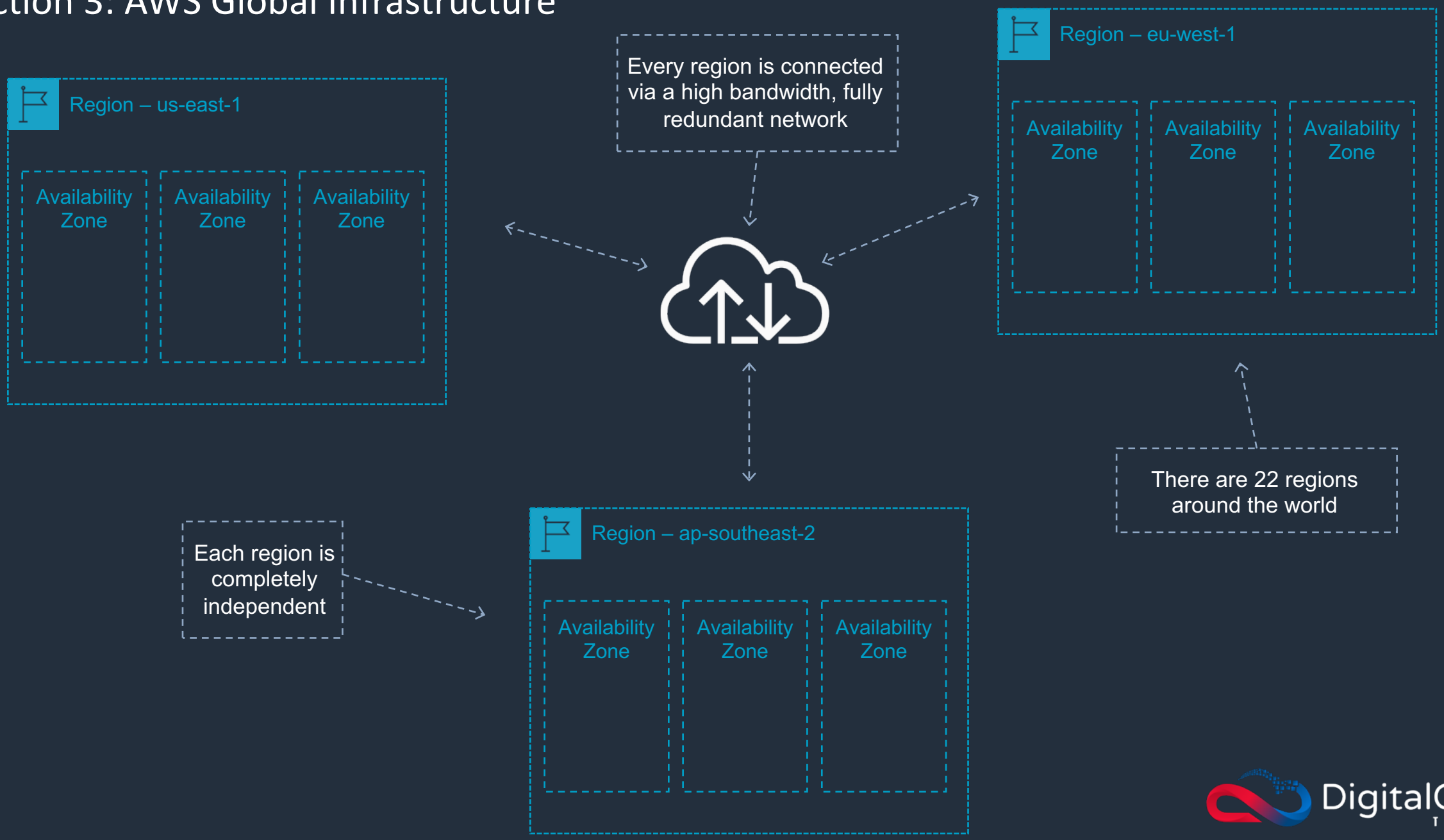
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Section 3: AWS Availability Zones

- Availability Zones (AZs) are locations into which you launch resources, such as Amazon EC2 instances
- AZs physically separate and isolated from each other
- AZs span one or more data centers and have direct, low-latency, high throughput and redundant network connections between each other
- Each AZ is designed as an independent failure zone
- AZs are physically separated within a typical metropolitan region, and use discrete power sources

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Section 3: AWS Global Infrastructure



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Section 3: AWS Services in Scope

Identity and Access Management

- AWS IAM

AWS Compute

- Amazon EC2
- Amazon ECS
- AWS Lambda
- Amazon LightSail

AWS Storage

- Amazon S3
- Amazon EBS
- Amazon EFS
- AWS Storage Gateway

AWS Networking

- Amazon VPC
- AWS Direct Connect

Databases

- Amazon RDS
- Amazon DynamoDB
- Amazon RedShift
- Amazon ElastiCache

Elastic Load Balancing and Auto Scaling

- Elastic Load Balancing
- Auto Scaling

Content Delivery and DNS Services

- Amazon Route 53
- Amazon CloudFront

Monitoring and Logging Services

- Amazon CloudWatch
- AWS CloudTrail

Notification Services

- Amazon SNS

Migration and Transfer

- Database Migration Service
- Server Migration Service
- AWS Snowball

Cloud Governance and Security

- AWS GuardDuty
- AWS KMS
- AWS WAF & Shield
- AWS CloudHSM
- AWS Artifact
- AWS Inspector
- AWS Trusted Advisor
- AWS Config
- AWS Service Catalog
- AWS Personal Health Dashboard

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Section 3: Global vs Regional Services



Identity and Access Management

- AWS IAM

AWS Compute

- Amazon EC2
- Amazon ECS
- AWS Lambda
- Amazon LightSail

AWS Storage

- Amazon S3
- Amazon EBS
- Amazon EFS
- AWS Storage Gateway

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Migration and Transfer

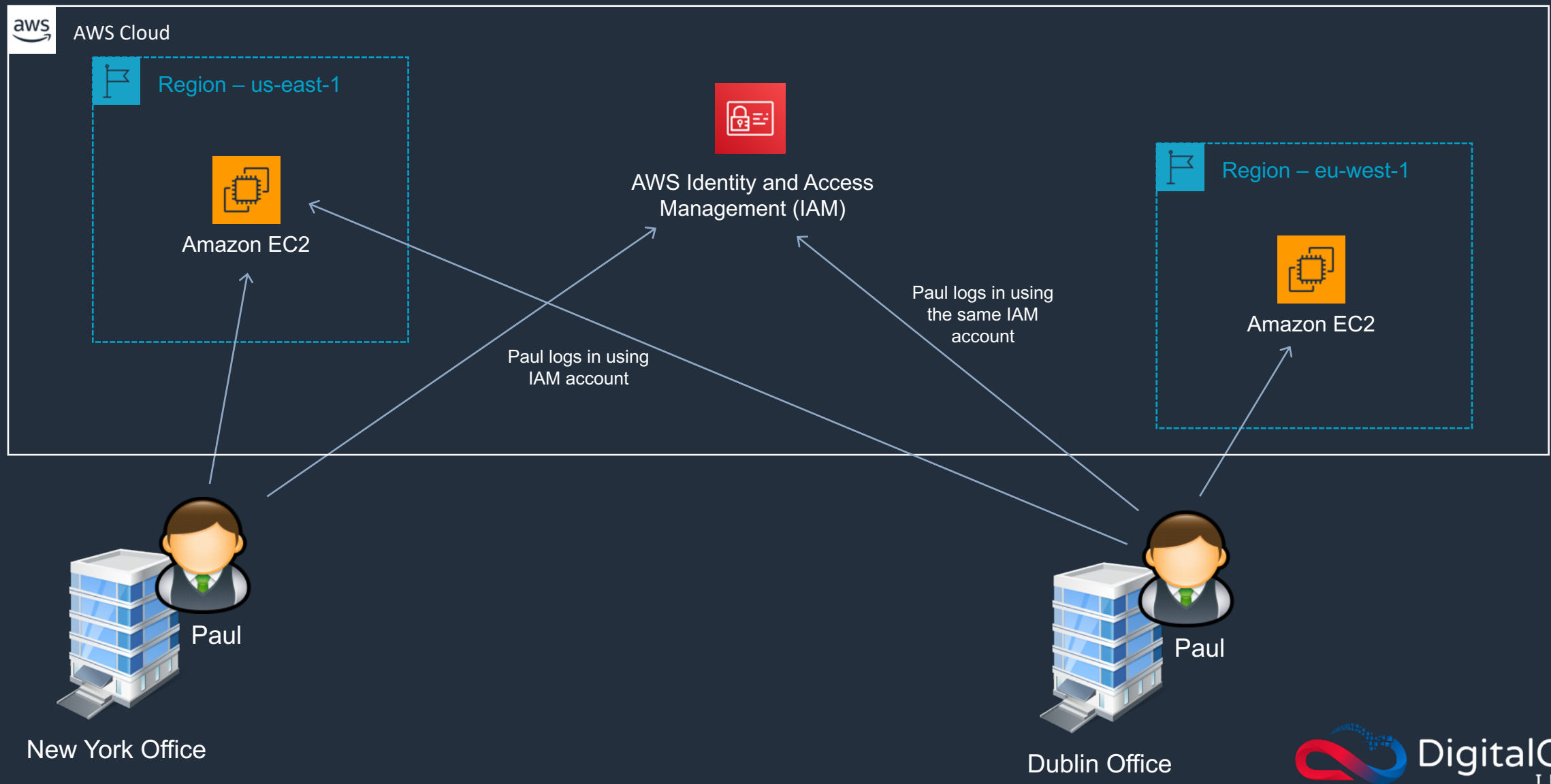
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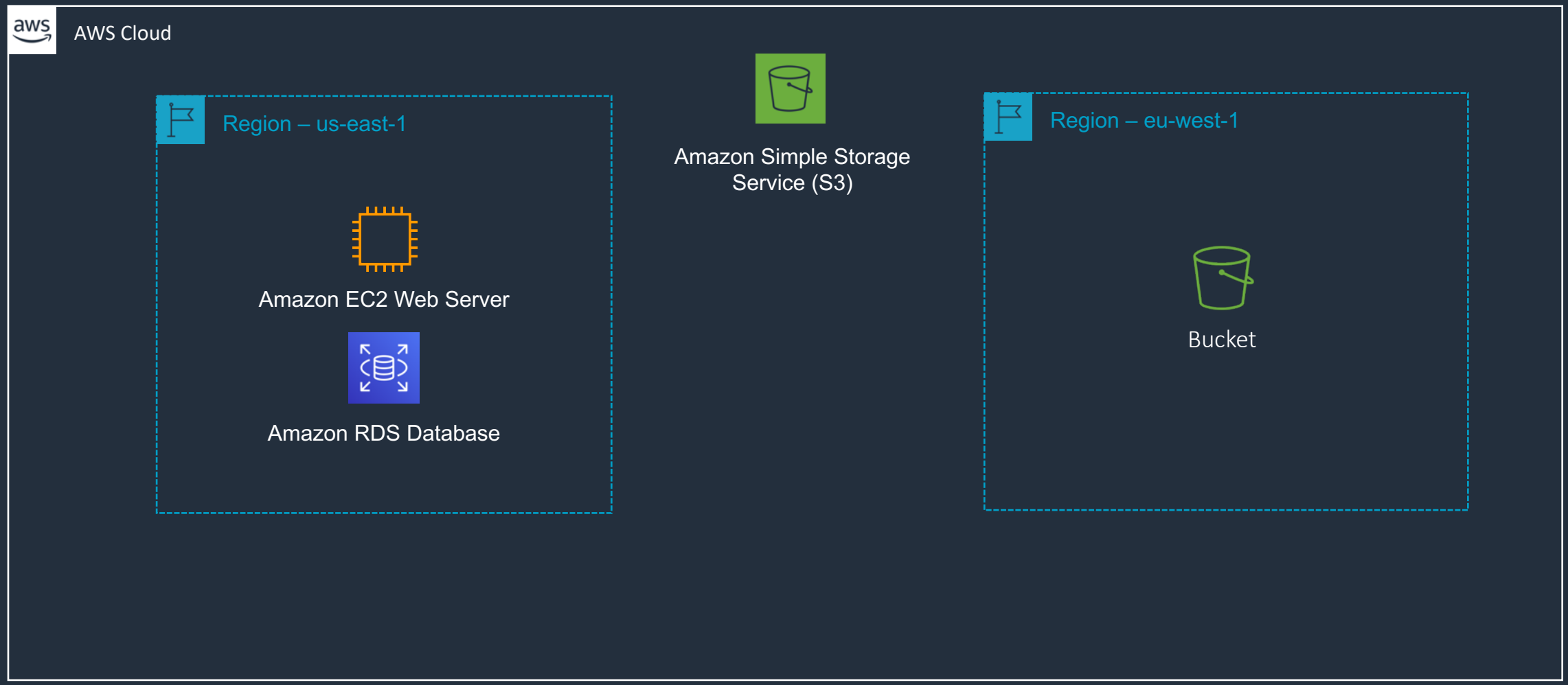
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Section 3: Global Service Example - IAM



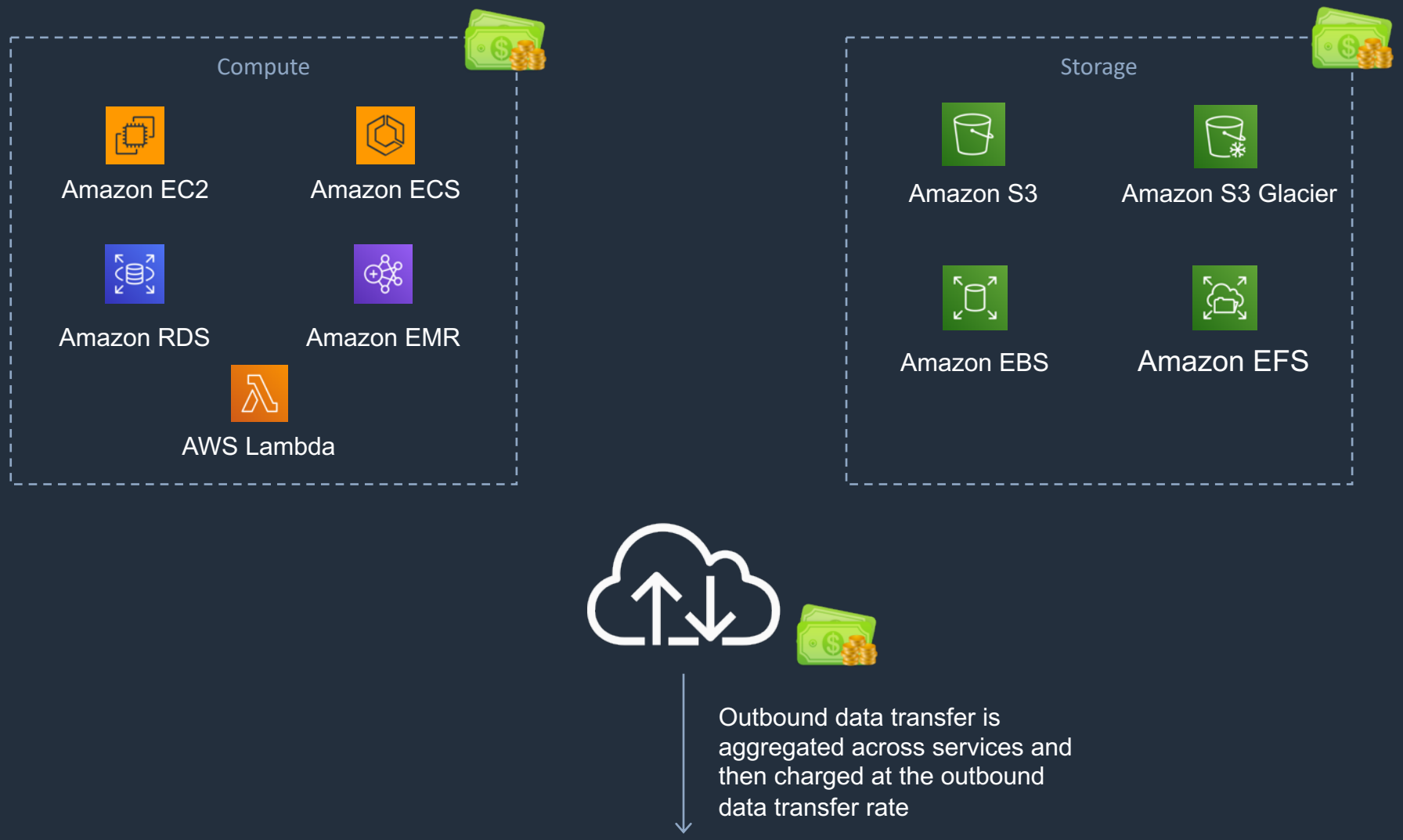
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Section 3: Regional Service Examples



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Section 3: Three Fundamentals of AWS Pricing



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Section 3: AWS Pricing Models

- On-demand
  - Used for Compute and Database capacity
  - No long-term commitments or upfront payments
- Dedicated Instances
  - Available for Amazon EC2
  - Hardware is dedicated to a single customer
- Spot Instances
  - Purchase spare capacity with no commitments
  - Great discounts from hourly rates
- Reservations
  - Up to 75% discount in exchange a term commitment

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Section 3: Reservations

- Options for 1 or 3 year term
- Options to pay:
  - No upfront
  - Partial upfront
  - All upfront
- Available for these services:
  - Amazon EC2 Reserved Instances
  - Amazon DynamoDB Reserved Capacity
  - Amazon ElastiCache Reserved Nodes
  - Amazon Relational Database Service (RDS) Reserved Instances
  - Amazon RedShift Reserved Nodes

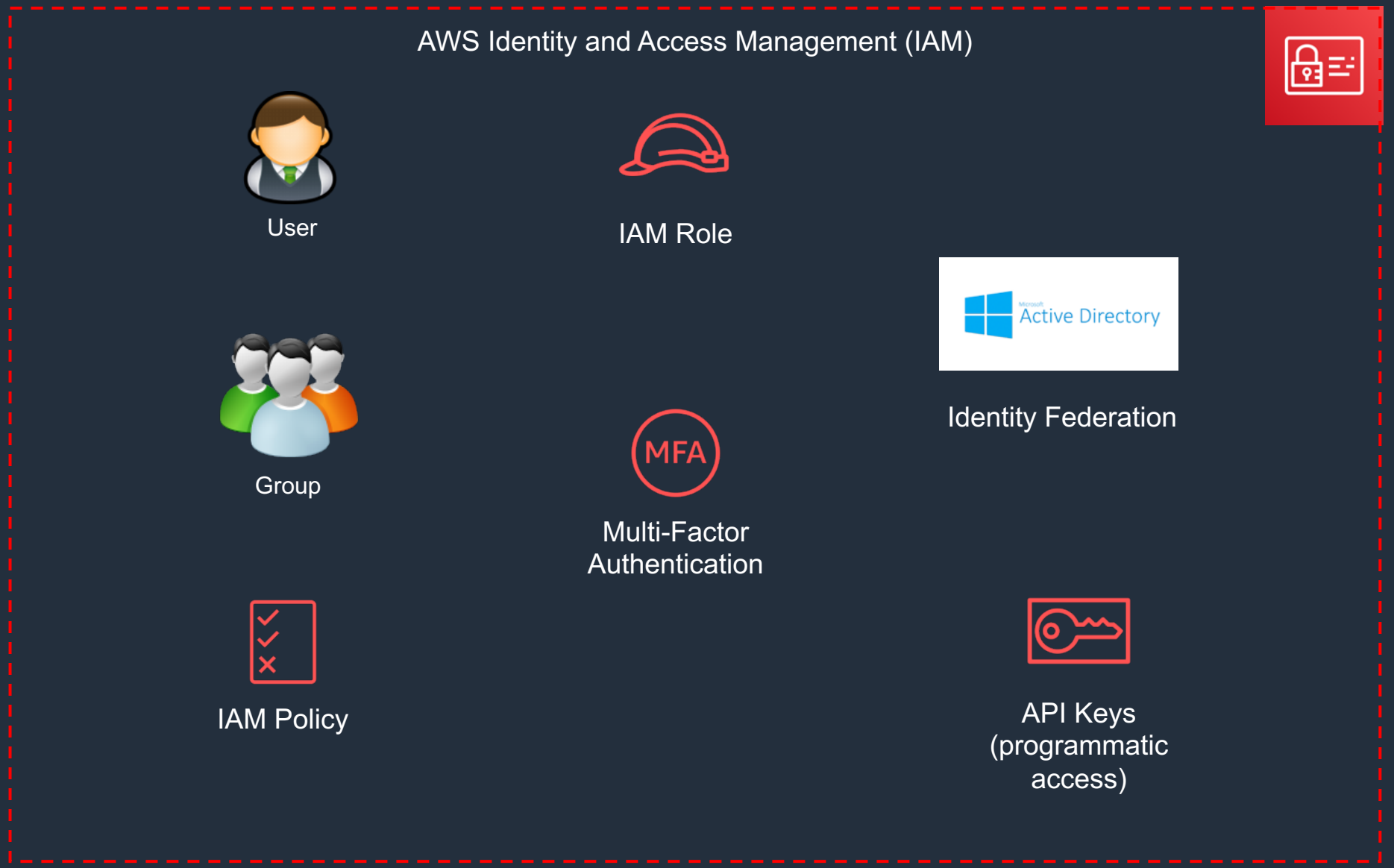
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SECTION 4

Identity and Access Management (IAM)

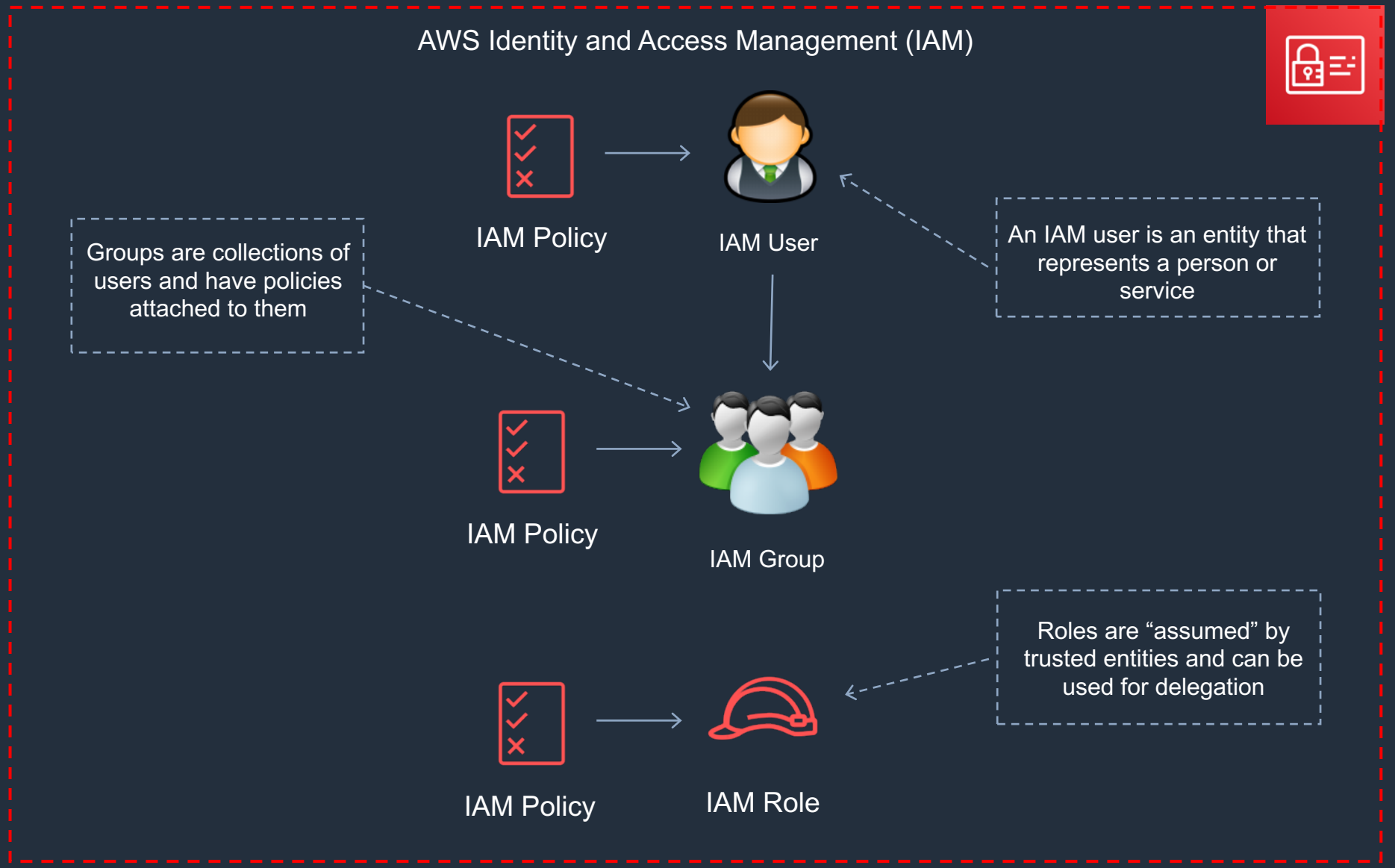


Section 4: Identity and Access Management (IAM) Overview



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Section 4: IAM Users, Groups, Roles and Policies



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Section 4: IAM Users

- An IAM user is an entity that represents a person or service
- Can be assigned:
  - An access key ID and secret access key for programmatic access to the AWS API, CLI, SDK, and other development tools
  - A password for access to the management console
- By default users cannot access anything in your account
- The account root user credentials are the email address used to create the account and a password
- The root account has full administrative permissions and these cannot be restricted
- Best practice for root accounts:
  - Don’t use the root user credentials
  - Don’t share the root user credentials
  - Create an IAM user and assign administrative permissions as required
  - Enable Multi-Factor Authentication (MFA)



Eric



Ethan



Andrea

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Section 4: IAM Users

- IAM users can be created to represent applications and these are known as “service accounts”
- You can have up to 5000 users per AWS account
- Each user account has a friendly name and an Amazon Resource Name (ARN) which uniquely identifies the user across AWS
- You should create individual IAM accounts for users (best practice not to share accounts)
- A password policy can be defined for enforcing password length, complexity etc. (applies to all users)



Eric



Ethan



Andrea

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Section 4: IAM Groups

- Groups are collections of users and have policies attached to them
- A group is not an identity and cannot be identified as a principal in an IAM policy
- Use groups to assign permissions to users
- Use the principal of least privilege when assigning permissions
- You cannot nest groups (groups within groups)



Developers



AWS Admins



Operations

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Section 4: IAM Roles

- Roles are created and then “assumed” by trusted entities and define a set of permissions for making AWS service requests
- With IAM Roles you can delegate permissions to resources for users and services without using permanent credentials (e.g. user name and password)
- IAM users or AWS services can assume a role to obtain temporary security credentials that can be used to make AWS API calls
- You can delegate using roles
- There are no credentials associated with a role (password or access keys)



S3 Full Access



DynamoDB Read-Only



AWSLambdaBasicExecutionRole

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Section 4: IAM Policies

- Policies are documents that define permissions and can be applied to users, groups and roles
- Policy documents are written in JSON (key value pair that consists of an attribute and a value)
- All permissions are implicitly denied by default
- The most restrictive policy is applied
- The IAM policy simulator is a tool to help you understand, test, and validate the effects of access control policies
- The Condition element can be used to apply further conditional logic



S3 Full Access



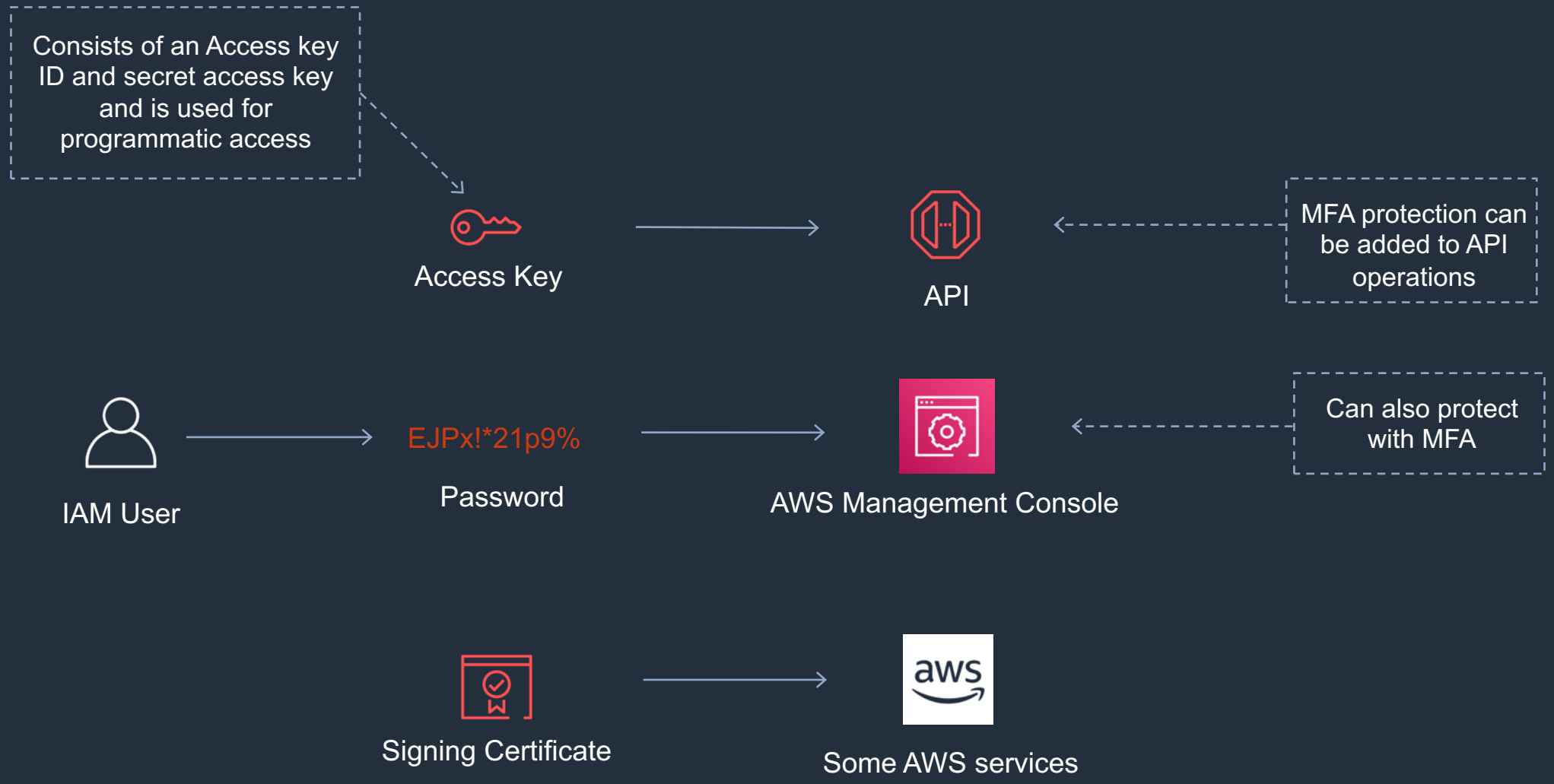
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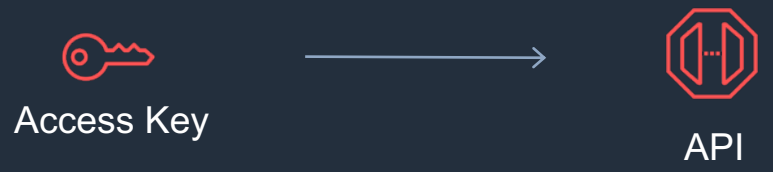
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Section 4: Authentication Methods



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Section 4: IAM Access Keys



- A combination of an access key ID and a secret access key
- These can be used to make programmatic calls to AWS when using the API in program code or at a command prompt when using the AWS CLI or the AWS PowerShell tools
- You can create, modify, view or rotate access keys
- When created IAM returns the access key ID and secret access key
- The secret access is returned only at creation time and if lost a new key must be created
- Ensure access keys and secret access keys are stored securely
- Users can be given access to change their own keys through IAM policy (not from the console)
- You can disable a user's access key which prevents it from being used for API calls

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Section 4: IAM Console Password

- A password that the user can enter to sign into interactive sessions such as the AWS Management Console
- You can allow users to change their own passwords
- You can allow selected IAM users to change their passwords by disabling the option for all users and using an IAM policy to grant permissions for the selected users



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Section 4: IAM Server Certificate / Signing Certificate

- SSL/TLS certificates that you can use to authenticate with some AWS services
- AWS recommends that you use the AWS Certificate Manager (ACM) to provision, manage and deploy your server certificates
- Use IAM only when you must support HTTPS connections in a region that is not supported by ACM



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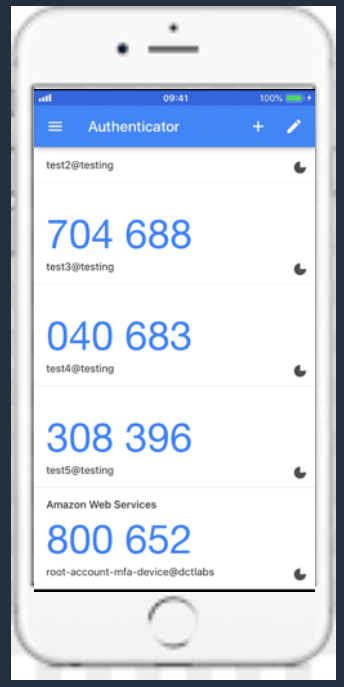
Section 4: Multi-Factor Authentication

Something you know:

EJPx!*21p9%

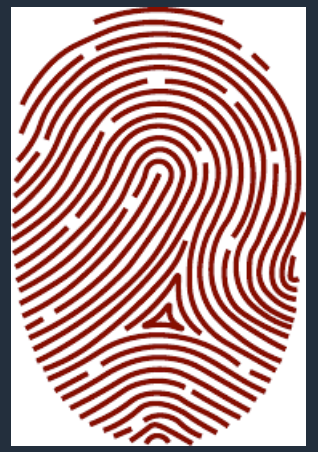
Password

Something you have:





Something you are:



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Section 4: Multi-Factor Authentication in AWS

Something you know:



IAM User

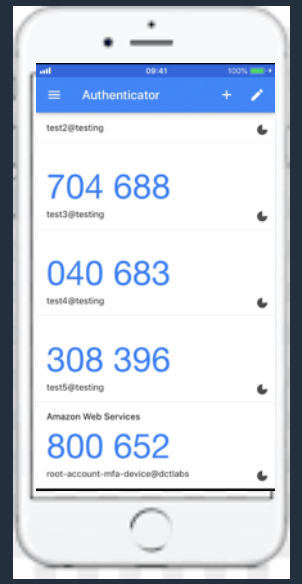
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Password

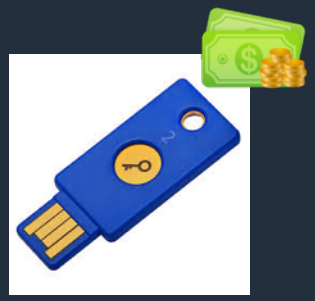
Something you have:

MFA Virtual MFA

e.g. Google Authenticator on your smart phone



MFA Physical MFA



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Section 4: AWS Security Token Service (STS)

- The AWS Security Token Service (STS) is a web service that enables you to request temporary, limited-privilege credentials for IAM users or for users that you authenticate (federated users)
- By default, AWS STS is available as a global service, and all AWS STS requests go to a single endpoint at https://sts.amazonaws.com
- All regions are enabled for STS by default but can be disabled
- The region in which temporary credentials are requested must be enabled
- Credentials will always work globally



Temporary security credential

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Section 4: IAM Best Practices

- Lock away the AWS root user access keys
- Create individual IAM users
- Use AWS defined policies to assign permissions whenever possible
- Use groups to assign permissions to IAM users
- Grant least privilege
- Use access levels to review IAM permissions
- Configure a strong password policy for users
- Enable MFA
- Use roles for applications that run on AWS EC2 instances
- Delegate by using roles instead of sharing credentials
- Rotate credentials regularly
- Remove unnecessary credentials
- Use policy conditions for extra security
- Monitor activity in your AWS account

DigitalCloud TRAINING

IAM Users

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DigitalCloud TRAINING

IAM Roles

- Roles are created and then “assumed” by trusted entities and define a set of permissions for making AWS service requests
- With IAM Roles you can delegate permissions to resources for users and services without using permanent credentials (e.g. user name and password)
- IAM users or AWS services can assume a role to obtain temporary security credentials that can be used to make AWS API calls
- You can delegate using roles
- There are no credentials associated with a role (password or access keys)



S3 Full Access

DynamoDB Read-Only

AWSLambdaBasicExecutionRole

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IAM Policies

- Policies are documents that define permissions and can be applied to users, groups and roles
- Policy documents are written in JSON (key value pair that consists of an attribute and a value)
- All permissions are implicitly denied by default
- The most restrictive policy is applied
- The IAM policy simulator is a tool to help you understand, test, and validate the effects of access control policies
- The Condition element can be used to apply further conditional logic





S3 Full Access



DynamoDB Read-Only



AWSLambdaBasicExecutionRole

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Authentication Methods



Consists of an Access key ID and secret access key and is used for programmatic access

Access Key

API

MFA protection can be added to API operations

IAM User

EJPx!*21p9%

Password

AWS Management Console

Can also protect with MFA

Signing Certificate

Some AWS services

DigitalCloud TRAINING

IAM Access Keys



- A combination of an access key ID and a secret access key
- These can be used to make programmatic calls to AWS when using the API in program code or at a command prompt when using the AWS CLI or the AWS PowerShell tools
- You can create, modify, view or rotate access keys
- When created IAM returns the access key ID and secret access key
- The secret access is returned only at creation time and if lost a new key must be created
- Ensure access keys and secret access keys are stored securely
- Users can be given access to change their own keys through IAM policy (not from the console)
- You can disable a user's access key which prevents it from being used for API calls

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IAM Console Password



- A password that the user can enter to sign into interactive sessions such as the AWS Management Console
- You can allow users to change their own passwords
- You can allow selected IAM users to change their passwords by disabling the option for all users and using an IAM policy to grant permissions for the selected users



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IAM Server Certificate / Signing Certificate



- SSL/TLS certificates that you can use to authenticate with some AWS services
- AWS recommends that you use the AWS Certificate Manager (ACM) to provision, manage and deploy your server certificates
- Use IAM only when you must support HTTPS connections in a region that is not supported by ACM



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Multi-Factor Authentication in AWS



Something you know:



IAM User

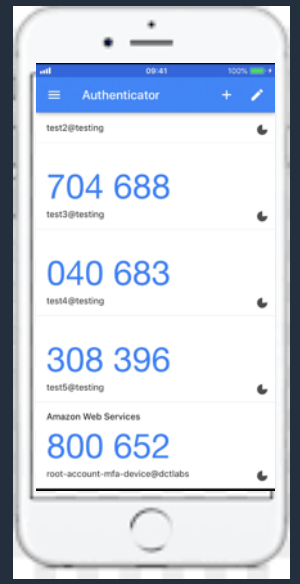
EJPx!*21p9%

Password

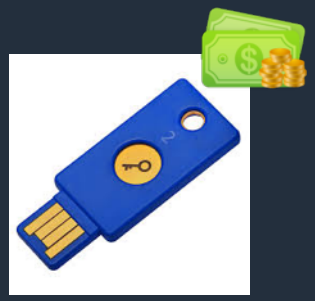
Something you have:

MFA Virtual MFA

e.g. Google Authenticator on your smart phone



MFA Physical MFA



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AWS Security Token Service (STS)

- The AWS Security Token Service (STS) is a web service that enables you to request temporary, limited-privilege credentials for IAM users or for users that you authenticate (federated users)
- By default, AWS STS is available as a global service, and all AWS STS requests go to a single endpoint at https://sts.amazonaws.com
- All regions are enabled for STS by default but can be disabled
- The region in which temporary credentials are requested must be enabled
- Credentials will always work globally



Temporary security credential

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IAM Best Practices



- Lock away the AWS root user access keys
- Create individual IAM users
- Use AWS defined policies to assign permissions whenever possible
- Use groups to assign permissions to IAM users
- Grant least privilege
- Use access levels to review IAM permissions
- Configure a strong password policy for users
- Enable MFA
- Use roles for applications that run on AWS EC2 instances
- Delegate by using roles instead of sharing credentials
- Rotate credentials regularly
- Remove unnecessary credentials
- Use policy conditions for extra security
- Monitor activity in your AWS account

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SECTION 5

Create AWS Free Tier Account (optional) – NO SLIDES

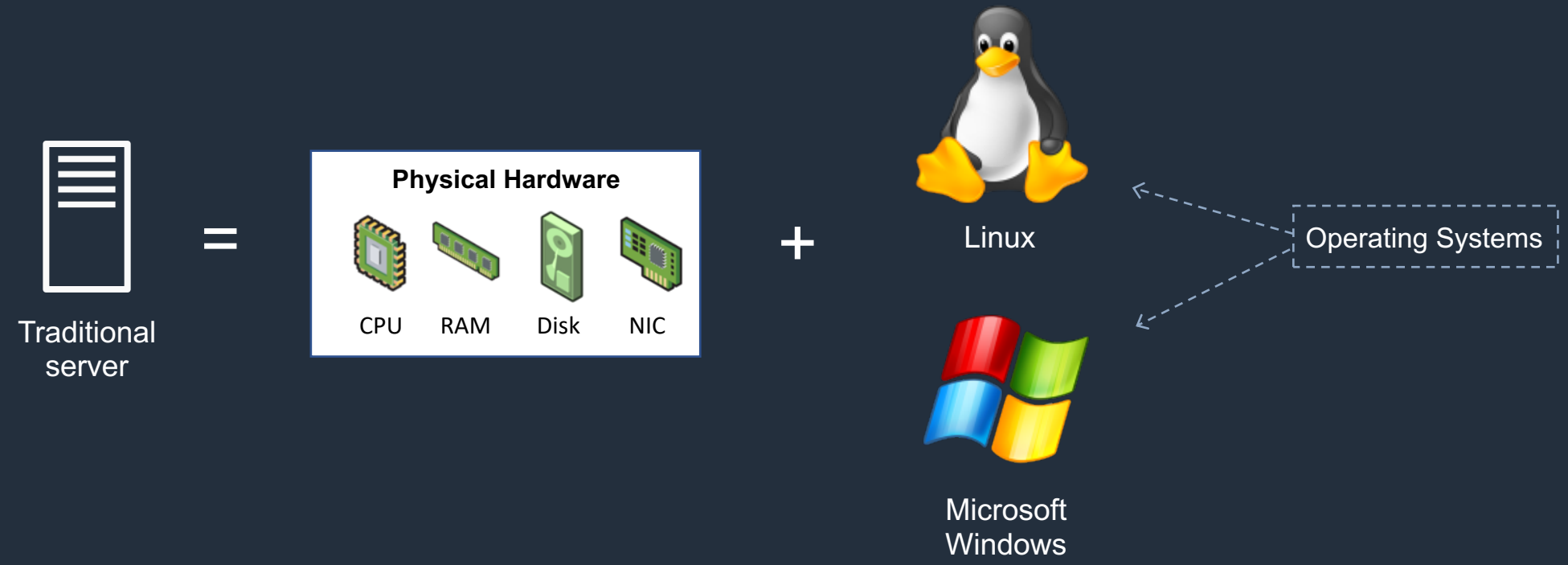


SECTION 6

AWS Compute

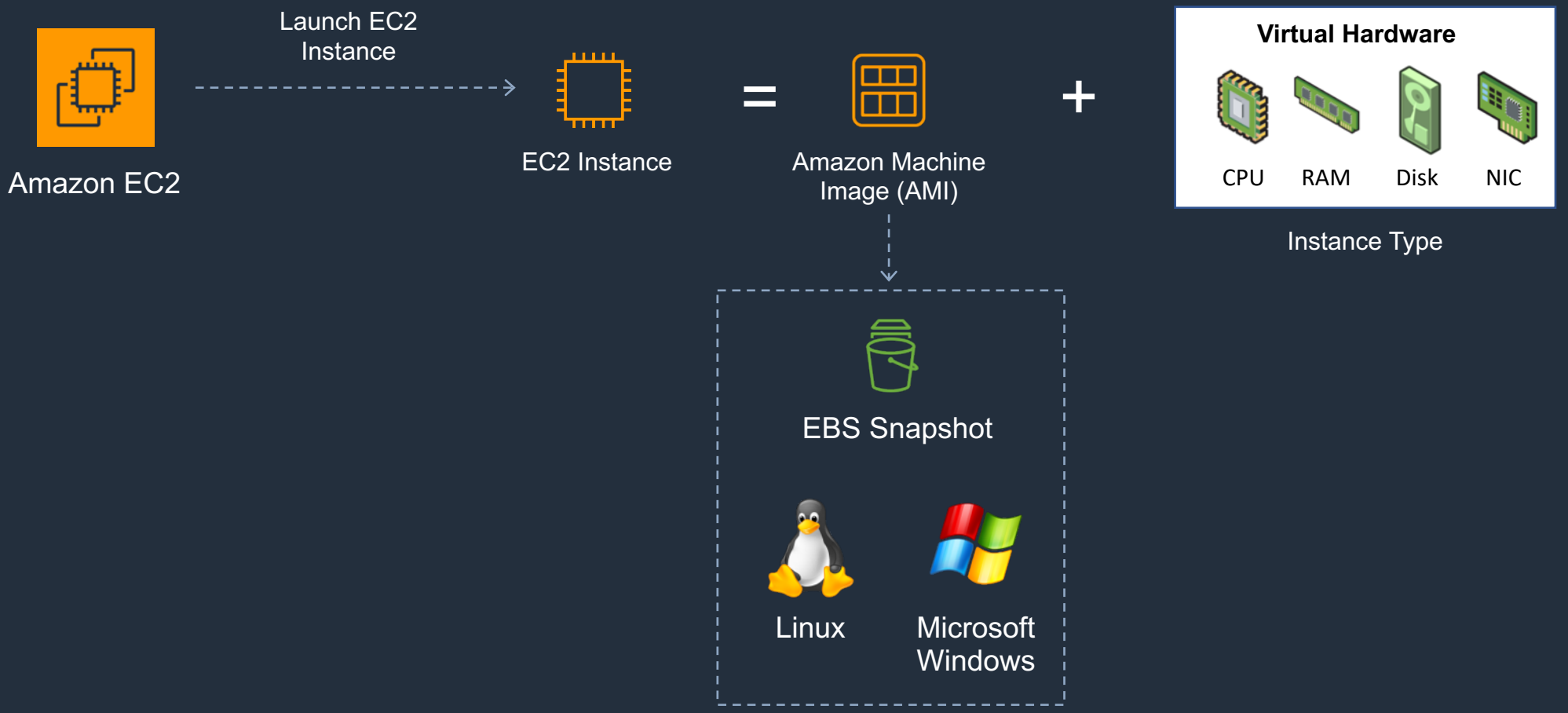


Section 6: Traditional Servers



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Section 6: Amazon Elastic Compute Cloud (EC2)



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Section 6: Amazon EC2 Overview



Amazon EC2

- Amazon Elastic Compute Cloud (Amazon EC2) is a web service in the AWS Compute suite of products that provides secure, resizable compute capacity in the cloud

- Elastic Web-Scale computing – you can increase or decrease capacity within minutes and commission one to thousands of instances simultaneously

- Completely controlled – You have complete control include root access to each instance and can stop and start instances without losing data and using web service APIs

- Flexible Cloud Hosting Services – you can choose from multiple instance types, operating systems, and software packages as well as instances with varying memory, CPU and storage configurations

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Section 6: Amazon Machine Images (AMI)



Amazon EC2

- An Amazon Machine Image (AMI) provides the information required to launch an instance
- An AMI includes the following:
  - One or more EBS snapshots, or, for instance-store-backed AMIs, a template for the root volume of the instance (for example, an operating system, an application server, and applications).
  - Launch permissions that control which AWS accounts can use the AMI to launch instances.
  - A block device mapping that specifies the volumes to attach to the instance when it's launched
- AMIs come in three main categories:
  - Community AMIs - free to use, generally you just select the operating system you want
  - AWS Marketplace AMIs - pay to use, generally come packaged with additional, licensed software
  - My AMIs - AMIs that you create yourself

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Section 6: Amazon EC2 Instance Types

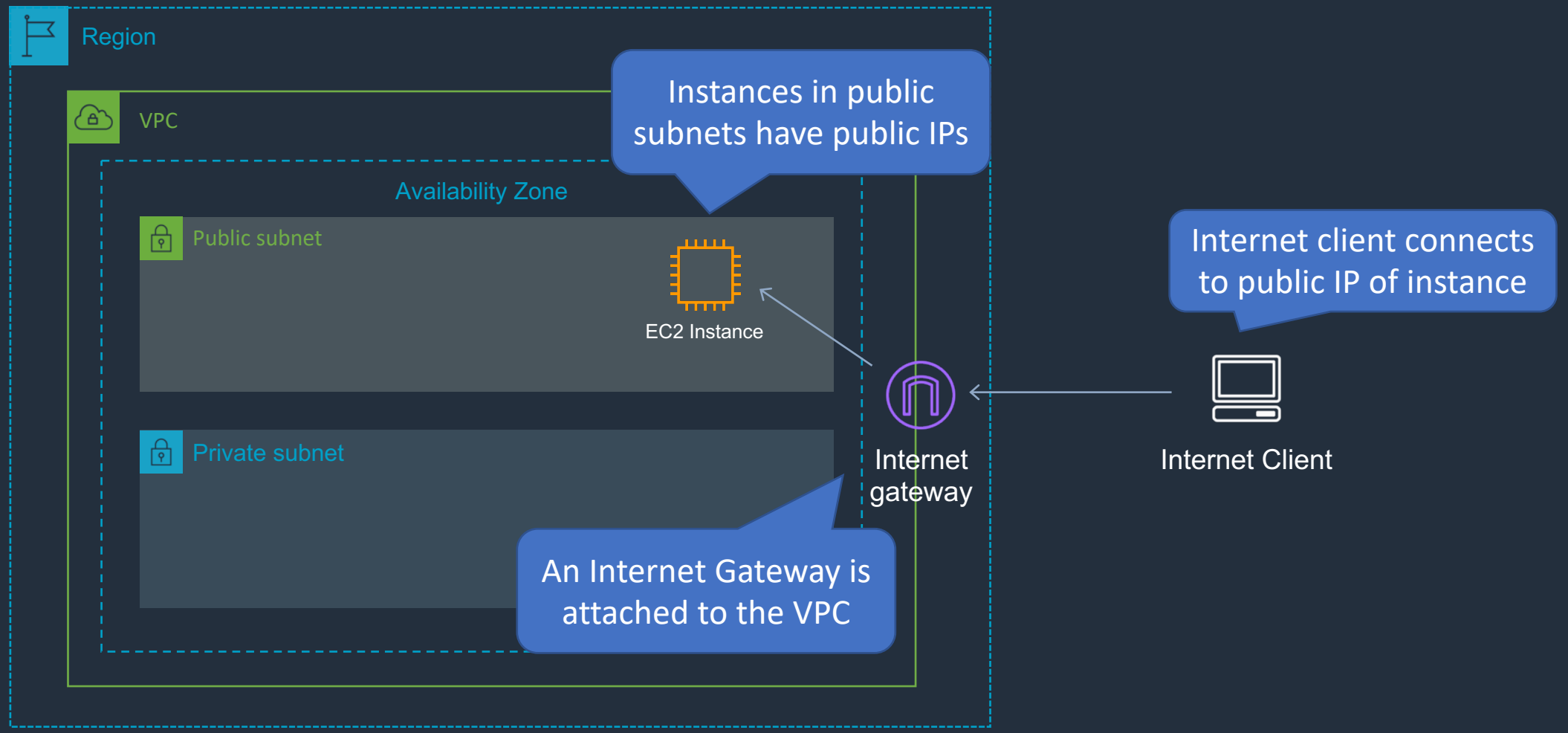


Amazon EC2

Category	Families	Purpose/Design
General Purpose	A1, T3, T3a, T2, M5, M5a, M4	General purpose instances provide a balance of compute, memory and networking resources, and can be used for a variety of diverse workloads
Compute Optimized	C5, C5n, C4	Compute Optimized instances are ideal for compute bound applications that benefit from high performance processors
Memory Optimized	R5, R5a, R4, X1e, X1, High Memory, z1d	Memory optimized instances are designed to deliver fast performance for workloads that process large data sets in memory
Accelerated Computing	P3, P2, G4, G3, F1	Accelerated computing instances use hardware accelerators, or co-processors, to perform functions, such as floating-point number calculations, graphics processing, or data pattern matching
Storage Optimized	I3, I3en, D2, H1	This instance family provides Non-Volatile Memory Express (NVMe) SSD-backed instance storage optimized for low latency, very high random I/O performance, high sequential read throughput and provide high IOPS at a low cost

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Amazon EC2 Instance in a Public Subnet



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Section 6: Instance User Data and Instance Metadata



Amazon EC2

User Data

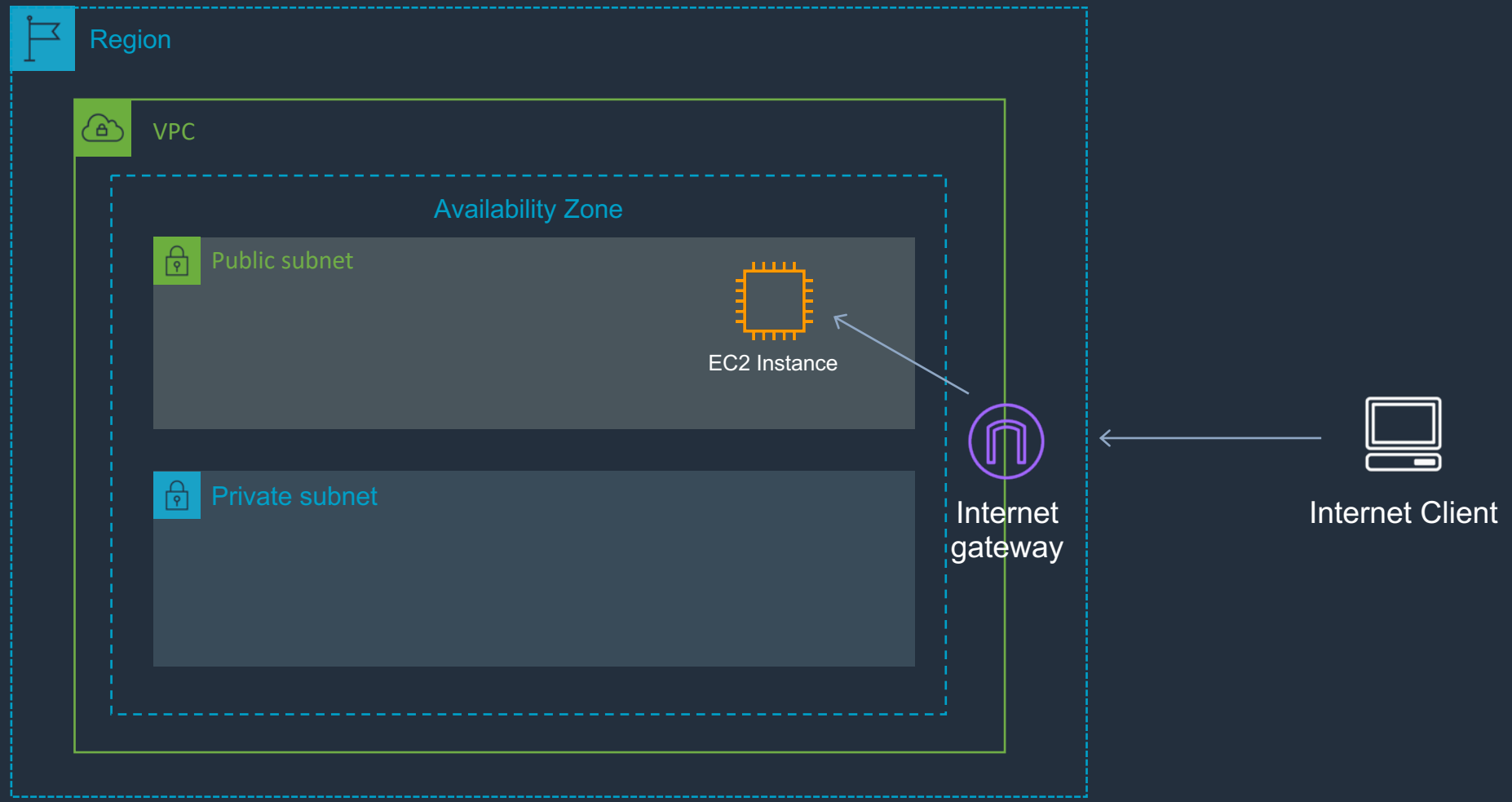
- User data is data that is supplied by the user at instance launch in the form of a script
- User data is limited to 16KB
- User data and metadata are not encrypted

Metadata

- Instance metadata is data about your instance that you can use to configure or manage the running instance
- Instance metadata is available at http://169.254.169.254/latest/meta-data
- The Instance Metadata Query tool allows you to query the instance metadata without having to type out the full URI or category names

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Amazon EC2 Instance with Apache Web Service



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Section 6: Amazon Elastic Container Service (ECS)



Amazon Elastic Container Service

- Amazon Elastic Container Service (ECS) provides a highly scalable, high performance container management service that supports Docker containers
- A container is similar to a virtual instance, but there's less to manage
- With containers the code, runtime, system tools, system libraries and settings are packaged up
- Containers run quickly and reliably from one computing environment to another
- Amazon ECS eliminates the need for you to install, operate, and scale your own cluster management infrastructure

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Section 6: Amazon Elastic Container Service (ECS)



Amazon Elastic Container Service

- An Amazon ECS launch type determines the type of infrastructure on which your tasks and services are hosted
- There are two launch types and the table below describes some of the differences between the two launch types:

Amazon EC2	Amazon Fargate
You explicitly provision EC2 instances	The control plane asks for resources and Fargate automatically provisions
You're responsible for upgrading, patching, care of EC2 pool	Fargate provisions compute as needed
You must handle cluster optimization	Fargate handles cluster optimization
More granular control over infrastructure	Limited control, as infrastructure is automated

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Section 6: ECS Terminology

Elastic Container Service (ECS)	Description
Cluster	Logical grouping of EC2 instances
Container instance	EC2 instance running the the ECS agent
Task Definition	Blueprint that describes how a docker container should launch
Task	A running container using settings in a Task Definition
Service	Defines long running tasks – can control task count with Auto Scaling and attach an ELB

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Section 6: AWS Lambda

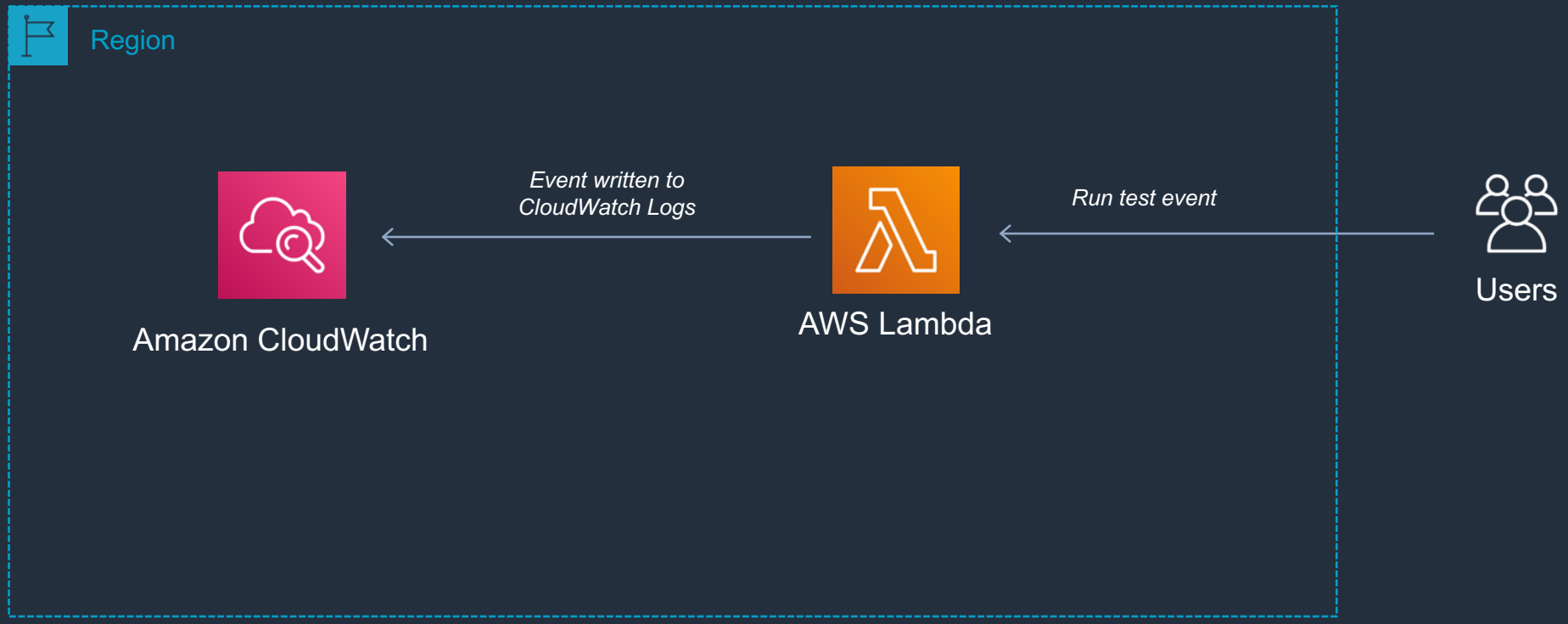


AWS Lambda

- AWS Lambda is a serverless computing technology that allows you to run code without provisioning or managing servers
- AWS Lambda executes code only when needed and scales automatically
- You pay only for the compute time you consume (you pay nothing when your code is not running)
- Benefits of AWS Lambda:
  - No servers to manage
  - Continuous scaling
  - Subsecond metering
  - Integrates with almost all other AWS services
- Primary use cases for AWS Lambda:
  - Data processing
  - Real-time file processing
  - Real-time stream processing
  - Build serverless backends for web, mobile, IOT, and 3rd party API requests

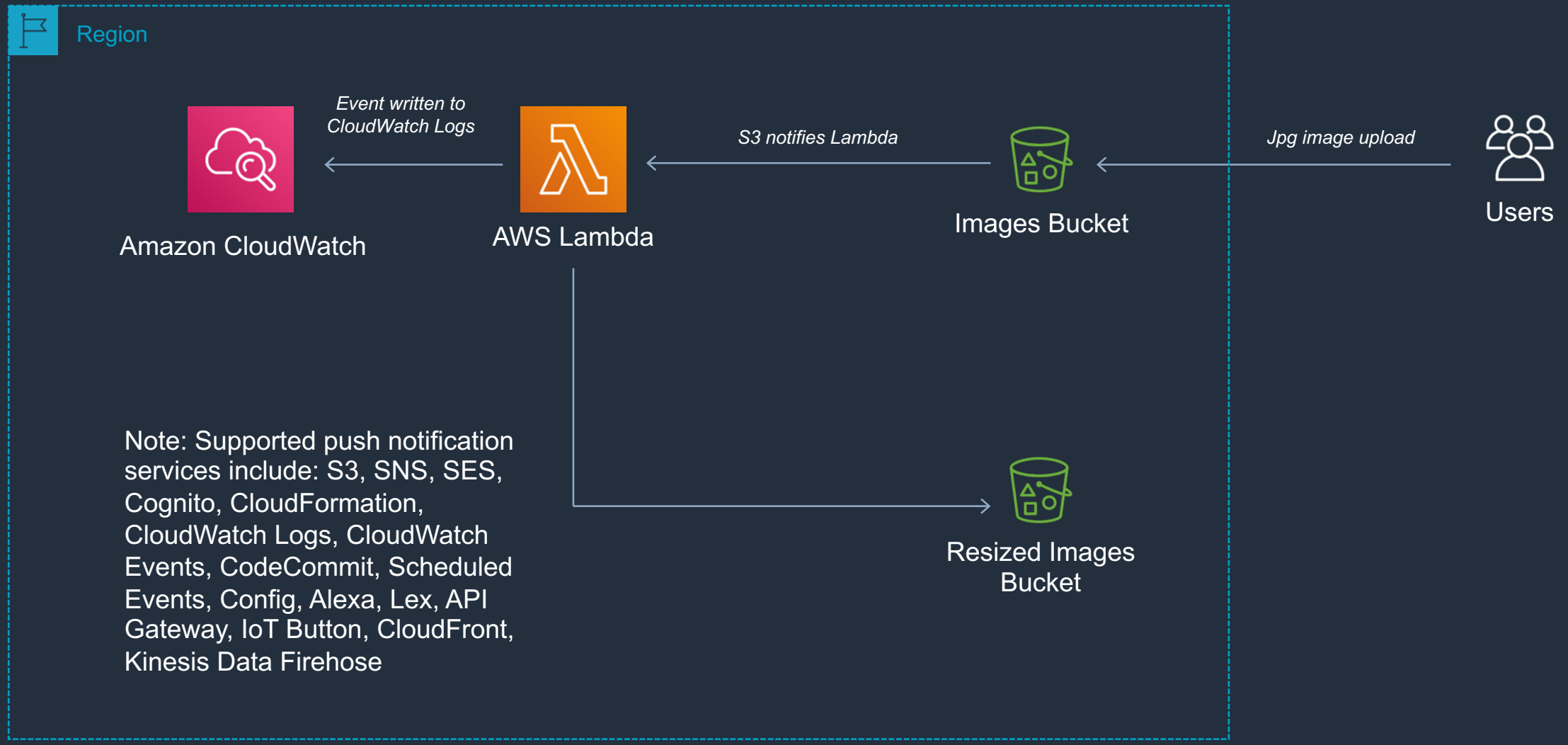
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Section 6: AWS Lambda – Hello World



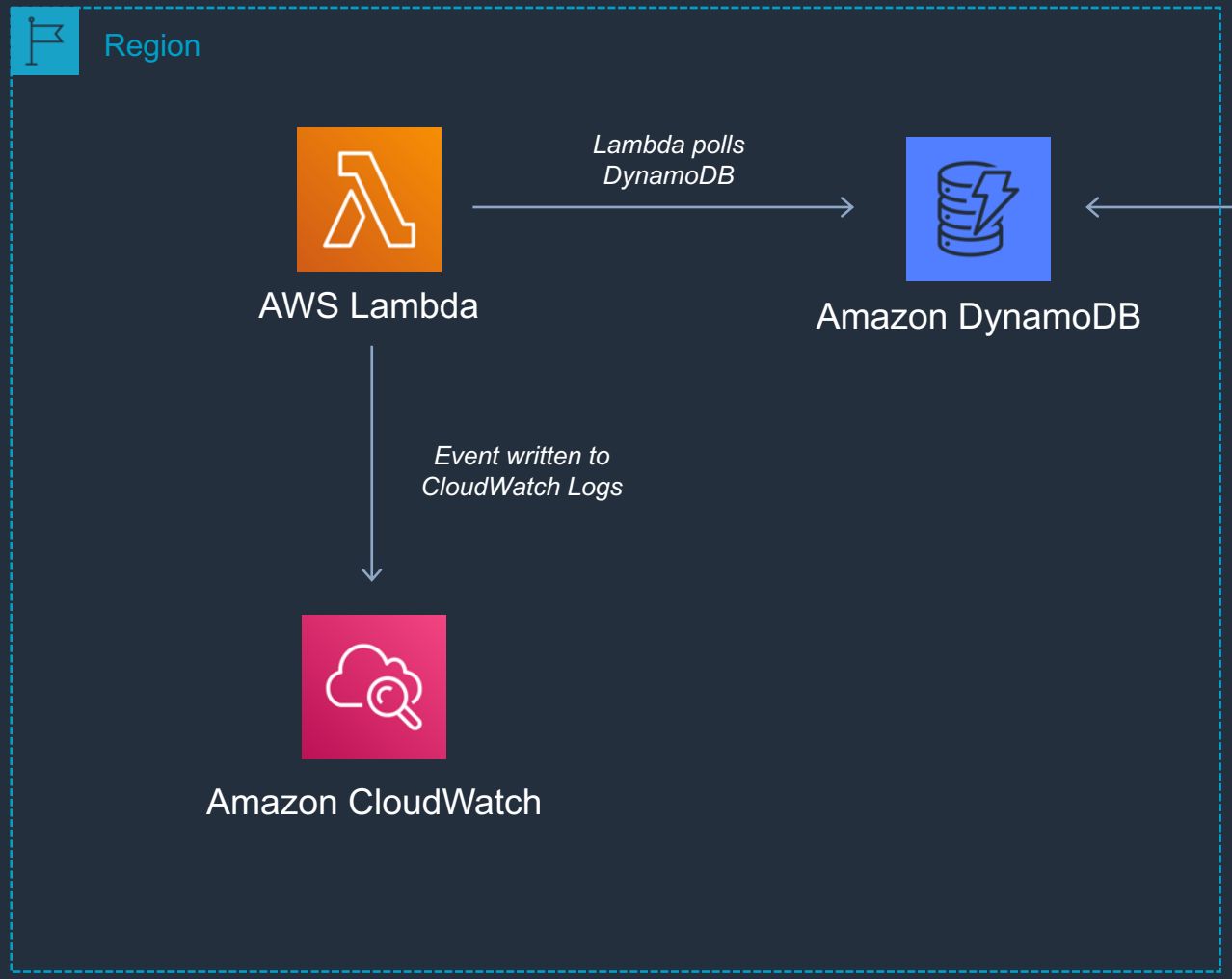
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Section 6: AWS Lambda – S3 Event Source Mapping



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Section 6: AWS Lambda – DynamoDB Event Source Mapping



Note: Supported poll-based services are DynamoDB, Kinesis, and SQS

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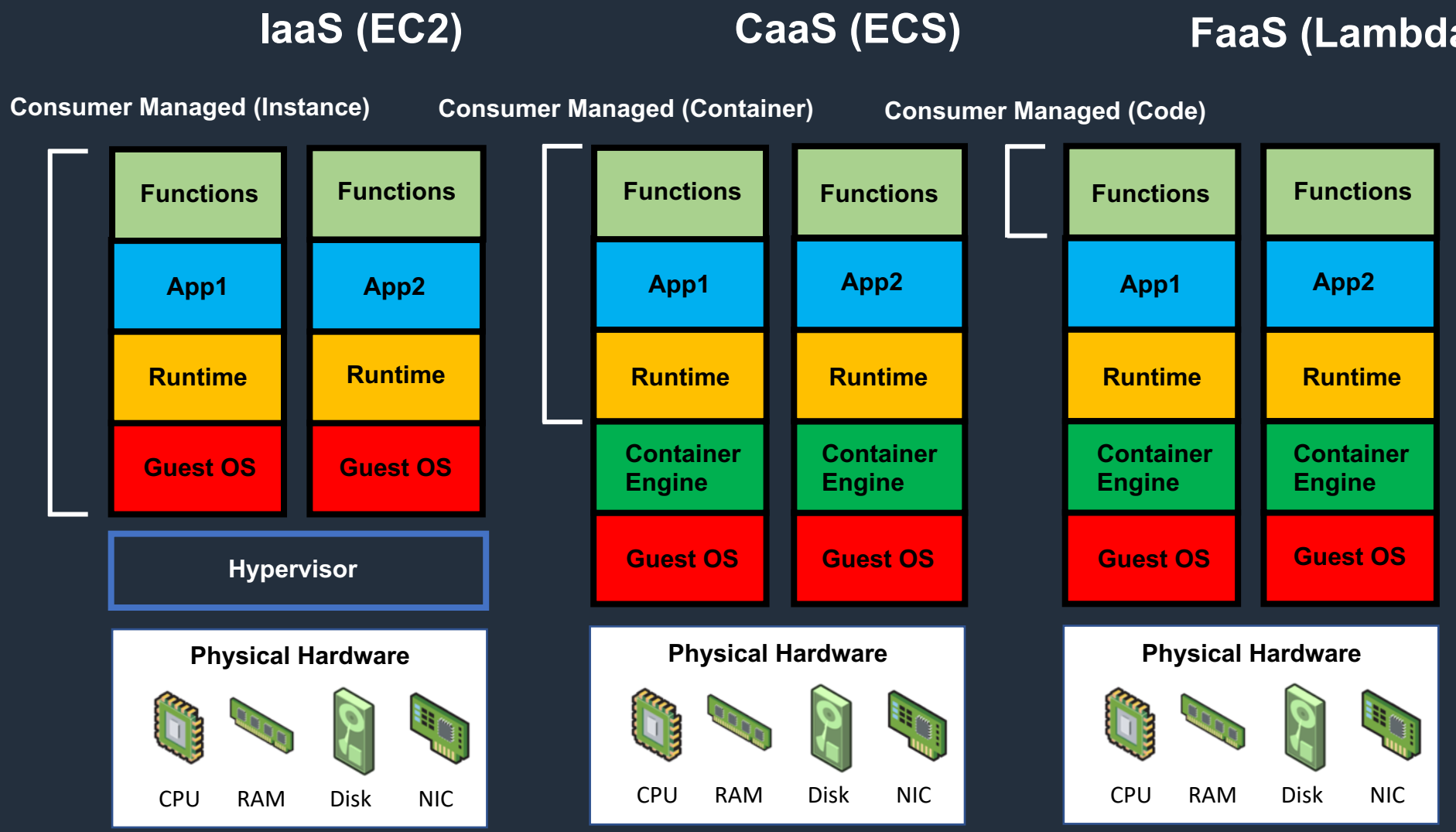
Section 6: Amazon LightSail



Amazon Lightsail

- Amazon Lightsail is great for users who do not have deep AWS technical expertise as it make it very easy to provision compute services
- Amazon Lightsail provides developers compute, storage, and networking capacity and capabilities to deploy and manage websites, web applications, and databases in the cloud
- Amazon Lightsail includes everything you need to launch your project quickly – a virtual machine, SSD-based storage, data transfer, DNS management, and a static IP
- Amazon Lightsail provides preconfigured virtual private servers (instances) that include everything required to deploy and application or create a database
- Can create Instances and Databases, and configure Static IP, DNS Zone, Load Balancers, storage, and snapshots

Section 6: IaaS, CaaS and FaaS



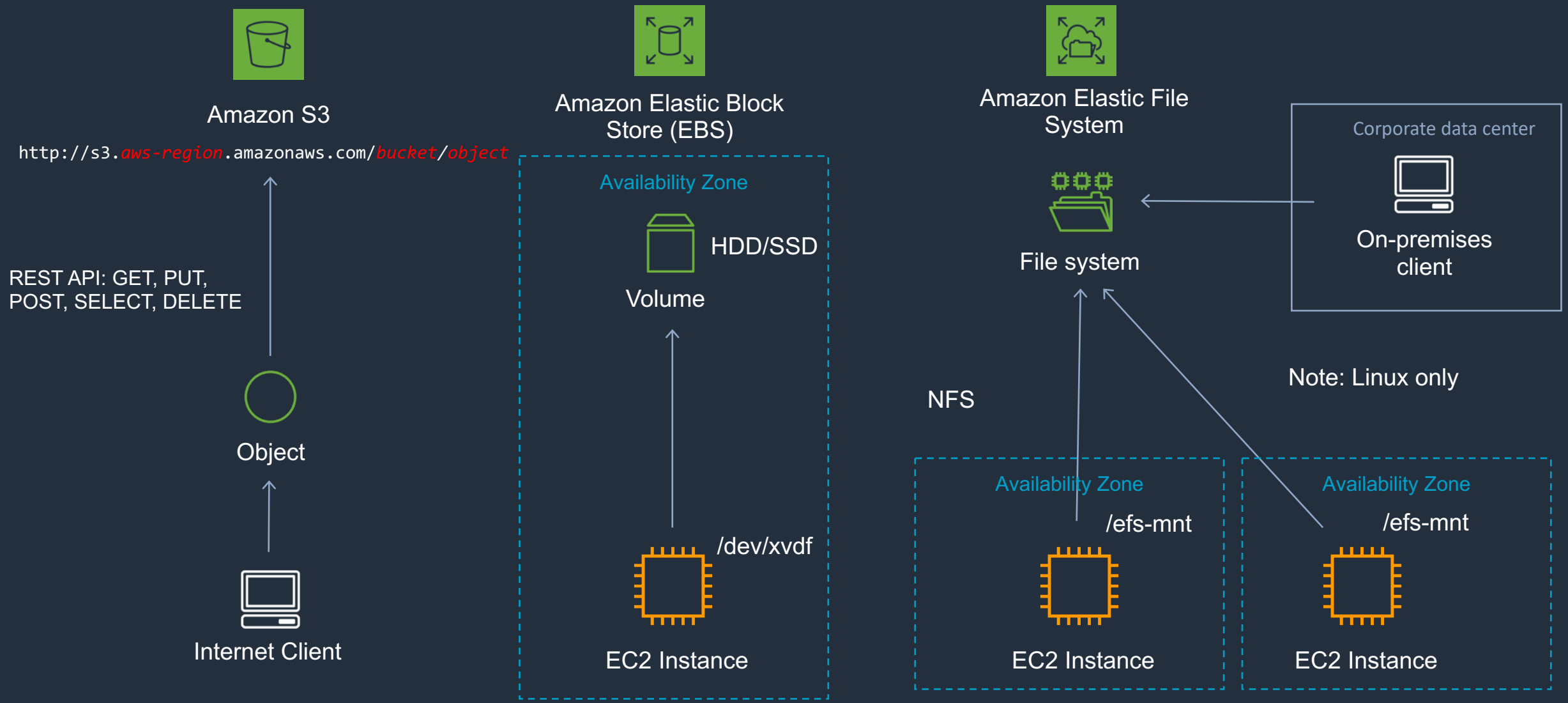
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Section 6: Comparing Compute Options

EC2	ECS (EC2 Launch Type)	ECS (Fargate Launch Type)	Lambda
You manage the operating system	You manage container instance (EC2) and the containers (tasks)	You manage the containers (tasks)	You manage the code
Scale vertically – more CPU/Mem/HDD or scale horizontally (automatic) with Auto Scaling	Manually add container instances or use ECS Services and EC2 Auto Scaling	AWS scales the cluster automatically	Lambda automatically scales concurrent executions up to default limit (1000)
Use for traditional applications and long running tasks	Use for microservices and batch use cases where you need containers and need to retain management of underlying platform	Use for microservices and batch use cases	Use for ETL, infrastructure automation, data validation, mobile backends
Pay for instance run time based on family/type	Pay for instance run time based on family/type	Pay for container run time based on allocated resources	Pay only for execution time based on memory allocation

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Section 7: Object, Block, and File Storage



DigitalCloud TRAINING

Section 6: Object, Block, and File Storage Systems

Object Storage

- Object-based storage systems manage data as individual objects, rather than as blocks and sectors (block-based) or a file hierarchy (file-based)
- Object-based storage is accessed using a REST API (URL with HTTP methods, e.g. GET, PUT)
- With object storage data is managed as individual objects rather than a file hierarchy (as with a traditional file system)
- Each object includes the data itself, metadata (data about the data), and a globally unique identifier
- Due to its flat file structure, object storage has virtually unlimited scalability and allows the retention of massive amounts of unstructured data

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Section 6: Object, Block, and File Storage Systems

Block Storage

- Data is stored and managed in blocks within sectors and tracks and is controlled by a server-based operating system
- Block storage volumes appear as local disks to the operating system and can be partitioned and formatted
- You can use block storage devices as a boot volume
- Common use cases for block storage are structured information such as file systems, databases, transactional logs, SQL databases and virtual machines (VMs)

File Storage

- File-based storage systems manage data in a file hierarchy
- A file system is mounted via the network to a client computer where it then becomes accessible for reading and writing data
- Protocols used for accessing file systems include NFS or CIFS/SMB

DigitalCloud TRAINING

SECTION 7

AWS Storage



Section 7: Amazon S3



Bucket

`http://bucket.s3.aws-region.amazonaws.com`

`http://s3.aws-region.amazonaws.com/bucket`



Object

- Key
- Version ID
- Value
- Metadata
- Subresources
- Access control information



Amazon S3

Private Connection



Public Internet

S3 Gateway Endpoint



Internet gateway

VPC

Public subnet



EC2 Instance

Private subnet



EC2 Instance



Internet Client

DigitalCloud TRAINING

Section 7: Amazon Simple Storage Service (S3)



Amazon Simple Storage Service (S3)

- Amazon S3 is object storage built to store and retrieve any amount of data from anywhere – web sites and mobile apps, corporate applications, and data from IoT sensors or devices
- You can store any type of file in S3
- S3 is designed to deliver 99.999999999% durability
- Typical use cases include:
  - Backup and Storage – Provide data backup and storage services for others
  - Application Hosting – Provide services that deploy, install, and manage web applications
  - Media Hosting – Build a redundant, scalable, and highly available infrastructure that hosts video, photo, or music uploads and downloads
  - Software Delivery – Host your software applications that customers can download
  - Static Website – you can configure a static website to run from an S3 bucket

DigitalCloud TRAINING

Section 7: Amazon Simple Storage Service (S3)



Amazon Simple Storage Service (S3)

- Files are stored in buckets
- Buckets are root level folders
- Files can be anywhere from 0 bytes to 5 TB
- There is unlimited storage available
- S3 is a universal namespace so bucket names must be unique globally
- However, you create your buckets within a REGION
- It is a best practice to create buckets in regions that are physically closest to your users to reduce latency
- Objects consist of:
  - Key (name of the object)
  - Value (data made up of a sequence of bytes)
  - Version ID (used for versioning)
  - Metadata (data about the data that is stored)

DigitalCloud TRAINING

Section 7: Amazon Simple Storage Service (S3)



Amazon Simple Storage Service (S3)

- Pricing:
  - Storage
  - Requests
  - Storage management pricing
  - Data transfer pricing
  - Transfer acceleration

DigitalCloud TRAINING

Section 7: Amazon Simple Storage Service (S3)



Amazon Simple Storage Service (S3)

- There are six S3 storage classes:
  - S3 Standard (durable, immediately available, frequently accessed)
  - S3 Intelligent-Tiering (automatically moves data to the most cost-effective tier)
  - S3 Standard-IA (durable, immediately available, infrequently accessed)
  - S3 One Zone-IA (lower cost for infrequently accessed data with less resilience)
  - S3 Glacier (archived data, retrieval times in minutes or hours)
  - S3 Glacier Deep Archive (lowest cost storage class for long term retention)

DigitalCloud TRAINING

Section 7: Amazon Simple Storage Service (S3)



Amazon Simple Storage Service (S3)

	S3 Standard	S3 Intelligent-Tiering*	S3 Standard-IA	S3 One Zone-IA†	S3 Glacier	S3 Glacier Deep Archive
Designed for durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Designed for availability	99.99%	99.9%	99.9%	99.5%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99.9%	99.9%
Availability Zones	≥3	≥3	≥3	1	≥3	≥3
Minimum capacity charge per object	N/A	N/A	128KB	128KB	40KB	40KB
Minimum storage duration charge	N/A	30 days	30 days	30 days	90 days	180 days
Retrieval fee	N/A	N/A	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved
First byte latency	milliseconds	millseconds	milliseconds	milliseconds	select minutes or hours	select hours
Storage type	Object	Object	Object	Object	Object	Object
Lifecycle transitions	Yes	Yes	Yes	Yes	Yes	Yes

DigitalCloud TRAINING

Section 7: Amazon Simple Storage Service (S3)



Amazon Simple Storage Service (S3)

S3 Capability	How it works
Transfer Acceleration	Speed up data uploads using CloudFront in reverse
Requester Pays	The requester rather than the bucket owner pays for requests and data transfer
Tags	Assign tags to objects to use in costing, billing, security etc.
Events	Trigger notifications to SNS, SQS, or Lambda when certain events happen in your bucket
Static Web Hosting	Simple and massively scalable static website hosting
BitTorrent	Use the BitTorrent protocol to retrieve any publicly available object by automatically generating a .torrent file

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Section 7: Amazon S3 –Versioning

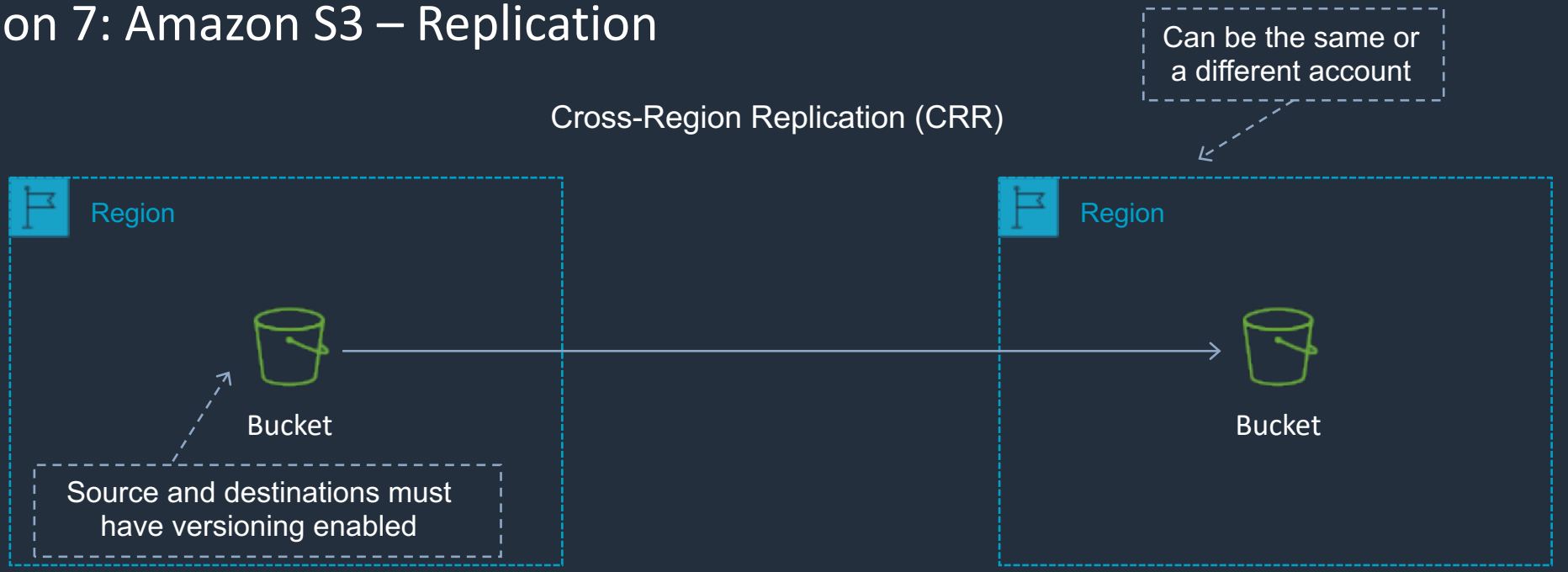


Amazon Simple Storage Service (S3)

- Amazon S3 versioning maintains multiple variants of an object in the same bucket
- Can be used to preserve, retrieve, and restore every version of every object in an S3 bucket
- Can be enabled at any time
- Once enabled can be suspended

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Section 7: Amazon S3 – Replication





Amazon Simple Storage Service (S3)



DigitalCloud TRAINING

Section 7: Amazon S3 –Replication



Amazon Simple Storage Service (S3)

- You can replicate objects between different AWS Regions or within the same AWS Region
  - Cross-Region replication (CRR) is used to copy objects across Amazon S3 buckets in different AWS Regions
  - Same-Region replication (SRR) is used to copy objects across Amazon S3 buckets in the same AWS Region
- Why use replication?
  - Meet compliance requirements for storing data at greater distances (CRR)
  - Minimize latency for users who are closer to another AWS Region (CRR)
  - Backup copy of your data in another AWS Region (CRR)
  - Copy the objects to another S3 storage class
  - Aggregate logs into a single bucket

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Section 7: Amazon S3 – Replication

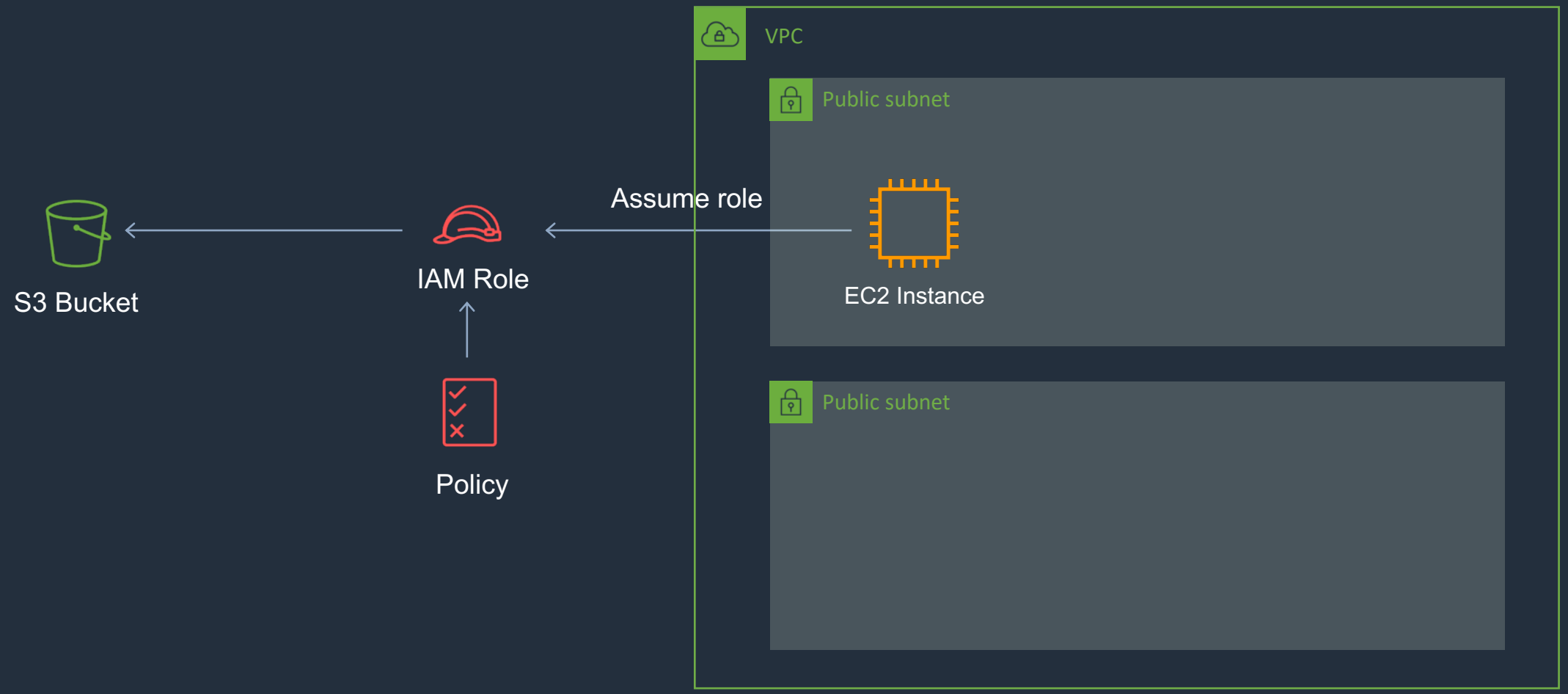
- How:
  - Enable the AWS Region in account
  - Enable versioning on source and destination buckets
  - Ensure S3 has permissions to both buckets
  - Configure replication



Amazon Simple Storage Service (S3)

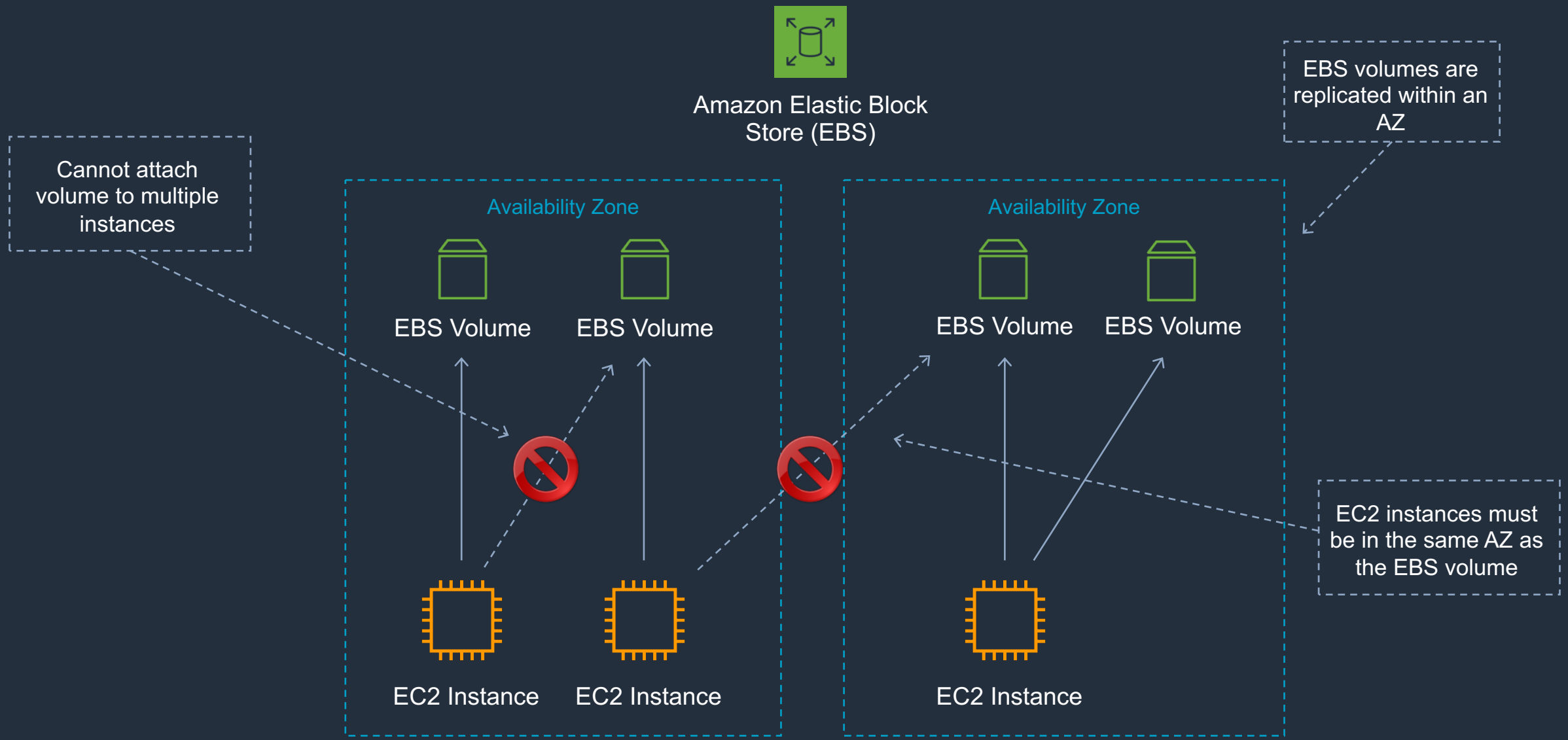
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Section 7: Access Amazon S3 Bucket from EC2 with IAM Role



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Section 7: Amazon Elastic Block Store (EBS)



DigitalCloud TRAINING

Section 7: Amazon Elastic Block Store (EBS)



Amazon Elastic Block Store (EBS)

- Amazon Elastic Block Store (Amazon EBS) provides persistent block storage volumes for use with Amazon EC2 instances in the AWS Cloud
- Each Amazon EBS volume is automatically replicated within its Availability Zone to protect you from component failure, offering high availability and durability
- EBS volume data persists independently of the life of the instance
- EBS volumes do not need to be attached to an instance
- You can attach multiple EBS volumes to an instance
- You cannot attach an EBS volume to multiple instances (use Elastic File Store instead)
- EBS volumes must be in the same AZ as the instances they are attached to
- Termination protection is turned off by default and must be manually enabled (keeps the volume/data when the instance is terminated)
- Root EBS volumes are deleted on termination by default
- Extra non-boot volumes are not deleted on termination by default
- The behaviour can be changed by altering the “DeleteOnTermination” attribute

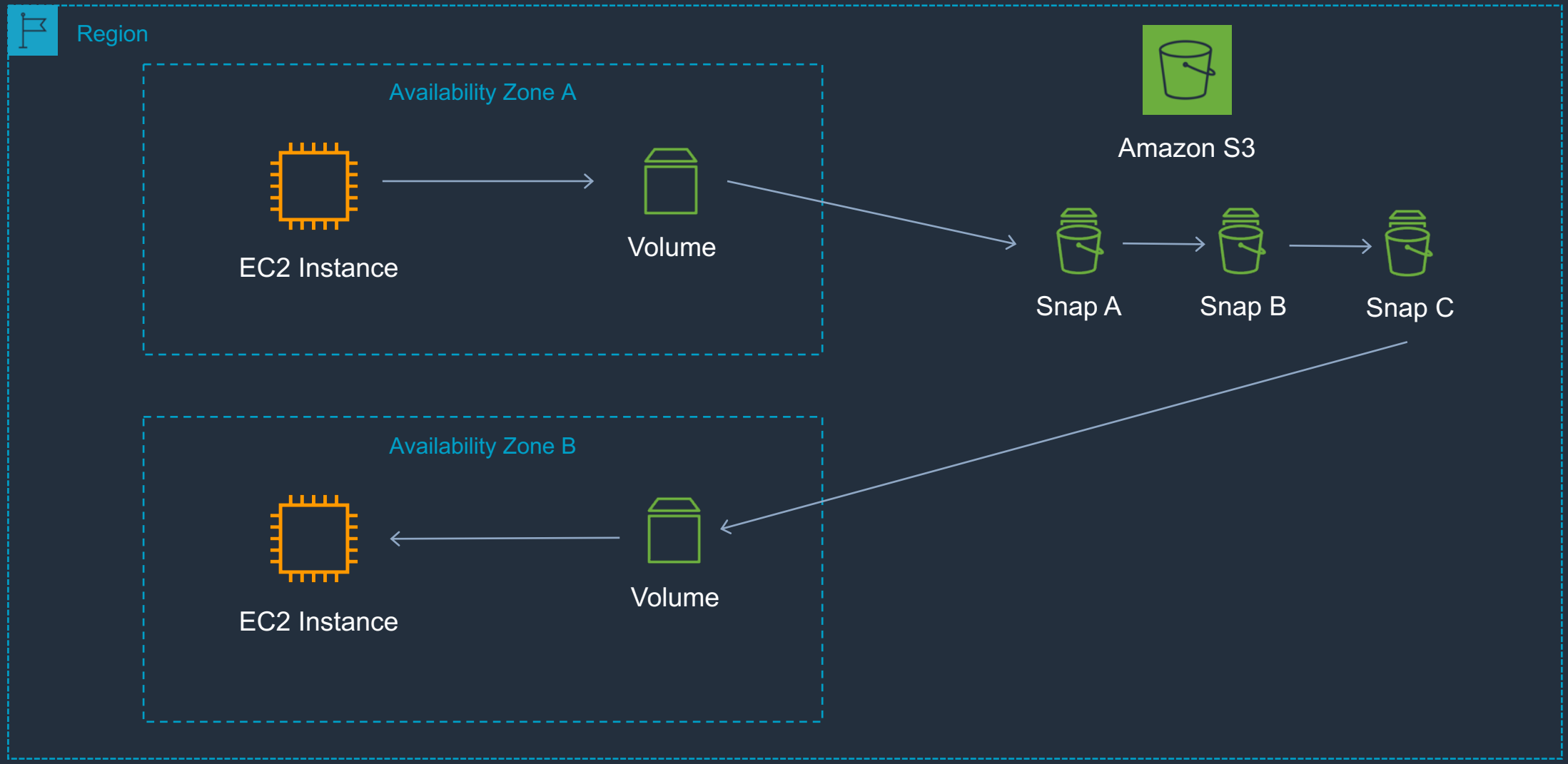
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Section 7: Amazon Elastic Block Store (EBS)

	Solid State Drives (SSD)		Hard Disk Drives (HDD)	
Volume Type	EBS Provisioned IOPS SSD (io1)	EBS General Purpose SSD (gp2)	Throughput Optimized HDD (st1)	Cold HDD (sc1)
Short Description	Highest performance SSD volume designed for latency-sensitive transactional workloads	General Purpose SSD volume that balances price performance for a wide variety of transactional workloads	Low cost HDD volume designed for frequently accessed, throughput intensive workloads	Lowest cost HDD volume designed for less frequently accessed workloads
Use Cases	I/O-Intensive NoSQL and relational databases	Boot volumes, low-latency interactive apps, dev & test	Big data, data warehouses, log processing	Colder data requiring fewer scans per day
Volume Size	4GB – 16TB	1 GB – 16 TB	500 GB – 16 TB	500 GB – 16 TB
Max IOPS/Volume	64,000	16,000	500	250
Max Throughput/Volume	1,000 MB/s	250 MB/s	500 MB/s	250 MB/s

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Section 7: EBS Snapshots



DigitalCloud TRAINING

Section 7: Amazon Elastic Block Store (EBS)



Amazon Elastic Block Store (EBS)



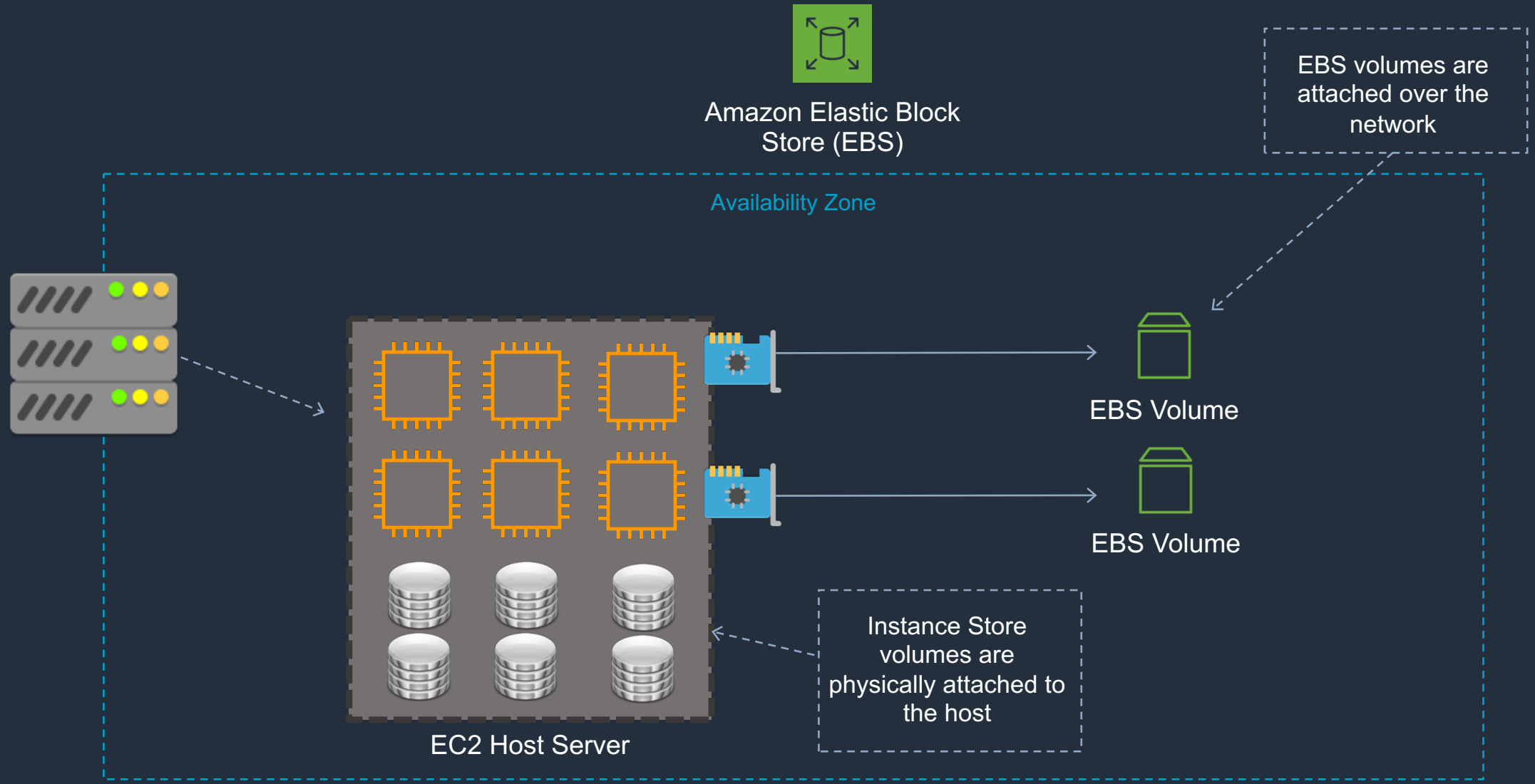
Snapshot

EBS Snapshots

- Snapshots capture a point-in-time state of an instance
- Snapshots are stored on S3
- Does not provide granular backup (not a replacement for backup software)
- If you make periodic snapshots of a volume, the snapshots are incremental, which means that only the blocks on the device that have changed after your last snapshot are saved in the new snapshot
- Even though snapshots are saved incrementally, the snapshot deletion process is designed so that you need to retain only the most recent snapshot in order to restore the volume
- Snapshots can only be accessed through the EC2 APIs
- EBS volumes are AZ specific but snapshots are region specific

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Section 7: Amazon Elastic Block Store (EBS) vs Instance Store



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Section 7: Amazon Elastic Block Store (EBS)



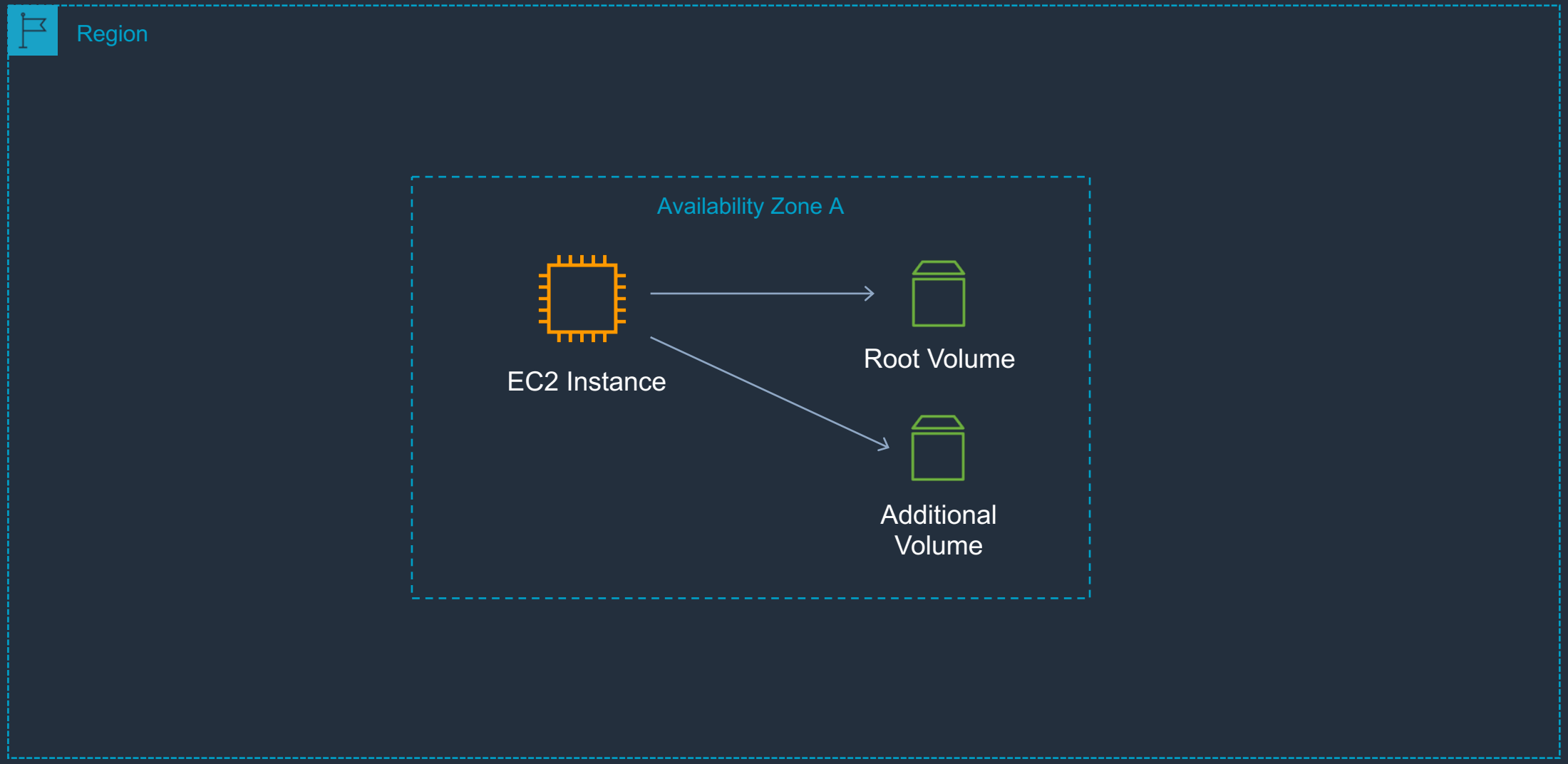
Amazon Elastic Block Store (EBS)

Instance Store Volumes

- Instance store volumes are high performance local disks that are physically attached to the host computer on which an EC2 instance runs
- Instance stores are ephemeral which means the data is lost when powered off (non-persistent)
- Instances stores are ideal for temporary storage of information that changes frequently, such as buffers, caches, or scratch data
- Instance store volume root devices are created from AMI templates stored on S3
- Instance store volumes cannot be detached/reattached

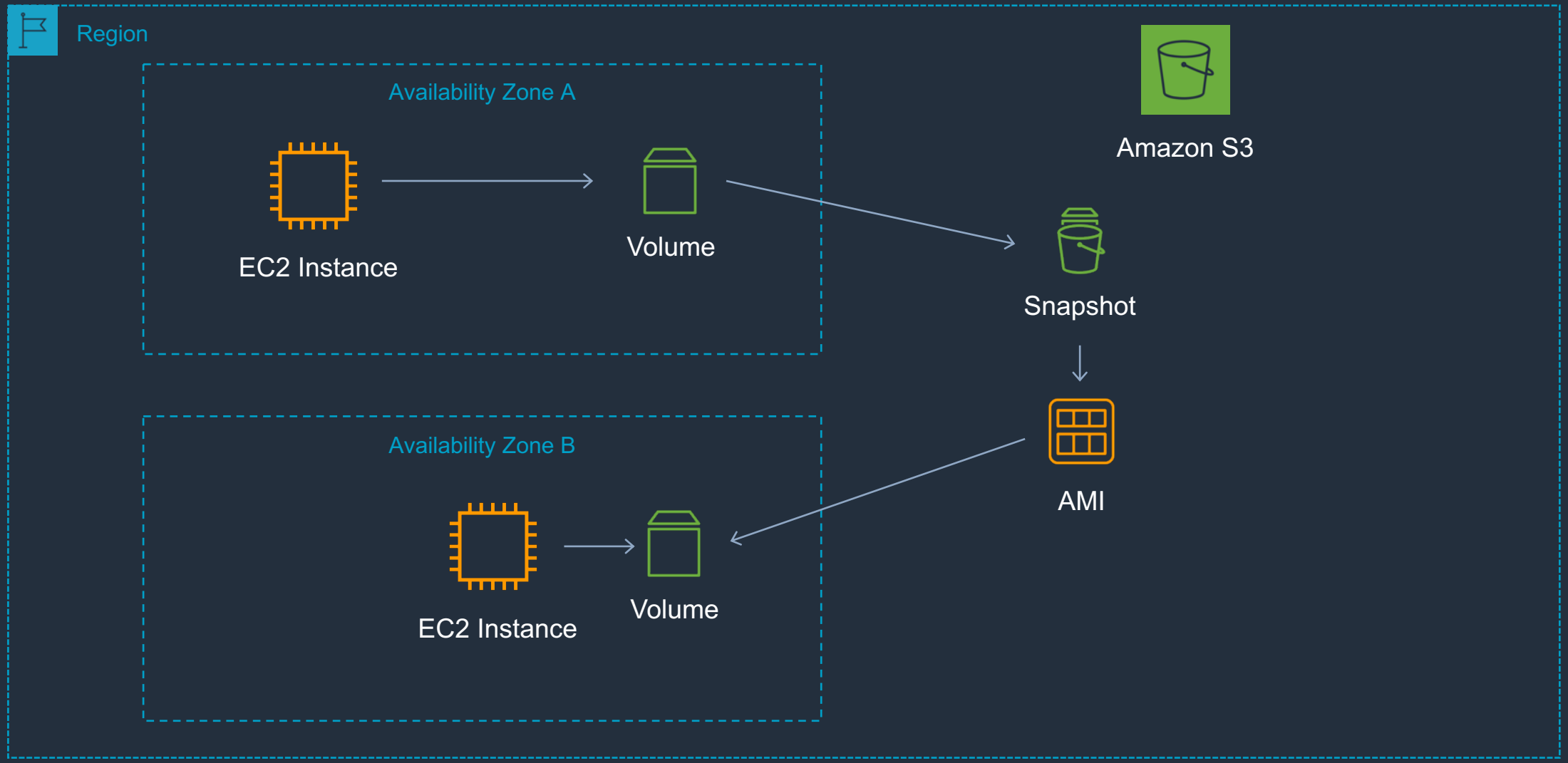
DigitalCloud TRAINING

Section 7: Launch Instance, create and add new EBS Volume



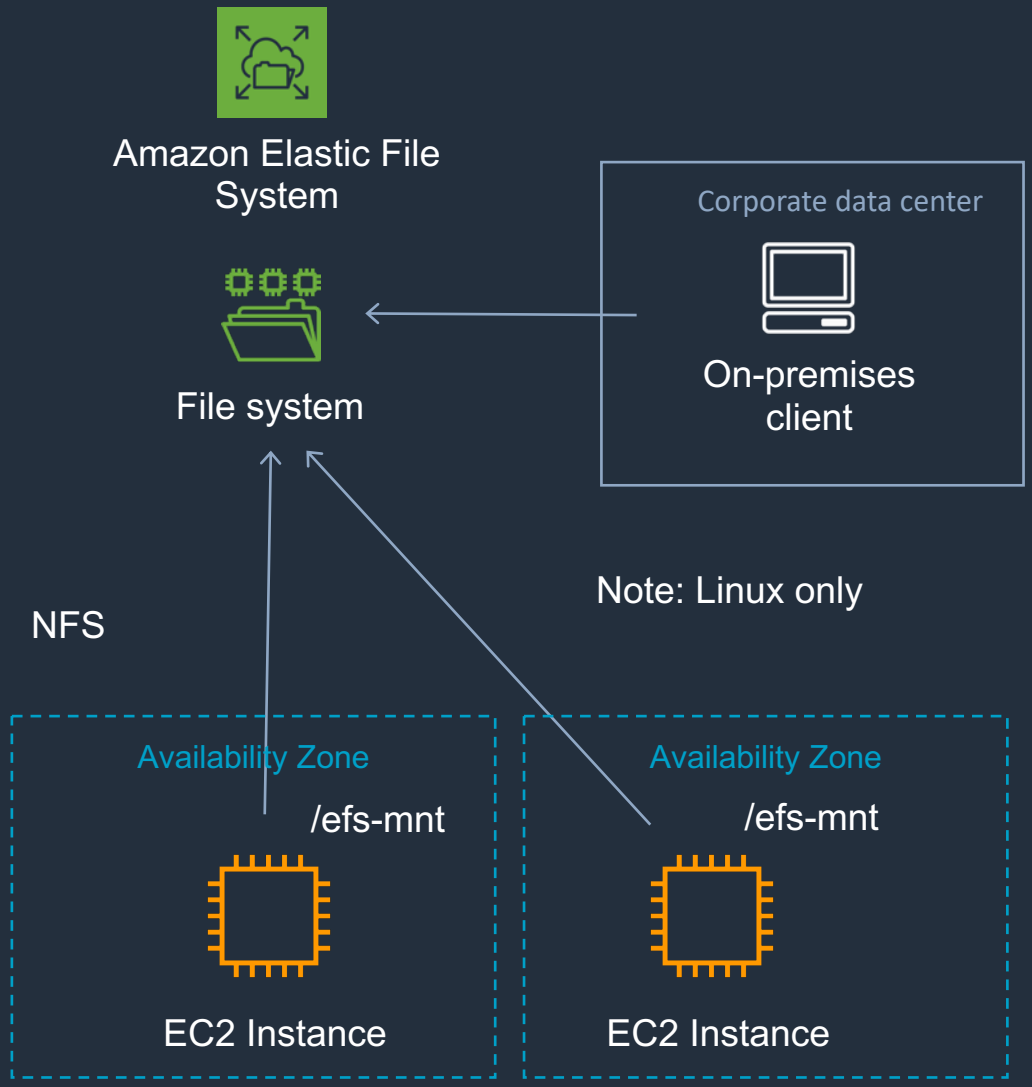
DigitalCloud TRAINING

Section 7: Take Snapshot, Create AMI, Launch New Instance



DigitalCloud TRAINING

Section 7: Amazon Elastic File System (EFS)



DigitalCloud TRAINING

Section 7: Amazon Elastic File System (EFS)



Amazon Elastic File System

- EFS is a fully-managed service that makes it easy to set up and scale file storage in the Amazon Cloud
- EFS provides a file system interface and uses the NFS protocol
- Good for big data and analytics, media processing workflows, content management, web serving, home directories etc.
- Data is stored across multiple AZ's within a region
- Read after write consistency
- Pay for what you use (no pre-provisioning required)
- Can scale up to petabytes
- EFS is elastic and grows and shrinks as you add and remove data
- Can concurrently connect 1 to 1000s of EC2 instances, from multiple AZs
- A file system can be accessed concurrently from all AZs in the region where it is located
- By default you can create up to 10 file systems per account
- On-premises access can be enabled via Direct Connect or AWS VPN

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Section 7: Amazon Elastic File System (EFS)

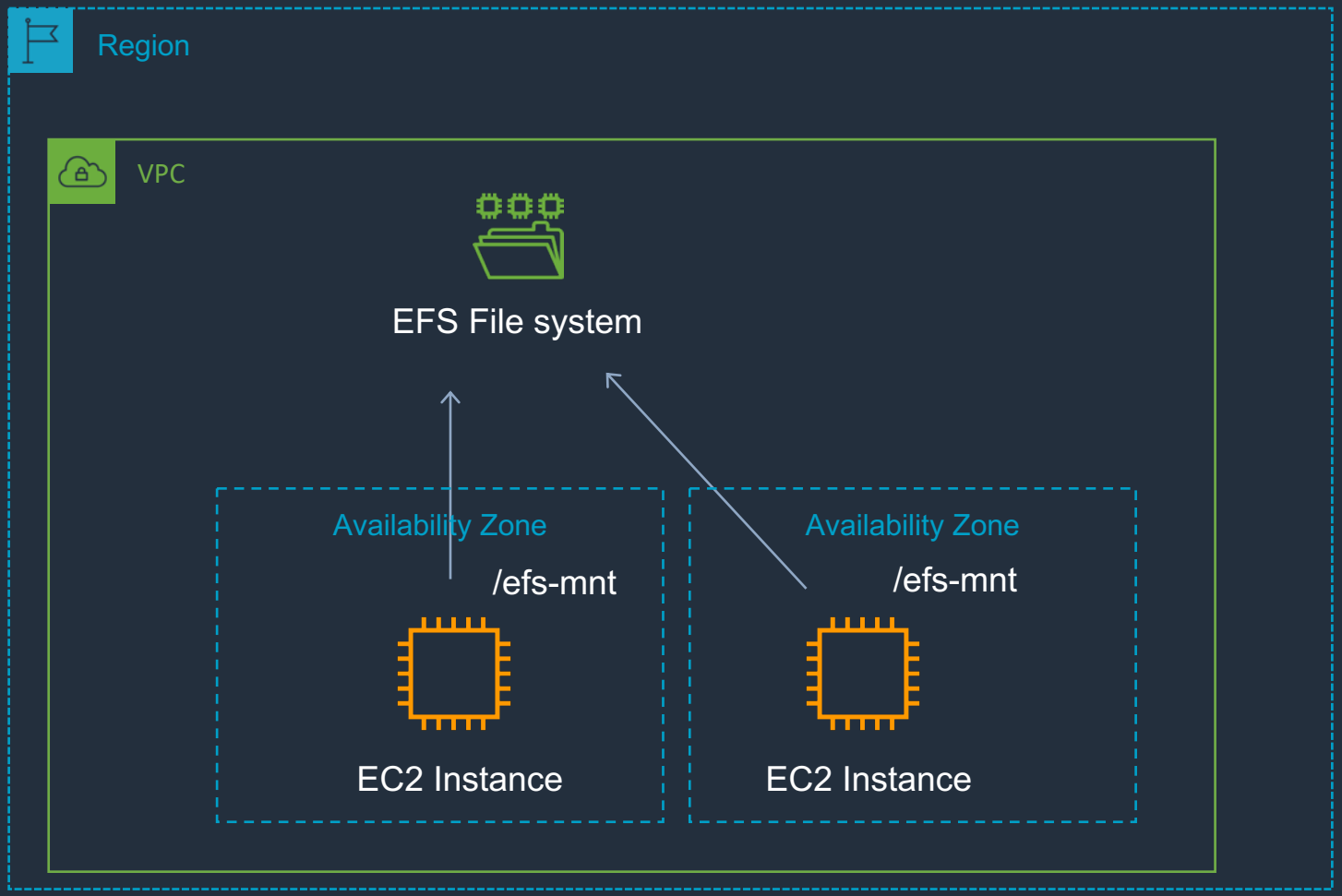


Amazon Elastic File System

- Instances can be behind an Amazon Elastic Load Balancer
- There are two performance modes:
  - "General Purpose" performance mode is appropriate for most file systems
  - "Max I/O" performance mode is optimized for applications where tens, hundreds, or thousands of EC2 instances are accessing the file system

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Amazon Elastic File System (EFS)



DigitalCloud TRAINING

Section 7: AWS Storage Gateway



AWS Storage Gateway

- AWS Storage Gateway is a hybrid cloud storage service that gives you on-premises access to virtually unlimited cloud storage
- Seamlessly connects on-premises applications to cloud storage, caching data locally for low-latency access
- There are three types of gateway:
  - Tape Gateway
  - File Gateway – accessed using SMB, NFS or S3 API
  - Volume Gateway – block device accessed using iSCSI

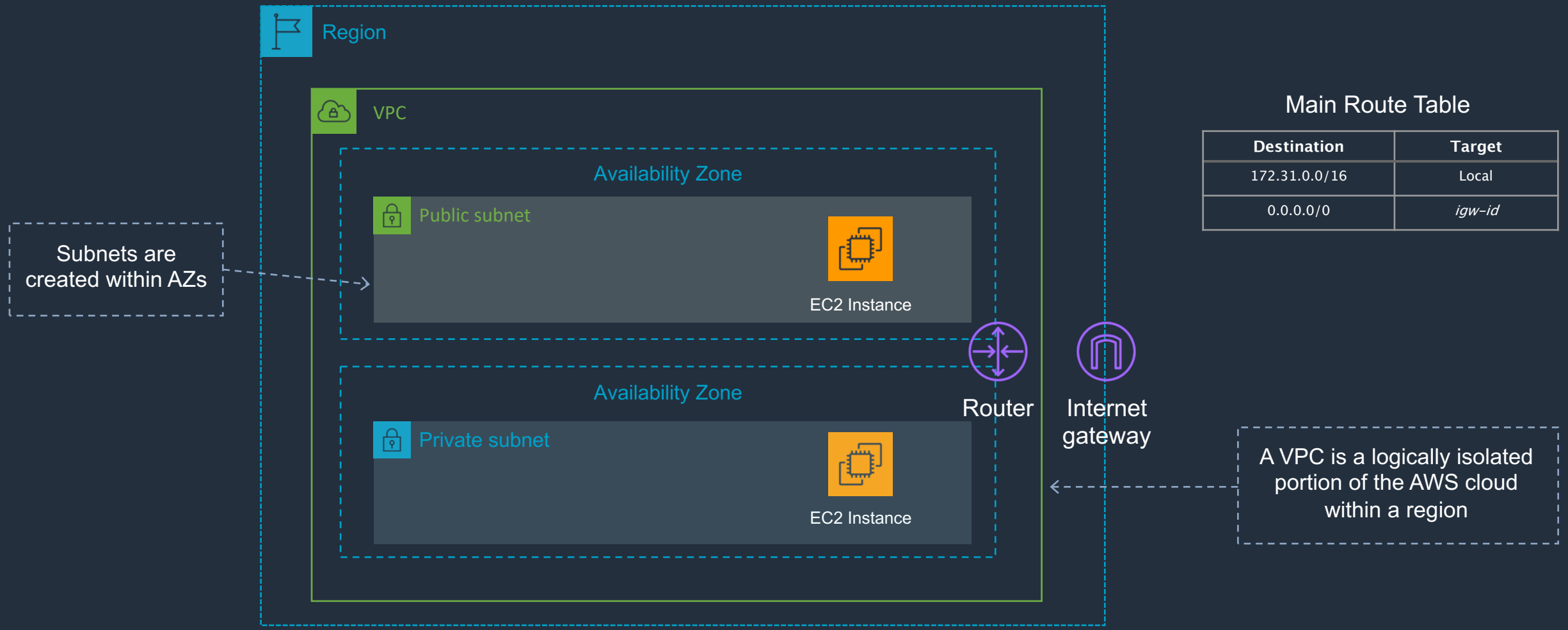
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SECTION 8

Amazon Virtual Private Cloud (VPC)



Section 8: Virtual Private Cloud (VPC)

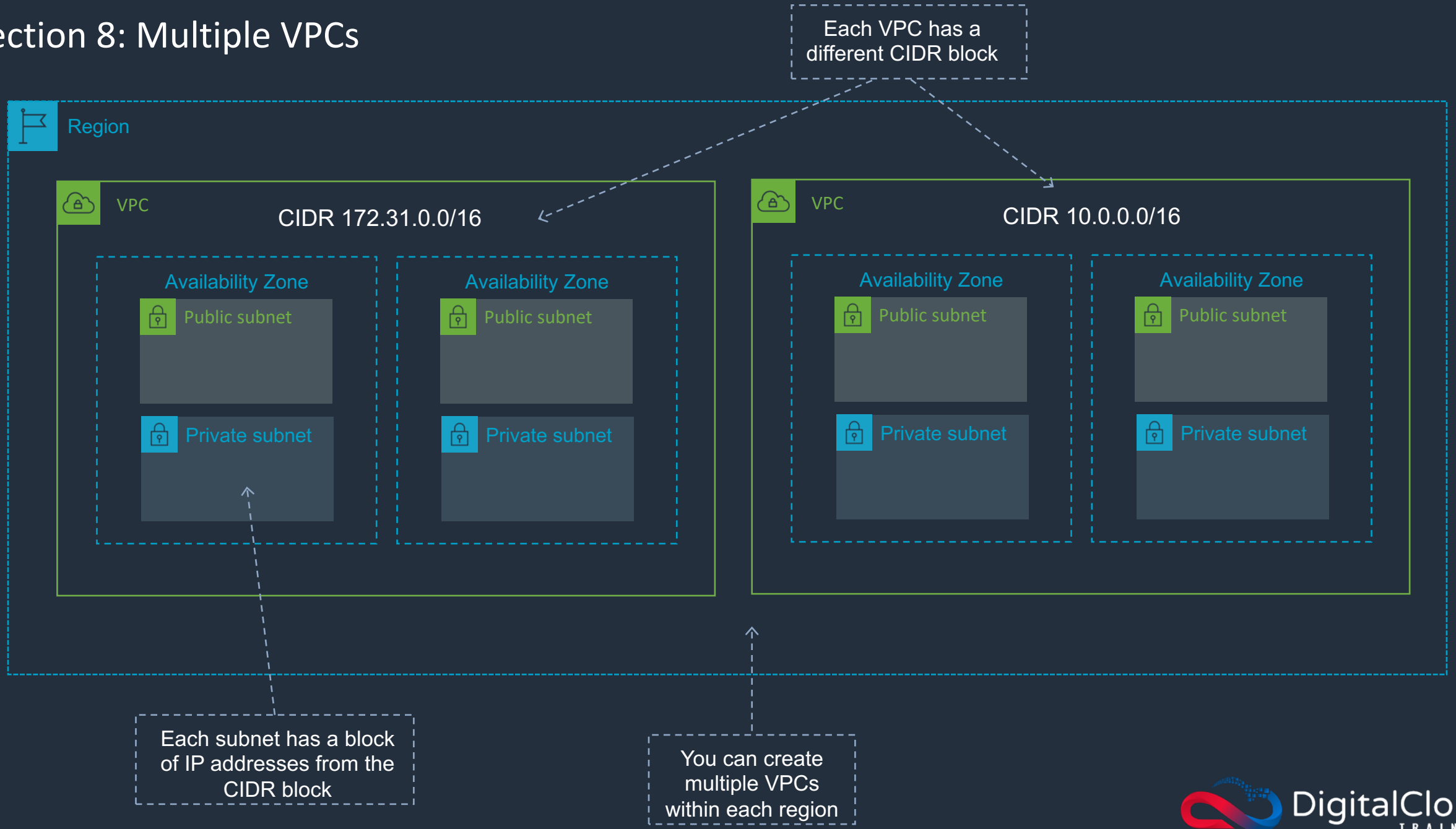


Main Route Table

Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*igw-id*

DigitalCloud TRAINING

Section 8: Multiple VPCs



DigitalCloud TRAINING

Section 8: VPC Components

VPC Component	What it is
Virtual Private Cloud (VPC)	A logically isolated virtual network in the AWS cloud
Subnet	A segment of a VPC's IP address range where you can place groups of isolated resources
Internet Gateway/Egress-only Internet Gateway	The Amazon VPC side of a connection to the public Internet for IPv4/IPv6
Router	Routers interconnect subnets and direct traffic between Internet gateways, virtual private gateways, NAT gateways, and subnets
Peering Connection	Direct connection between two VPCs
VPC Endpoints	Private connection to public AWS services
NAT Instance	Enables Internet access for EC2 instances in private subnets managed by you)
NAT Gateway	Enables Internet access for EC2 instances in private subnets (managed by AWS)
Virtual Private Gateway	The Amazon VPC side of a Virtual Private Network (VPN) connection
Customer Gateway	Customer side of a VPN connection
AWS Direct Connect	High speed, high bandwidth, *private* network connection from customer to aws
Security Group	Instance-level firewall
Network ACL	Subnet-level firewall

DigitalCloud TRAINING

Section 8: Virtual Private Cloud

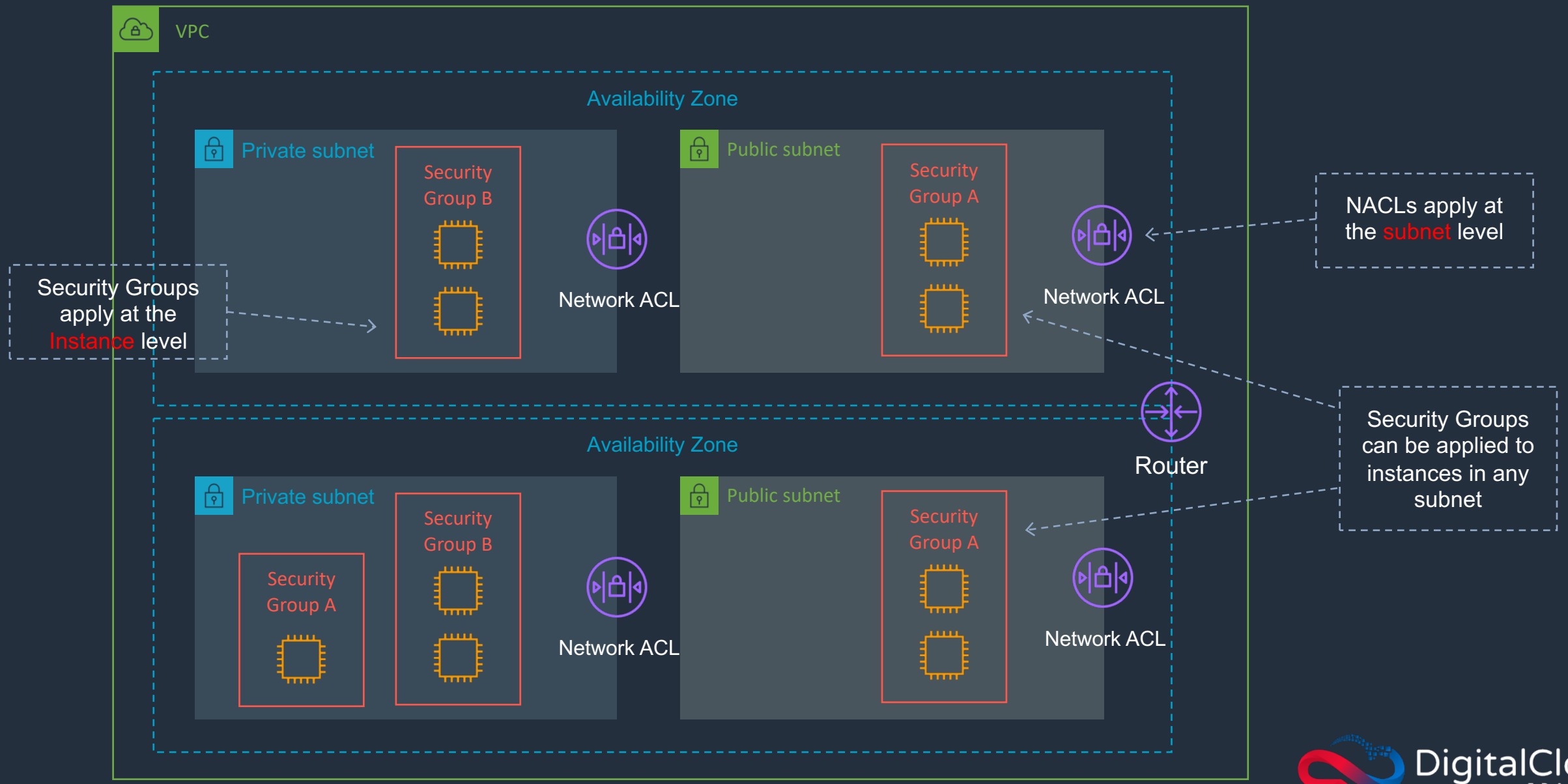


Amazon VPC

- A virtual private cloud (VPC) is a virtual network dedicated to your AWS account
- Analogous to having your own DC inside AWS
- It is logically isolated from other virtual networks in the AWS Cloud
- Provides complete control over the virtual networking environment including selection of IP ranges, creation of subnets, and configuration of route tables and gateways
- You can launch your AWS resources, such as Amazon EC2 instances, into your VPC
- When you create a VPC, you must specify a range of IPv4 addresses for the VPC in the form of a Classless Inter-Domain Routing (CIDR) block; for example, 10.0.0.0/16
- A VPC spans all the Availability Zones in the region
- You have full control over who has access to the AWS resources inside your VPC
- By default you can create up to 5 VPCs per region
- A default VPC is created in each region with a subnet in each AZ

DigitalCloud TRAINING

Section 8: Security Groups & Network Access Control Lists (NACLs)



DigitalCloud TRAINING

Section 8: Security Groups vs NACLs

Security Group	Network ACL
Operates at the instance (interface) level	Operates at the subnet level
Supports allow rules only	Supports allow and deny rules
Stateful	Stateless
Evaluates all rules	Processes rules in order
Applies to an instance only if associated with a group	Automatically applies to all instances in the subnets its associated with

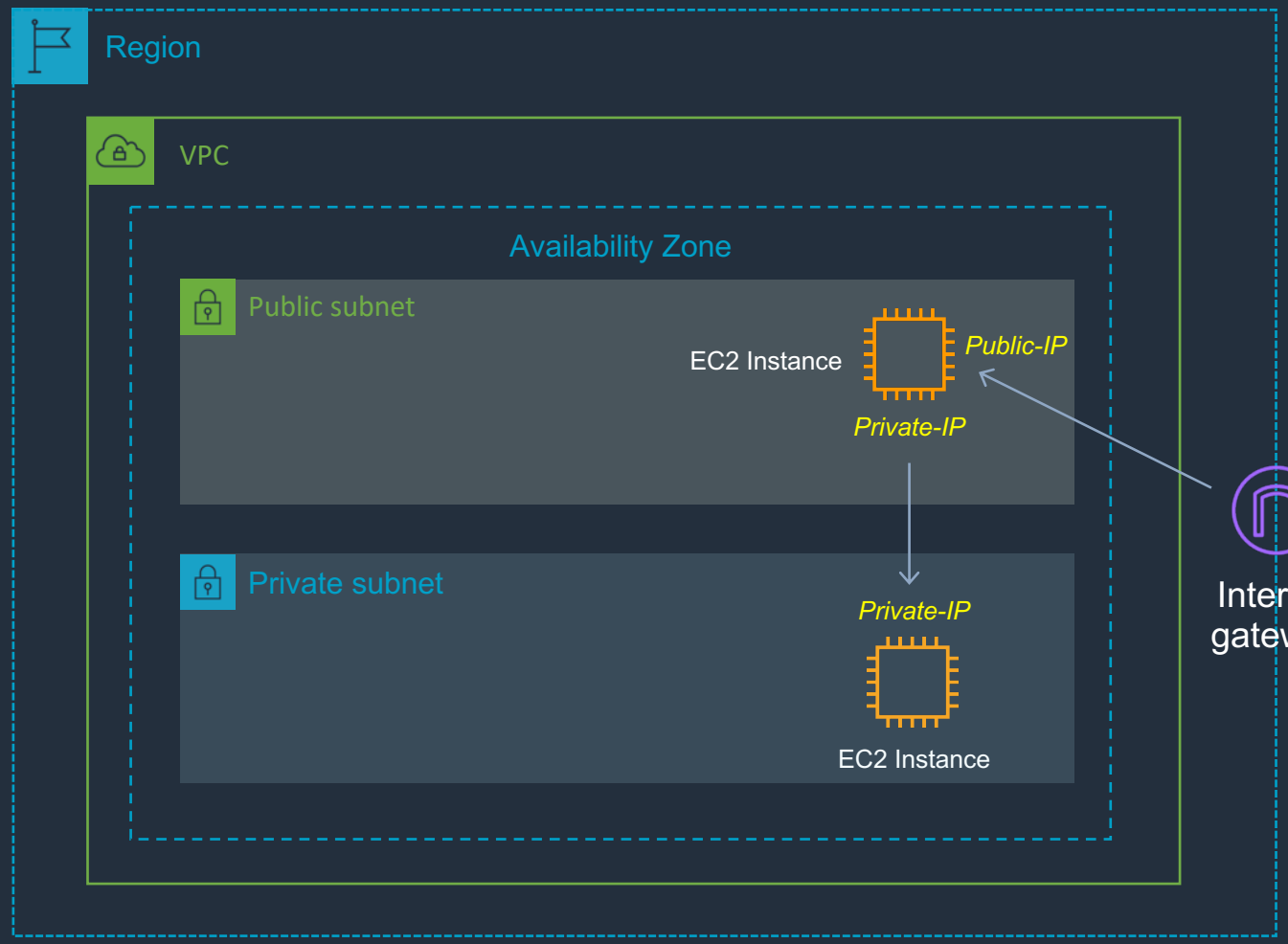
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Section 8: Public, Private, and Elastic IP addresses

Name	Description
Public IP address	Lost when the instance is stopped Used in Public Subnets No charge Associated with a private IP address on the instance Cannot be moved between instances
Private IP address	Retained when the instance is stopped Used in Public and Private Subnets
Elastic IP address	Static Public IP address You are charged if not used Associated with a private IP address on the instance Can be moved between instances and Elastic Network Adapters

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Section 8: Connect to instance in a private subnet





Internet gateway



Linux Client

Public Subnet Route Table

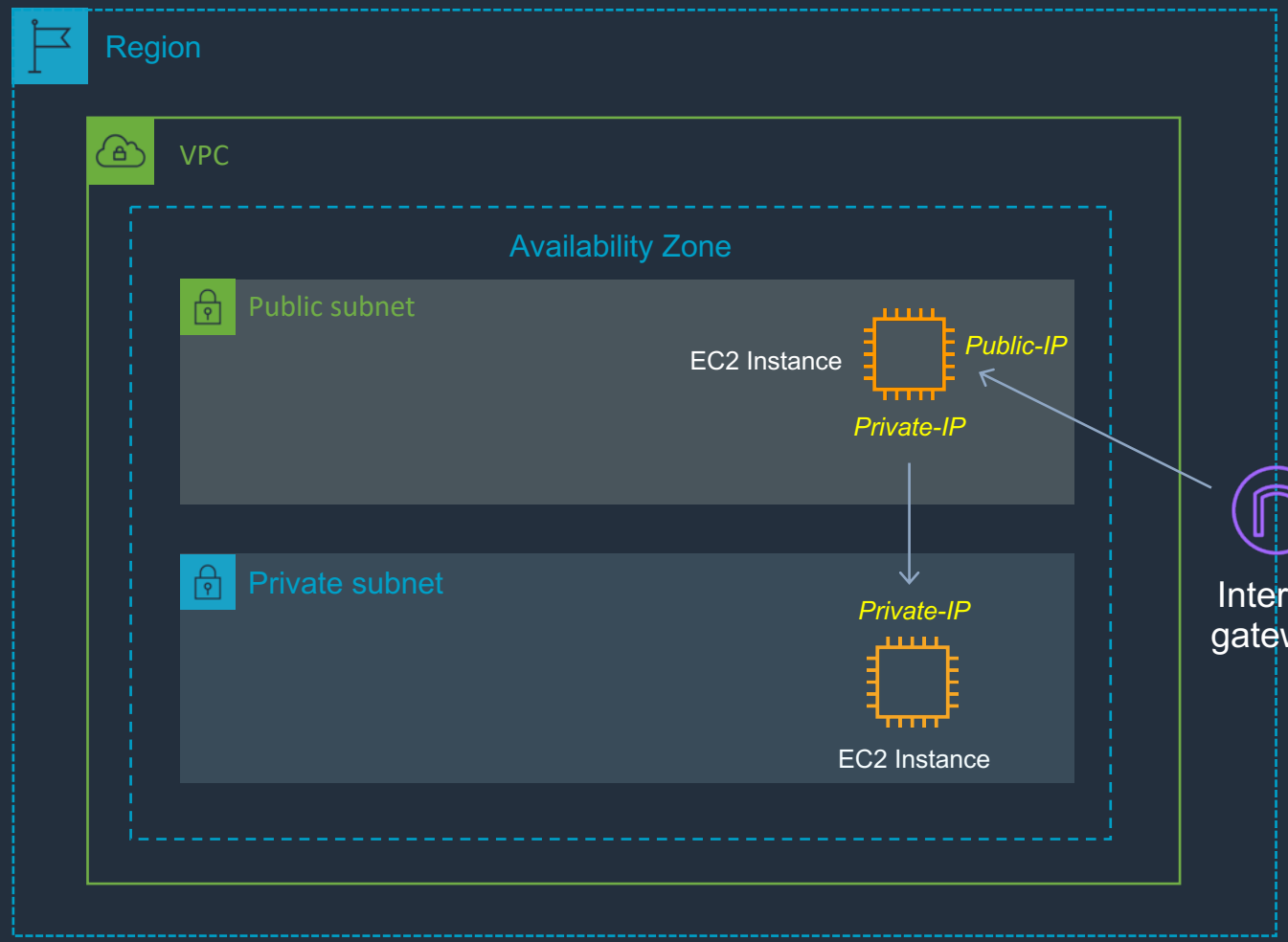
Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*igw-id*

Private Subnet Route Table

Destination	Target
172.31.0.0/16	Local

DigitalCloud TRAINING

Section 8: Connect to instance in a private subnet





Internet gateway



Windows Client

Public Subnet Route Table

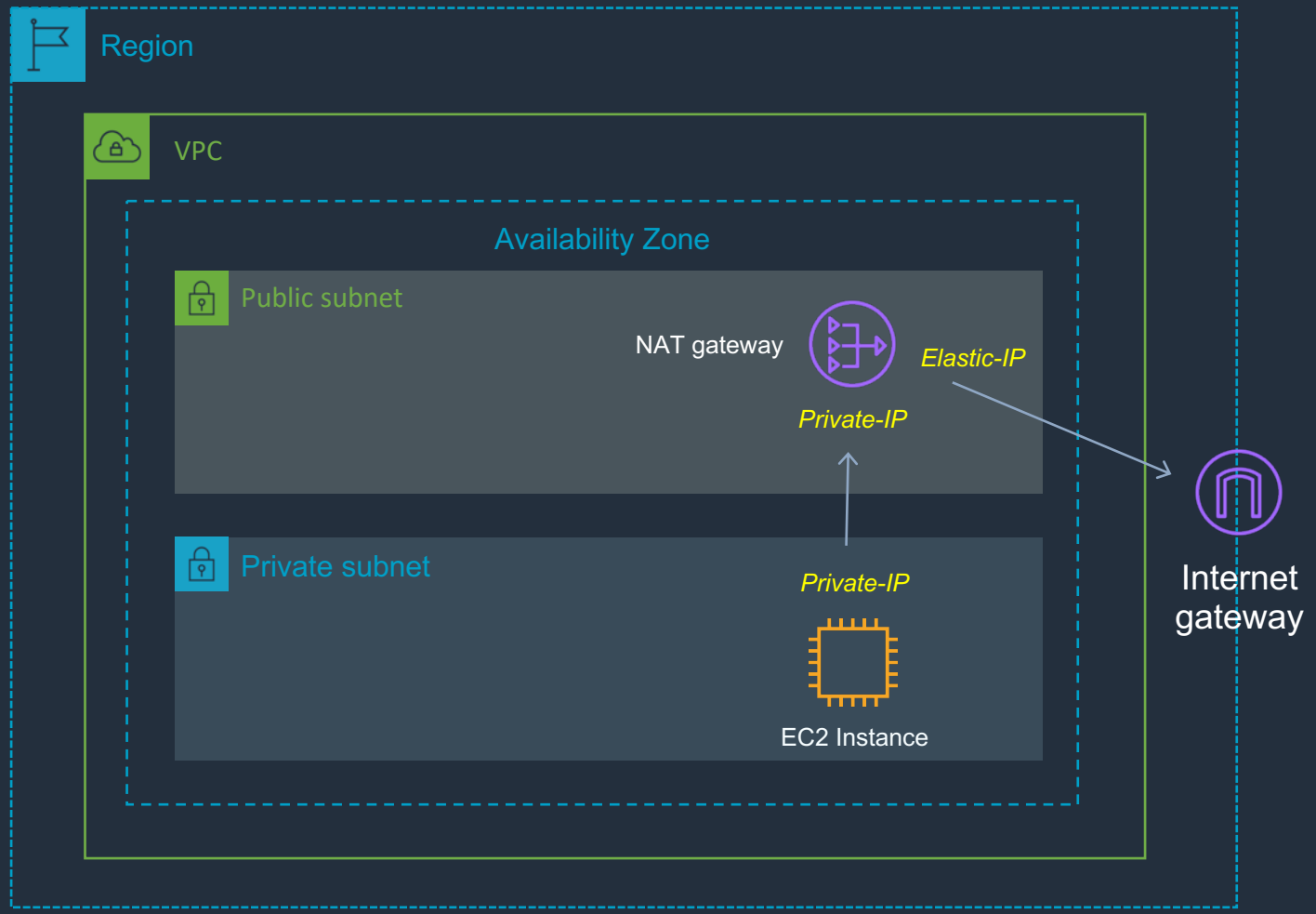
Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*igw-id*

Private Subnet Route Table

Destination	Target
172.31.0.0/16	Local

DigitalCloud TRAINING

Section 8: Private Subnet with NAT Gateway



Public Subnet Route Table

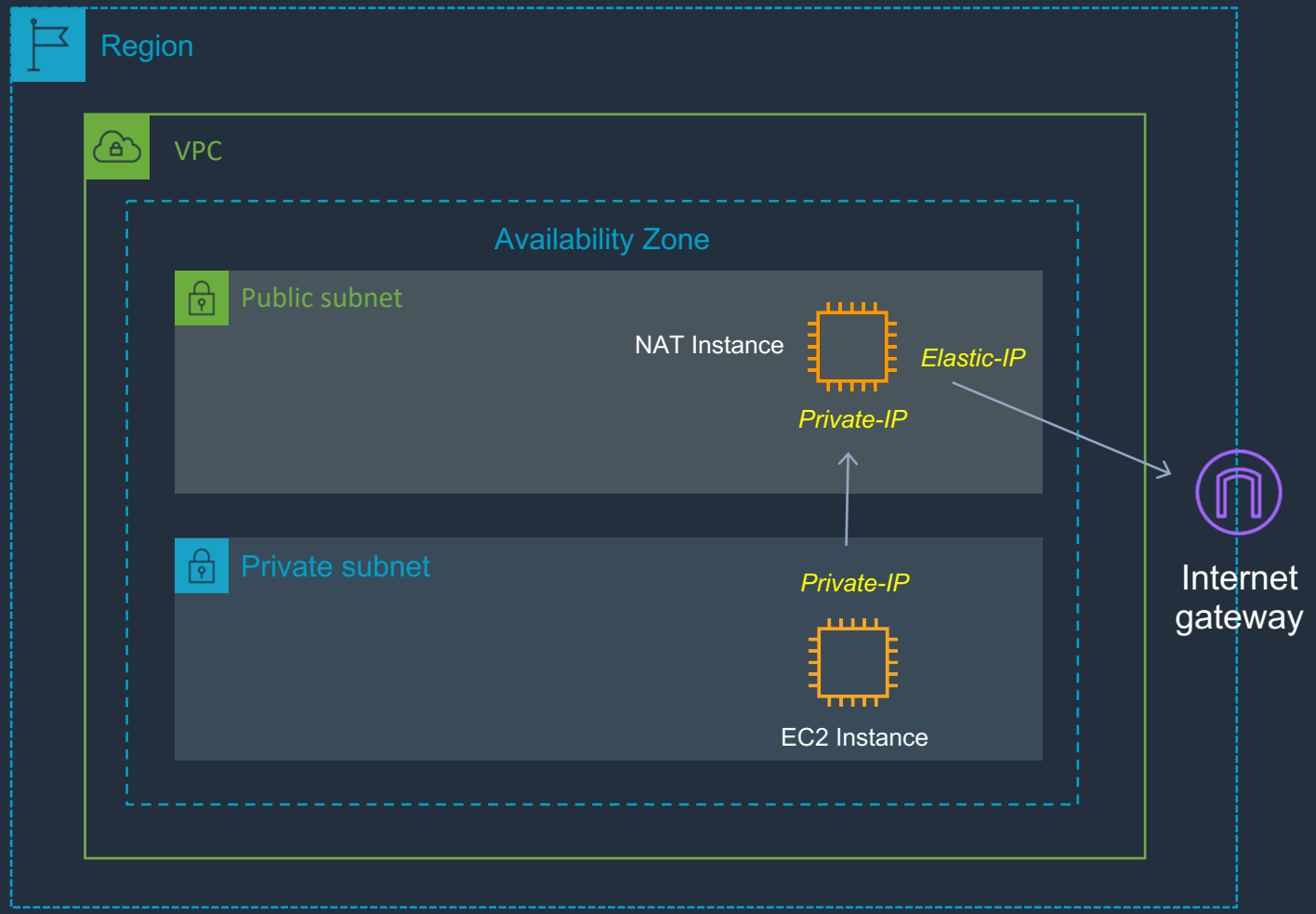
Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*igw-id*

Private Subnet Route Table

Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*nat-gateway-id*

DigitalCloud TRAINING

Section 8: Private Subnet with NAT Instance



Public Subnet Route Table

Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*igw-id*

Private Subnet Route Table

Destination	Target
172.31.0.0/16	Local
0.0.0.0/0	*nat-instance-id*

DigitalCloud TRAINING

Section 8: NAT Instance vs NAT Gateway

NAT Instance	NAT Gateway
Managed by you (e.g. software updates)	Managed by AWS
Scale up (instance type) manually and use enhanced networking	Elastic scalability up to 45 Gbps
No high availability – scripted/auto-scaled HA possible using multiple NATs in multiple subnets	Provides automatic high availability within an AZ and can be placed in multiple AZs
Need to assign Security Group	No Security Groups
Can use as a bastion host	Cannot access through SSH
Use an Elastic IP address or a public IP address with a NAT instance	Choose the Elastic IP address to associate with a NAT gateway at creation
Can implement port forwarding through manual customisation	Does not support port forwarding

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Section 8: Connecting securely to a VPC

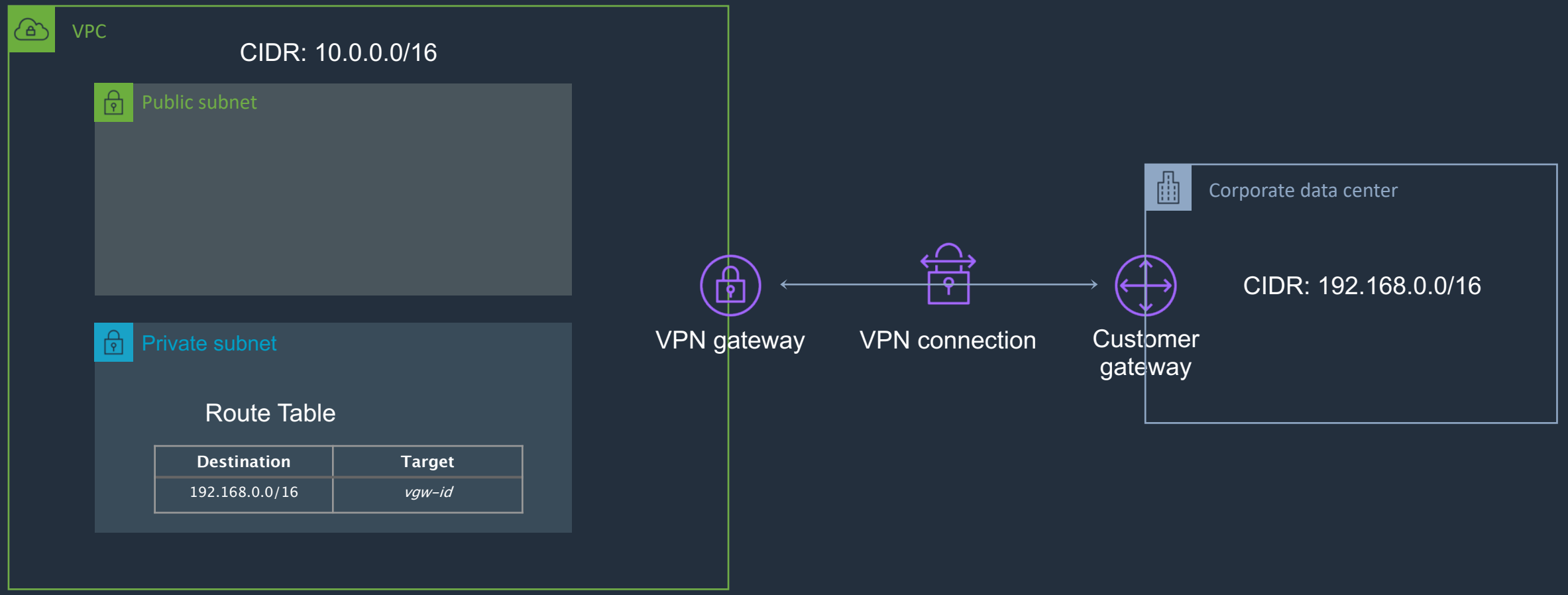


Amazon VPC

- Options for securely connecting to a VPC are:
  - AWS managed VPN – fast to setup
  - AWS Direct Connect – high bandwidth, low-latency but takes weeks to months to setup
  - VPN CloudHub – used for connecting multiple sites to AWS
  - Software VPN – use 3rd party software

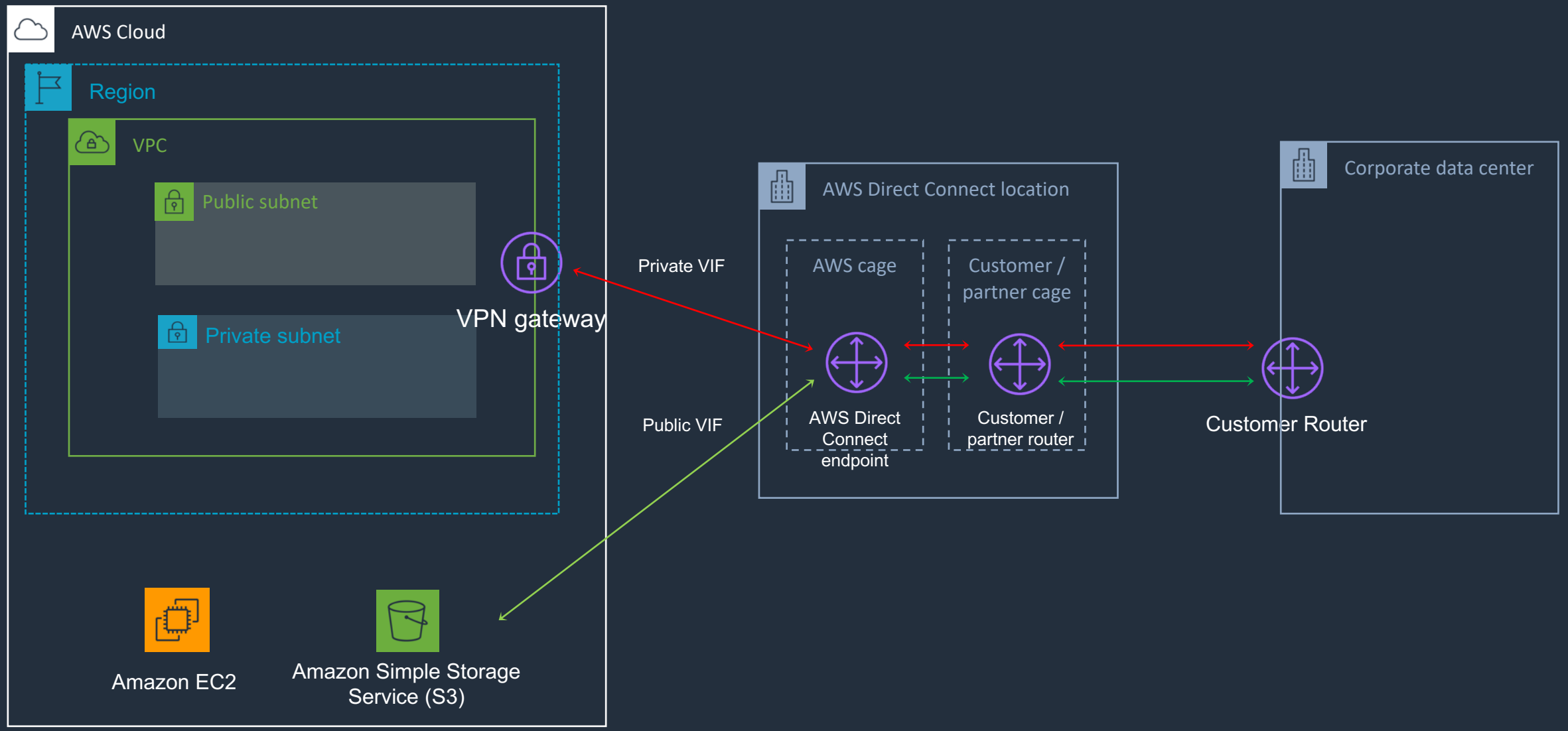
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Section 8: Virtual Private Networks (VPN)



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Section 8: AWS Direct Connect



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Section 8: AWS Direct Connect

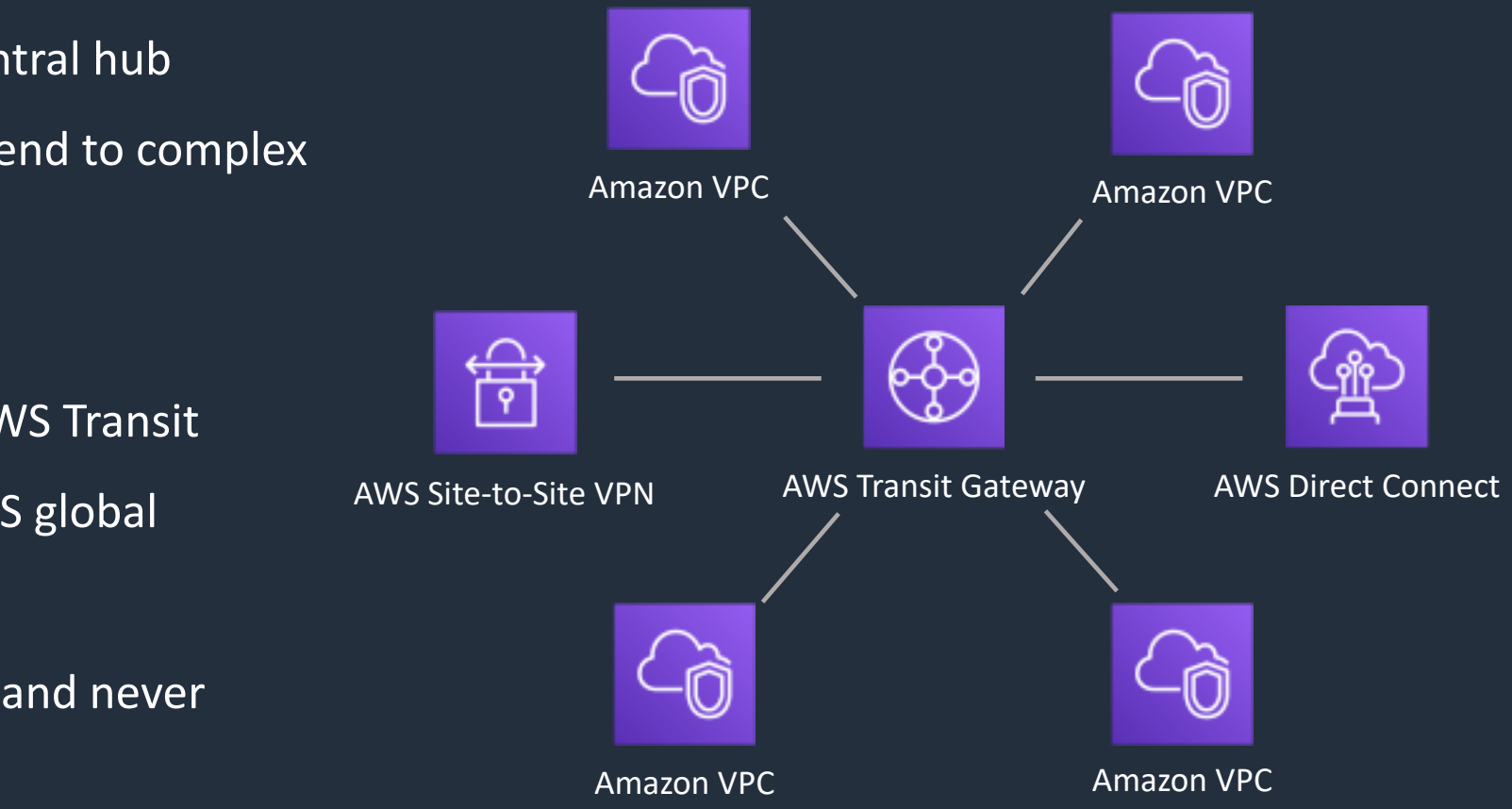


- AWS Direct Connect is a network service that provides an alternative to using the Internet to connect a customer's on premise sites to AWS
- Can be used to create a hybrid cloud
- Data is transmitted through a private network connection between AWS and a customer's datacenter or corporate network
- Benefits:
  - Reduce cost when using large volumes of traffic
  - Increase reliability (predictable performance)
  - Increase bandwidth (predictable bandwidth)
  - Decrease latency
- Direct Connect is charged by port hours and data transfer
- Available in 1Gbps and 10Gbps
- Speeds of 50Mbps, 100Mbps, 200Mbps, 300Mbps, 400Mbps, and 500Mbps can be purchased through AWS Direct Connect Partners

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Section 8: AWS Transit Gateway

- AWS Transit Gateway connects VPCs and on-premises networks through a central hub
- Simplifies networks and puts an end to complex peering relationships.
- Acts as a cloud router
- Inter-Region peering connects AWS Transit Gateways together using the AWS global network
- Data is automatically encrypted, and never travels over the public internet



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SECTION 9

AWS Databases



Section 9: Database Types – Relational vs Non-Relational

Key differences are how data are ***managed*** and how data are ***stored***

Relational	Non-Relational
Organized by tables, rows and columns	Varied data storage models
Rigid schema (SQL)	Flexible schema (NoSQL) – data stored in key-value pairs, columns, documents or graphs
Rules enforced within database	Rules can be defined in application code (outside database)
Typically scaled vertically	Scales horizontally
Supports complex queries and joins	Unstructured, simple language that supports any kind of schema
Amazon RDS, Oracle, MySQL, IBM DB2, PostgreSQL	Amazon DynamoDB, MongoDB, Redis, Neo4j

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Section 9: Relational Database

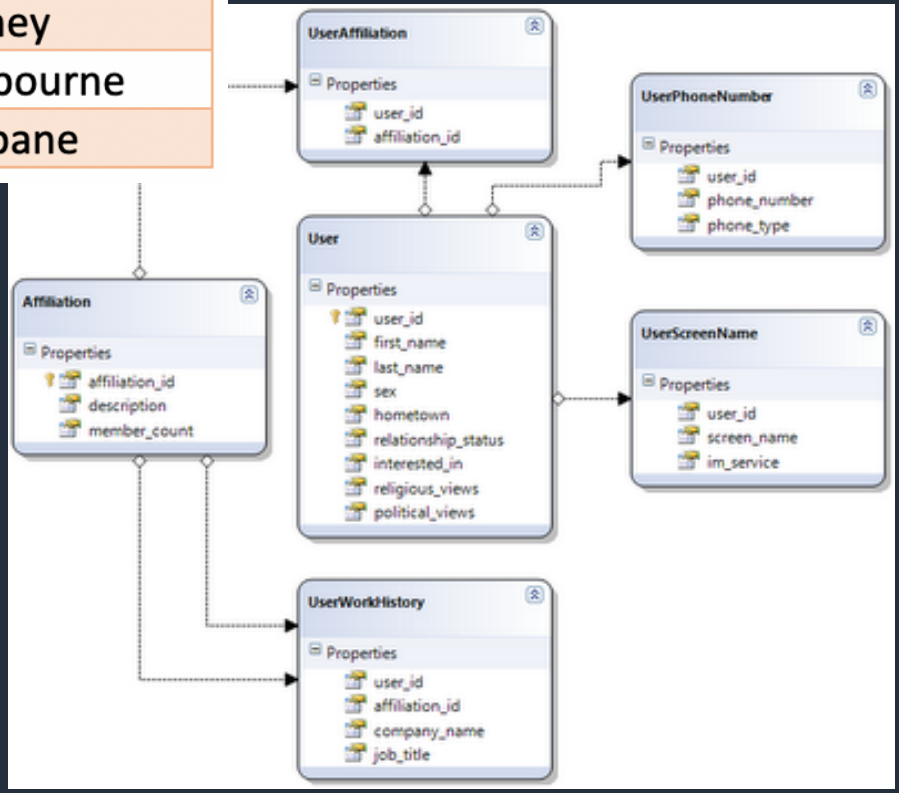
EmployeeID	FirstName	LastName	JobRole	Location
00001	Paul	Peterson	Senior Developer	Sydney
00002	Kaleigh	Annette	Assistant Manager	Brisbane
00003	Carl	Wood	Sales Support	Sydney
00004	Vinni	Jones	Customer Services	Melbourne
00005	Stefanie	Howard	IT Architect	Brisbane

Relational Database

Structured Query Language (SQL) query:

```
SELECT FirstName
FROM employees
WHERE Location = Sydney
```

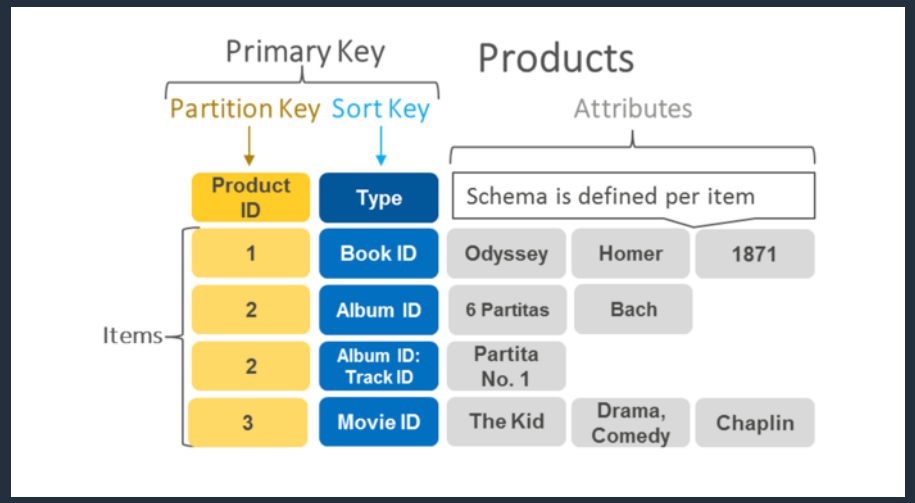
Relational Database – Multiple Tables



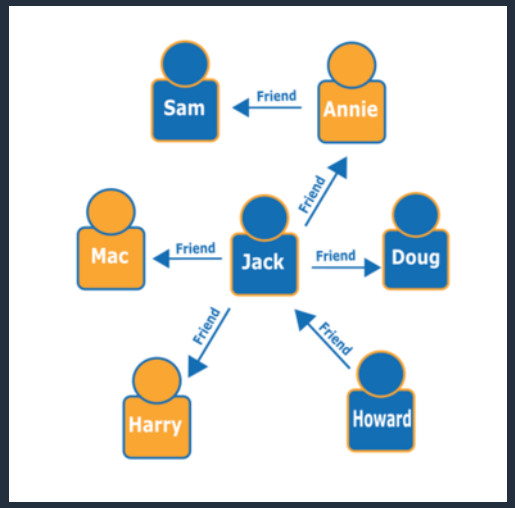
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Section 9: Types of Non-Relational DB (NoSQL)

Key-value – e.g. Amazon DynamoDB



Graph – e.g. Amazon Neptune



Document – e.g. MongoDB



```json
[
    {
        "year" : 2013,
        "title" : "Turn It Down, Or Else!",
        "info" : {
            "directors" : [ "Alice Smith", "Bob Jones"],
            "release_date" : "2013-01-18T00:00:00Z",
            "rating" : 6.2,
            "genres" : ["Comedy", "Drama"],
            "image_url" : "http://ia.media-imdb.com/images/N/O9ERWAU7FS797AJ7LU8HN09AMUP908RLlo5JF90EWR7LJKQ7@@._V1_SX400_.jpg",
            "plot" : "A rock band plays their music at high volumes, annoying the neighbors.",
            "actors" : ["David Matthewman", "Jonathan G. Neff"]
        }
    },
    {
        "year": 2015,
        "title": "The Big New Movie",
        "info": {
            "plot": "Nothing happens at all.",
            "rating": 0
        }
    }
]
```

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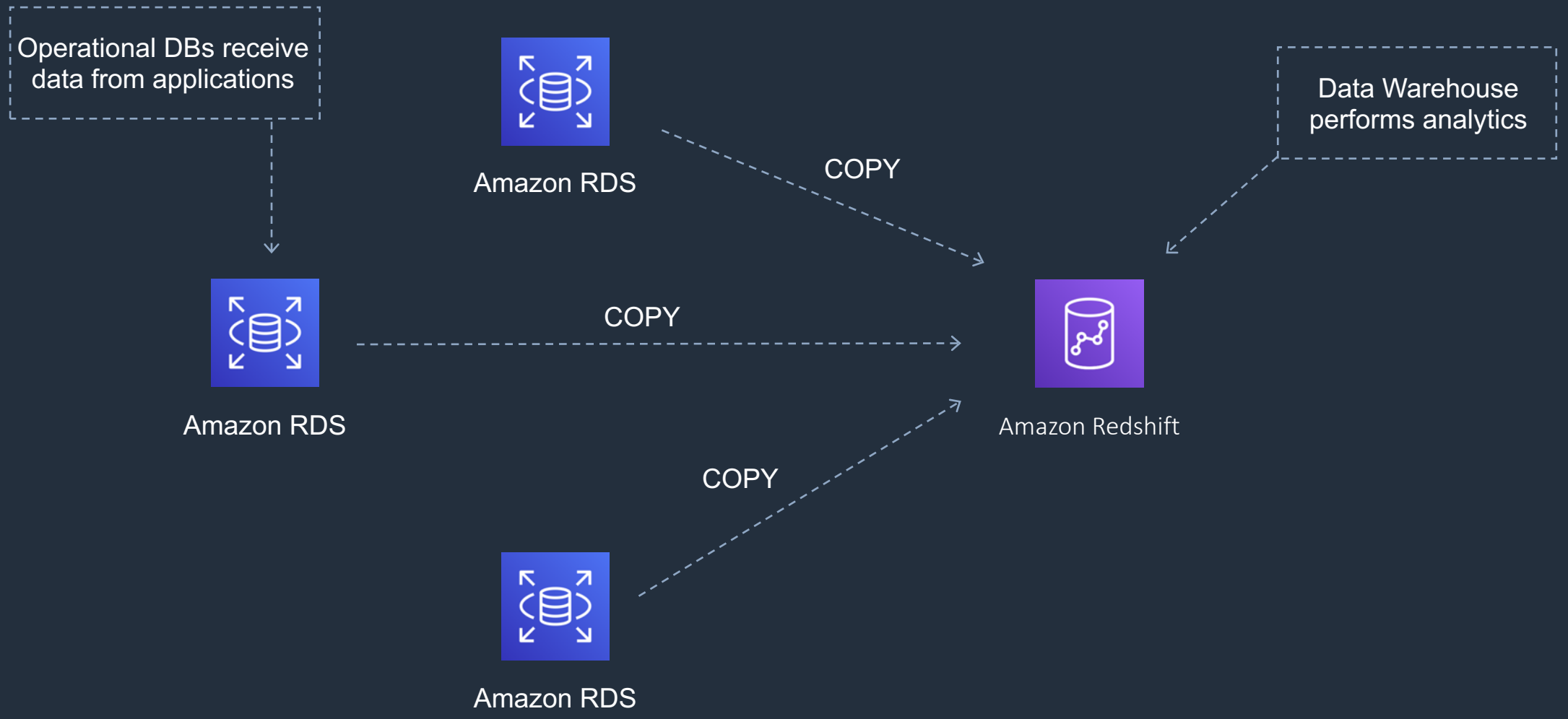
Section 9: Database Types – Operational vs Analytical

Key differences are ***use cases*** and how the database is ***optimized***

Operational / transactional	Analytical
Online Transaction Processing (OLTP)	Online Analytics Processing (OLAP) – the source data comes from OLTP DBs
Production DBs that process transactions. E.g. adding customer records, checking stock availability (INSERT, UPDATE, DELETE)	Data warehouse. Typically, separated from the customer facing DBs. Data is extracted for decision making
Short transactions and simple queries	Long transactions and complex queries
Relational examples: Amazon RDS, Oracle, IBM DB2, MySQL	Relational examples: Amazon RedShift, Teradata, HP Vertica
Non-relational examples: MongoDB, Cassandra, Neo4j, HBase	Non-relational examples: Amazon EMR, MapReduce

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Section 9: Database Types – Operational vs Analytical



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Section 9: Databases in AWS

Data Store	Use Case
Database on EC2	• Need full control over instance and database • Third-party database engine (not available in RDS)
Amazon RDS	• Need traditional relational database • e.g. Oracle, PostgreSQL, Microsoft SQL, MariaDB • Data is well-formed and structured
Amazon DynamoDB	• NoSQL database • In-memory performance • High I/O needs • Dynamic scaling
Amazon RedShift	• Data warehouse for large volumes of aggregated data
Amazon ElastiCache	• Fast temporary storage for small amounts of data

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Section 9: Amazon Relational Database Service (RDS)



Amazon RDS

- Amazon Relational Database Service (Amazon RDS) is a managed service that makes it easy to set up, operate, and scale a relational database in the cloud
- RDS uses EC2 instances, so you must choose an instance family/type
- Relational databases are known as Structured Query Language (SQL) databases
- RDS is an Online Transaction Processing (OLTP) type of database
- Easy to setup, highly available, fault tolerant, and scalable
- Common use cases include online stores and banking systems
- You can encrypt your Amazon RDS instances and snapshots at rest by enabling the encryption option for your Amazon RDS DB instance
- Encryption uses AWS Key Management Service (KMS)

DigitalCloud TRAINING

Section 9: Amazon Relational Database Service (RDS)



Amazon RDS

- Amazon RDS supports the following database engines:
  - SQL Server
  - Oracle
  - MySQL Server
  - PostgreSQL
  - Aurora
  - MariaDB
- Scalability: Can only be scaled up by increasing instance size (compute and storage)
- Fault tolerance / disaster recovery with Multi-AZ option
- Automatic failover for Multi-AZ option
- Read replicas option for read heavy workloads

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Section 9: Amazon Relational Database Service (RDS)



Amazon RDS

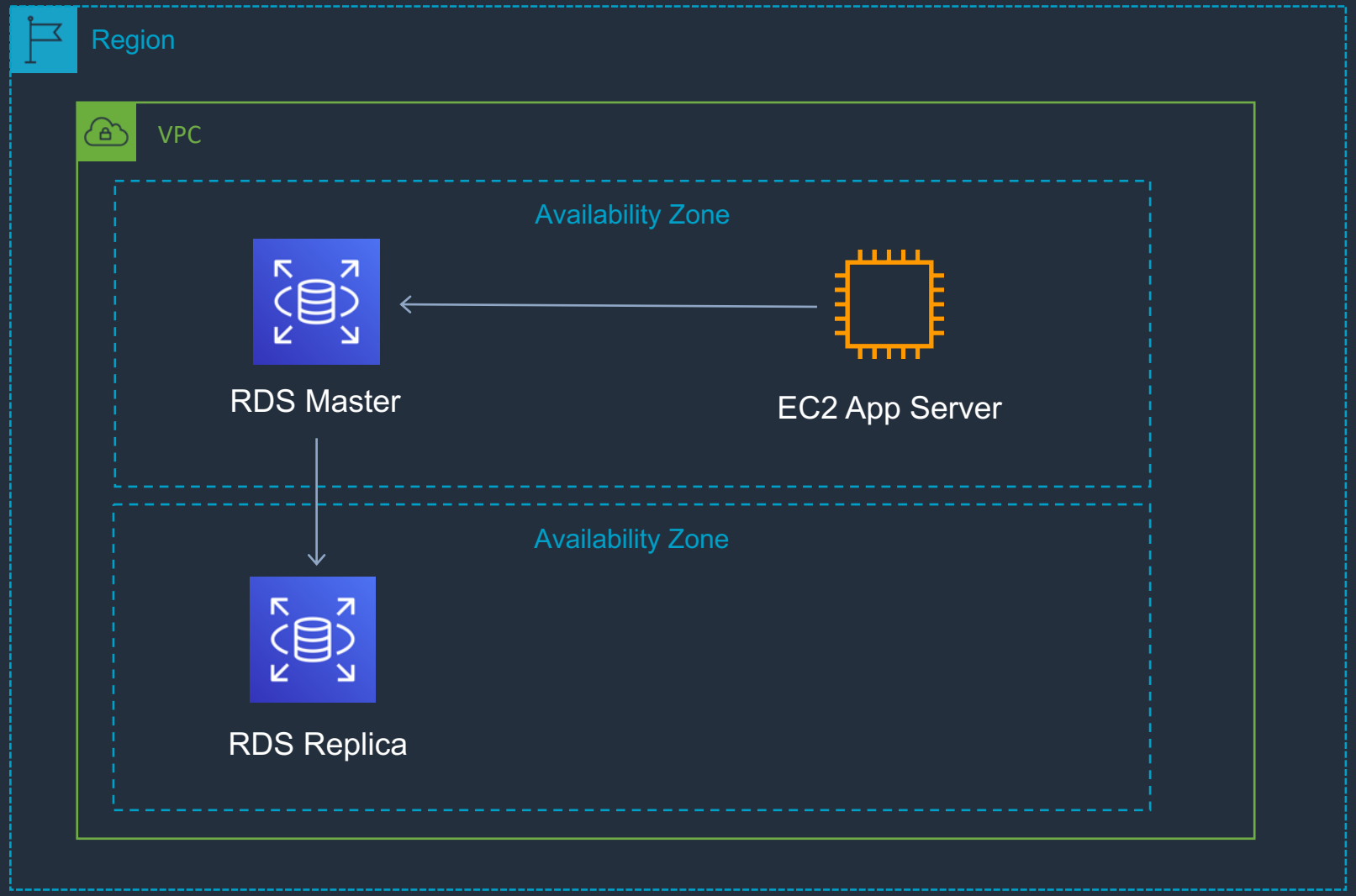
Amazon Aurora

- Amazon Aurora is an AWS database offering in the RDS family
- Amazon Aurora is a MySQL and PostgreSQL-compatible relational database built for the cloud
- Amazon Aurora is up to five times faster than standard MySQL databases and three times faster than standard PostgreSQL databases
- Amazon Aurora features a distributed, fault-tolerant, self-healing storage system that auto-scales up to 64TB per database instance

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Section 9: Amazon RDS Multi-AZ

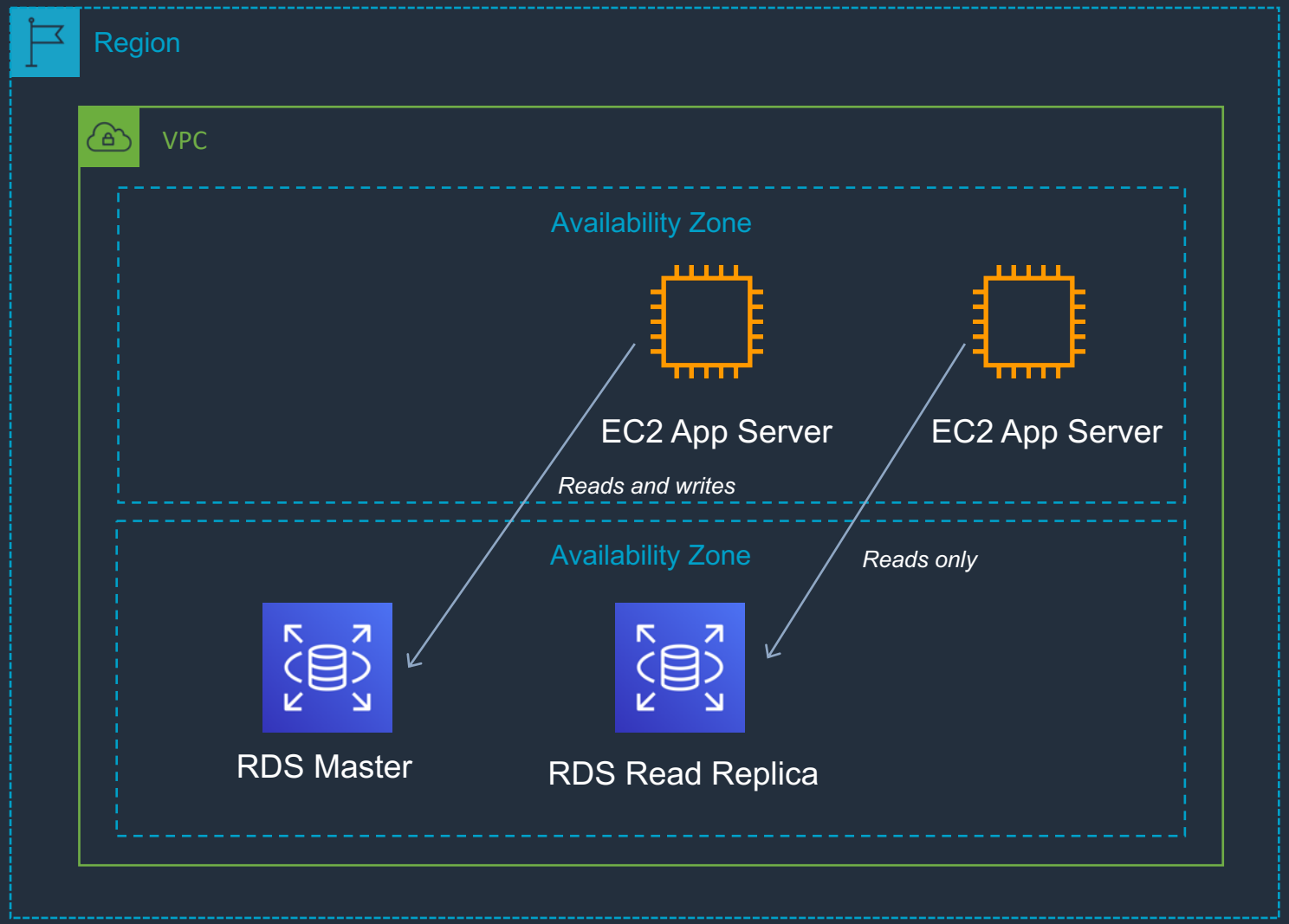
Used for Fault Tolerance and Disaster Recovery



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Section 9: Amazon RDS Read Replicas

Used for scaling read performance



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Section 9: DynamoDB Overview

DynamoDB Feature	Benefit
Serverless	Fully managed, fault tolerant, service
Highly available	99.99% availability SLA
NoSQL type of database with Name / Value structure	Flexible schema, good for when data is not well structured or unpredictable
Horizontal scaling	Seamless scalability to any scale with push button scaling or Auto Scaling
DynamoDB Accelerator (DAX)	Fully managed in-memory cache for DynamoDB that increases performance (microsecond latency)
Backup	Point-in-time recovery down to the second in last 35 days; On-demand backup and restore

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Section 9: Amazon DynamoDB

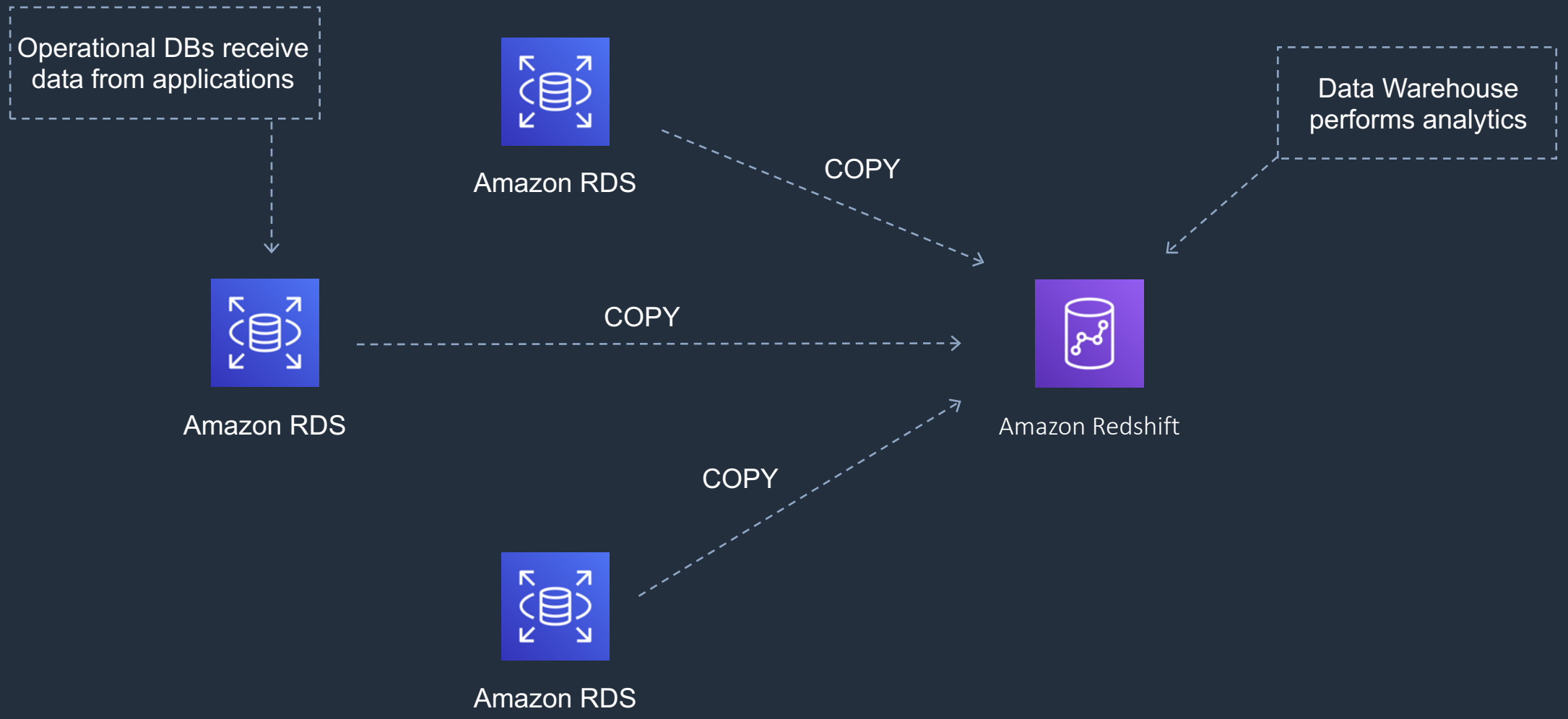


Amazon DynamoDB

- Amazon DynamoDB is a fully managed NoSQL database service that provides fast and predictable performance with seamless scalability
- Push button scaling means that you can scale the DB at any time without incurring downtime
- Data is synchronously replicated across 3 facilities (AZs) in a region
- Amazon DynamoDB global tables provides a fully managed solution for deploying a multi-region, multi-master database
- Amazon DynamoDB Accelerator (DAX) is a fully managed, highly available, in-memory cache for DynamoDB that delivers up to a 10x performance improvement

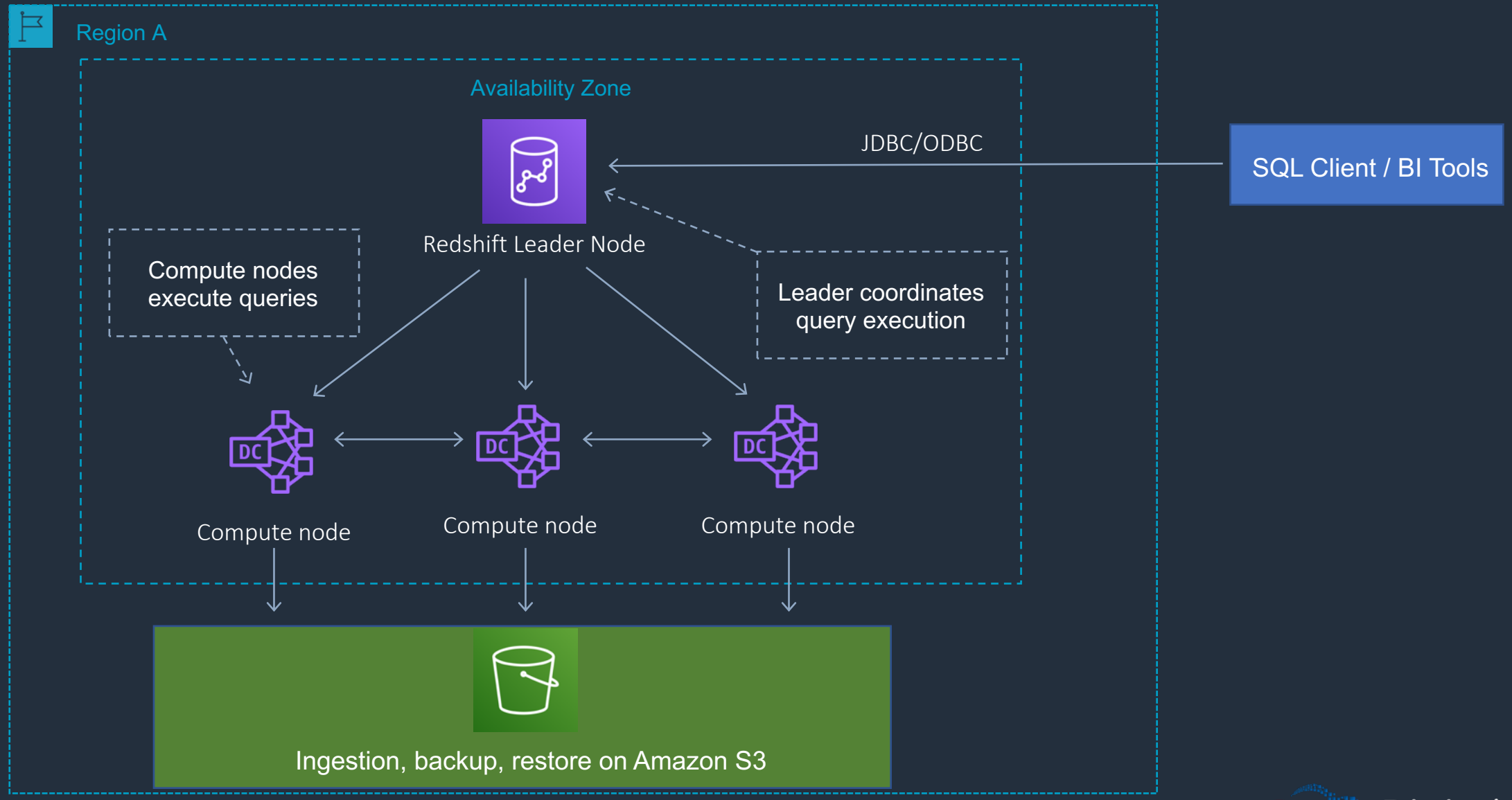
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Section 9: Amazon RedShift



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Section 9: Amazon RedShift



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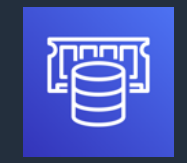
Section 9: Amazon RedShift



- Amazon Redshift is a fast, fully managed data warehouse that makes it simple and cost-effective to analyze all your data using standard SQL and existing Business Intelligence (BI) tools
- RedShift is a SQL based data warehouse used for analytics applications
- RedShift is a relational database that is used for Online Analytics Processing (OLAP) use cases
- RedShift is ideal for processing large amounts of data for business intelligence
- RedShift uses Amazon EC2 instances, so you must choose an instance family/type
- RedShift uses columnar data storage
- RedShift always keeps three copies of your data
- RedShift provides continuous/incremental backups

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Section 9: Amazon ElastiCache



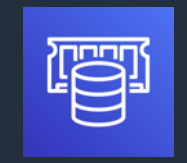
Amazon ElastiCache

- ElastiCache is a web service that makes it easy to deploy and run Memcached or Redis protocol-compliant server nodes in the cloud
- The in-memory caching provided by ElastiCache can be used to significantly improve latency and throughput for many read-heavy application workloads or compute-intensive workloads
- ElastiCache nodes run on Amazon EC2 instances, so you must choose an instance family/type

Use Case	Benefit
Web session store	In cases with load-balanced web servers, store web session information in Redis so if a server is lost, the session info is not lost, and another web server can pick it up
Database caching	Use Memcached in front of AWS RDS to cache popular queries to offload work from RDS and return results faster to users
Leaderboards	Use Redis to provide a live leaderboard for millions of users of your mobile app
Streaming data dashboards	Provide a landing spot for streaming sensor data on the factory floor, providing live real-time dashboard displays

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Section 9: Amazon ElastiCache



Amazon ElastiCache

- There are two types of ElastiCache engine:
  - Memcached – simplest model, can run large nodes with multiple cores/threads, can be scaled in and out, can cache objects such as DBs
  - Redis – complex model, supports encryption, master / slave replication, cross AZ (HA), automatic failover and backup/restore

Memcached	Redis
Simple, no-frills	**You need encryption**
You need to elasticity (scale out and in)	You need HIPAA compliance
You need to run multiple CPU cores and threads	Support for clustering
You need to cache objects (e.g. database queries)	You need complex data types
	You need HA (replication
	Pub/Sub capability

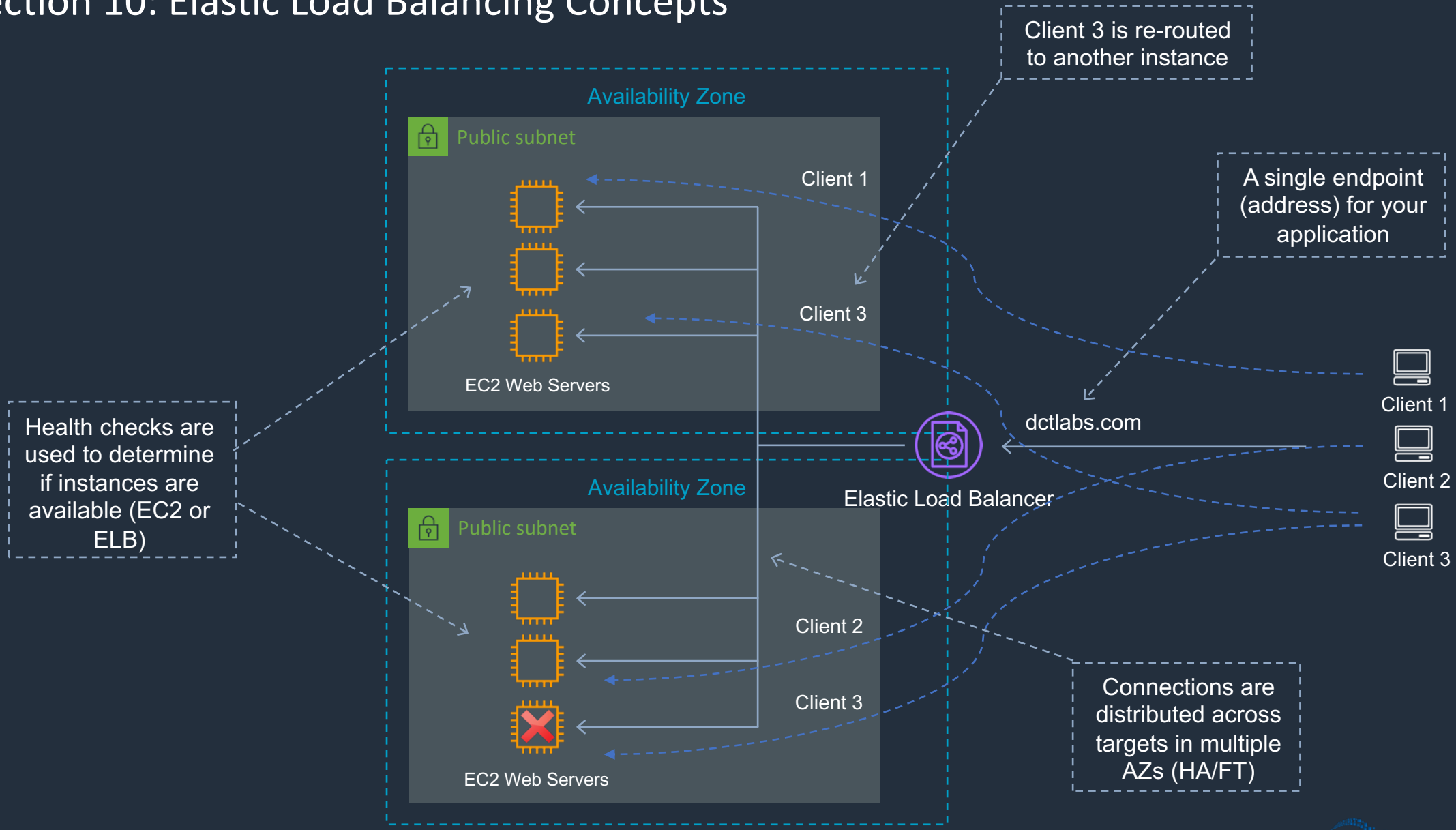
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SECTION 10

Elastic Load Balancing and Auto Scaling

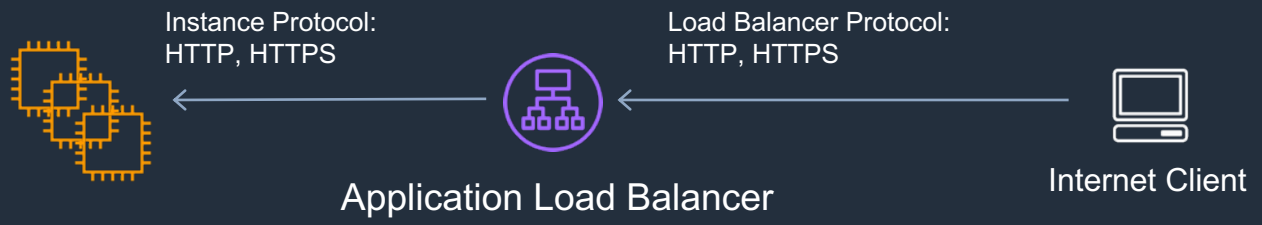


Section 10: Elastic Load Balancing Concepts



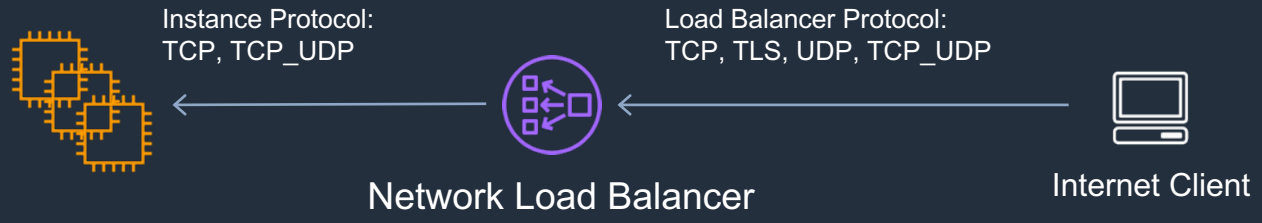
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Section 10: Elastic Load Balancing (ELB) Types



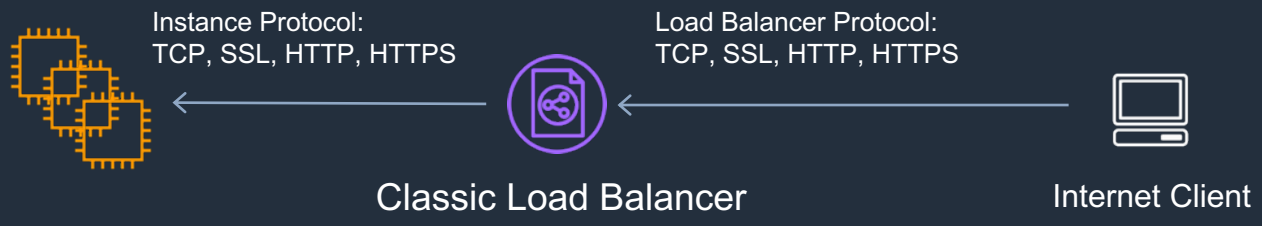
Application Load Balancer

- Operates at the request level
- Routes based on the content of the request (layer 7)
- Supports path-based routing, host-based routing, query string parameter-based routing, and source IP address-based routing
- Supports IP addresses, Lambda Functions and containers as targets



Network Load Balancer

- Operates at the connection level
- Routes connections based on IP protocol data (layer 4)
- Offers ultra high performance, low latency and TLS offloading at scale
- Can have static IP / Elastic IP
- Supports UDP and static IP addresses as targets



Classic Load Balancer

- Old generation; not recommended for new applications
- Performs routing at Layer 4 and Layer 7
- Use for existing applications running in EC2-Classic

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Section 10: Elastic Load Balancing



Elastic Load Balancing

- ELB automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, and IP addresses
- ELB can handle the varying load of your application traffic in a single Availability Zone or across multiple Availability Zones
- ELB features high availability, automatic scaling, and robust security necessary to make your applications fault tolerant
- There are three types of Elastic Load Balancer (ELB) on AWS:
  - Application Load Balancer (ALB) – layer 7 load balancer that routes connections based on the content of the request
  - Network Load Balancer (NLB) – layer 4 load balancer that routes connections based on IP protocol data
  - Classic Load Balancer (CLB) – this is the oldest of the three and provides basic load balancing at both layer 4 and layer 7

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Section 10: Elastic Load Balancing



Elastic Load Balancing

Application Load Balancer (ALB)

- ALB is best suited for load balancing of HTTP and HTTPS traffic and provides advanced request routing targeted at the delivery of modern application architectures, including microservices and containers
- Operating at the individual request level (Layer 7), Application Load Balancer routes traffic to targets within Amazon Virtual Private Cloud (Amazon VPC) based on the content of the request

Network Load Balancer (NLB)

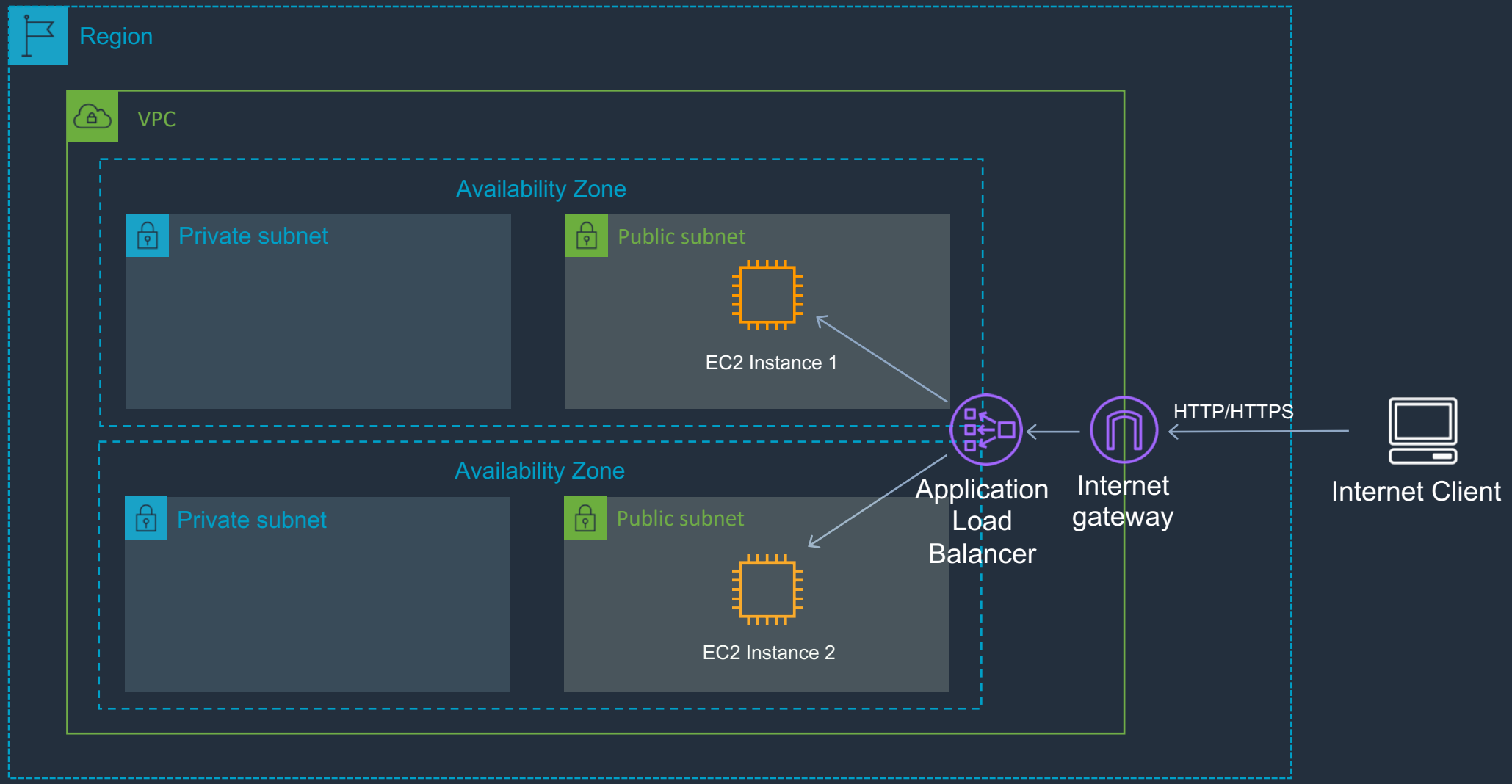
- NLB is best suited for load balancing of TCP traffic where extreme performance is required
- Operating at the connection level (Layer 4), NLB is capable of handling millions of requests per second while maintaining ultra-low latencies

Classic Load Balancer (CLB)

- CLB provides basic load balancing across multiple Amazon EC2 instances and operates at both the request level and connection level

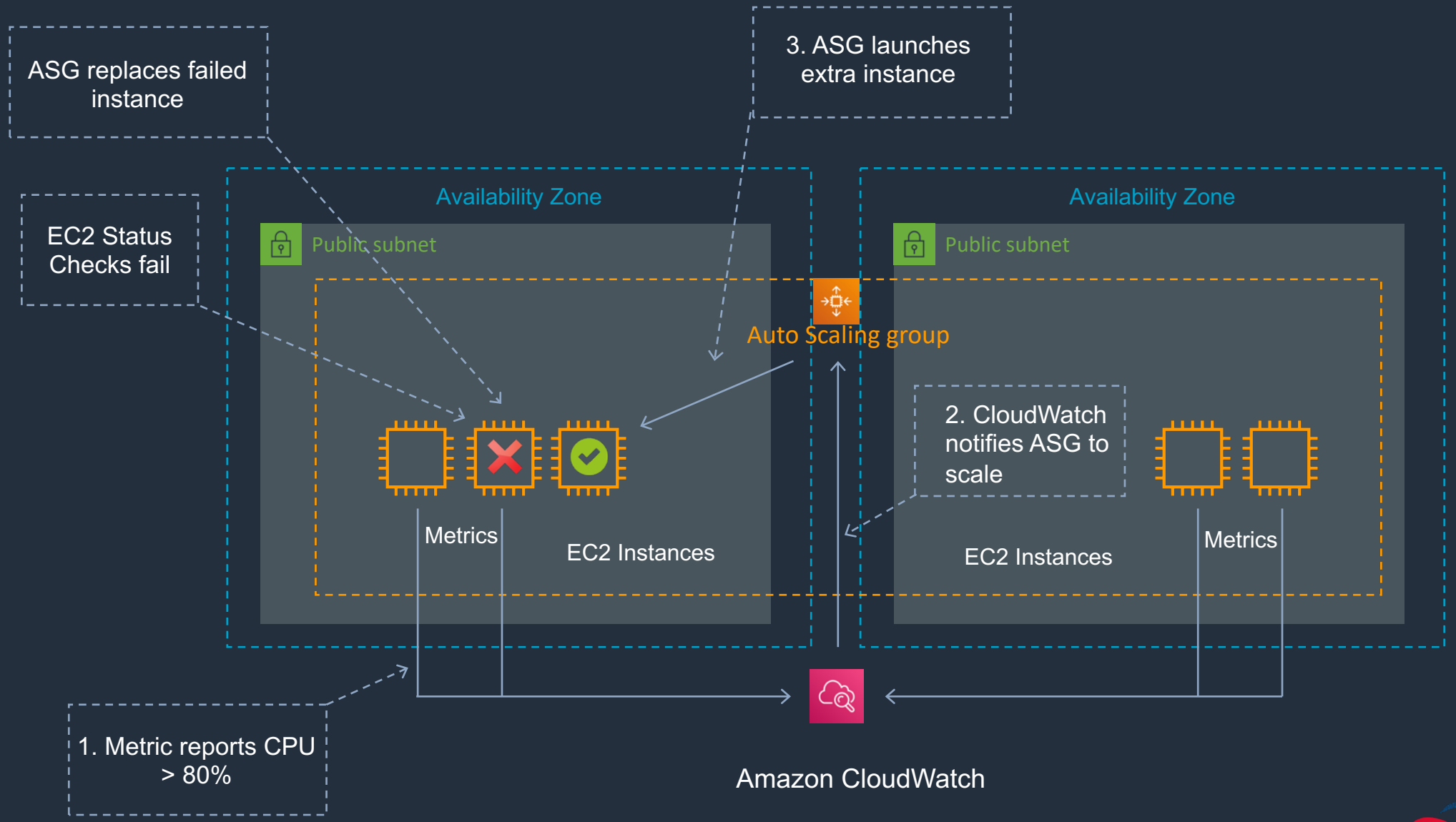
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Application Load Balancer (Internet-Facing)



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Section 10: Amazon EC2 Auto Scaling



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Section 10: Amazon EC2 Auto Scaling



- Amazon EC2 Auto Scaling provides *horizontal* scaling
- Auto Scaling provides elasticity and scalability
- AWS Auto Scaling automates the process of adding (scaling up) OR removing (scaling down) EC2 instances based on the traffic demand for your application
- Auto Scaling helps to ensure that you have the correct number of EC2 instances available to handle the application load
- You create collections of EC2 instances, called Auto Scaling Group (ASG)
- You can specify the minimum number of instances in each ASG, and AWS Auto Scaling will ensure the group never goes beneath this size
- You can also specify the maximum number of instances in each ASG and the group will never go above this size
- A desired capacity can be configured and AWS Auto Scaling will ensure the group has this number of instances

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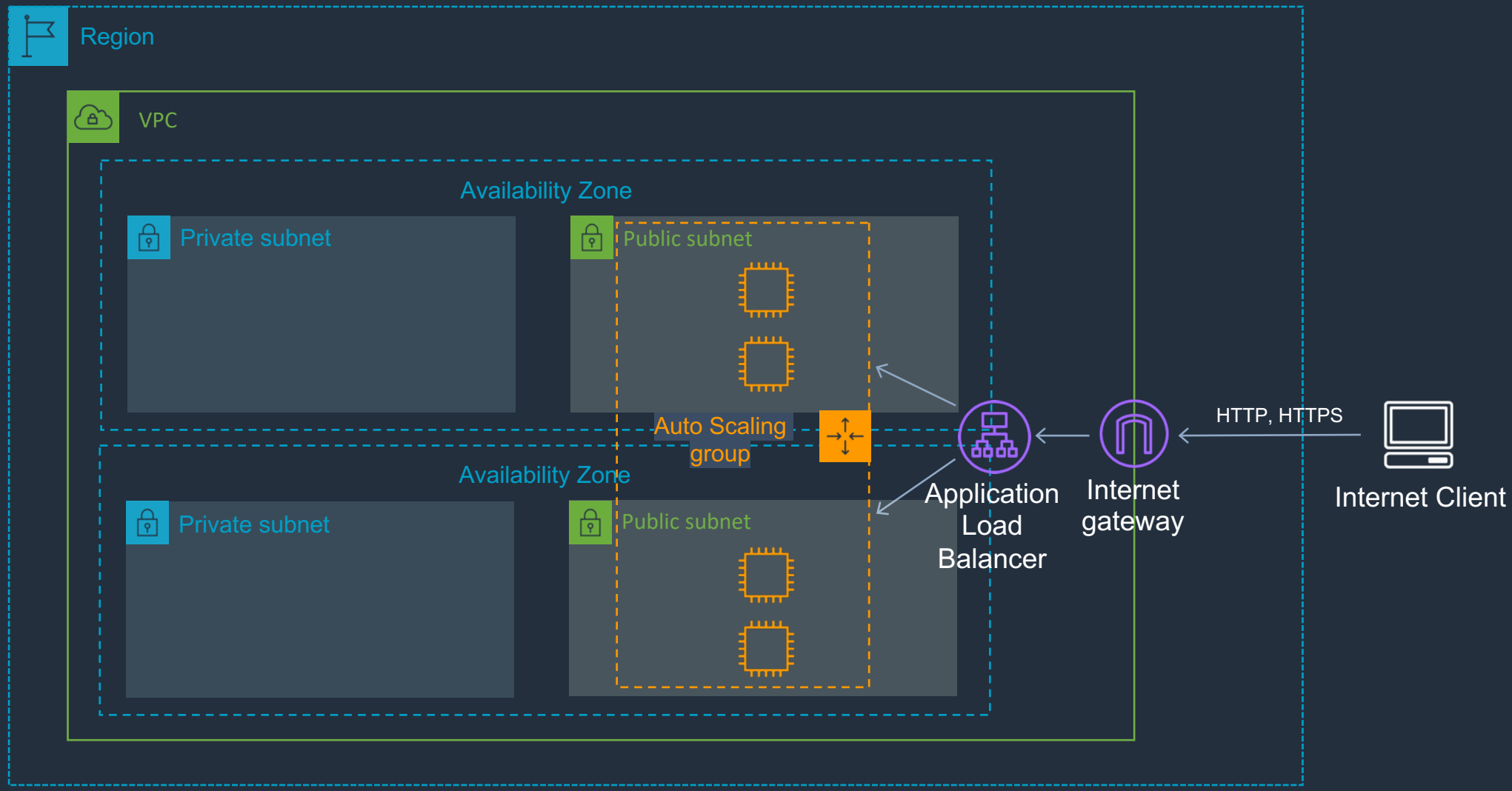
Section 10: Amazon EC2 Auto Scaling



- You can also specify scaling policies that control when Auto Scaling launches or terminates instances
- Scaling policies determine when, if, and how the ASG scales and shrinks (on-demand/dynamic scaling, cyclic/scheduled scaling)
- Scaling Plans define the triggers and when instances should be provisioned/de-provisioned
- A launch configuration is the template used to create new EC2 instances and includes parameters such as instance family, instance type, AMI, key pair and security groups

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Auto Scaling Group with ALB



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SECTION 11

Content Delivery and DNS Services



Section 11: Amazon Route 53

Amazon Route 53

Domain Registration

.net
.com
.org

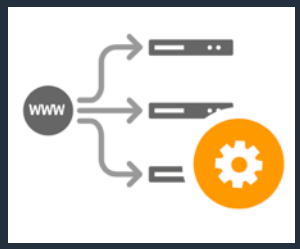
Hosted zone

example.com
dctlabs.com

Health Checks

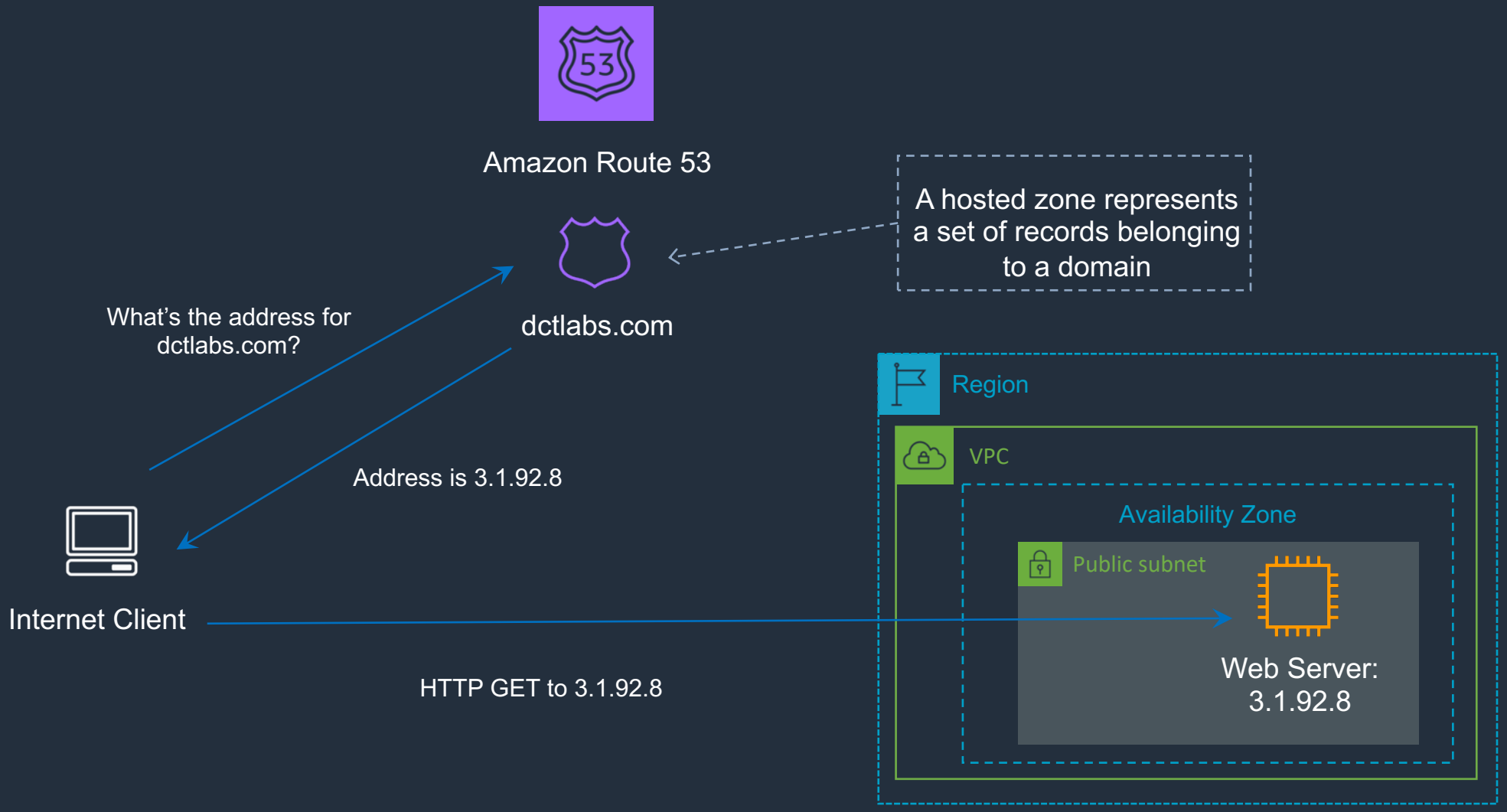
EC2 Instances

Traffic Flow



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Section 11: DNS Resolution with AWS Route 53



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Section 11: Route 53 DNS Record Types

Supported DNS records

- A (address record)
- AAAA (IPv6 address record)
- CNAME (canonical name record)
- Alias (an Amazon Route 53-specific virtual record)
- CAA (certification authority authorization)
- MX (mail exchange record)
- NAPTR (name authority pointer record)
- NS (name server record)
- PTR (pointer record)
- SOA (start of authority record)
- SPF (sender policy framework)
- SRV (service locator)
- TXT (text record)

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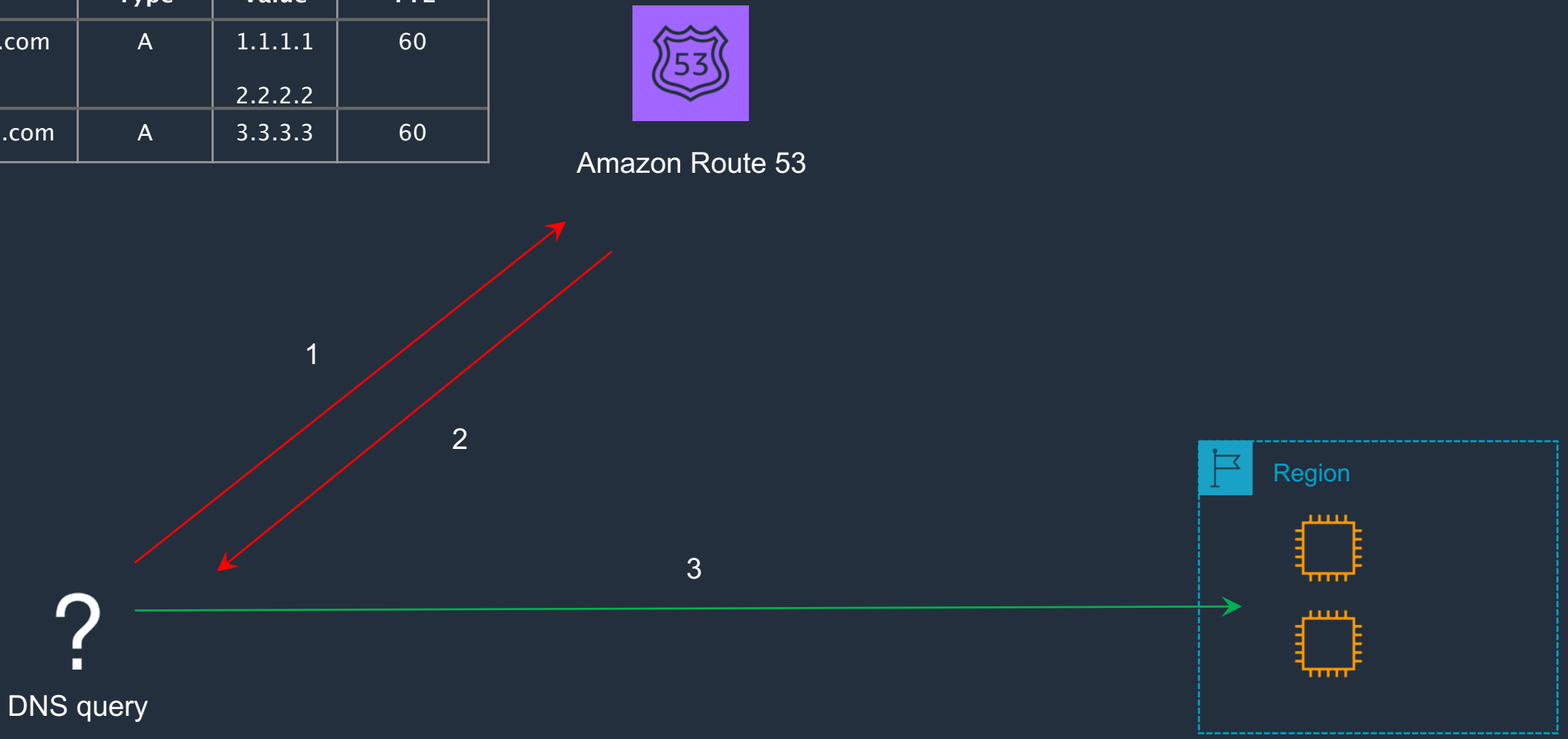
Section 11: Route 53 DNS Record Types

Routing Policy	What it does
Simple	Simple DNS response providing the IP address associated with a name
Failover	If primary is down (based on health checks), routes to secondary destination
Geolocation	Uses geographic location you're in (e.g. Europe) to route you to the closest region
Geoproximity	Routes you to the closest region within a geographic area
Latency	Directs you based on the lowest latency route to resources
Multivalue answer	Returns several IP addresses and functions as a basic load balancer
Weighted	Uses the relative weights assigned to resources to determine which to route to

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Section 11: Route 53 - Simple Routing Policy

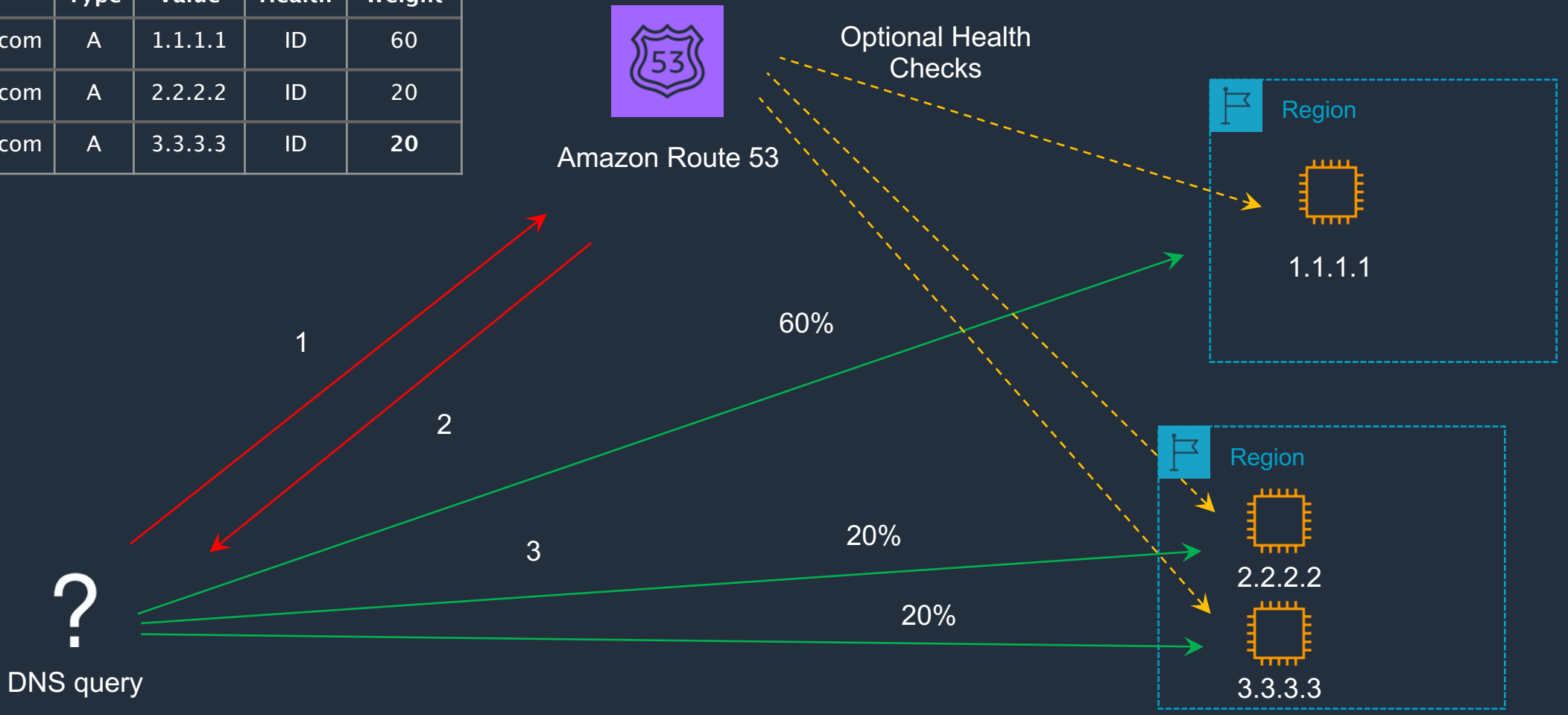
Name	Type	Value	TTL
simple.dctlabs.com	A	1.1.1.1 2.2.2.2	60
simpler.dctlabs.com	A	3.3.3.3	60



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Section 11: Route 53 - Weighted Routing Policy

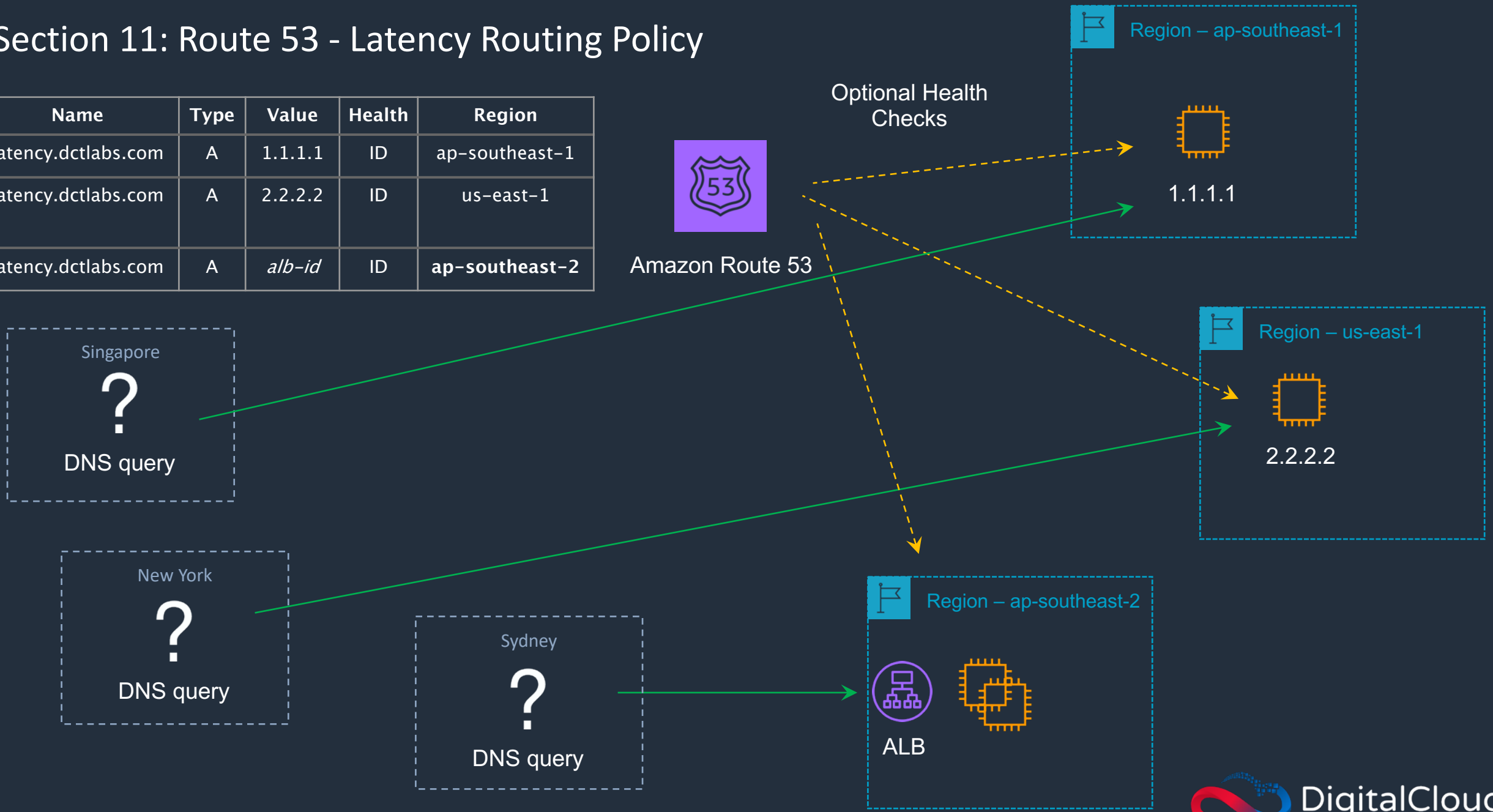
Name	Type	Value	Health	Weight
weighted.dctlabs.com	A	1.1.1.1	ID	60
weighted.dctlabs.com	A	2.2.2.2	ID	20
weighted.dctlabs.com	A	3.3.3.3	ID	**20**



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Section 11: Route 53 - Latency Routing Policy

Name	Type	Value	Health	Region
latency.dctlabs.com	A	1.1.1.1	ID	ap-southeast-1
latency.dctlabs.com	A	2.2.2.2	ID	us-east-1
latency.dctlabs.com	A	*alb-id*	ID	**ap-southeast-2**



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Section 11: Route 53 - Failover Routing Policy

Name	Type	Value	Health	Record Type
failover.dctlabs.com	A	1.1.1.1	ID	Primary
failover.dctlabs.com	A	*alb-id*		Secondary



Amazon Route 53

Health Check required on Primary

Corporate data center

Traditional server

Region – us-east-1

1.1.1.1

?

DNS query

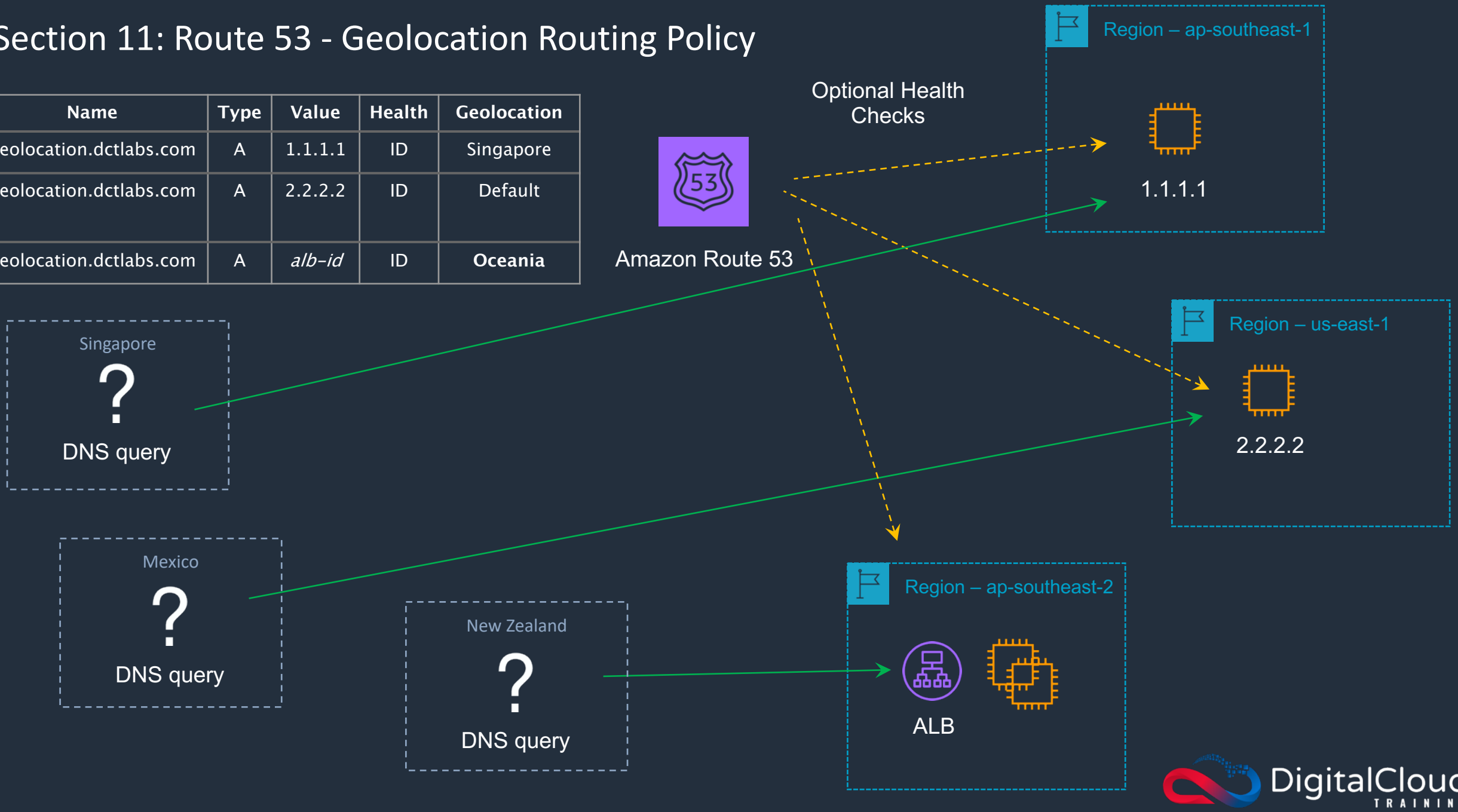
Region – ap-southeast-2

ALB

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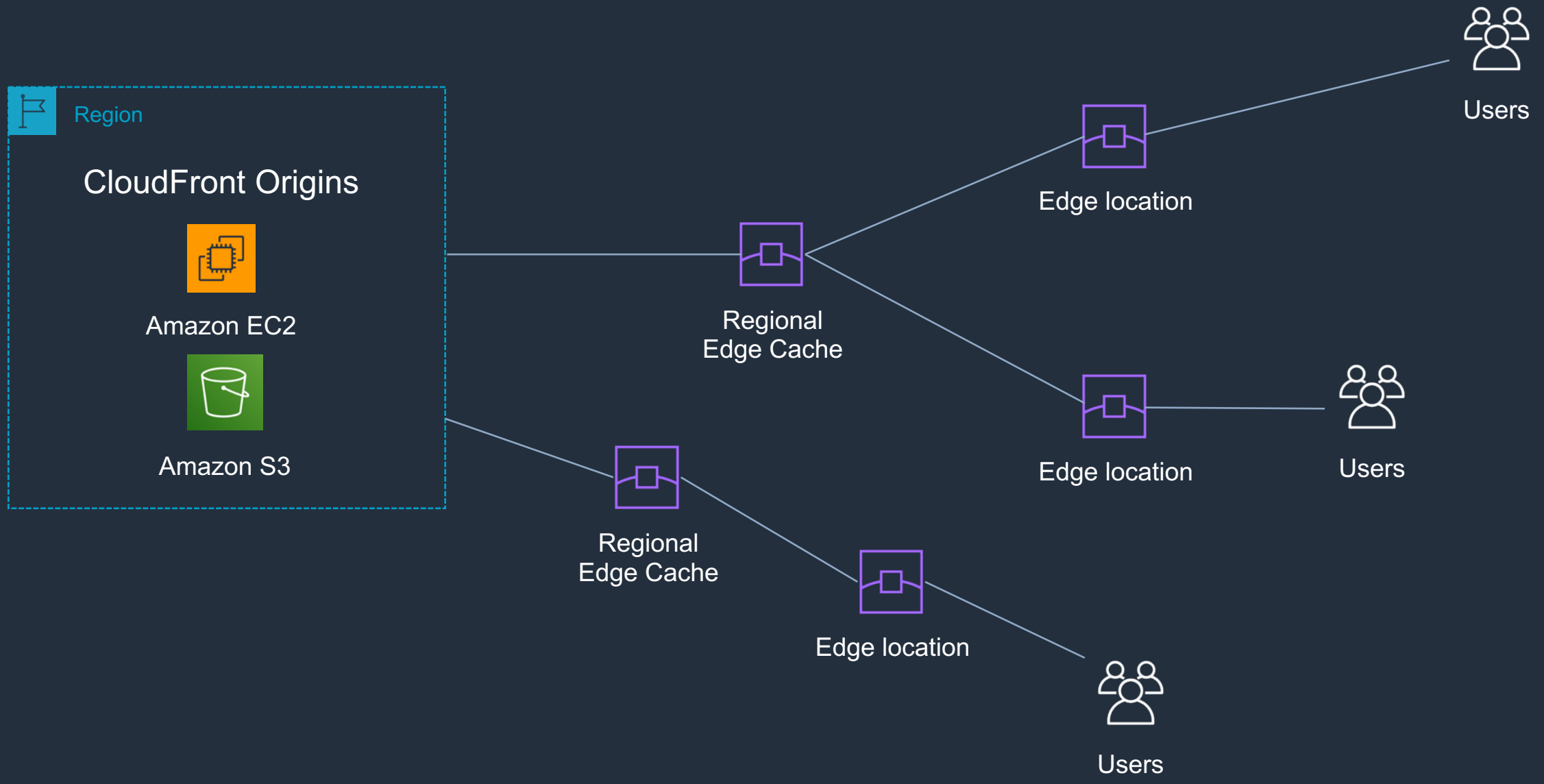
Section 11: Route 53 - Geolocation Routing Policy

Name	Type	Value	Health	Geolocation
geolocation.dctlabs.com	A	1.1.1.1	ID	Singapore
geolocation.dctlabs.com	A	2.2.2.2	ID	Default
geolocation.dctlabs.com	A	*alb-id*	ID	**Oceania**



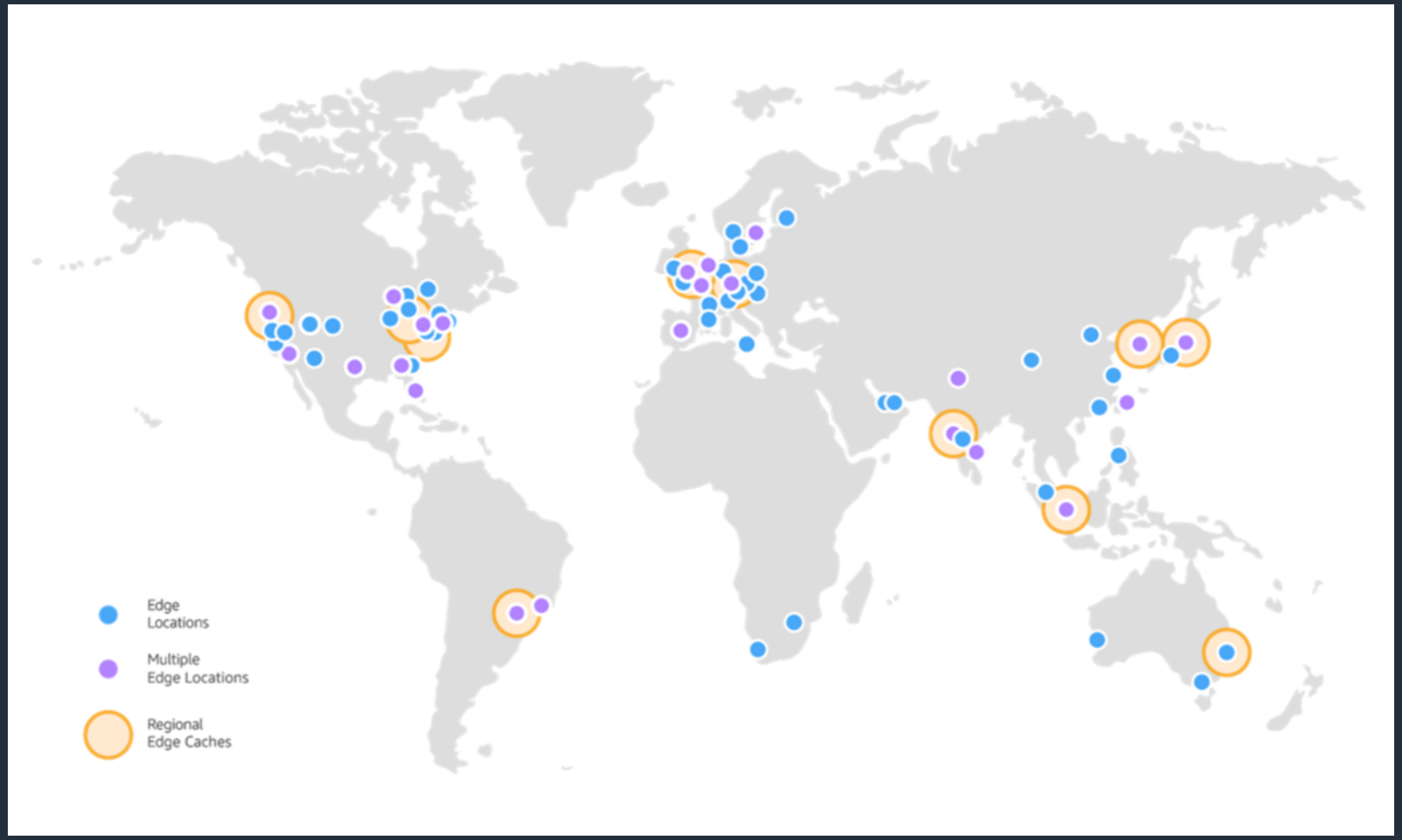
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Section 11: Amazon CloudFront Overview



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Section 11: CloudFront – Points of Presence



Points of Presence:

- 176 Edge Locations
- 11 Regional Edge Caches
- 69 cities
- 30 countries

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Section 11: Amazon CloudFront



Amazon Route 53

- CloudFront is a content delivery network (CDN) that allows you to store (cache) your content at "edge locations" located around the world
- This allows customers to access content more quickly and provides security against DDoS attacks
- CloudFront can be used for data, videos, applications, and APIs
- CloudFront benefits:
  - Cache content at Edge Location for fast distribution to customers
  - Built-in Distributed Denial of Service (DDoS) attack protection
  - Integrates with many AWS services (S3, EC2, ELB, Route 53, Lambda)

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Section 11: Amazon CloudFront



Amazon Route 53

Origins and Distributions:

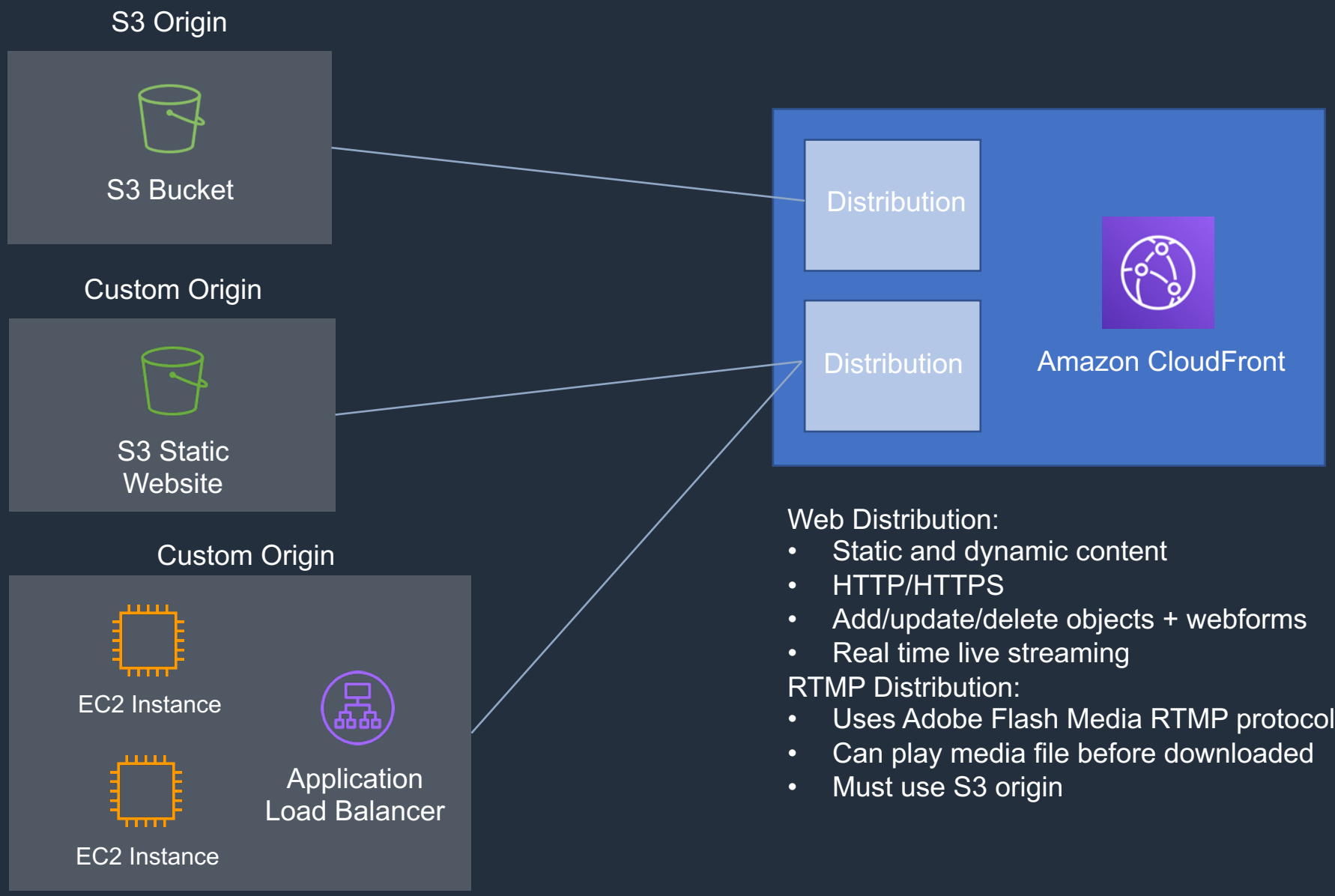
- An origin is the origin of the files that the CDN will distribute
- Origins can be either an S3 bucket, an EC2 instance, an Elastic Load Balancer, or Route 53 – can also be external (non-AWS)
- To distribute content with CloudFront you need to create a distribution
- There are two types of distribution: Web Distribution and RTMP Distribution

CloudFront uses Edge Locations and Regional Edge Caches:

- An edge location is the location where content is cached (separate to AWS regions/AZs)
- Requests are automatically routed to the nearest edge location
- Regional Edge Caches are located between origin web servers and global edge locations and have a larger cache
- Regional Edge caches aim to get content closer to users

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Section 11: CloudFront Distribution and Origins



Web Distribution:

- Static and dynamic content
- HTTP/HTTPS
- Add/update/delete objects + webforms
- Real time live streaming

RTMP Distribution:

- Uses Adobe Flash Media RTMP protocol
- Can play media file before downloaded
- Must use S3 origin

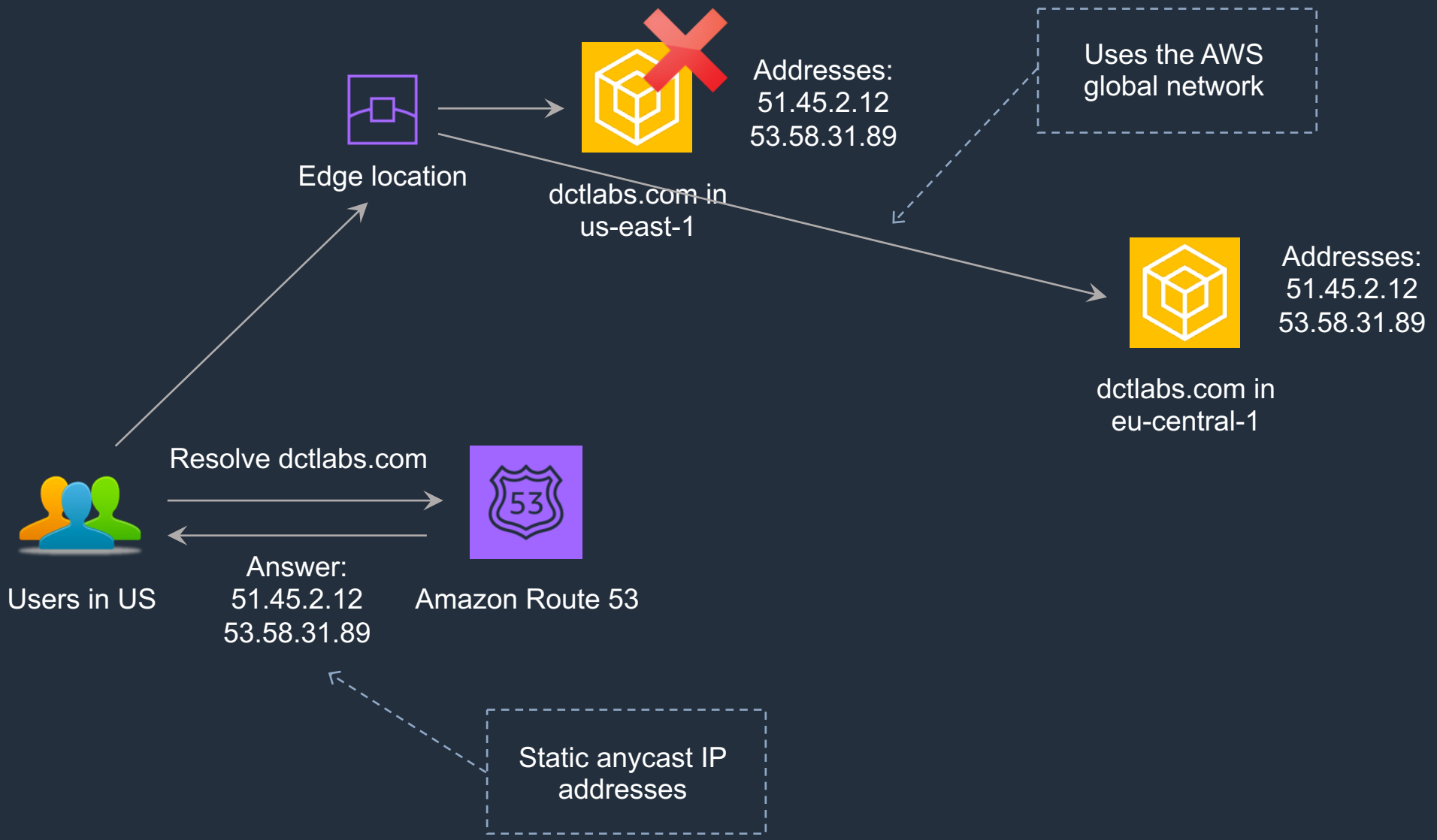
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Section 11: AWS Global Accelerator

- AWS Global Accelerator improves the availability and performance of applications with local or global users
- It provides static IP addresses that act as a fixed entry point to application endpoints in a single or multiple AWS Regions,
- Applications endpoints can be Application Load Balancers, Network Load Balancers or Amazon EC2 instances.
- AWS Global Accelerator uses the AWS global network to optimize the path from your users to your applications, improving performance.
- AWS Global Accelerator continually monitors the health of application endpoints and will detect an unhealthy endpoint and redirect traffic to healthy endpoints in less than 1 minute.

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Section 11: AWS Global Accelerator



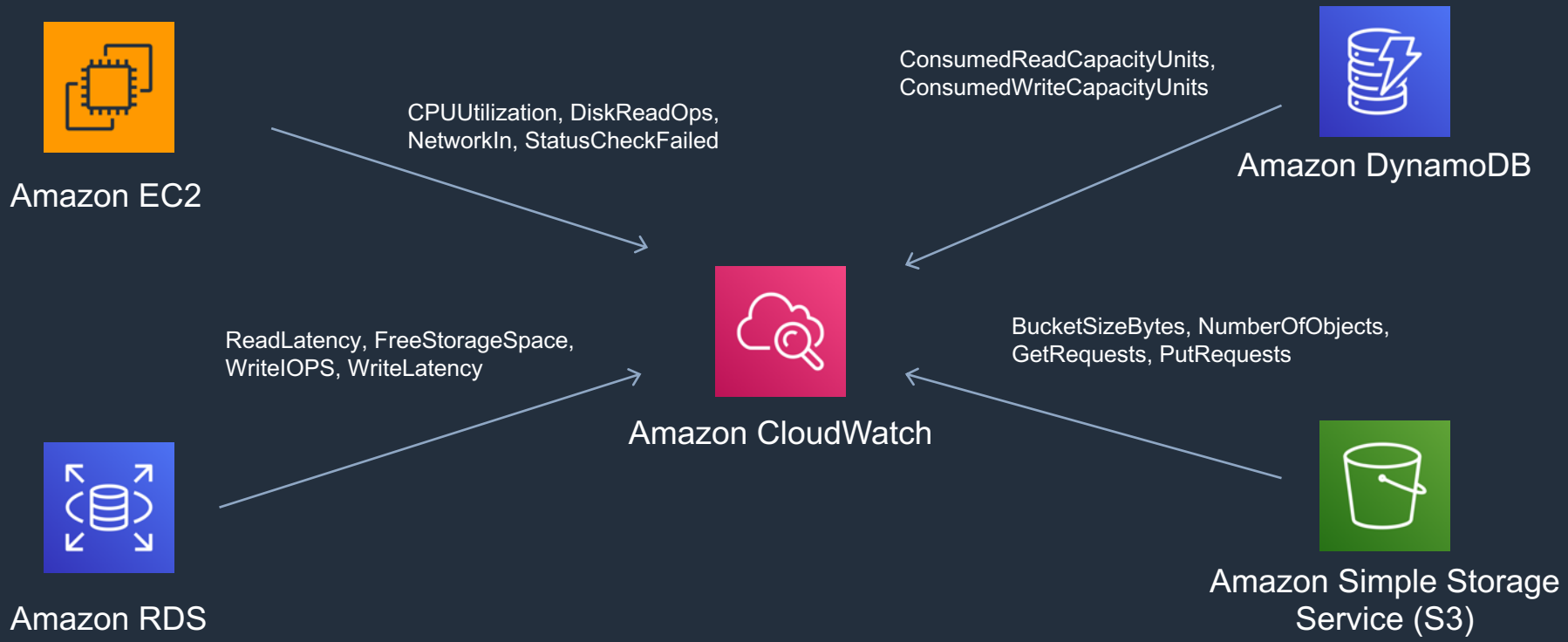
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SECTION 12

Monitoring and Logging Services



Section 12: Monitoring with Amazon CloudWatch



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Section 12: Amazon CloudWatch

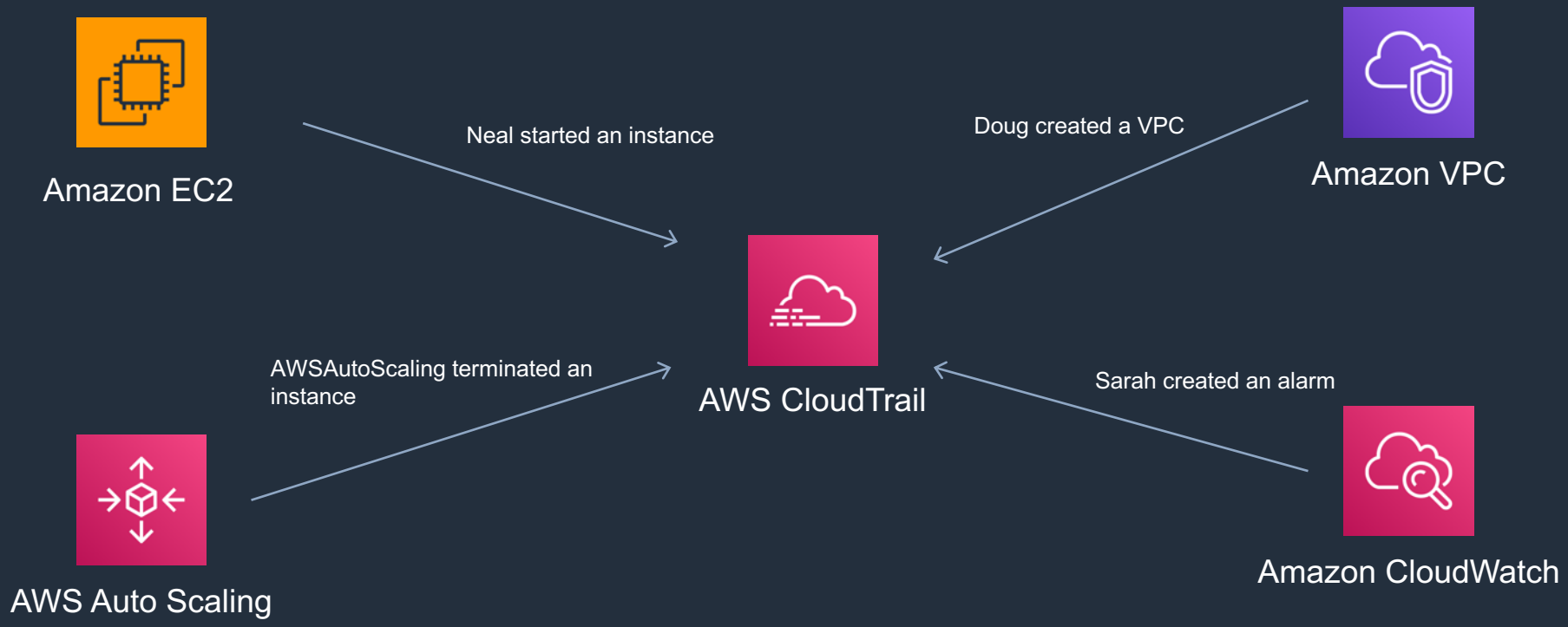


Amazon CloudWatch

- Amazon CloudWatch is a monitoring service for AWS cloud resources and the applications you run on AWS
- CloudWatch is for performance monitoring (CloudTrail is for auditing)
- Used to collect and track metrics, collect and monitor log files, and set alarms
- CloudWatch is a regional service
- CloudWatch Alarms can be set to react to changes in your resources
- CloudWatch Events generates events when resource states change and delivers them to targets for processing
- CloudWatch Logs collects and centralizes logs from AWS resources
- Any log files generated by your applications
- Gain system-wide visibility into resource utilization
- CloudWatch monitoring includes application performance

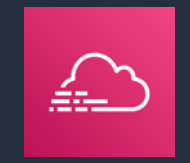
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Section 12: Auditing with Amazon CloudTrail



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Section 12: Amazon CloudTrail



AWS CloudTrail

- AWS CloudTrail is a web service that records activity made on your account and delivers log files to an Amazon S3 bucket
- CloudTrail is for auditing (CloudWatch is for performance monitoring)
- CloudTrail is about logging and saves a history of API calls for your AWS account
- Provides visibility into user activity by recording actions taken on your account
- API history enables security analysis, resource change tracking, and compliance auditing
- Logs API calls made via:
  - AWS Management Console
  - AWS SDKs
  - Command line tools
  - Higher-level AWS services (such as CloudFormation)

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SECTION 13

Automation and Platform Services

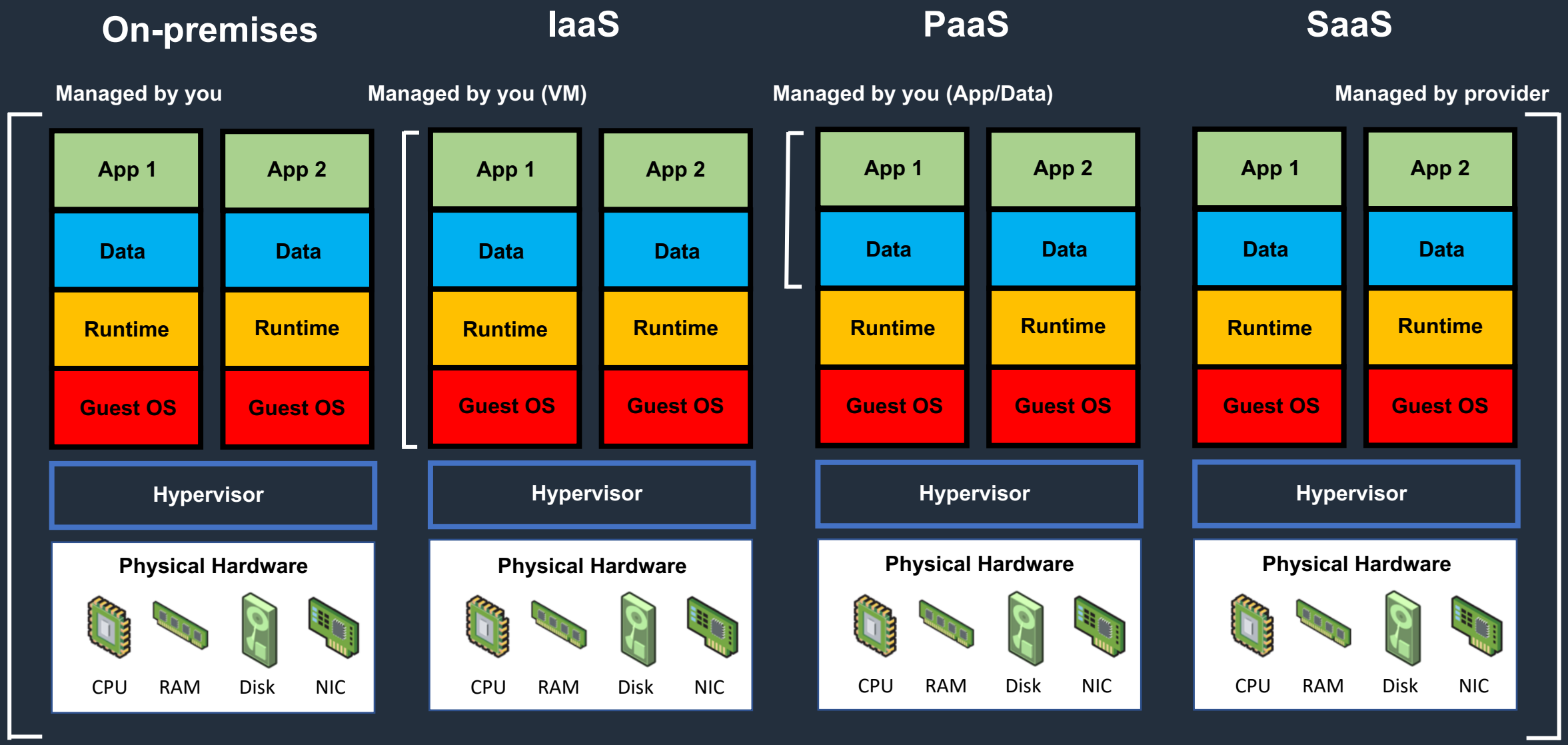


Section 13: Automation and Platform Services

CloudFormation	Elastic Beanstalk
“Template-driven provisioning”	“Web apps made easy"
Deploys infrastructure using code	Deploys applications on EC2 (PaaS)
Can be used to deploy almost any AWS service	Deploys web applications based on Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker
Uses JSON or YAML template files	Uses ZIP or WAR files (or Git)
CloudFormation can deploy Elastic Beanstalk environments	Elastic Beanstalk cannot deploy using CloudFormation
Similar to Terraform	Similar to Google App Engine

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Section 13: Cloud Computing Service Models



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Section 13: AWS CloudFormation

- AWS CloudFormation provides a common language for you to describe and provision all the infrastructure resources in your cloud environment
- CloudFormation can be used to provision a broad range of AWS resources
- Think of CloudFormation as deploying infrastructure as code



template1

Choose template lang

```
{
"AWSTemplateFormatVersion": "2010-09-09",
"Description": "AWS CloudFormation Sample Template LAMP_Single_Instance: Create a LAMP stacl
"Parameters": {
    "KeyName": {
        "Description": "Name of an existing EC2 KeyPair to enable SSH access to the instanc
        "Type": "AWS::EC2::KeyPair::KeyName",
        "ConstraintDescription": "must be the name of an existing EC2 KeyPair."
    },
    "DBName": {
        "Default": "MyDatabase",
        "Description": "MySQL database name",
        "Type": "String",
        "MinLength": "1",
        "MaxLength": "64",
        "AllowedPattern": "[a-zA-Z][a-zA-Z0-9]*",
        "ConstraintDescription": "must begin with a letter and contain only alphanumeric characters."
    },
    "DBUser": {
        "NoEcho": "true",
```

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Section 13: AWS CloudFormation



AWS CloudFormation

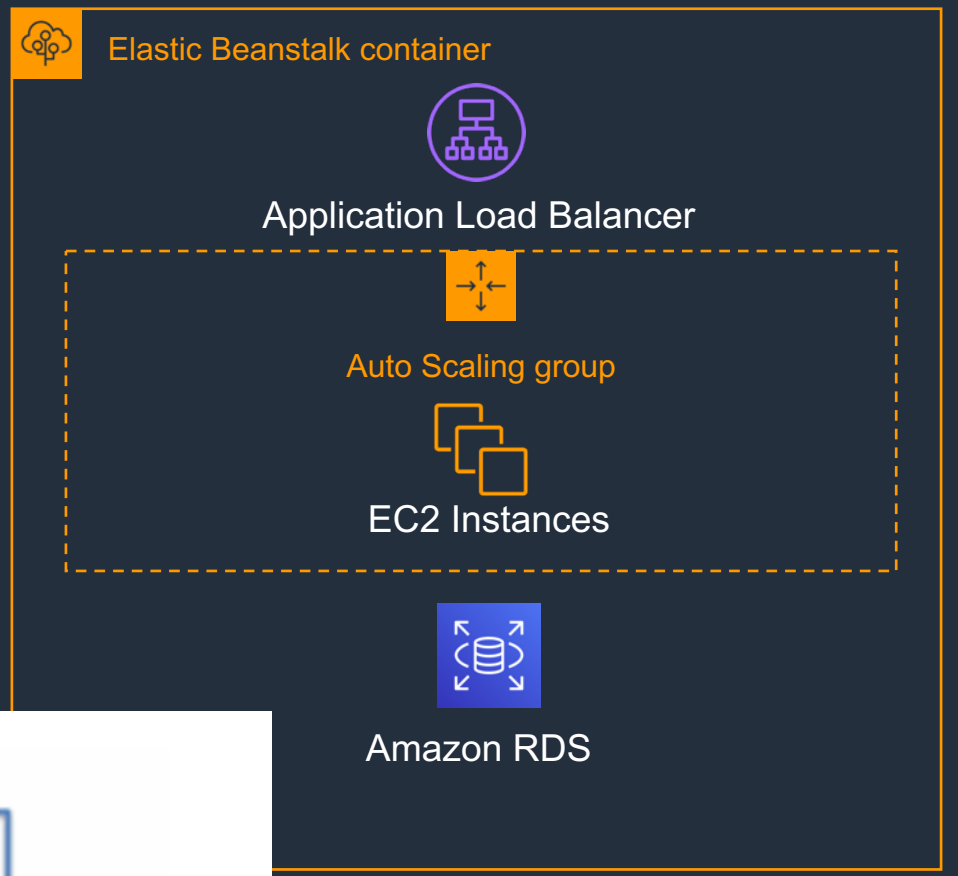
- Think of CloudFormation as deploying infrastructure as code
- Elastic Beanstalk is more focussed on deploying applications on EC2 (PaaS)
- CloudFormation can deploy Elastic Beanstalk-hosted applications however the reverse is not possible

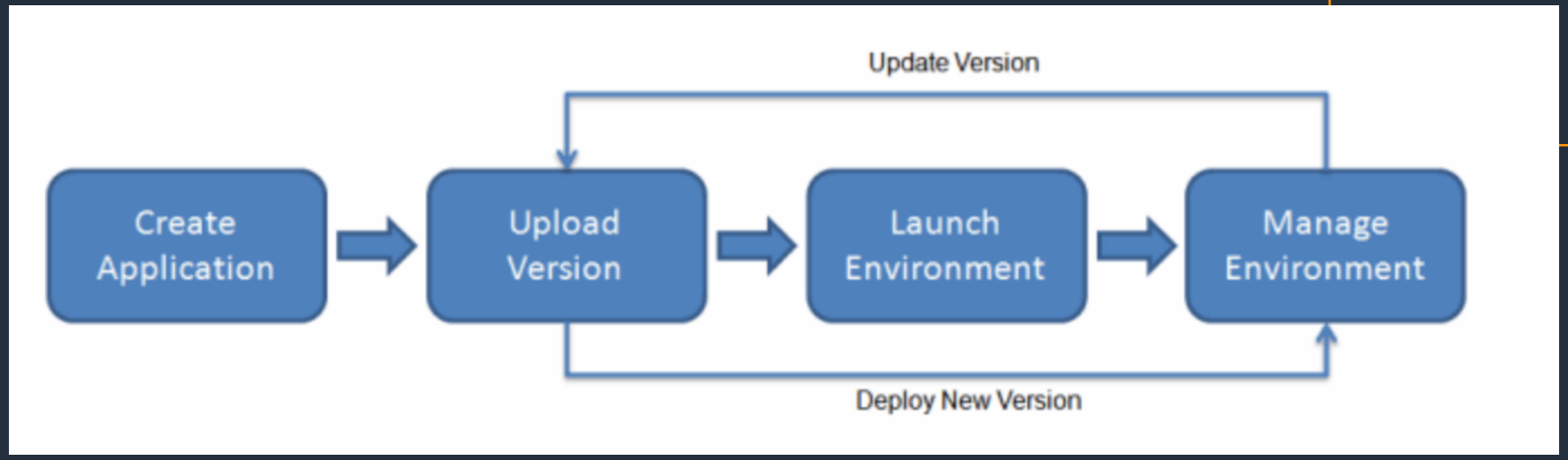
Component	Description
Templates	The JSON or YAML text file that contains the instructions for building out the AWS environment
Stacks	The entire environment described by the template and created, updated, and deleted as a single unit
Change Sets	A summary of proposed changes to your stack that will allow you to see how those changes might impact your existing resources before implementing them

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Section 13: AWS Elastic Beanstalk

- AWS Elastic Beanstalk can be used to quickly deploy and manage applications in the AWS Cloud
- Developers upload applications and Elastic Beanstalk handles the deployment details of capacity provisioning, load balancing, auto-scaling, and application health monitoring
- Considered a Platform as a Service (PaaS) solution





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Section 13: AWS Elastic Beanstalk



AWS Elastic Beanstalk

- Supports Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker web applications
- Integrates with VPC
- Integrates with IAM
- Can provision most database instances
- Allows full control of the underlying resources
- Code is deployed using a WAR file or Git repository

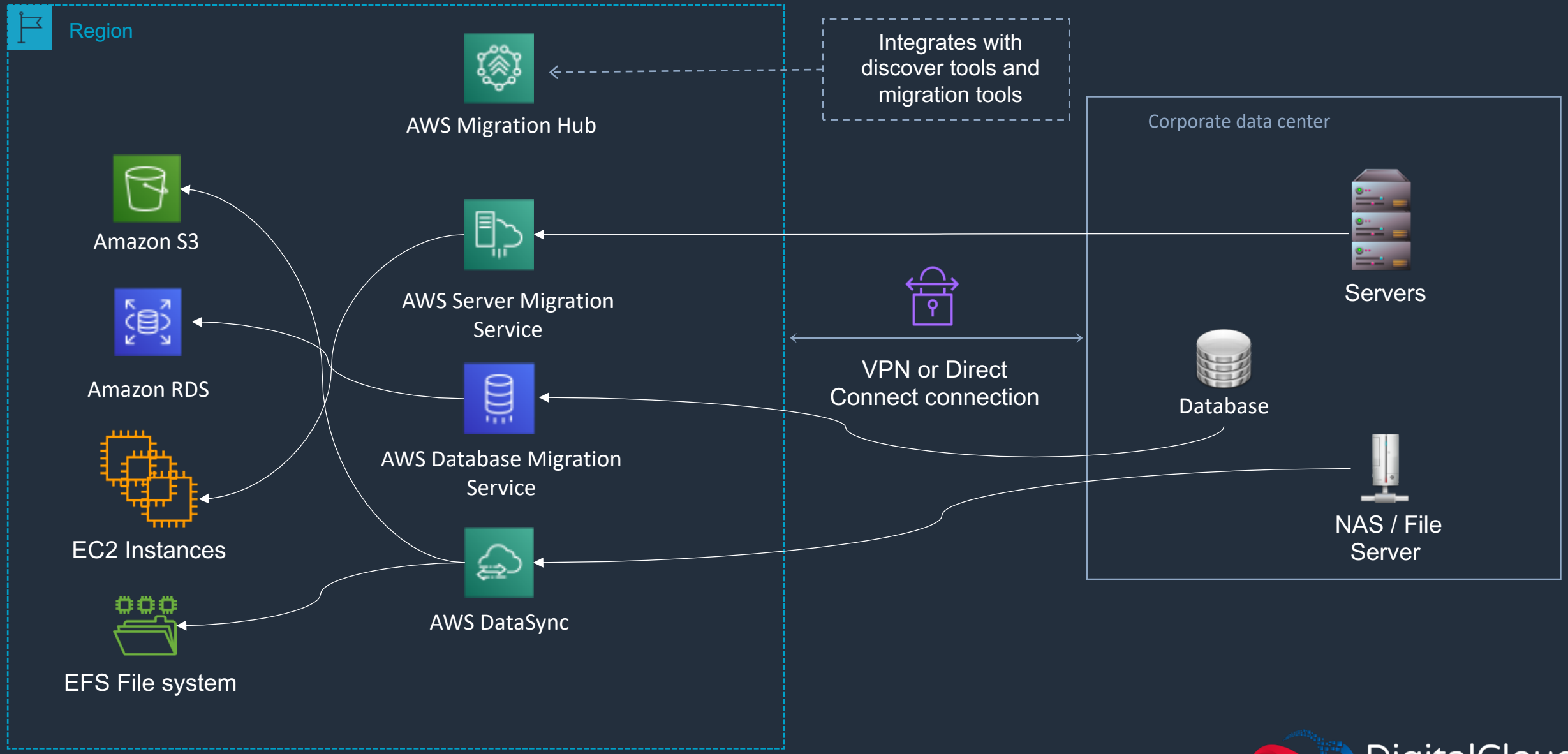
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SECTION 14

Migration and Transfer Services

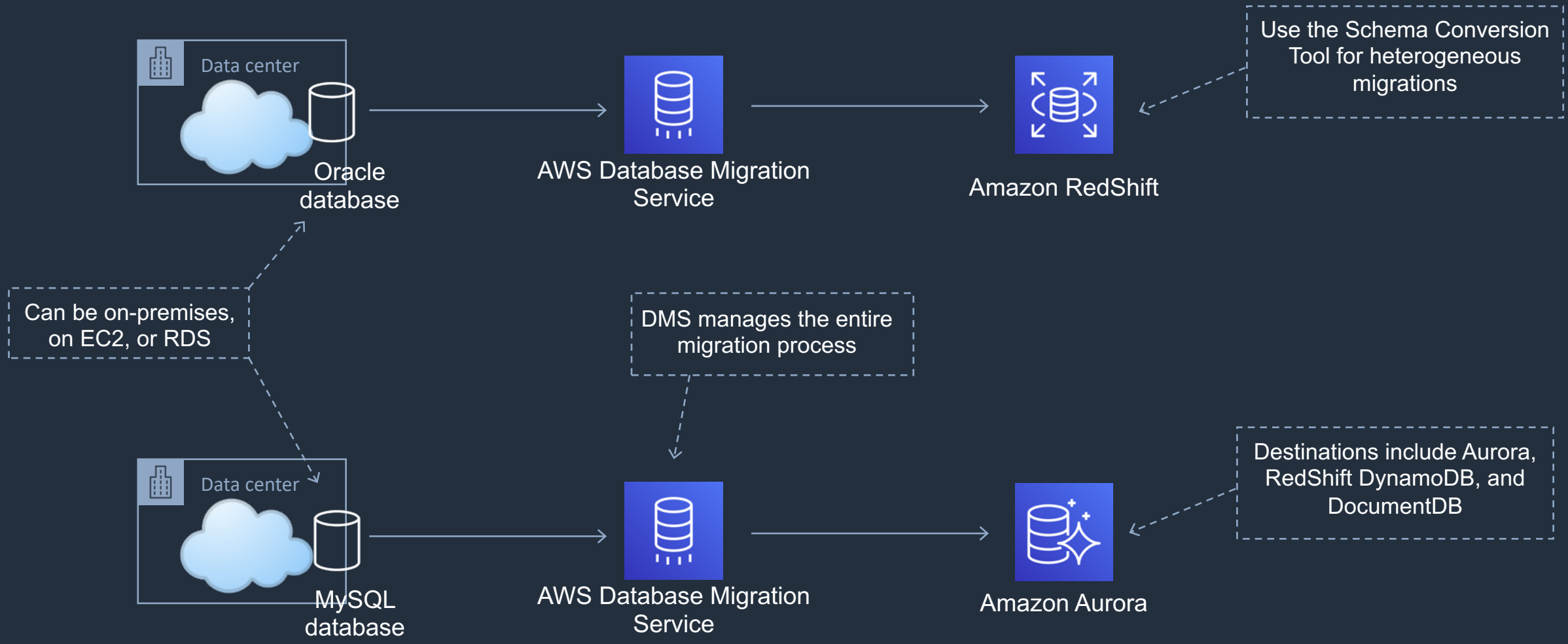


Migration and Transfer



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AWS Database Migration Service (DMS)



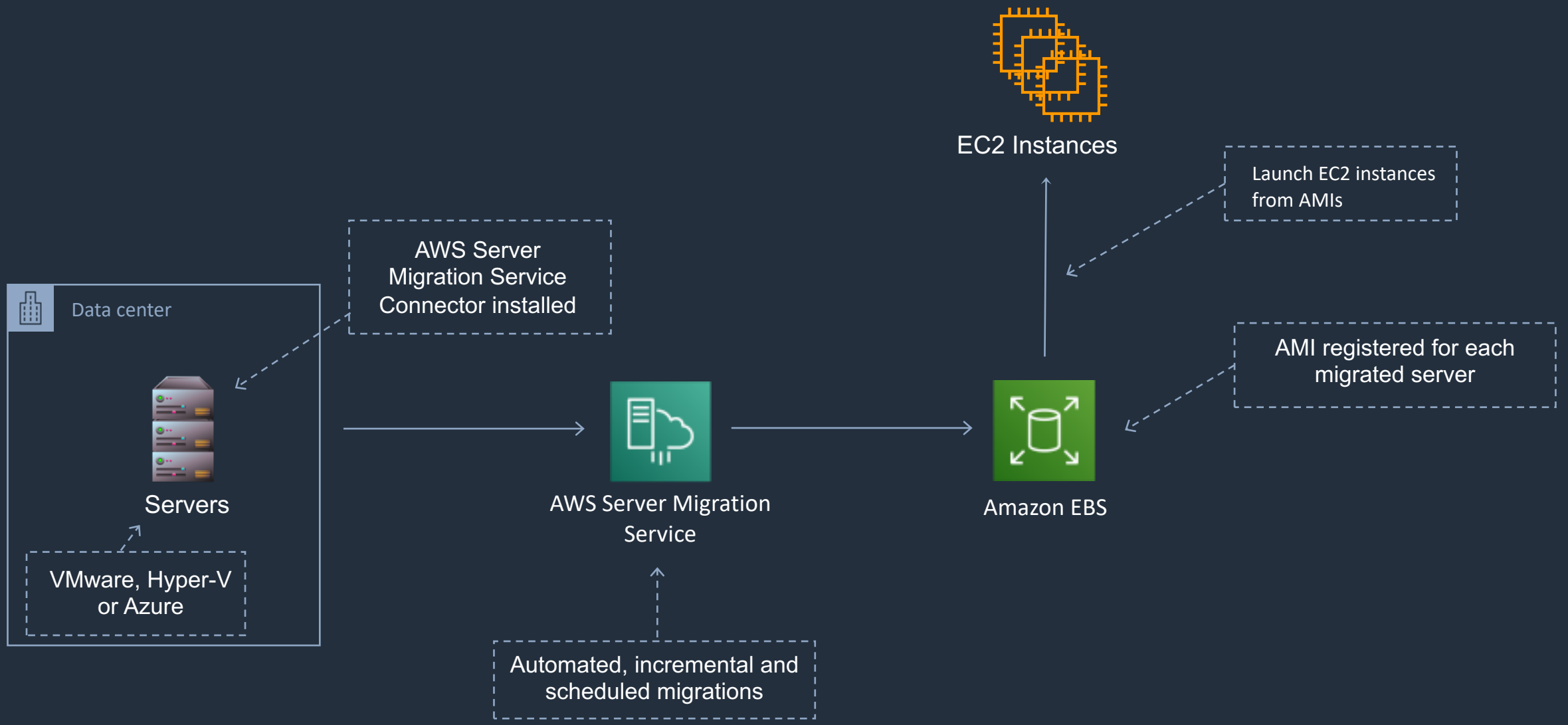
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Database Migration Service (DMS)

- Use to migrate databases from on-premises, Amazon EC2 or Amazon RDS.
- Supports homogenous (e.g. Oracle to Oracle) as well as heterogenous (e.g. Oracle to Amazon Aurora).
- Data is continuously replicated while the application is live, minimizing downtime.
- Pay based on compute resources used during the migration and log storage.
- Fully managed migration process.

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AWS Database Server Migration Service (SMS)



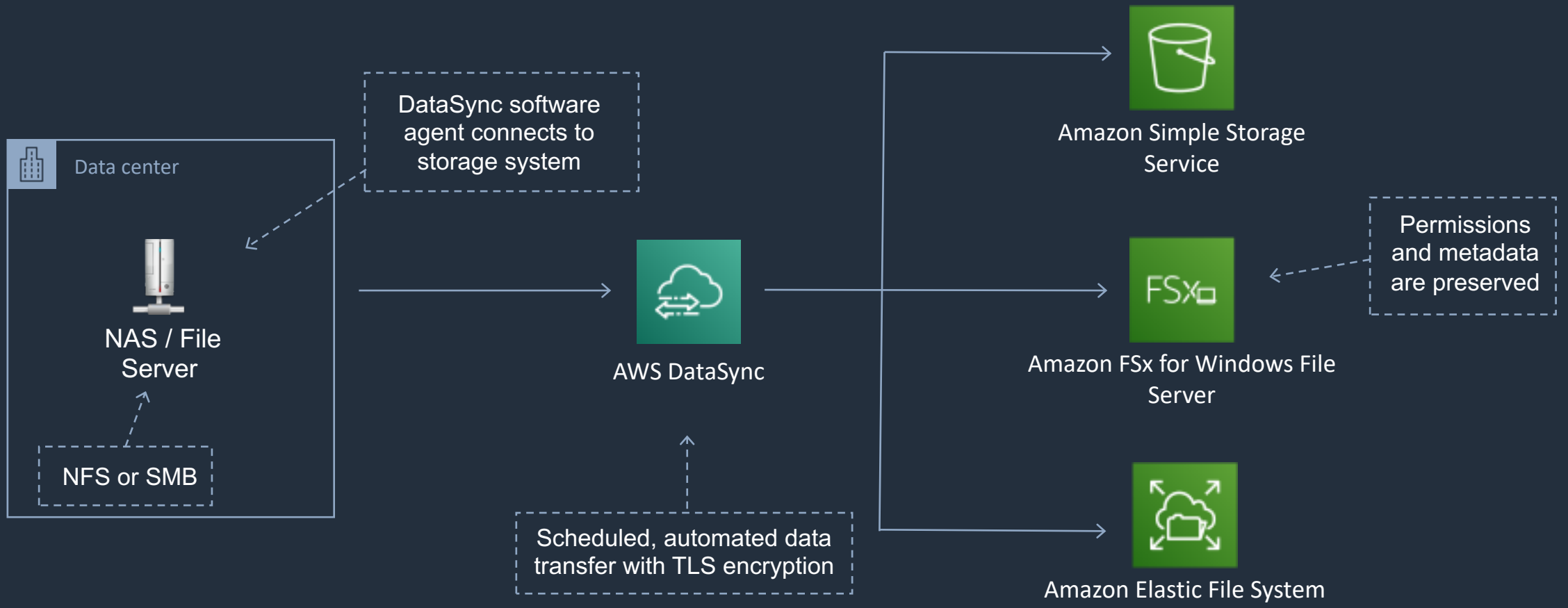
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Server Migration Service (SMS)

- Agentless service for migrating on-premises and cloud-based VMs to AWS.
- Source platforms can be VMware, Hyper-V or Azure.
- AWS Server Migration Service Connector is installed on the source platform.
- Server volumes are replicated (encrypted with TLS) and saved as AMIs which can then be launched as EC2 instances.
- Can use application groupings and SMS will launch servers in a CloudFormation stack.
- You can replicate your on-premises servers to AWS for up to 90 days (per server).
- Provides automated, live incremental server replication and AWS Console support.

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AWS DataSync



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AWS DataSync

- DataSync software agent connects to on-premises NAS storage systems.
- The NAS can use NFS or SMB protocols.
- Synchronizes data into AWS using a Scheduled, automated data transfer with TLS encryption.
- Destination can be Amazon S3, Amazon EFS or Amazon FSx for Windows File Server.
- Can improve performance for data transfers up to 10x faster than traditional tooling.
- Permissions and metadata are preserved.
- Pay per-GB transferred.

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AWS Snowball and Snowmobile



AWS Snowball



AWS Snowmobile

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AWS Snowball and Snowmobile



AWS Snowball

- With AWS Snowball (Snowball), you can transfer hundreds of terabytes or petabytes of data between your on-premises data centers and Amazon Simple Storage Service
- Uses a secure storage device for physical transportation
- AWS Snowball Client is software that is installed on a local computer and is used to identify, compress, encrypt, and transfer data
- Uses 256-bit encryption (managed with the AWS KMS) and tamper-resistant enclosures with TPM
- Snowball (80TB) (50TB) “petabyte scale”
- Snowball Edge (100TB) “petabyte scale” comes with onboard storage and compute capabilities
- Snowmobile – “exabyte scale” with up to 100PB per Snowmobile

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AWS Snowball Edge



AWS Snowball

- AWS Snowball Edge is a data migration and edge computing device that comes in two options:

1. **Snowball Edge Storage Optimized devi**ces provide both block storage and Amazon S3-compatible object storage

Use for: local storage and large scale-data transfer.

1. **Snowball Edge Compute Optimized devices** provide 52 vCPUs, block and object storage, and an optional GPU for use cases like advanced machine learning and full motion video analysis in disconnected environments.

Use for: machine learning and processing, and storage in environments with intermittent connectivity (like manufacturing, industrial, and transportation) or in extremely remote locations (like military or maritime operations) before shipping them back to AWS.

DigitalCloud TRAINING

SECTION 15

AWS Billing and Pricing



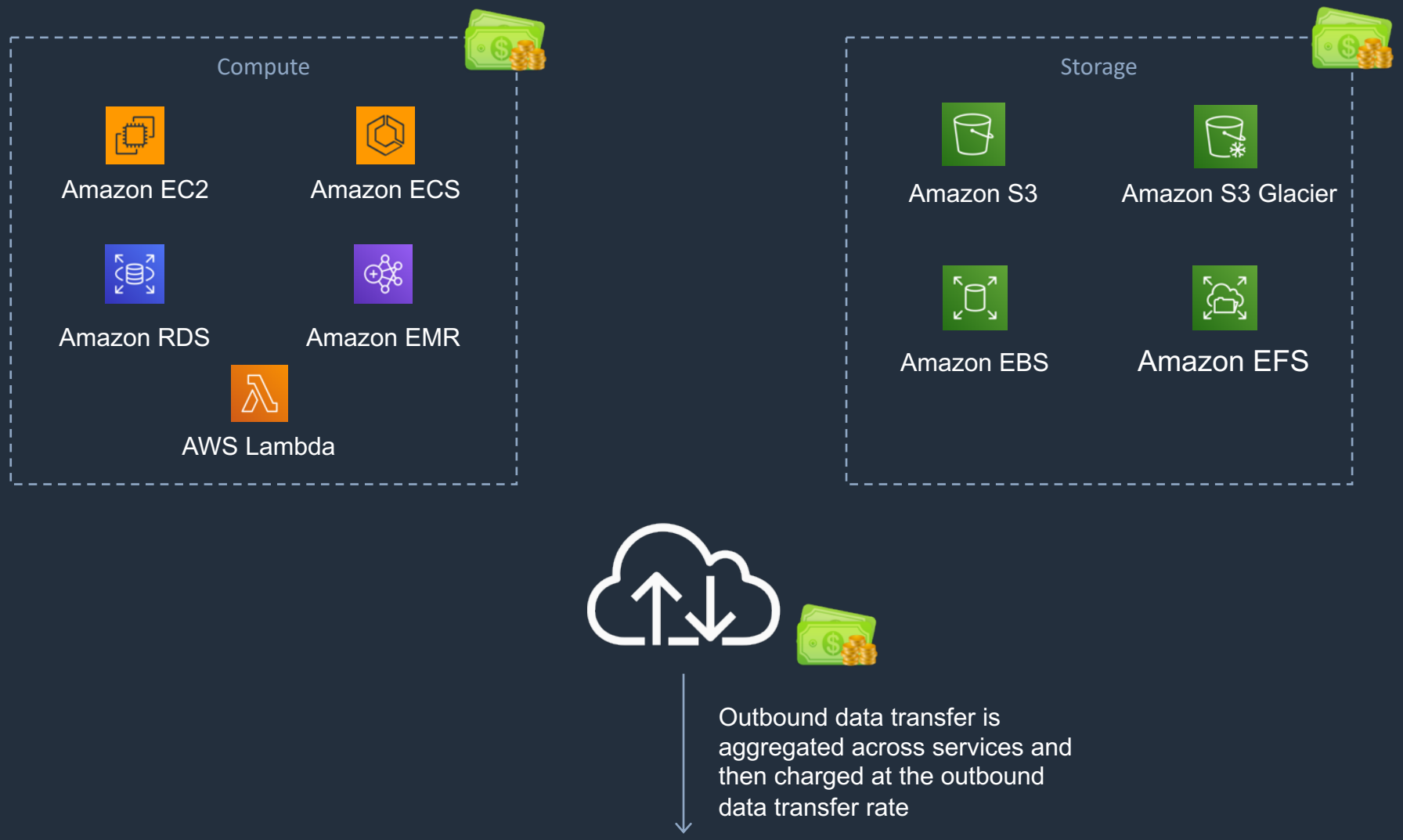
Section 15: Pricing: What you need to know



- You need to know:
  - What you get for free
  - What you get charged for
  - How you are charged – e.g. per GB, per instance hour etc.
  - Special payment options - .e.g. reservations
- You don’t need to know:
  - Exactly how much you get charged

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Section 15: Three Fundamentals of AWS Pricing



DigitalCloud TRAINING

Section 15: AWS Pricing Models



- On-demand
  - Used for Compute and Database capacity
  - No long-term commitments or upfront payments
- Dedicated Instances
  - Available for Amazon EC2
  - Hardware is dedicated to a single customer
- Spot Instances
  - Purchase spare capacity with no commitments
  - Great discounts from hourly rates
- Reservations
  - Up to 75% discount in exchange a term commitment

DigitalCloud TRAINING

Section 15: Reservations



- Options for 1 or 3 year term
- Options to pay:
  - No upfront
  - Partial upfront
  - All upfront
- Available for these services:
  - Amazon EC2 Reserved Instances
  - Amazon DynamoDB Reserved Capacity
  - Amazon ElastiCache Reserved Nodes
  - Amazon Relational Database Service (RDS) Reserved Instances
  - Amazon RedShift Reserved Nodes

DigitalCloud TRAINING

Section 15: Reservations



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  - Amazon RedShift Reserved Nodes

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Section 15: Free Stuff



- Amazon VPC
- Elastic Beanstalk (but not the resources created)
- CloudFormation (but not the resources created)
- Identity Access Management (IAM)
- Auto Scaling (but not the resources created)
- Consolidated Billing

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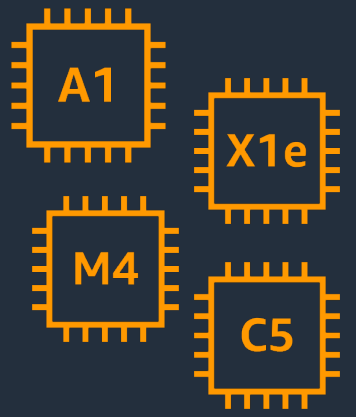
Section 15: Compute Pricing



Amazon EC2 Pricing Components:

- Clock hours of server uptime
- Instance configuration
- Instance type
- Number of instances
- Load balancing
- Detailed monitoring (CloudWatch)
- Auto Scaling (resources created)
- Elastic IP addresses (charged if allocated but not used)
- Operating systems and software packages





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Section 15: Compute Pricing



Amazon EC2 Pricing Models

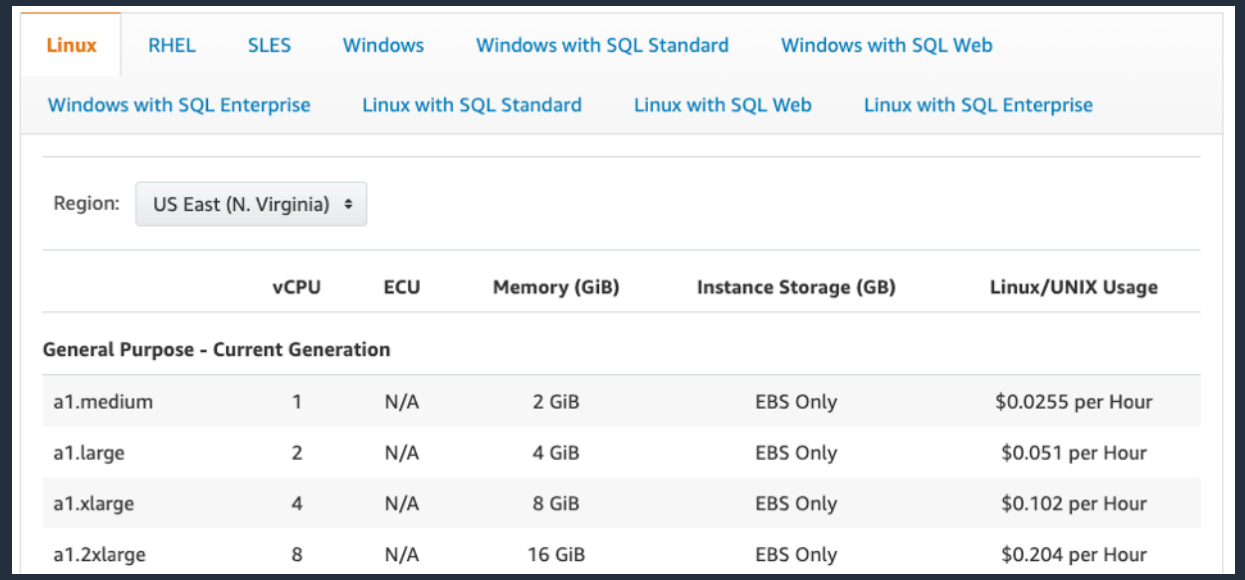
- 5 Pricing models available:
  - On-demand
  - Reserved Instances
  - Spot Instances
  - Dedicated Hosts
  - Savings Plans (new November 2019, not yet on the exam)

DigitalCloud TRAINING

Section 15: Compute Pricing

On-Demand

- Means you pay for compute or database capacity with no long-term commitments or upfront payments
- You pay for the compute capacity per hour or per second (per second is Linux only, and applies to On-Demand, Reserved and Spot instances)
- Recommended for users who prefer low cost and flexibility without upfront payment or long-term commitments
- Good for applications with short-term, spiky, or unpredictable workloads that cannot be interrupted



	vCPU	ECU	Memory (GiB)	Instance Storage (GB)	Linux/UNIX Usage
General Purpose - Current Generation					
a1.medium	1	N/A	2 GiB	EBS Only	$0.0255 per Hour
a1.large	2	N/A	4 GiB	EBS Only	$0.051 per Hour
a1.xlarge	4	N/A	8 GiB	EBS Only	$0.102 per Hour
a1.2xlarge	8	N/A	16 GiB	EBS Only	$0.204 per Hour



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Section 15: Compute Pricing



Reserved Instances

- Reserved instances provide significant discounts, up to 75% compared to On-Demand pricing, by paying for capacity ahead of time
- Provide a capacity reservation when applied to a specific Availability Zone

STANDARD 3-YEAR TERM					
Payment Option	**Upfront**	**Monthly***	**Effective Hourly****	**Savings over On-Demand**	**On-Demand Hourly**
No Upfront	$0	$8.03	$0.011	57%	$0.0255
Partial Upfront	$134	$3.72	$0.010	60%	
All Upfront	$252	$0.00	$0.010	62%	

- Good for applications that have predictable usage, that need reserved capacity, and for customers who can commit to a 1 or 3-year term
- Reservation options include no upfront, partial upfront and all upfront

DigitalCloud TRAINING

Section 15: Compute Pricing



Spot Instances

- Purchase spare computing capacity with no upfront commitment at discounted hourly rates
- Provides up to 90% off the On-Demand price
- Recommended for applications that have flexible start and end times, applications that are only feasible at very low compute prices, and users with urgent computing needs for a lot of additional capacity
- In the old model Spot instances were terminated because of higher competing bids, in the new model this does not happen but instances still may be terminated (with a 2 minute warning) when EC2 needs the capacity back – note: the exam may not be updated to reflect this yet

DigitalCloud TRAINING

Section 15: Compute Pricing



Dedicated Hosts

- A dedicated host is an EC2 server dedicated to a single customer
- Runs in your VPC
- Good for when you want to leverage existing server-bound software licences such as Windows Server, SQL Server, and SUSE Linux Enterprise Server
- Also good for meeting compliance requirements

Dedicated Instances

- Dedicated Instances are Amazon EC2 instances that run in a VPC on hardware that's dedicated to a single customer
- Dedicated instances are physically isolated at the host hardware level from instances that belong to other AWS accounts
- Dedicated instances may share hardware with other instances from the same AWS account that are not Dedicated instances

DigitalCloud TRAINING

Section 15: Compute Pricing



Characteristic	Dedicated Instances	Dedicated Hosts
Enables the use of dedicated physical servers	**X**	**X**
Per instance billing	X	
Per host billing		X
Visibility of sockets, cores, host ID		X
Affinity between a host and instance		X
Targeted instance placement		X
Automatic instance placement	X	X

DigitalCloud TRAINING

Section 15: Compute Pricing



AWS Lambda

- Pay only for what you use and charged based on the number of requests for functions and the time it takes to execute the code
- Price is dependent on the amount of memory allocated to the function

Requests

1M REQUESTS FREE

First 1M requests per month are free.

$0.20 PER 1M REQUESTS THEREAFTER

$0.0000002 per request.

Duration

400,000 GB-SECONDS PER MONTH FREE

First 400,000 GB-seconds per month, up to 3.2M seconds of compute time, are free.

$0.0000166667 FOR EVERY GB-SECOND USED THEREAFTER

The price depends on the amount of memory you allocate to your function.

Amazon ECS

- With EC2 launch type you pay for the compute instances in your cluster
- With Fargate launch type you pay for the vCPU and memory resources that your containerized application requests

DigitalCloud TRAINING

Section 15: Types of Reserved Instance



- Standard Reserved Instance
  - Some attributes, such as instance size, can be modified during the term
  - The instance family cannot be modified
  - You cannot exchange a Standard Reserved Instance, only modify it
  - Can be sold in the Reserved Instance Marketplace
- Convertible Reserved Instance
  - Can be exchanged during the term for another Convertible Reserved Instance with new attributes including instance family, instance type, platform, scope, or tenancy
  - Cannot be sold in the Reserved Instance Marketplace

DigitalCloud TRAINING

Section 15: Types of Reserved Instance



- Scheduled Reserved Instance
  - Enable you to purchase capacity reservations that recur on a daily, weekly, or monthly basis, with a specified start time and duration, for a one-year term
  - You reserve the capacity in advance, so that you know it is available when you need it
  - You pay for the time that the instances are scheduled, even if you do not use them

DigitalCloud TRAINING

Section 15: Savings Plans



- Flexible pricing model that offer low prices on EC2, Lambda, and Fargate usage
- Commit to a consistent amount of usage (measured in $/hour) for a 1 or 3 year term
- Compute Savings Plans:
  - Provides the most flexibility and helps to reduce costs by up to 66%
  - Apply to EC2 instance usage regardless of instance family, size, AZ, region, OS or tenancy, and also apply to Fargate or Lambda usage
- EC2 Instance Savings Plans:
  - Provide the lowest prices, with savings up to 72%
  - Commit to usage of individual instance families in a region
  - Can change your usage between instances within a family in that region

DigitalCloud TRAINING

Section 15: Storage Pricing

Amazon S3



- Storage - pricing per class, e.g. Standard or IA
- Storage quantity – data volume stored in your buckets on a per GB basis

S3 Standard Storage	
First 50 TB / Month	$0.023 per GB
Next 450 TB / Month	$0.022 per GB
Over 500 TB / Month	$0.021 per GB

- Number of requests – the number and type of requests, e.g. GET, PUT, POST, LIST, COPY
- Lifecycle transitions requests – moving data between storage classes
- Data transfer – data transferred out of an S3 region is charged
- Transfer Acceleration
- Storage management pricing for some additional features

S3 Inventory††	$0.0025 per million objects listed
S3 Analytics Storage Class Analysis††	$0.10 per million objects monitored per month
S3 Object Tagging	$0.01 per 10,000 tags per month

DigitalCloud TRAINING

Section 15: Storage Pricing

Amazon S3 Glacier



- Extremely low cost and you pay only for what you need with no commitments of upfront fees
- Charged for requests and data transferred out of Glacier
- "Amazon Glacier Select" pricing allows queries to run directly on data stored on Glacier without having to retrieve the archive. Priced on amount of data scanned, returned, and number of requests initiated
- Three options for access to archives, listed in the table below:

	Expedited	Standard	Bulk
Data access time	**1-5 minutes**	**3-5 hours**	**5-12 hours**
Data retrievals	$0.03 per GB	$0.01 per GB	$0.0025 per GB
Retrieval requests	On-Demand: $0.01 per request Provisioned: $100 per Provisioned Capacity Unit	$0.050 per 1,000 requests	$0.025 per 1,000 requests

DigitalCloud TRAINING

Section 15: Storage Pricing



Amazon Elastic Block Store (EBS)

- Volumes – volume storage for all EBS volumes type is charged by the amount of GB provisioned (not used) per month
- Snapshots – based on the amount of space consumed by snapshots in S3. Copying snapshots is charged on the amount of data copied across regions
- Data transfer – inbound data transfer is free, outbound data transfer charges are tiered



Volume



Snapshot

DigitalCloud TRAINING

Section 15: Storage Pricing



Amazon Elastic File System (EFS)

- Storage classes:
  - Standard
  - Infrequent Access Storage (EFS IA)
- Pay for the amount of storage you use
- Lower cost for EFS IA
- EFS Provisioned throughput



File system

DigitalCloud TRAINING

Section 15: Networking Pricing



Amazon Virtual Private Cloud (VPC)

- VPC itself is free, however the following are chargeable:
  - AWS Site-to-Site VPN Connection
  - AWS PrivateLink
  - Amazon VPC Traffic Mirroring
  - NAT Gateways



VPN connection



AWS PrivateLink



NAT gateway



AWS Direct Connect

AWS Direct Connect

- Pay for Port Hours
- Outbound data transfer

Capacity	Port-Hour rate (All AWS Direct Connect locations except in Japan)
1G	$0.30/hour
10G	$2.25/hour

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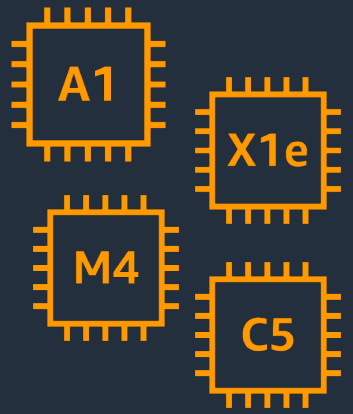
Section 15: Database Pricing



Amazon RDS

- Clock hours of server uptime - amount of time the DB instance is running
- Database engine, instance size and memory
- Database purchase type - e.g. On-Demand, Reserved
- Provisioned storage - backup is included up to 100% of the size of the DB
- Requests - the number of input and output requests to the DB
- Deployment type - single AZ or multi-AZ
- Data transfer - inbound is free, outbound data transfer costs are tiered





DigitalCloud TRAINING

Section 15: Database Pricing



Amazon DynamoDB

- Provisioned throughput (write)
- Provisioned throughput (read)
- Indexed data storage
- On-demand capacity mode:
  - Pay for the data reads and writes you application makes on the tables
  - No need to specify read and write throughput
- Provisioned capacity mode:
  - Specify the number of reads and writes per second you expect your application to require
  - Can use with Auto Scaling to automatically adjust
- Data transfer
- Global tables
- Reserved Capacity

DigitalCloud TRAINING

Section 15: Content Delivery and DNS Services Pricing



Amazon CloudFront

- Traffic distribution – data transfer and request pricing, varies across regions, and is based on the edge location from which the content is served
- Requests – the number and type of requests (HTTP or HTTPS) and the geographic region in which they are made
- Data transfer out – quantity of data transferred out of CloudFront edge locations
- There are additional chargeable items such as invalidation requests, field-level encryption requests, and custom SSL certificates

Amazon Route 53

- Monthly charge for managing hosted zones
- DNS queries
- Domain names

DigitalCloud TRAINING

Section 15: Tools for estimating costs and TCO



AWS Cost Explorer

- The AWS Cost Explorer is a free tool that allows you to view charts of your costs
- You can view cost data for the past 13 months and forecast how much you are likely to spend on a daily level over the next three months
- Cost Explorer can be used to discover patterns in how much you spend on AWS resources over time and to identify cost problem areas
- Cost Explorer can help you to identify service usage statistics such as:
  - Which services you use the most
  - View metrics for which AZ has the most traffic
  - Which linked account is used the most



AWS Cost Explorer

DigitalCloud TRAINING

Section 15: Tools for estimating costs and TCO



AWS Simple Monthly Calculator

- The AWS Simple Monthly Calculator helps customers and prospects estimate their monthly AWS bill more efficiently
- With the AWS Simple Monthly Calculator you can add services in different regions
- The calculator includes support for most AWS services and you can include additional costs such as data ingress/egress charges, data storage charges, and retrieval fees
- It is possible to select EC2 dedicated hosts and reserved instances with various pricing models
- Support can also be added

DigitalCloud TRAINING

Section 15: Tools for estimating costs and TCO



AWS Total Cost of Ownership (TCO) Calculator

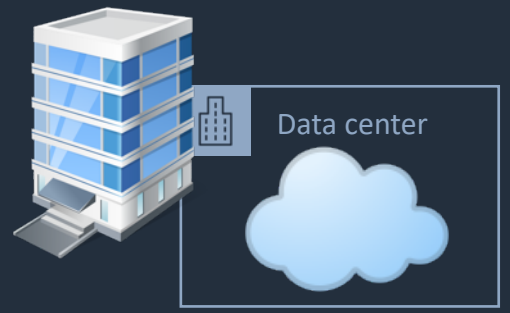
- The TCO calculator is a free tool provided by AWS that allows you to estimate the cost savings of using the AWS Cloud vs. using an on-premise data center
- The TCO calculator therefore helps you to reduce Total Cost of Ownership (TCO) by avoiding large capital expenditures on hardware and infrastructure
- The TCO calculator can also provide directional guidance on cost savings
- The TCO calculator works by you inputting cost elements of your current/or estimated on-premises data center, and comparing those cost requirements with how much it would cost on AWS
- Elements can be added/modified as you move through the process to best estimate the cost savings

DigitalCloud TRAINING

Section 15: Tools for estimating costs and TCO

What else is included in a TCO calculation?

- Rack infrastructure costs
- Virtualization software costs
- Facilities costs (data center space, power, and cooling)
- Networking costs
- Data transfer costs
- IT Labor costs - some labor costs are also required on AWS





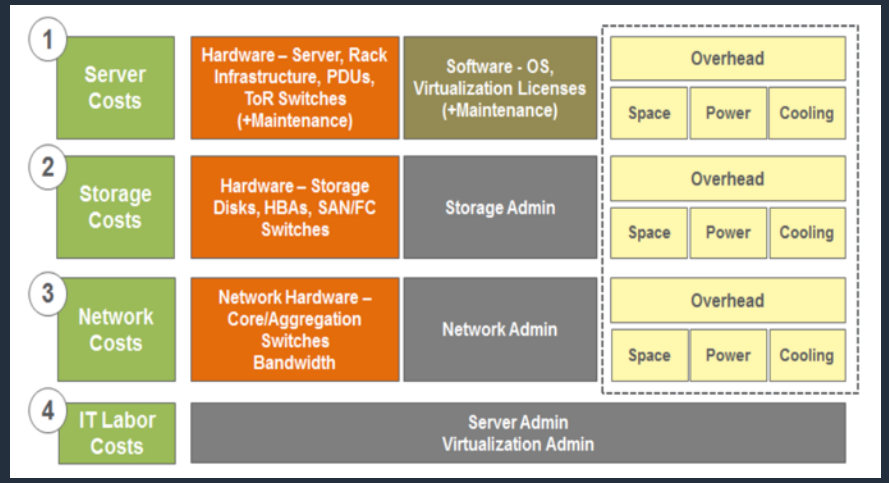
Operations

- Data center operations and security
- Equipment installation and maintenance
- System monitoring and operations activities
- Backup/recovery
- Etc.



Costs:

- Data center
- Power
- Computing Equipment
- Software licenses
- Maintenance contracts
- Staff wages
- Etc.



DigitalCloud TRAINING

Section 15: AWS Support Plans

There are four AWS support plans available:

- Basic – billing and account support only (access to forums only)
- Developer – business hours support via email
- Business – 24×7 email, chat and phone support
- Enterprise – 24×7 email, chat and phone support

- Enterprise support comes with a Technical Account Manager (TAM)
- Developer allows one person to open unlimited cases
- Business and Enterprise allow unlimited contacts to open unlimited cases
- Study the table at: https://aws.amazon.com/premiumsupport/plans/

DigitalCloud TRAINING

Section 15: AWS Organizations and Consolidated Billing

- AWS organizations allows you to consolidate multiple AWS accounts into an organization that you create and centrally manage
- Available in two feature sets:
  - Consolidated Billing
  - All features
- Includes root accounts and organizational units
- Policies are applied to root accounts or OUs
- Consolidated billing includes:
  - Paying Account – independent and cannot access resources of other accounts.
  - Linked Accounts – all linked accounts are independent

DigitalCloud TRAINING

Section 15: AWS Organizations and Consolidated Billing

- Consolidated billing has the following benefits:
- One bill – You get one bill for multiple accounts
  - Easy tracking – You can track the charges across multiple accounts and download the combined cost and usage data
  - Combined usage – You can combine the usage across all accounts in the organization to share the volume pricing discounts and Reserved Instance discounts. This can result in a lower charge for your project, department, or company than with individual standalone accounts
  - No extra fee – Consolidated billing is offered at no additional cost

DigitalCloud TRAINING

Section 15: Resource Groups and Tagging

- Tags are key / value pairs that can be attached to AWS resources
- Tags contain metadata (data about data)
- Tags can sometimes be inherited – e.g. resources created by Auto Scaling, CloudFormation or Elastic Beanstalk
- Resource groups make it easy to group resources using the tags that are assigned to them. You can group resources that share one or more tags
- Resource groups contain general information, such as:
  - Region
  - Name
  - Health Checks
- And also specific information, such as:
  - Public & private IP addresses (for EC2)
  - Port configurations (for ELB)
  - Database engine (for RDS)

DigitalCloud TRAINING

Section 15: Cost Allocation Tags

- Cost allocation tags track your AWS costs on a detailed level
- Must be enabled
- AWS uses the cost allocation tags to organize your resource costs on your cost allocation report

Total Cost	user:Owner	user:Stack	user:Cost Center	user:Application
0.95	DbAdmin	Test	80432	Widget2
0.01	DbAdmin	Test	80432	Widget2
3.84	DbAdmin	Prod	80432	Widget2
6.00	DbAdmin	Test	78925	Widget1
234.63	SysEng	Prod	78925	Widget1
0.73	DbAdmin	Test	78925	Widget1
0.00	DbAdmin	Prod	80432	Portal
2.47	DbAdmin	Prod	78925	Portal

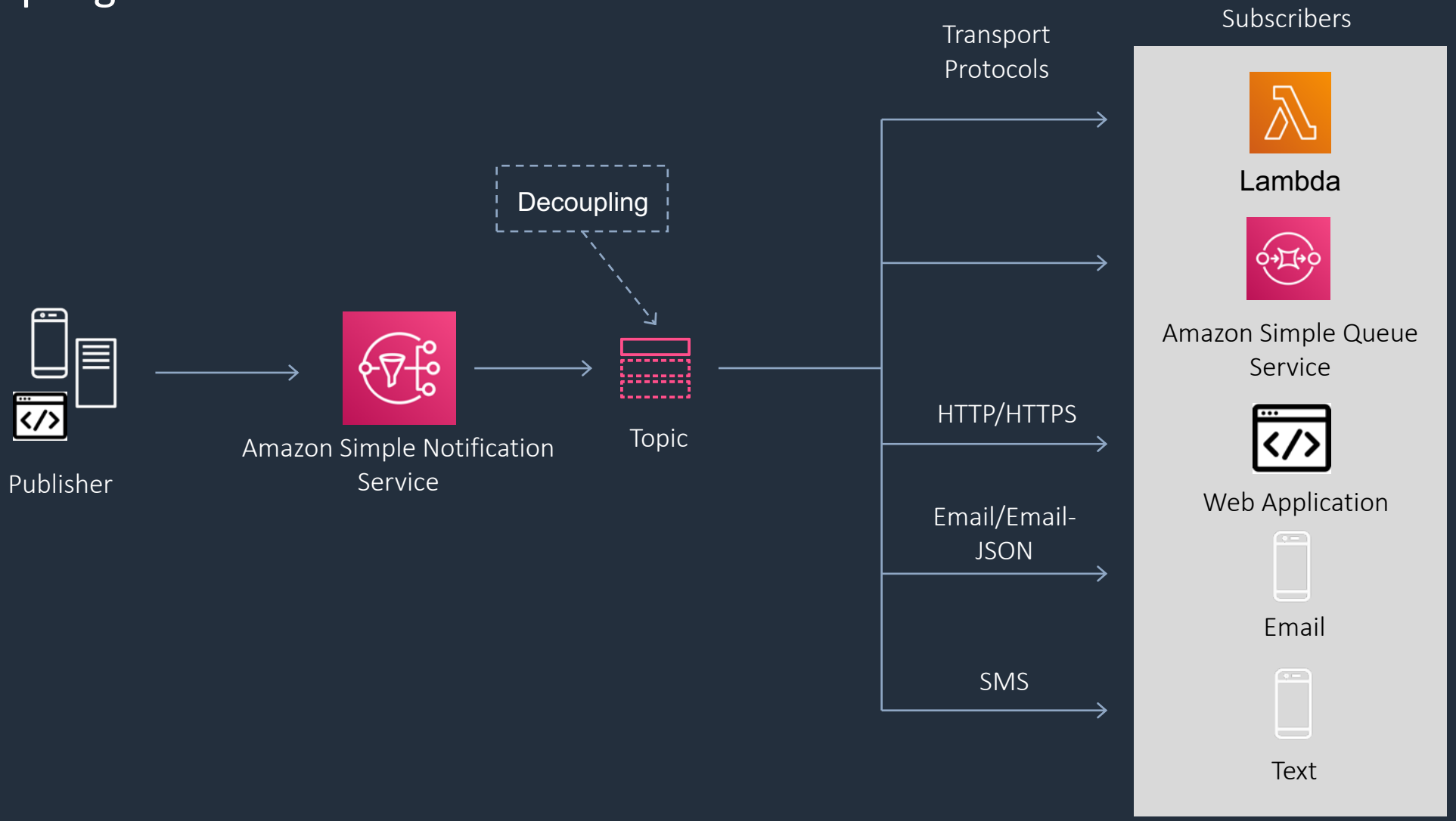
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SECTION 16

Application Integration

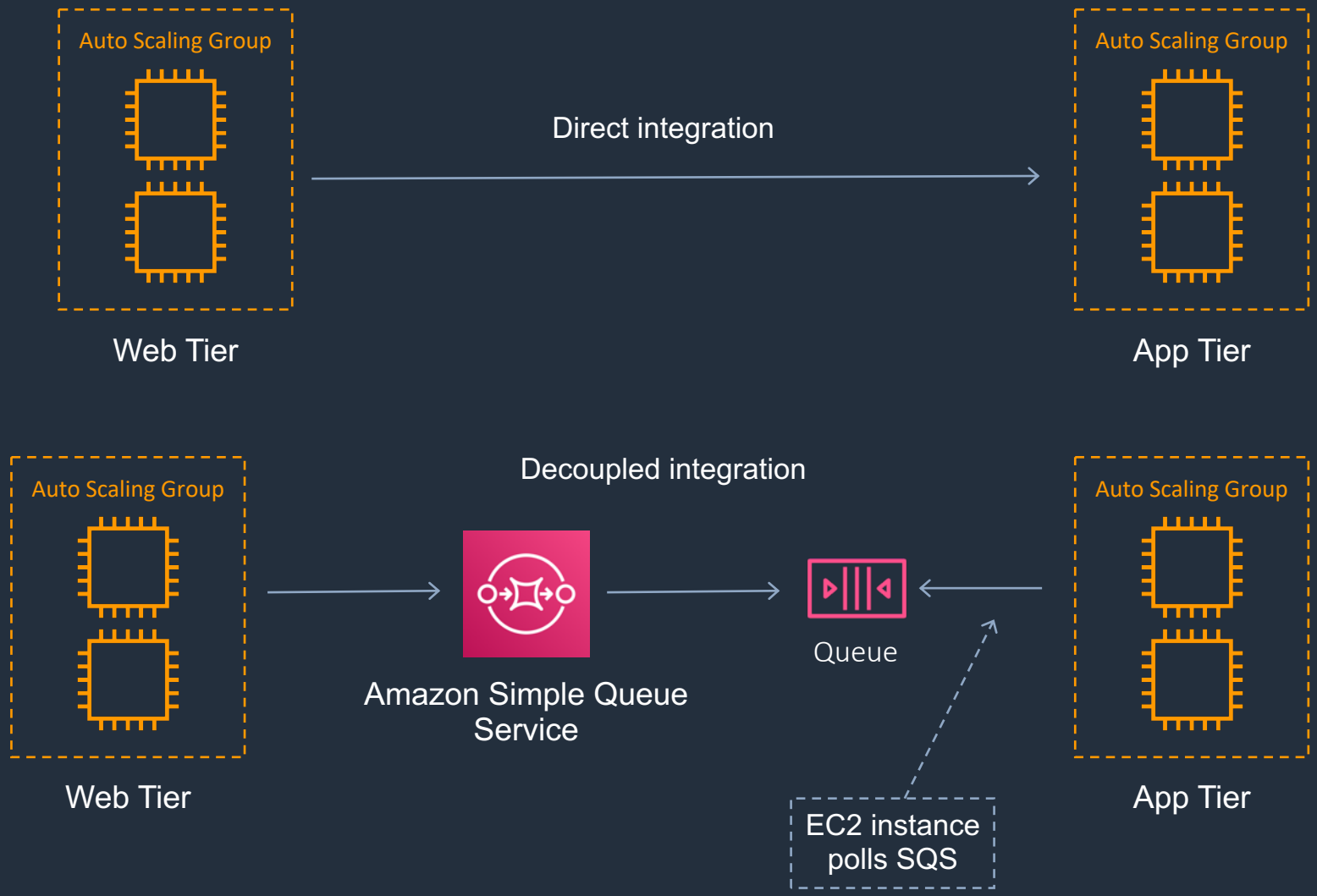


Decoupling with a Notification Service



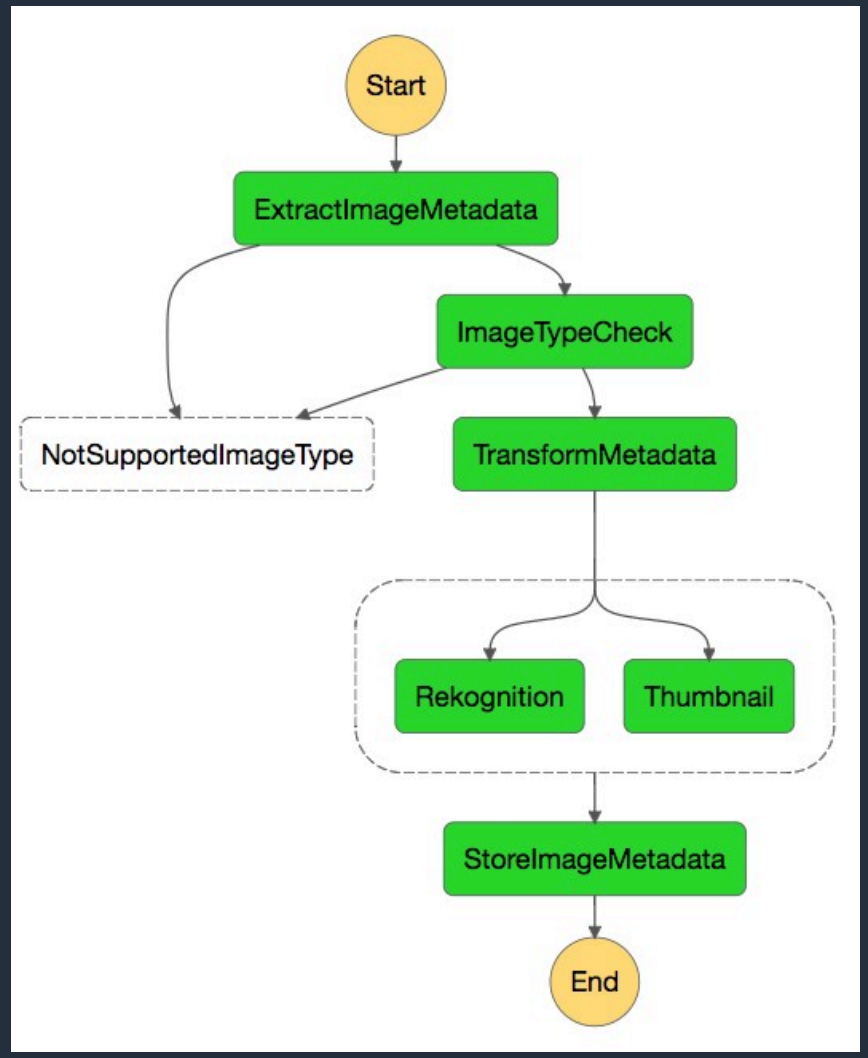
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Decoupling with a Message Queue



DigitalCloud TRAINING

Serverless Workflow Services



AWS Step Functions State Machine

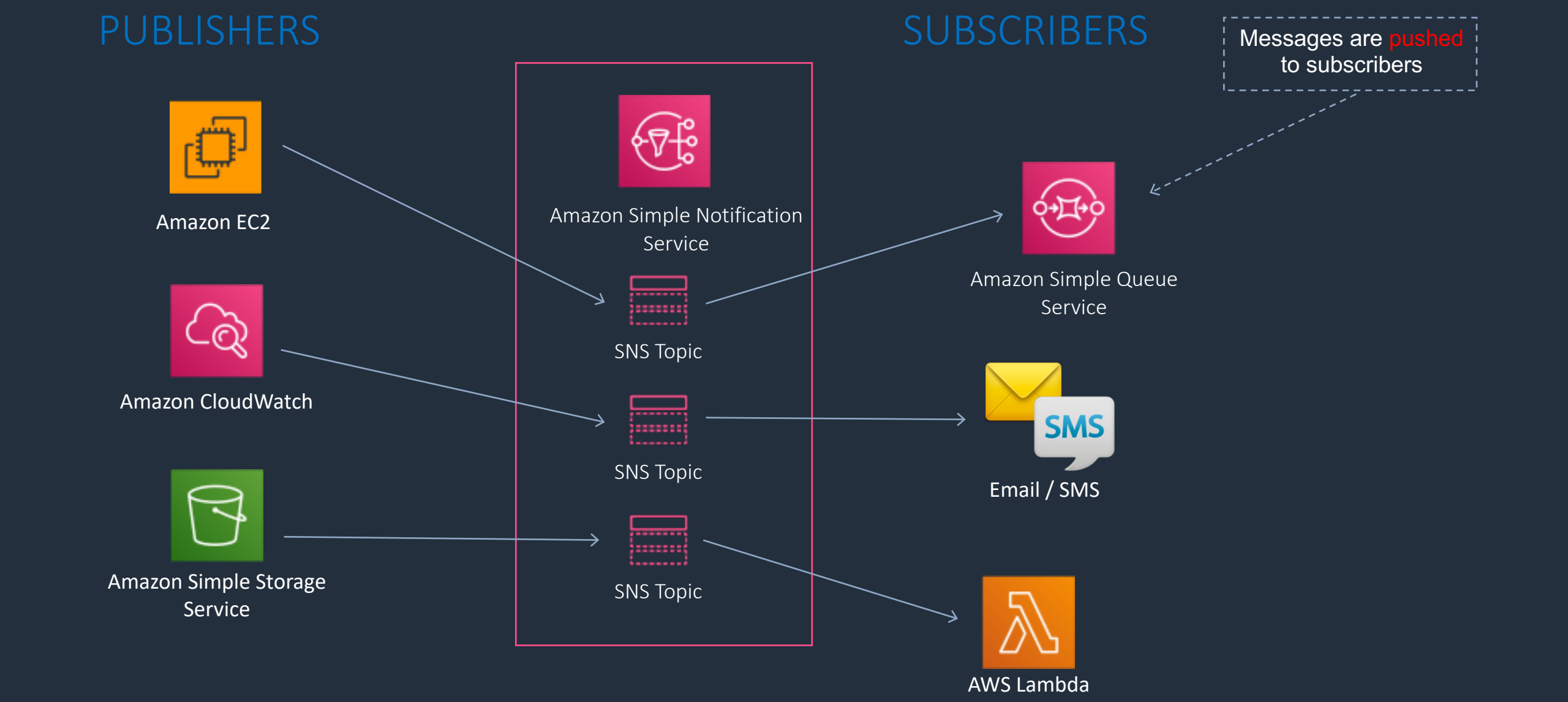
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Application Integration Services

Service	What it does	Example use cases
Simple Notification Service	Set up, operate, and send notifications from the cloud	Send email notification when CloudWatch alarm is triggered
Simple Queue Service	Messaging queue; store and forward patterns	Building distributed / decoupled applications
Simple Workflow Service	Need to support external processes or specialized execution logic	Human-enabled workflows like an order fulfilment system or for procedural requests Note: AWS recommends that for new applications customers consider Step Functions instead of SWF
Step Functions	Out-of-the-box coordination of AWS service components with visual workflow	Order processing workflow

DigitalCloud TRAINING

Amazon Simple Notification Service (SNS)



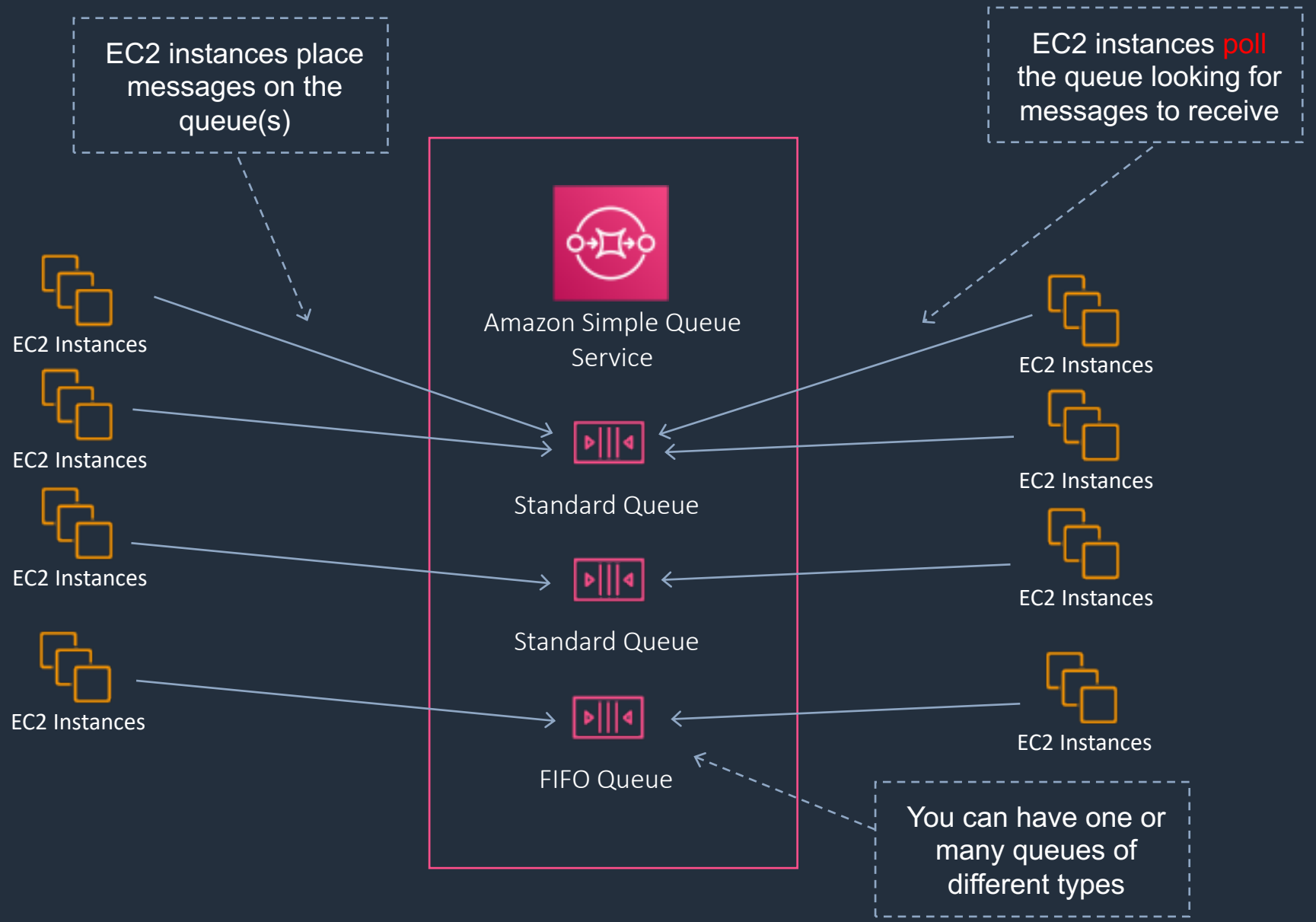
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Amazon Simple Notification Service (SNS)

- Amazon Simple Notification Service (Amazon SNS) is a web service that makes it easy to set up, operate, and send notifications from the cloud.
- Amazon SNS is used for building and integrating loosely-coupled, distributed applications.
- Provides instantaneous, push-based delivery (no polling).
- Uses simple APIs and easy integration with applications.
- Offered under an inexpensive, pay-as-you-go model with no up-front costs.

DigitalCloud TRAINING

Amazon Simple Queue Service (SQS)



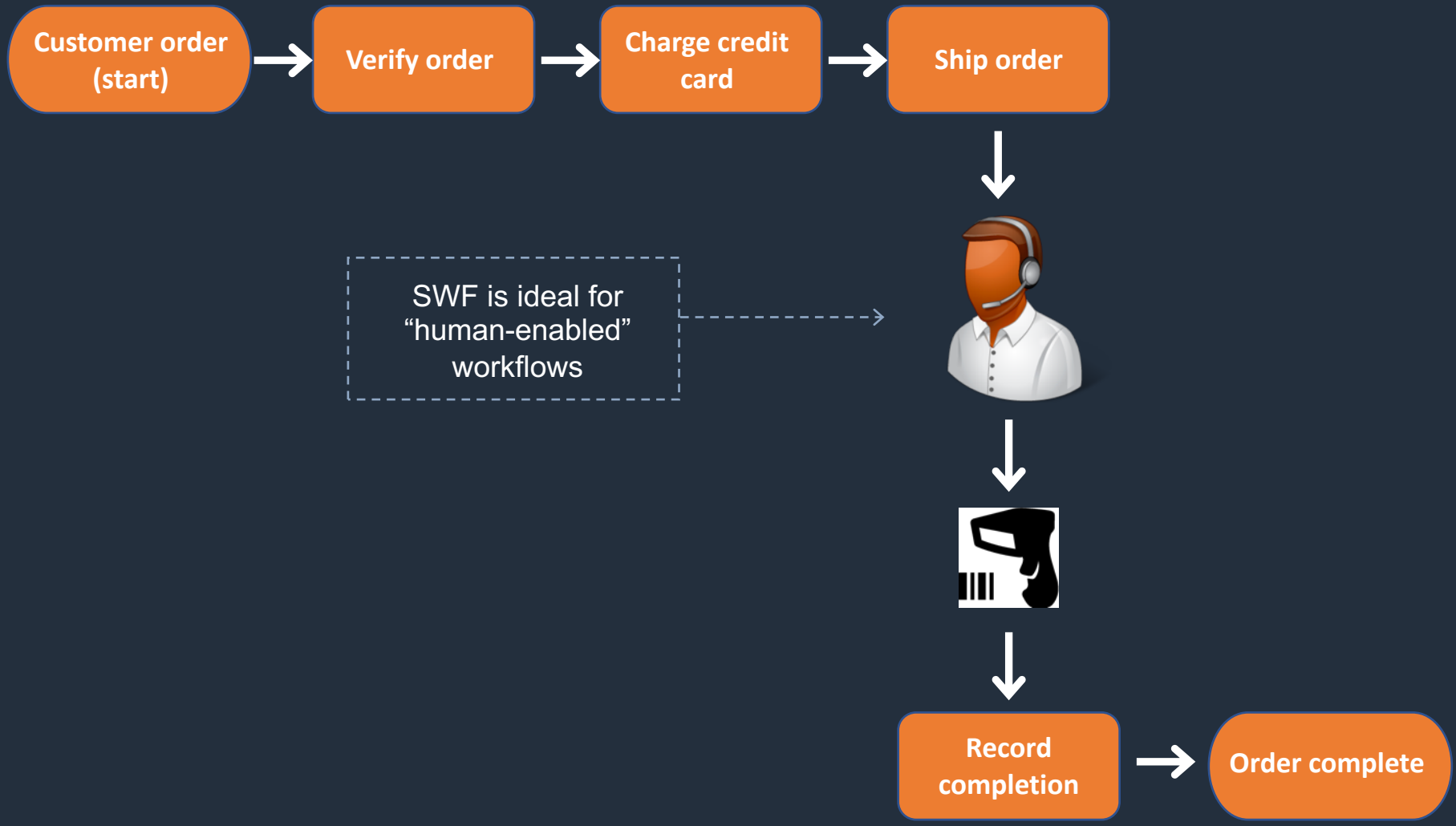
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Amazon Simple Queue Service (SQS)

- Amazon Simple Queue Service (Amazon SQS) is a web service that gives you access to message queues that store messages waiting to be processed.
- SQS offers a reliable, highly-scalable, hosted queue for storing messages in transit between computers.
- SQS is used for distributed/decoupled applications.
- SQS uses a message-oriented API.
- SQS uses pull based (polling) not push based.

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Amazon Simple Workflow Service (SWF)



DigitalCloud TRAINING

Amazon Simple Workflow Service (SWF)

- Amazon Simple Workflow Service (SWF) is a web service that makes it easy to coordinate work across distributed application components.
- Create distributed asynchronous systems as workflows.
- Supports both sequential and parallel processing.
- Tracks the state of your workflow which you interact and update via API.
- Best suited for human-enabled workflows like an order fulfilment system or for procedural requests.
- AWS recommends that for new applications customers consider Step Functions instead of SWF.
- SWF enables applications for a range of use cases, including media processing, web application back-ends, business process workflows, and analytics pipelines, to be designed as a coordination of tasks.

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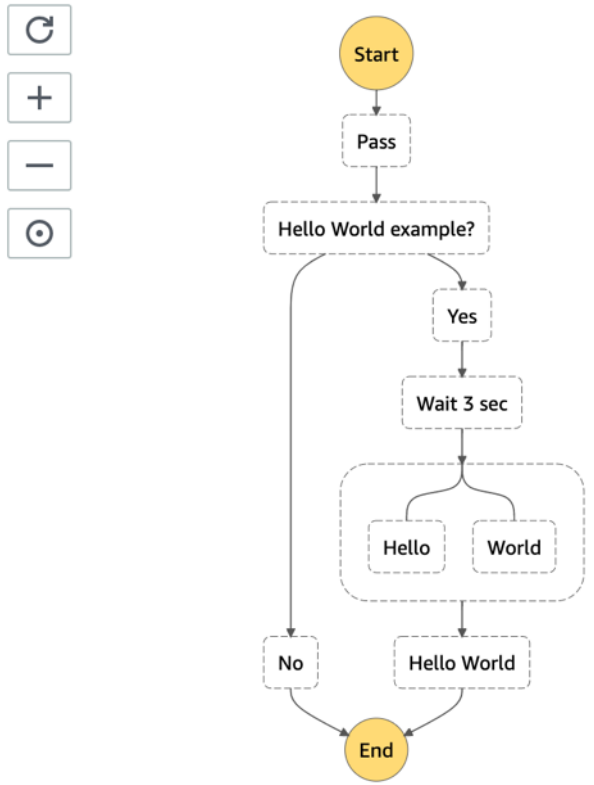
AWS Step Functions

Definition

Export ▼ Layout ▼

Generate code snippet ▼ Format JSON

```
{
  "Comment": "A Hello World example demonstrating various state types of the Amazon States Language",
  "StartAt": "Pass",
  "States": {
    "Pass": {
      "Comment": "A Pass state passes its input to its output, without performing work. Pass states are useful when constructing and debugging state machines.",
      "Type": "Pass",
      "Next": "Hello World example?"
    },
    "Hello World example?": {
      "Comment": "A Choice state adds branching logic to a state machine. Choice rules can implement 16 different comparison operators, and can be combined using And, Or, and Not",
      "Type": "Choice",
      "Choices": [
        {
          "Variable": "$.IsHelloWorldExample",
          "BooleanEquals": true,
          "Next": "Yes"
        },
        {
          "Variable": "$.IsHelloWorldExample",
```



DigitalCloud TRAINING

AWS Step Functions

- AWS Step Functions makes it easy to coordinate the components of distributed applications as a series of steps in a visual workflow.
- You can quickly build and run state machines to execute the steps of your application in a reliable and scalable fashion.
- How it works:

1. Define the steps of your workflow in the JSON-based Amazon States Language. The visual console automatically graphs each step in the order of execution.
2. Start an execution to visualize and verify the steps of your application are operating as intended. The console highlights the real-time status of each step and provides a detailed history of every execution.
3. AWS Step Functions operates and scales the steps of your application and underlying compute for you to help ensure your application executes reliably under increasing demand.

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SECTION 17

Management and Governance



AWS Config

AWS Config evaluates the configuration against desired configurations

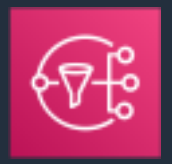
Example Services:



Amazon EC2



AWS Config



Amazon Simple Notification Service



Elastic Load Balancing



Amazon Simple Storage Service



Amazon RDS

Evaluations are displayed on a dashboard and can also be sent via SNS

Configuration changes occur and information is sent to AWS Config

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AWS Config

- AWS Config is a fully managed service that provides you with an AWS resource inventory, configuration history, and configuration change notifications to enable security and governance.
- You can discover existing AWS resources, export a complete inventory of your AWS resources with all configuration details, and determine how a resource was configured at any point in time.
- These capabilities enable compliance auditing, security analysis, resource change tracking, and troubleshooting.
- Allow you to assess, audit and evaluate configurations of your AWS resources.
- Very useful for Configuration Management as part of an ITIL program.

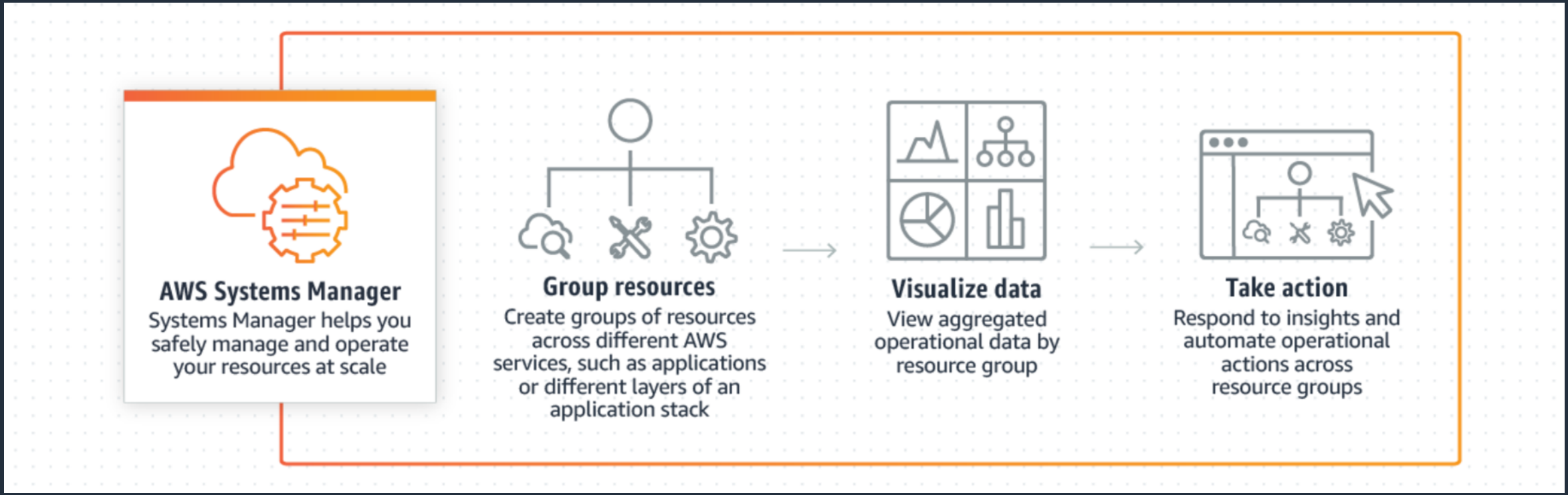
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AWS Config vs AWS CloudTrail

- AWS CloudTrail records user API activity on your account and allows you to access information about this activity.
- AWS Config records point-in-time configuration details for your AWS resources as Configuration Items (CIs).
- You can use an AWS Config to answer “What did my AWS resource look like?” at a point in time.
- You can use AWS CloudTrail to answer “Who made an API call to modify this resource?”.

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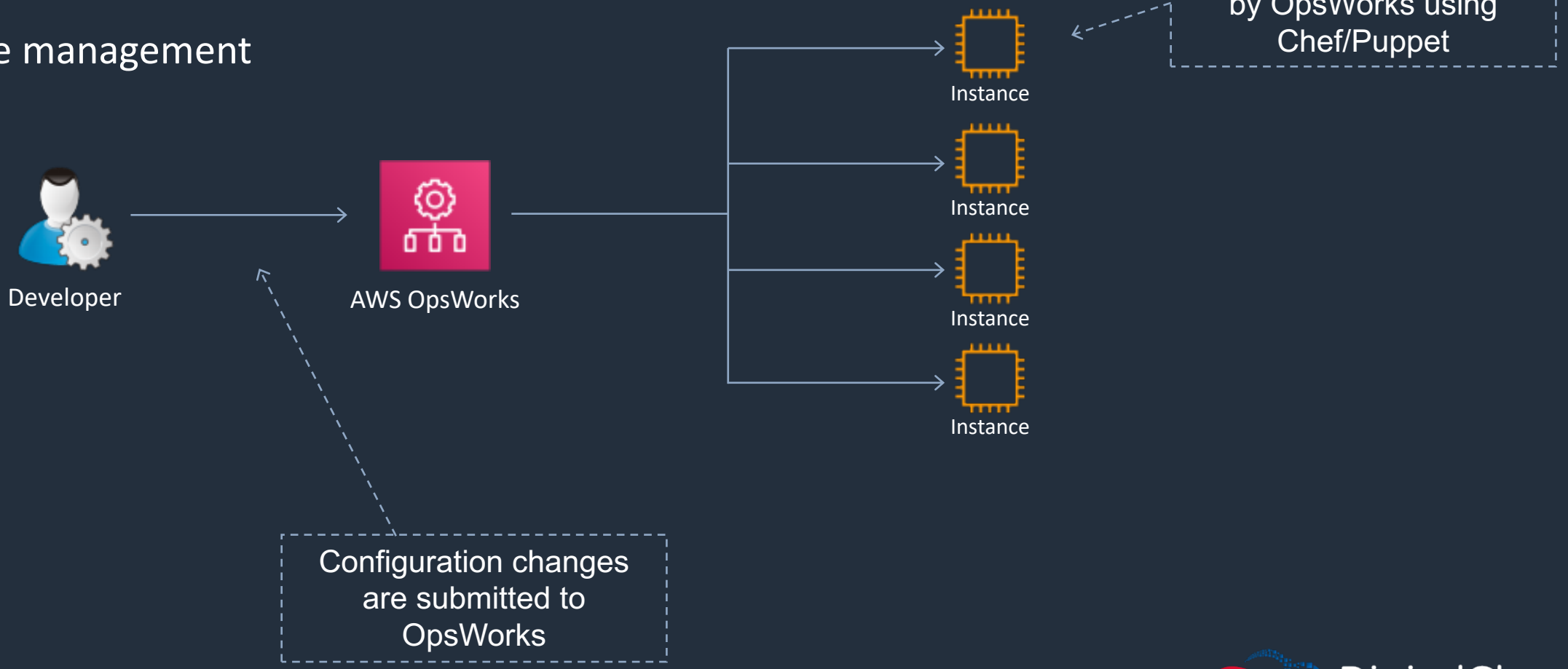
AWS Systems Manager



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AWS OpsWorks

- AWS OpsWorks is a configuration management service that provides managed instances of Chef and Puppet
- Updates include patching, updating, backup, configuration and compliance management



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AWS Service Catalog

- AWS Service Catalog allows organizations to create and manage catalogs of IT services that are approved for use on AWS
- These IT services can include everything from virtual machine images, servers, software, and databases to complete multi-tier application architectures
- AWS Service Catalog allows you to centrally manage commonly deployed IT services
- Helps to achieve consistent governance and meet compliance requirements
- Enables users to quickly deploy only the approved IT services they need

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AWS Trusted Advisor



AWS Trusted Advisor

- Trusted Advisor is an online resource that helps to reduce cost, increase performance and improve security by optimizing your AWS environment
- Trusted Advisor provides real time guidance to help you provision your resources following best practices
- Advisor will advise you on Cost Optimization, Performance, Security, and Fault Tolerance

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AWS Personal Health Dashboard



AWS Personal Health Dashboard

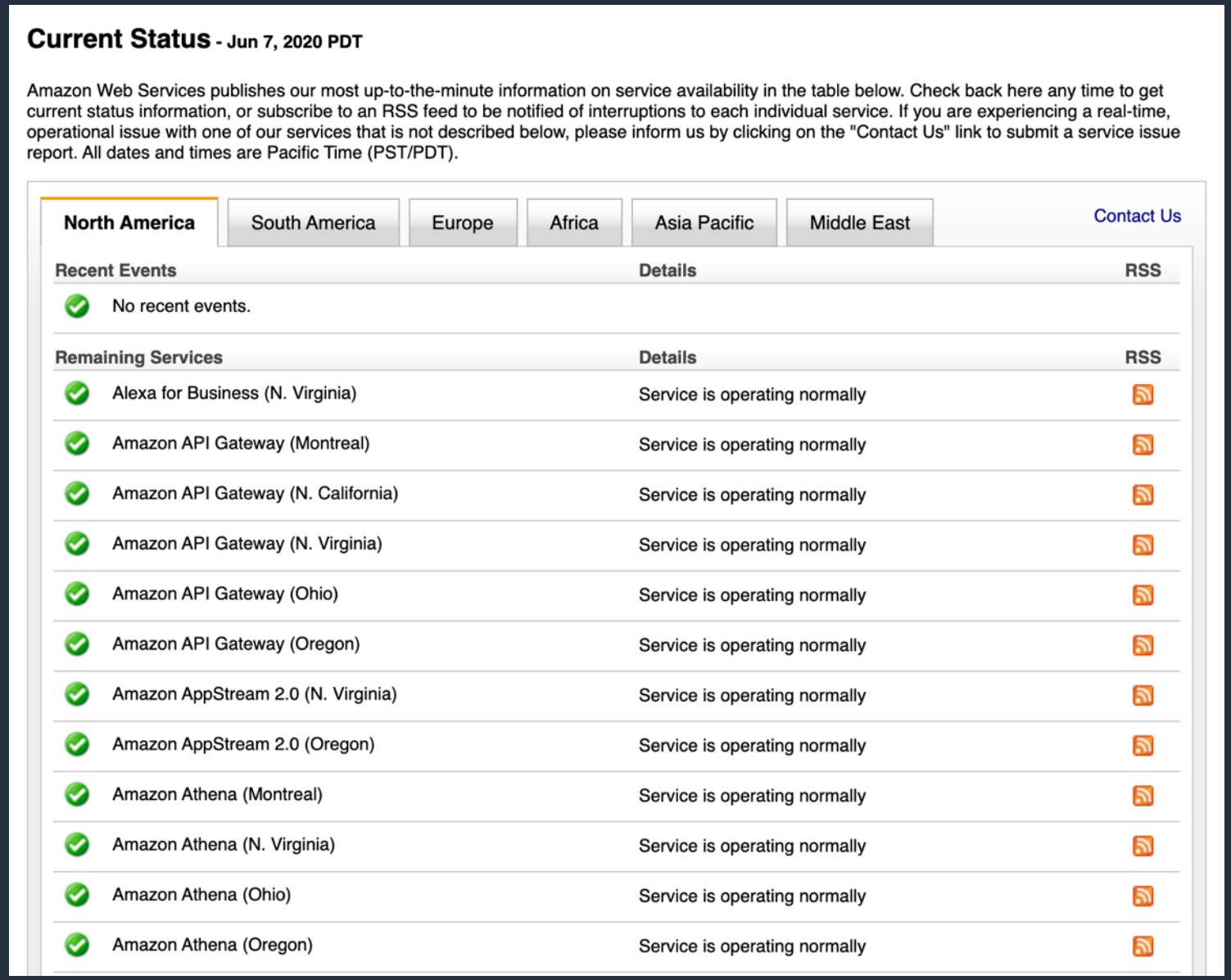
- AWS Personal Health Dashboard provides alerts and remediation guidance when AWS is experiencing events that may impact you
- Personal Health Dashboard gives you a personalized view into the performance and availability of the AWS services underlying your AWS resources
- The dashboard displays relevant and timely information to help you manage events in progress
- Also provides proactive notification to help you plan for scheduled activities
- Alerts are triggered by changes in the health of AWS resources, giving you event visibility, and guidance to help quickly diagnose and resolve issues
- You get a personalized view of the status of the AWS services that power your applications, enabling you to quickly see when AWS is experiencing issues that may impact you

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Service Health Dashboard

Not personalized information so may not be relevant to you

No proactive notification of scheduled activities



Current Status - Jun 7, 2020 PDT

Amazon Web Services publishes our most up-to-the-minute information on service availability in the table below. Check back here any time to get current status information, or subscribe to an RSS feed to be notified of interruptions to each individual service. If you are experiencing a real-time, operational issue with one of our services that is not described below, please inform us by clicking on the "Contact Us" link to submit a service issue report. All dates and times are Pacific Time (PST/PDT).

North America | South America | Europe | Africa | Asia Pacific | Middle East | Contact Us

Recent Events	Details	RSS
No recent events.		

Remaining Services	Details	RSS
Alexa for Business (N. Virginia)	Service is operating normally	
Amazon API Gateway (Montreal)	Service is operating normally	
Amazon API Gateway (N. California)	Service is operating normally	
Amazon API Gateway (N. Virginia)	Service is operating normally	
Amazon API Gateway (Ohio)	Service is operating normally	
Amazon API Gateway (Oregon)	Service is operating normally	
Amazon AppStream 2.0 (N. Virginia)	Service is operating normally	
Amazon AppStream 2.0 (Oregon)	Service is operating normally	
Amazon Athena (Montreal)	Service is operating normally	
Amazon Athena (N. Virginia)	Service is operating normally	
Amazon Athena (Ohio)	Service is operating normally	
Amazon Athena (Oregon)	Service is operating normally	

Shows current status information on service availability

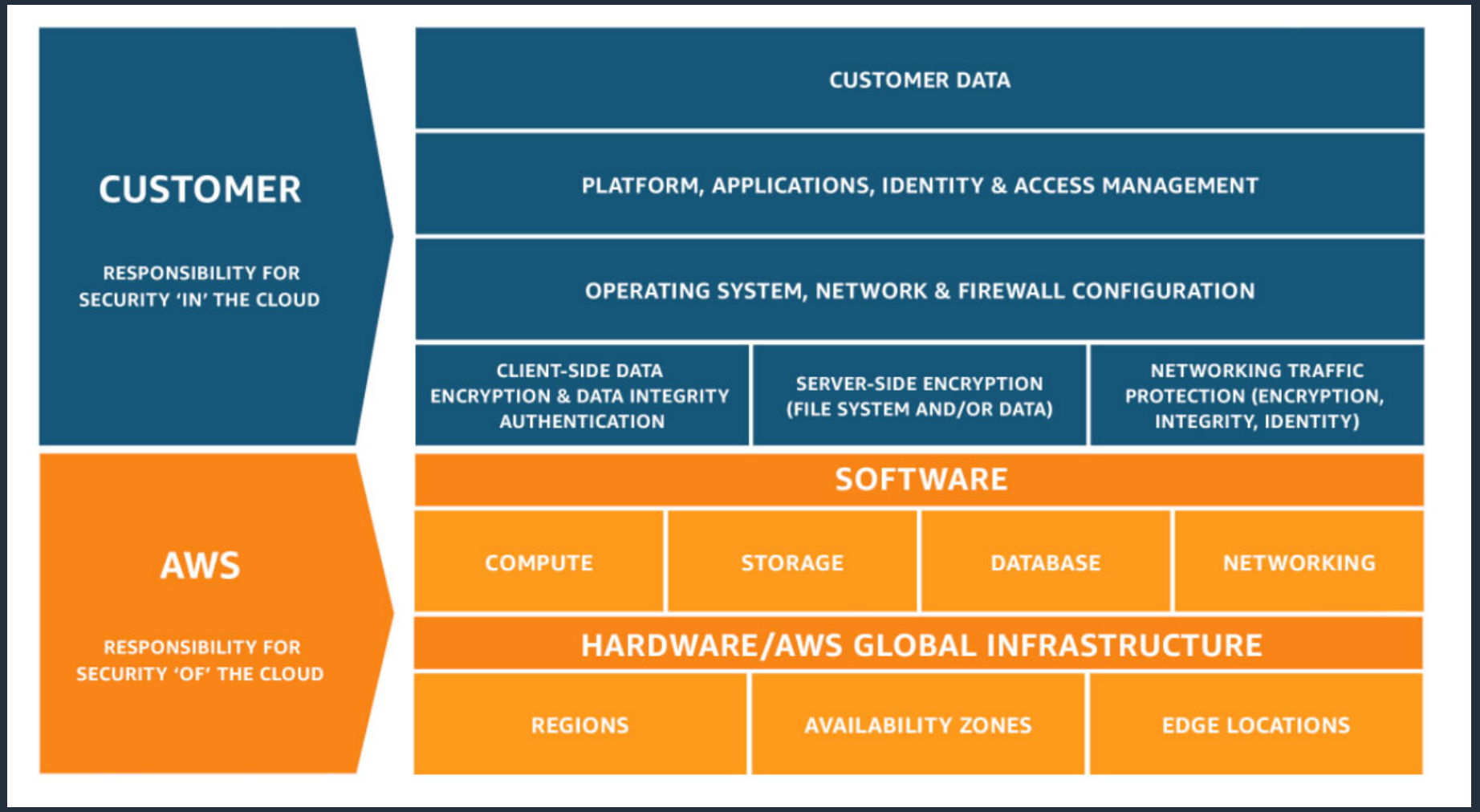
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SECTION 18

AWS Cloud Security and Identity



The Shared Responsibility Model



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The Shared Responsibility Model

- The AWS shared responsibility model defines what you (as an AWS account holder/user) and AWS are responsible for when it comes to security and compliance
- Security and Compliance is a shared responsibility between AWS and the customer
- AWS are responsible for “Security of the Cloud”
  - AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud
  - This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services
- Customers are responsible for “Security in the Cloud”
  - For EC2 this includes network level security (NACLs, security groups), operating system patches and updates, IAM user access management, and client and server-side data encryption

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The Shared Responsibility Model

- Inherited Controls – Controls which a customer fully inherits from AWS.
  - Physical and Environmental controls
- Shared Controls – Controls which apply to both the infrastructure layer and customer layers, but in completely separate contexts or perspectives.
  - Patch Management – AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest OS and applications.
  - Configuration Management – AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.
  - Awareness & Training - AWS trains AWS employees, but a customer must train their own employees.

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The Shared Responsibility Model

- Customer Specific – Controls which are solely the responsibility of the customer based on the application they are deploying within AWS services. Examples include:
  - Service and Communications Protection or Zone Security which may require a customer to route or zone data within specific security environments.

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The Shared Responsibility Model - Examples

CUSTOMER RESPONSIBILITY


Bucket with objects


Role



Multi-Factor Authentication


Security Group


Patch management


Auto Scaling

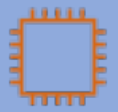

Staff training


Data encryption


IAM User

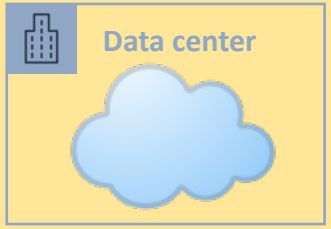

Network ACL


SSL encryption


EC2 Instance


Elastic load balancer

AWS RESPONSIBILITY




Data center security


Network router


Network switch


Server


Storage


Disk drive


Database Server

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AWS Compliance

- AWS Cloud Compliance enables you to understand the robust controls in place at AWS to maintain security and data protection in the cloud
- As systems are built on top of AWS Cloud infrastructure, compliance responsibilities will be shared
- Compliance programs include:
  - Certifications / attestations
  - Laws, regulations, and privacy
  - Alignments / frameworks

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AWS Artifact

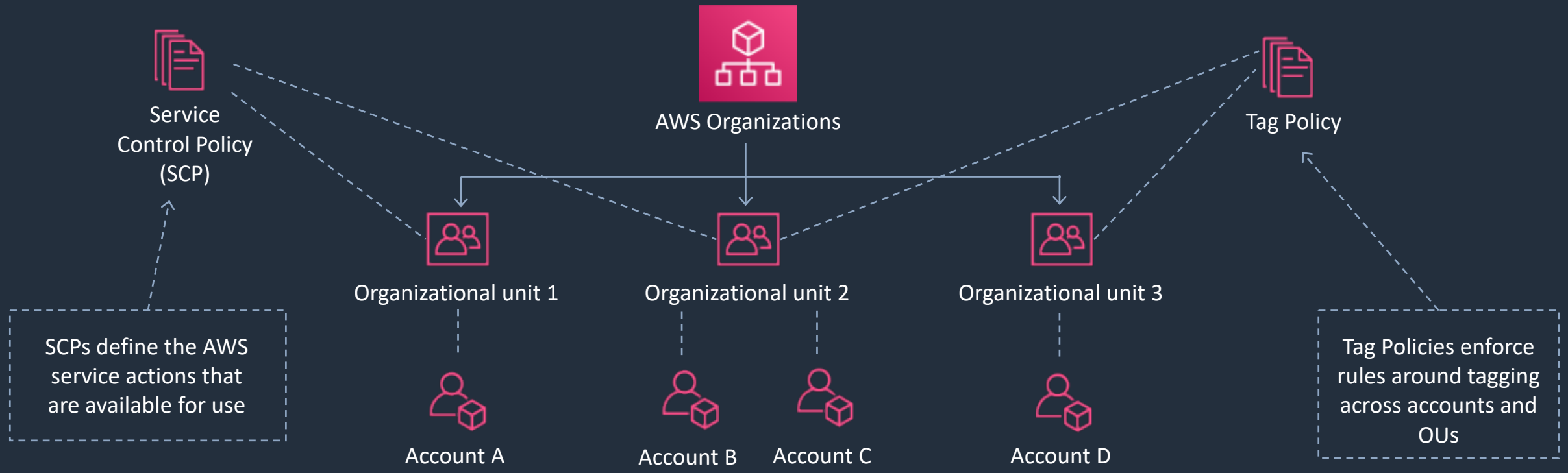


AWS Artifact

- AWS Artifact is your go-to, central resource for compliance-related information that matters to you
- It provides on-demand access to AWS' security and compliance reports and select online agreements
- Reports available in AWS Artifact include our Service Organization Control (SOC) reports, Payment Card Industry (PCI) reports, and certifications from accreditation bodies across geographies and compliance verticals that validate the implementation and operating effectiveness of AWS security controls
- Agreements available in AWS Artifact include the Business Associate Addendum (BAA) and the Nondisclosure Agreement (NDA)

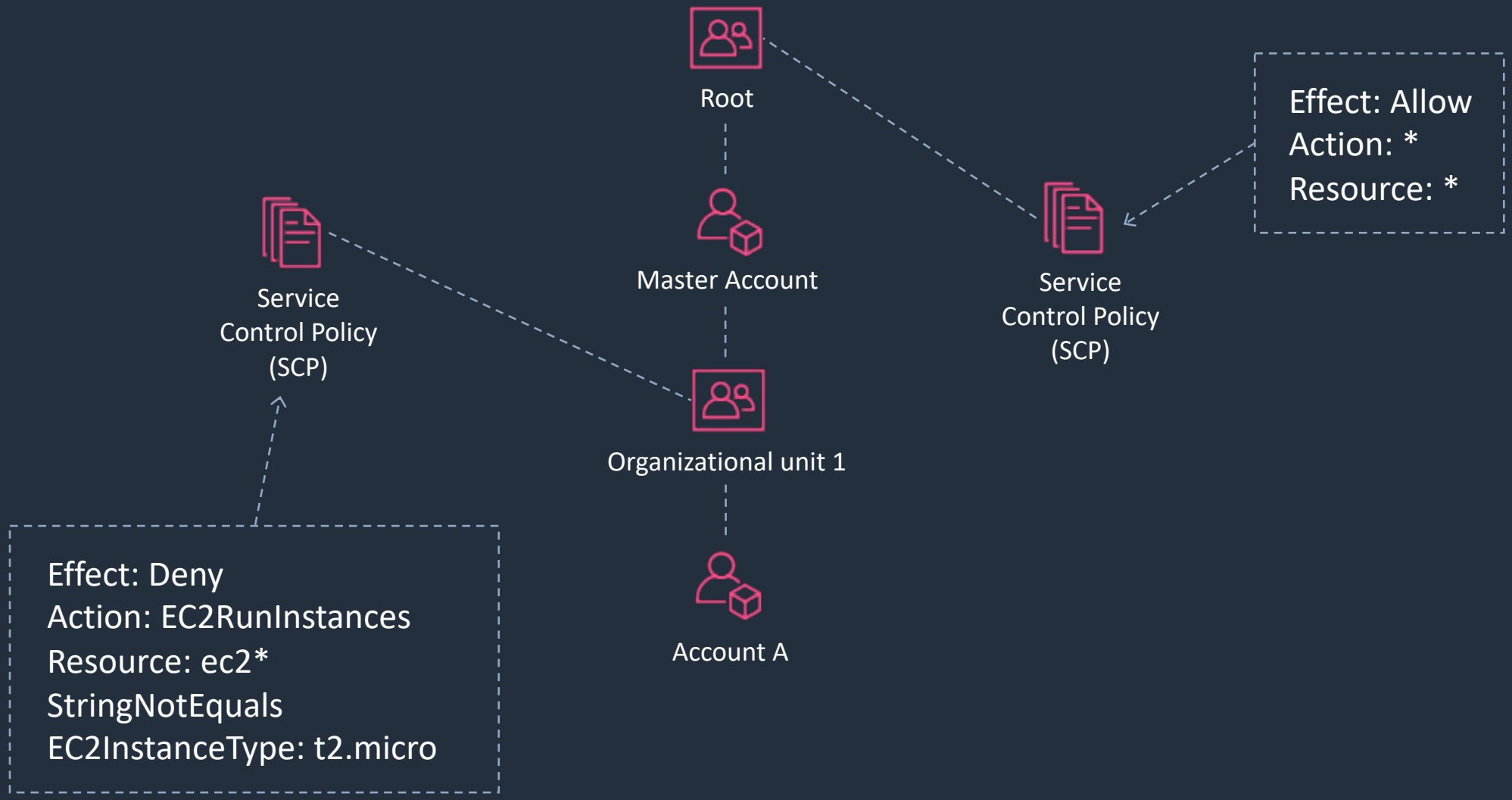
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AWS Organizations – Service Control Policies (SCP) and Tag Policies



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AWS Organizations – SCP to Restrict EC2 Instance Types



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Amazon Inspector

- Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS
- Inspector automatically assesses applications for vulnerabilities or deviations from best practices
- Uses an agent installed on EC2 instances
- Instances must be tagged



Amazon Inspector

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AWS WAF & Shield



AWS WAF

AWS Web Application Firewall (WAF)

- AWS WAF is a web application firewall
- Protects against common exploits that could compromise application availability, compromise security or consume excessive resources



AWS Shield

AWS Shield

- AWS Shield is a managed Distributed Denial of Service (DDoS) protection service
- Safeguards web application running on AWS with always-on detection and automatic inline mitigations
- Helps to minimize application downtime and latency
- Two tiers – Standard and Advanced
- Integrated with Amazon CloudFront

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Amazon Macie



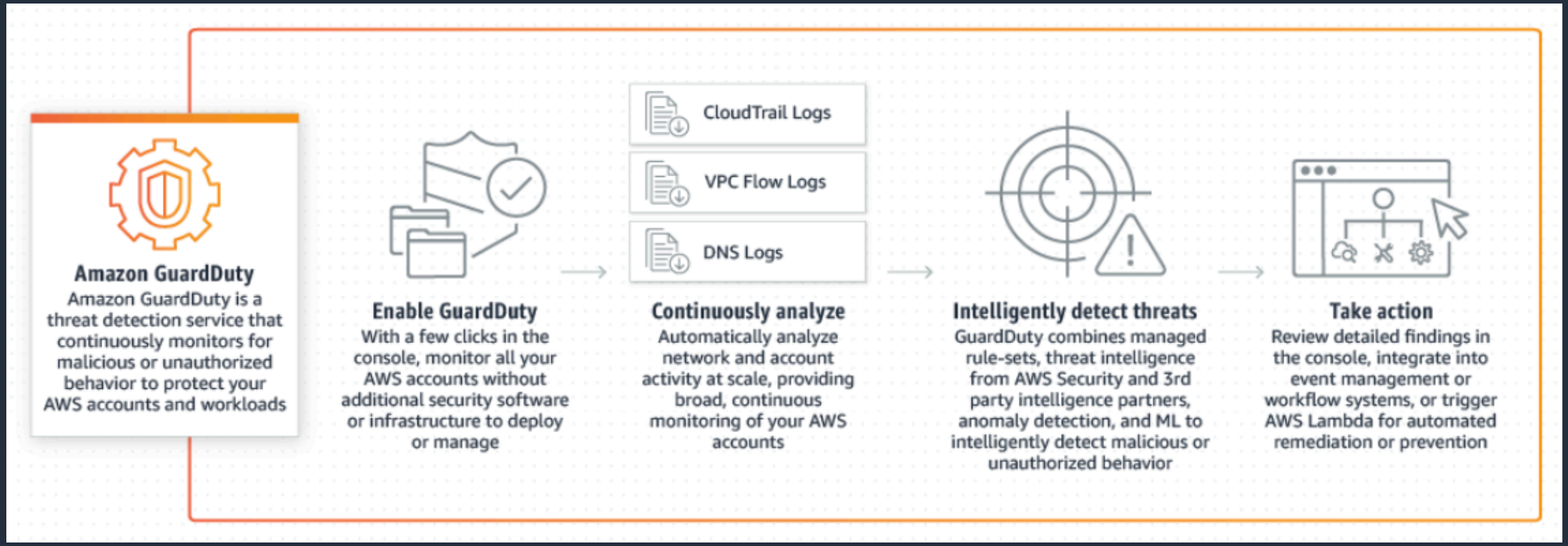
Amazon Macie

- Amazon Macie is a fully managed data security and data privacy service
- It uses machine learning and pattern matching to discover, monitor, and help you protect your sensitive data on Amazon S3
- Macie enables security compliance and preventive security as follows:
  - Identify a variety of data types, including PII, Protected Health Information (PHI), regulatory documents, API keys, and secret keys
  - Identify changes to policy and access control lists
  - Continuously monitor the security posture of Amazon S3
  - Generate security findings that you can view using the Macie console, AWS Security Hub, or Amazon EventBridge.
  - Manage multiple AWS accounts using AWS Organizations

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Amazon GuardDuty

- Amazon GuardDuty is a threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect your AWS accounts and workloads

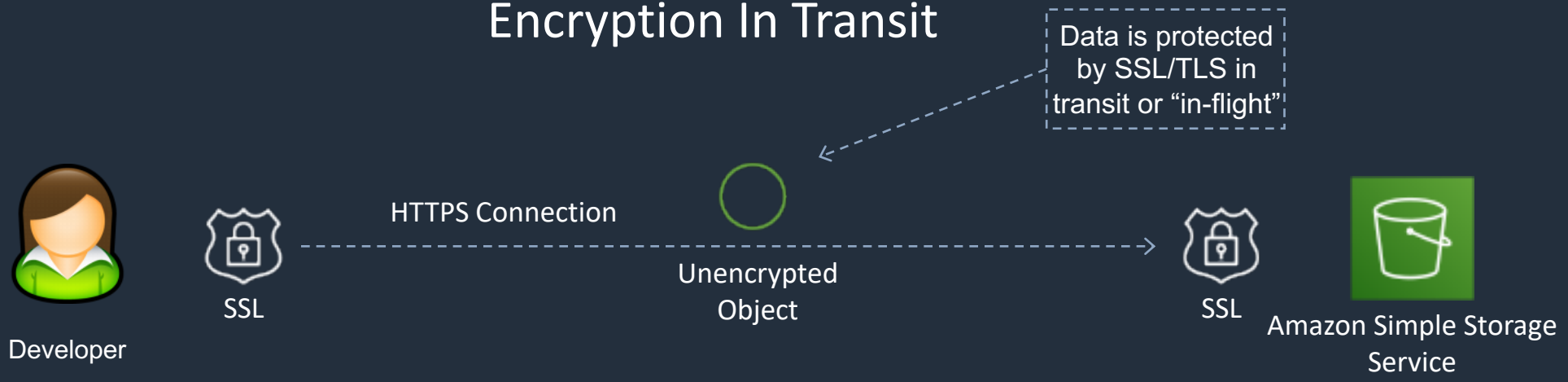


Amazon GuardDuty

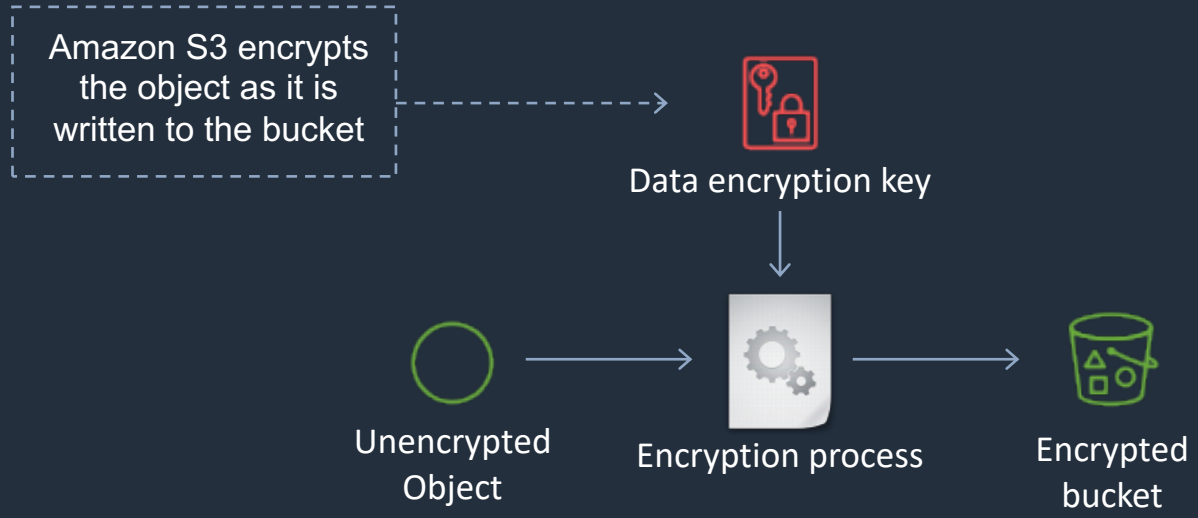
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Encryption – In Transit vs At Rest

Encryption In Transit



Encryption At Rest



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Encryption Key Management: KMS



AWS Key Management Service

AWS Key Management Service (KMS)

- AWS Key Management Service gives you centralized control over the encryption keys used to protect your data
- You can create, import, rotate, disable, delete, define usage policies for, and audit the use of encryption keys used to encrypt your data
- AWS Key Management Service is integrated with most other AWS services making it easy to encrypt the data you store in these services with encryption keys you control

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Encryption Key Management: CloudHSM



AWS CloudHSM

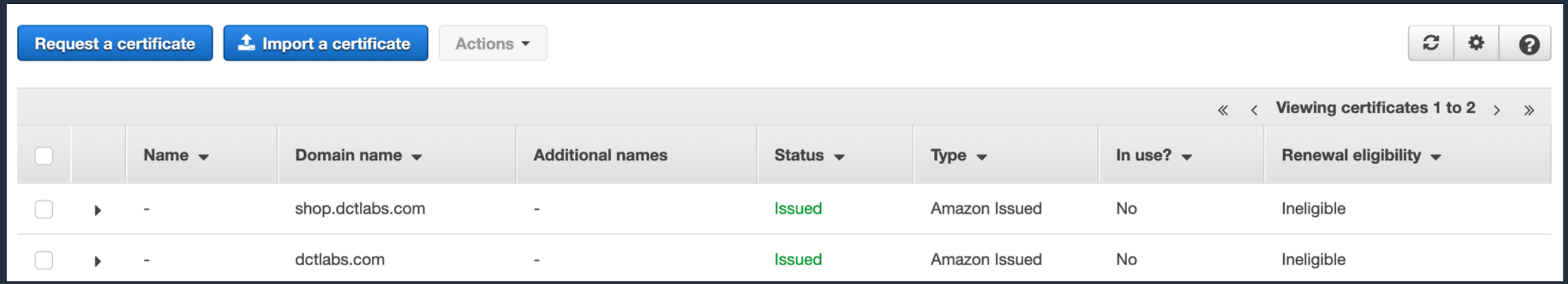
AWS CloudHSM

- AWS CloudHSM is a cloud-based hardware security module (HSM) that enables you to easily generate and use your own encryption keys on the AWS Cloud
- With CloudHSM, you can manage your own encryption keys using FIPS 140-2 Level 3 validated HSMs
- CloudHSM offers you the flexibility to integrate with your applications using industry-standard APIs, such as PKCS#11, Java Cryptography Extensions (JCE), and Microsoft CryptoNG (CNG) libraries

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AWS Certificate Manager (ACM)

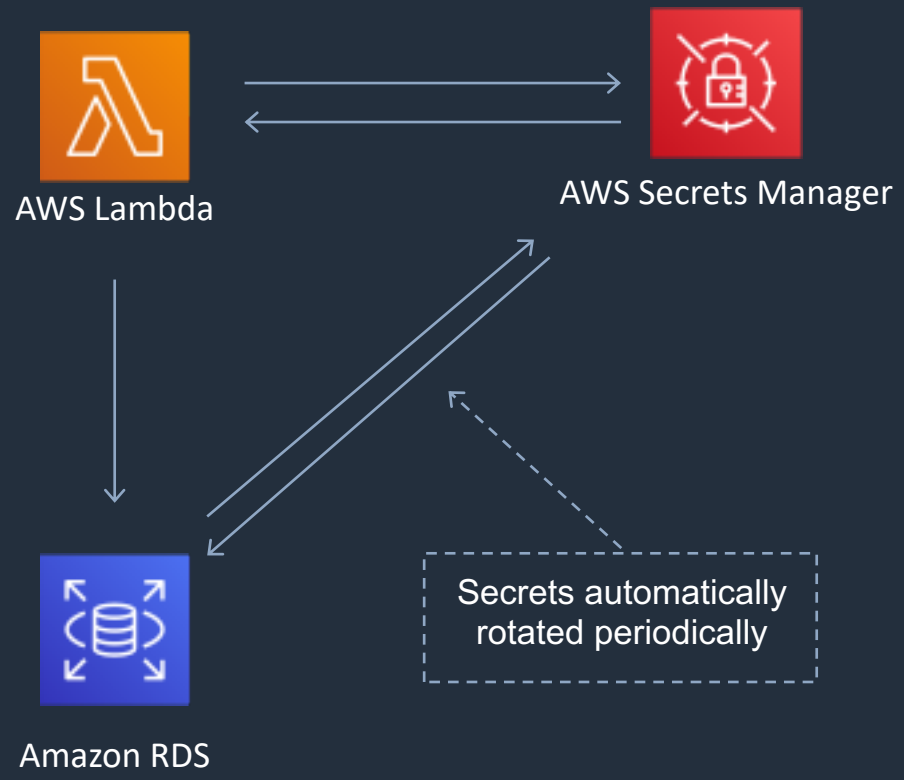
- AWS Certificate Manager is used for creating and managing public SSL/TLS certificates.
- You can use public certificates provided by ACM (ACM certificates) or certificates that you import into ACM.
- Can also request private certificates from a private certificate authority (CA) created using AWS Certificate Manager Private Certificate Authority.
- ACM certificates can secure multiple domain names and multiple names within a domain.
- You can also use ACM to create wildcard SSL certificates that can protect an unlimited number of subdomains.



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AWS Secrets Manager

- AWS Secrets Manager helps you meet your security and compliance requirements by enabling you to rotate secrets safely without the need for code deployments.
- Secrets Manager offers built-in integration for Amazon RDS, Amazon Redshift, and Amazon DocumentDB and rotates these database credentials on your behalf automatically.
- You can customize Lambda functions to extend Secrets Manager rotation to other secret types, such as API keys and OAuth tokens.



DigitalCloud TRAINING

Penetration Testing

- Penetration testing is the practice of testing one's own application's security for vulnerabilities by simulating an attack
- AWS allows penetration testing. There is a limited set of resources on which penetration testing can be performed
- You do not need permission to perform penetration testing against the following services:
  - Amazon EC2 instances, NAT Gateways, and Elastic Load Balancers
  - Amazon RDS and Amazon Aurora
  - Amazon CloudFront
  - Amazon API Gateways
  - AWS Lambda and Lambda Edge functions
  - Amazon Lightsail resources
  - Amazon Elastic Beanstalk environments

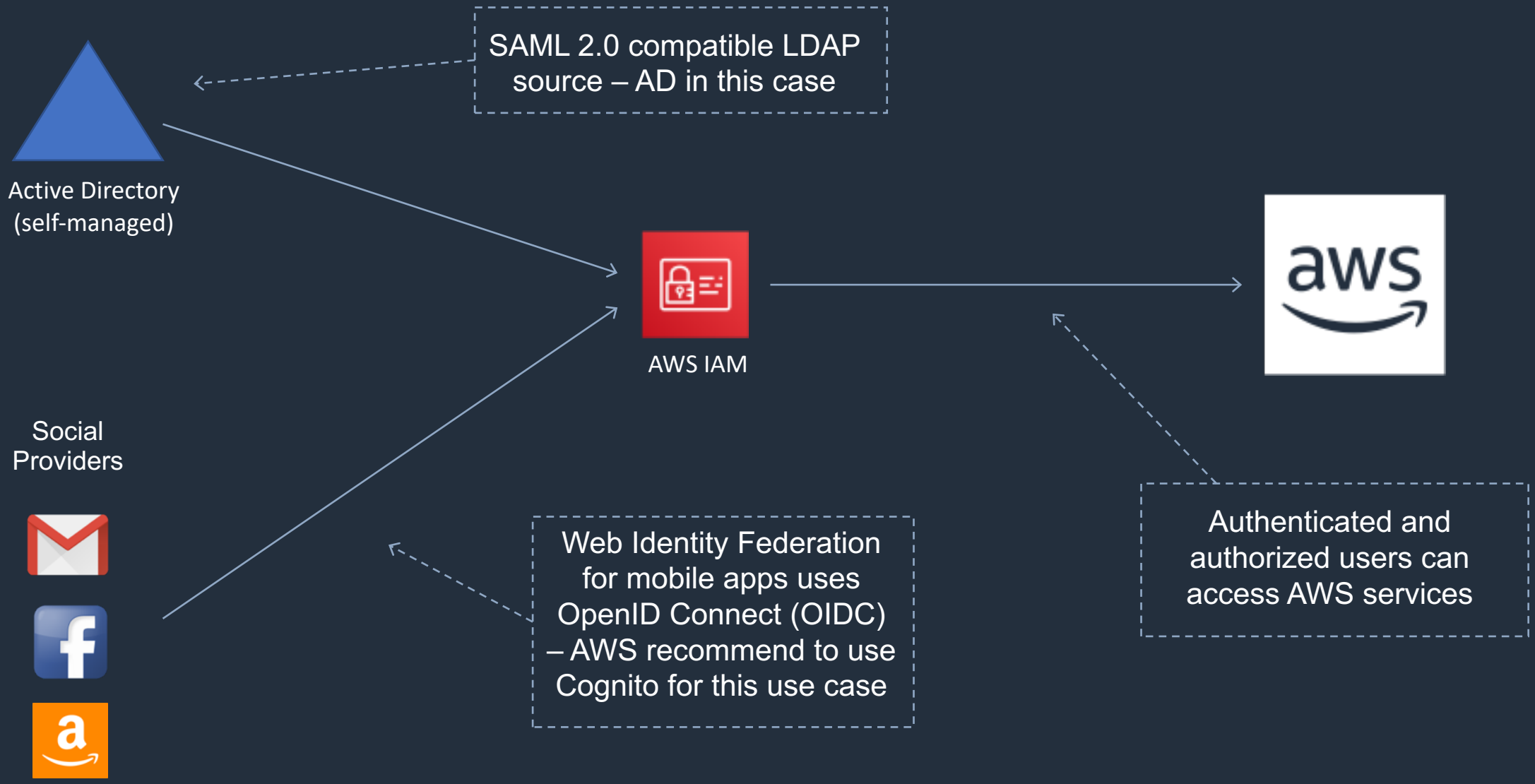
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Penetration Testing

- In case an account is or may be compromised, AWS recommend that the following steps are taken:
  - Change your AWS root account password
  - Change all IAM user's passwords
  - Delete or rotate all programmatic (API) access keys
  - Delete any resources in your account that you did not create
  - Respond to any notifications you received from AWS through the AWS Support Center and/or contact AWS Support to open a support case

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Identity Providers and Federation



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AWS Single Sign-on (SSO)

Identity sources can be AWS SSO, Active Directory and standard providers using SAML 2.0

Active Directory (self-managed)

AWS Directory Service

AWS Single Sign-On

AWS Organizations

AWS Account A

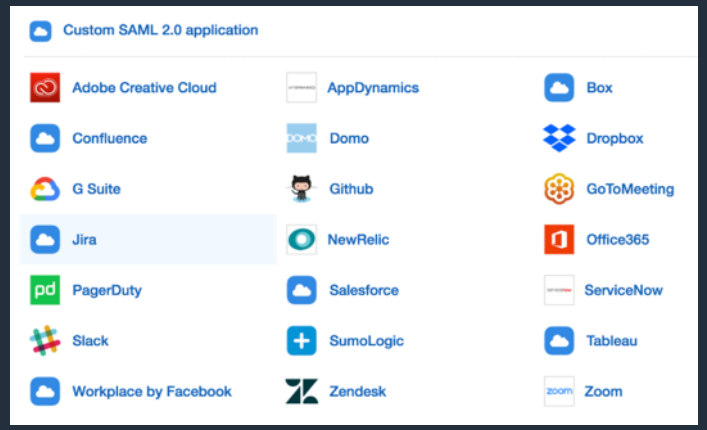
AWS Account B

Azure AD (self-managed)

AWS Account C

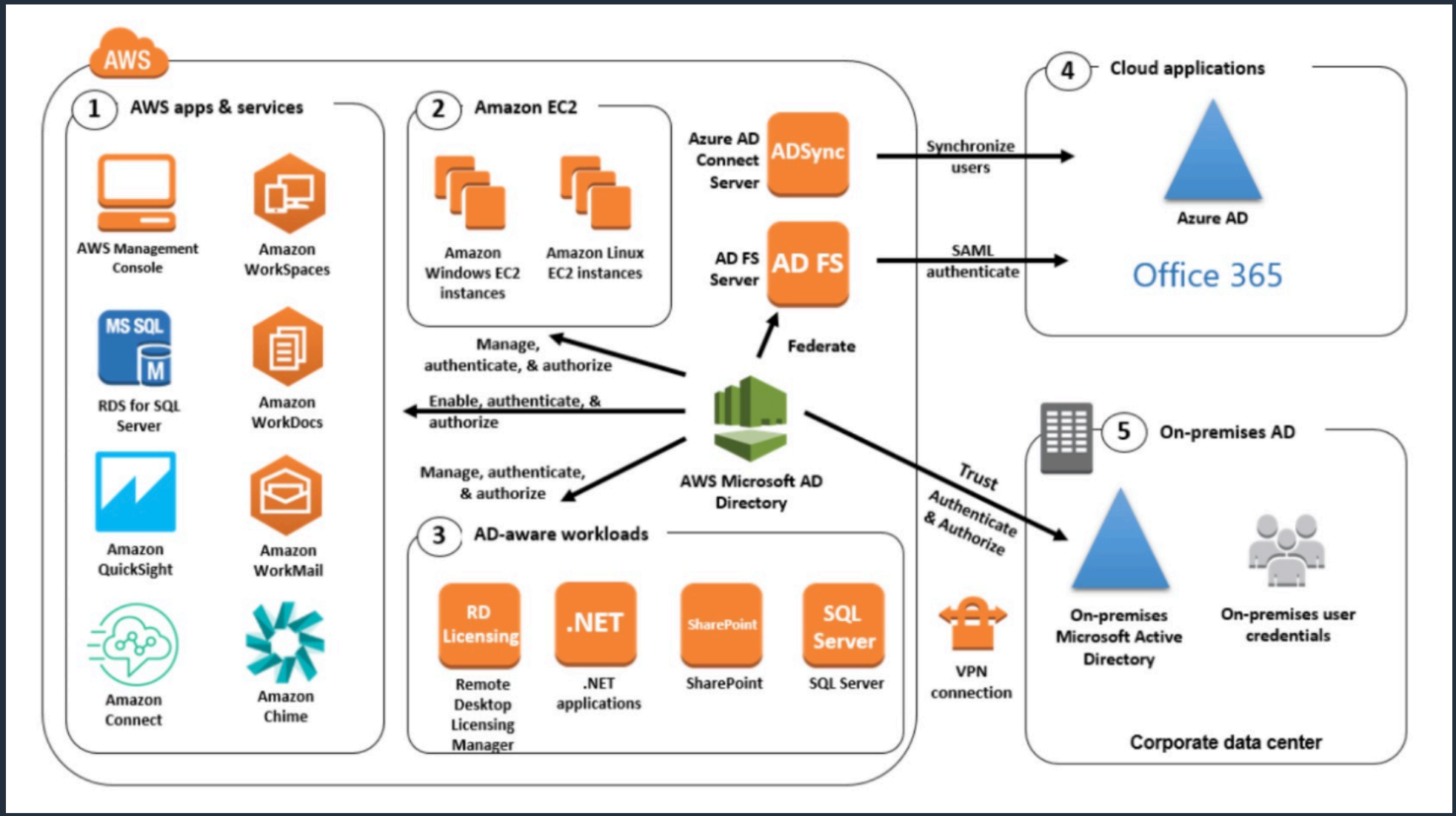
AWS Account D

Built-in SSO integrations to business applications



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AWS Directory Service - AWS Managed Microsoft AD



DigitalCloud TRAINING

AWS Directory Service - AWS Managed Microsoft AD



AWS Directory Service

- Fully managed AWS services on AWS infrastructure
- Best choice if you have more than 5000 users and/or need a trust relationship set up
- Runs on a Windows Server
- You can setup trust relationships to extend authentication from on-premises Active Directories into the AWS cloud
- On-premise users and groups can access resources in either domain using SSO
- Requires a VPN or Direct Connect connection
- Can be used as a standalone AD in the AWS cloud

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AWS Directory Service - Simple AD

- An inexpensive Active Directory-compatible service with common directory features.
- Standalone, fully managed, directory on the AWS cloud
- Simple AD is generally the least expensive option
- Best choice for less than 5000 users and don't need advanced AD features



Simple AD

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AWS Directory Service - AD Connector

Sign in to AWS applications such as Amazon WorkSpaces, Amazon WorkDocs, and Amazon WorkMail by using your Active Directory credentials.



Amazon WorkSpaces

Seamlessly join Windows EC2 instances to on-premise AD domain

Connection over VPN or Direct Connect.



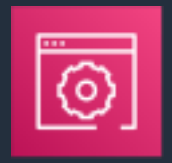
Active Directory (self-managed)



AD Connector



Amazon EC2



AWS Management Console

Provides federated sign-in to the AWS Management Console by mapping Active Directory identities to AWS Identity and Access Management (IAM) roles.

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AWS Directory Service – AD Connector



AD Connector

- AD Connector is a directory gateway for redirecting directory requests to your on-premise Active Directory
- Connects your existing on-premise AD to AWS
- Best choice when you want to use an existing Active Directory with AWS services
- You can also join EC2 instances to your on-premise AD through AD Connecto.
- You can also login to the AWS Management Console using your on-premise AD DCs for authentication

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AWS Directory Service – AD Connector vs Simple AD

Directory Service	Service Description	Use Case
AWS Directory Service for Microsoft Active Directory	AWS-managed full Microsoft AD running on Windows Server 2012 R2	Enterprises that want hosted Microsoft AD or you need LDAP for Linux apps
AD Connector	Allows on-premises users to log into AWS services with their existing AD credentials. Also allows EC2 instances to join AD domain	Single sign-on for on-premises employees and for adding EC2 instances to the domain
Simple AD	Low scale, low cost, AD implementation based on Samba. Can also join EC2 instances to the domain	Simple user directory, or you need LDAP compatibility.

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SECTION 19

Machine Learning



AWS Rekognition

Identify objects



Perform facial analysis

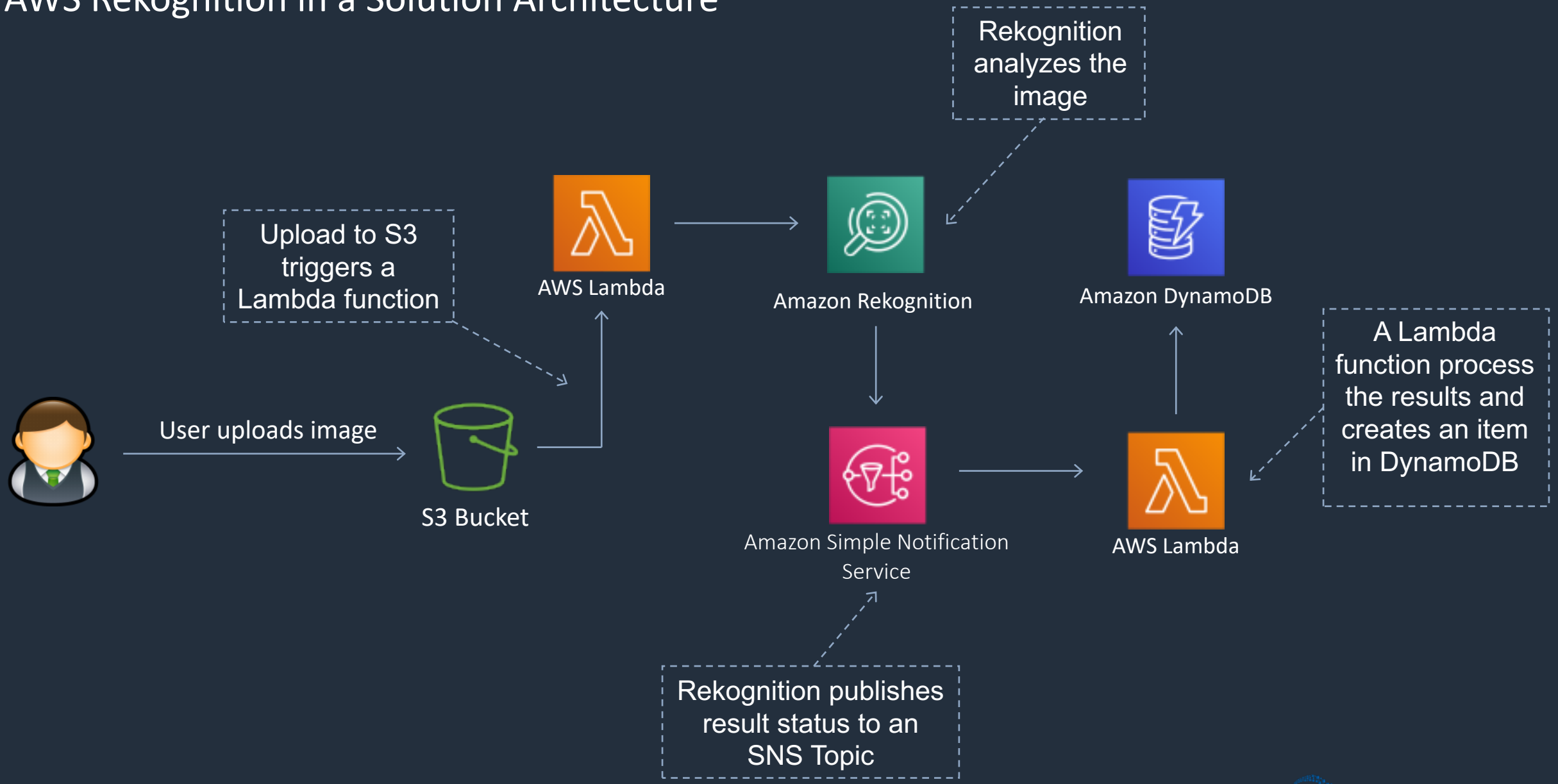


Celebrity recognition



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AWS Rekognition in a Solution Architecture



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AWS Rekognition

- Amazon Rekognition makes it easy to add image and video analysis to your applications using proven, highly scalable, deep learning technology that requires no machine learning expertise to use.
- Can identify objects, people, text, scenes, and activities in images and videos, as well as detect any inappropriate content.
- Also provides highly accurate facial analysis and facial search capabilities that you can use to detect, analyze, and compare faces for a wide variety of user verification, people counting, and public safety use cases.

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AWS Rekognition Video

- Rekognition Video processes a video stored in an Amazon S3 bucket.
- The completion status of the request is published to an Amazon Simple Notification Service topic.
- With Rekognition Video, you can track each person within a shot and through the video across shots.
- In streaming mode, you can search faces against a collection with tens of millions of faces in real time.
- Rekognition Video uses a Kinesis Video Stream as input, to process a video stream.

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Amazon Transcribe

- Amazon Transcribe enables you to add speech-to-text capability to applications
- Audio data is virtually impossible for computers to search and analyze. Therefore, recorded speech needs to be converted to text before it can be used in applications
- Amazon Transcribe uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately.
- Amazon Transcribe can be used to transcribe customer service calls, to automate closed captioning and subtitling, and to generate metadata for media assets to create a fully searchable archive.
- You can use Amazon Transcribe Medical to add medical speech to text capabilities to clinical documentation applications.

DigitalCloud TRAINING

Amazon Translate

- Amazon Translate is a neural machine translation service that delivers fast, high-quality, and affordable language translation
- Neural machine translation is a form of language translation automation that uses deep learning models to deliver more accurate and more natural sounding translation than traditional statistical and rule-based translation algorithms
- Amazon Translate allows you to localize content - such as websites and applications - for international users, and to easily translate large volumes of text efficiently

DigitalCloud TRAINING

Amazon Polly

- Amazon Polly is a service that turns text into lifelike speech, allowing you to create applications that talk, and build entirely new categories of speech-enabled products.
- Polly's Text-to-Speech (TTS) service uses advanced deep learning technologies to synthesize natural sounding human speech.

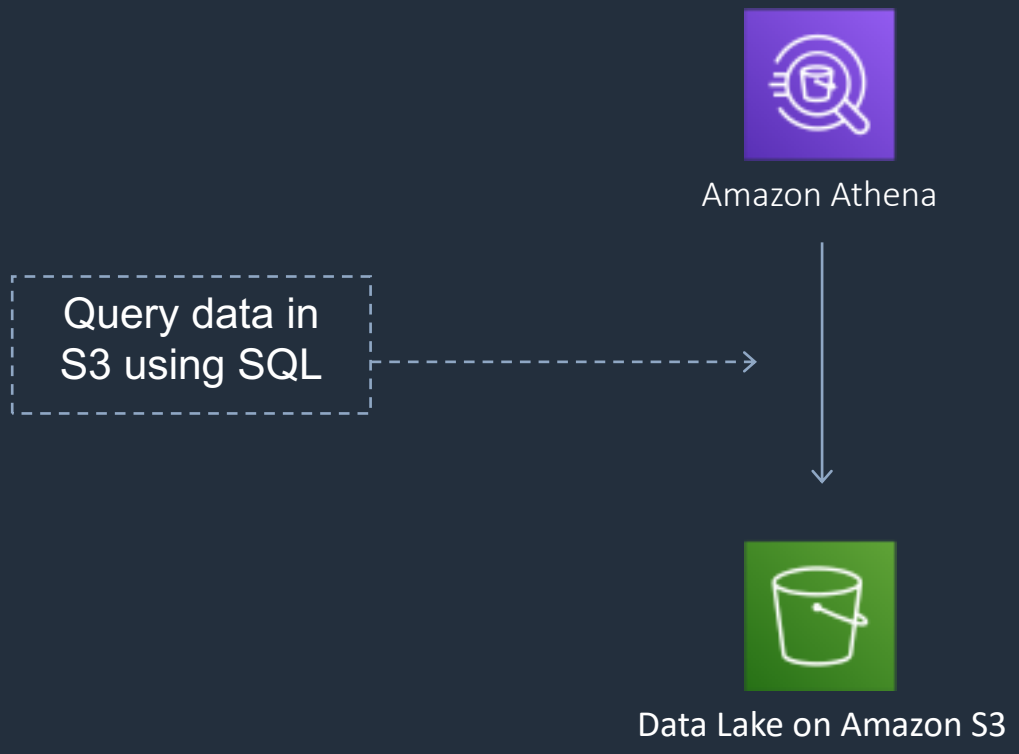
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SECTION 20

Analytics



Amazon Athena



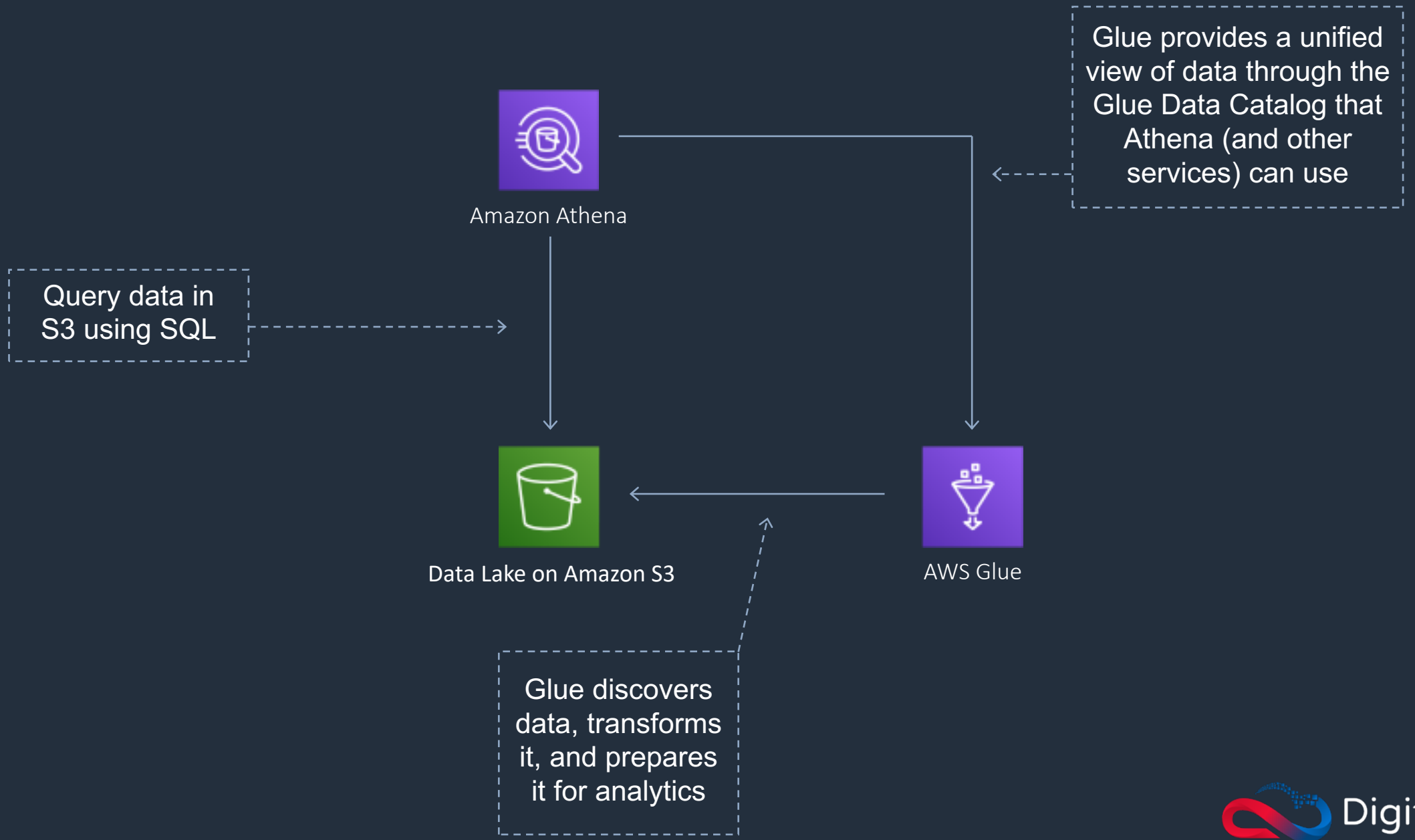
DigitalCloud TRAINING

Amazon Athena

- Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL.
- Serverless, so there is no infrastructure to manage, and you pay only for the queries you run
- Athena uses a managed Data Catalog (AWS Glue) to store information and schemas about the databases and tables that you create for your data stored in Amazon S3.
- Athena is optimized for fast performance with Amazon S3.
- With Athena, there's no need for complex ETL jobs to prepare data for analysis.
- This makes it easy for anyone with SQL skills to quickly analyze large-scale datasets.

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Amazon Athena with AWS Glue



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AWS Glue

- AWS Glue is a fully-managed, pay-as-you-go, extract, transform, and load (ETL) service that automates the time-consuming steps of data preparation for analytics
- AWS Glue automatically discovers and profiles data via the Glue Data Catalog, recommends and generates ETL code to transform your source data into target schemas
- Simply point AWS Glue to your data stored on AWS, and AWS Glue discovers data and stores the associated metadata (e.g. table definition and schema) in the AWS Glue Data Catalog
- Once cataloged, data is immediately searchable, queryable, and available for ETL
- Works with data lakes (e.g. data on S3), data warehouses (including RedShift), and data stores (including RDS or Amazon EC2 databases)

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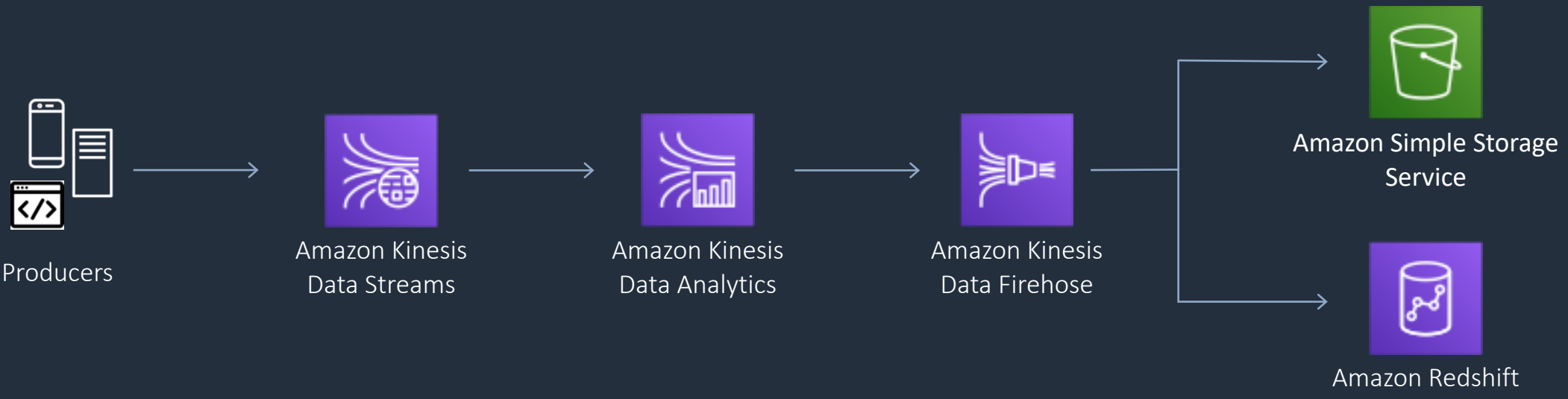
Amazon Kinesis - Overview

- Examples of streaming data use cases include:
  - Purchases from online stores.
  - Stock prices.
  - Game data (statistics and results as the gamer plays).
  - Social network data.
  - Geospatial data (think uber.com).
  - IoT sensor data.

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Amazon Kinesis - Overview

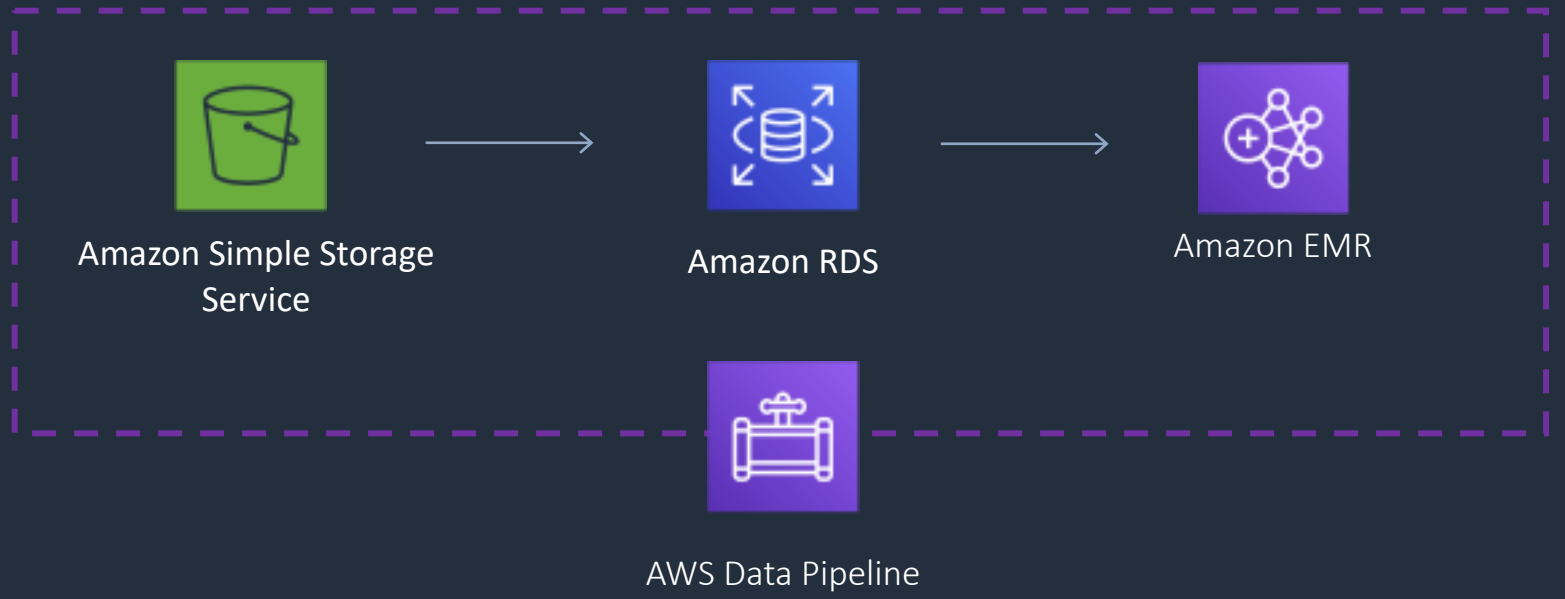
- Amazon Kinesis makes it easy to collect, process, and analyze real-time, streaming data so you can get timely insights and react quickly to new information
- Collection of services for processing streams of various data
- The Kinesis family includes Kinesis Data Streams, Kinesis Data Firehose, Kinesis Data Analytics and Kinesis Video Streams



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AWS Data Pipeline

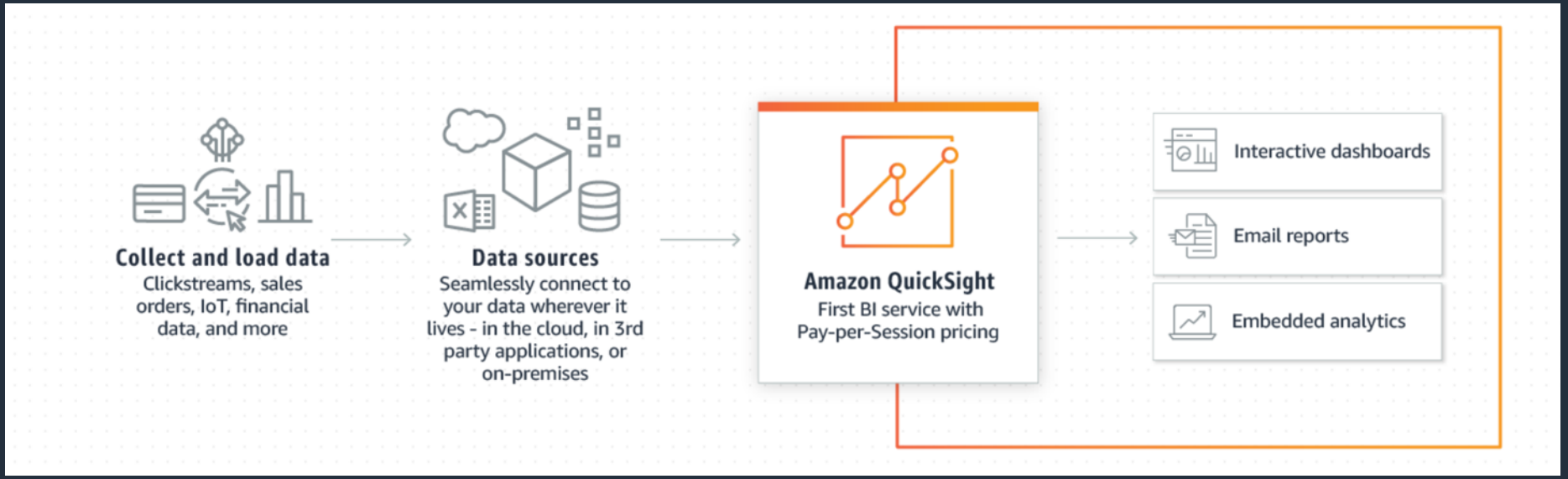
- AWS Data Pipeline is a web service that helps you reliably process and move data between different AWS compute and storage services, as well as on-premises data sources, at specified intervals
- With AWS Data Pipeline, you can regularly access your data where it's stored, transform and process it at scale, and transfer the results to services such as Amazon S3, Amazon RDS, Amazon DynamoDB, and Amazon EMR



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Amazon QuickSight

- Amazon QuickSight is a fast, cloud-powered business intelligence service that makes it easy to deliver insights to everyone in your organization
- As a fully managed service, QuickSight lets you easily create and publish interactive dashboards that include ML Insights. Dashboards can then be accessed from any device, and embedded into your applications, portals, and websites



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SECTION 21

Architecting for the Cloud

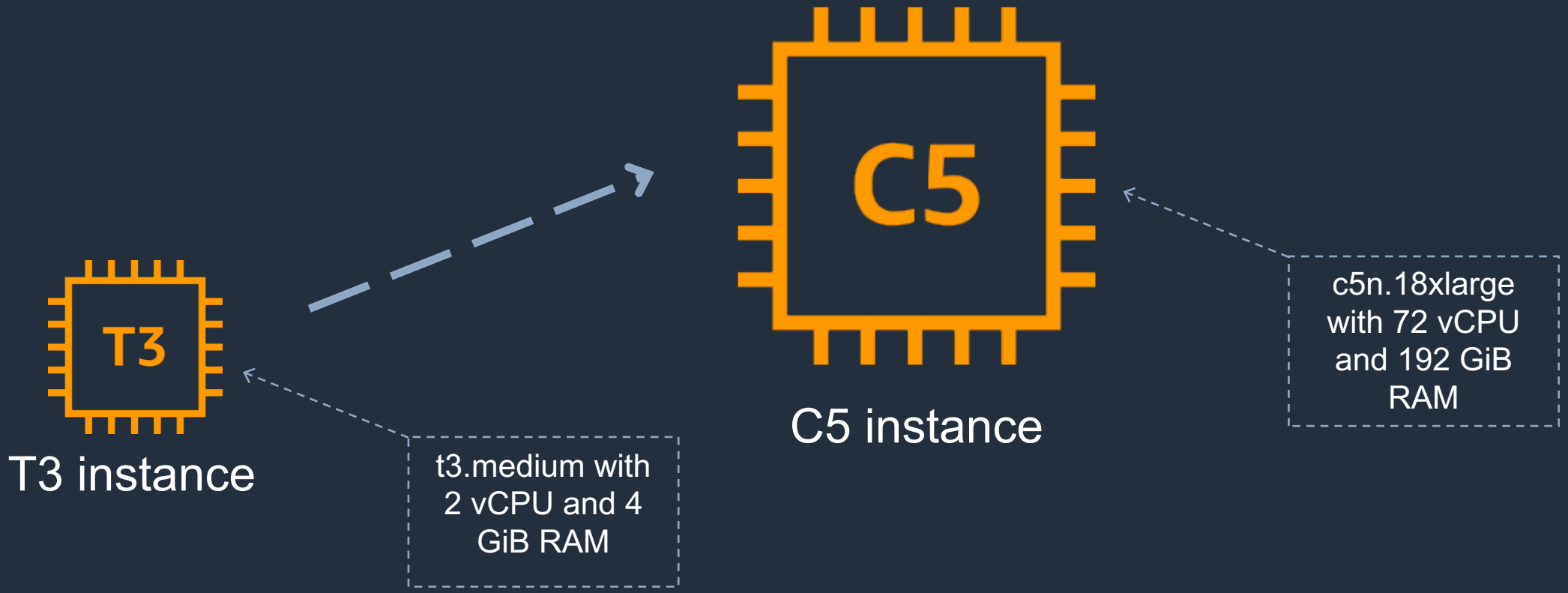


The Difference Between Traditional and Cloud Computing Environments

- IT Assets as Provisioned Resources
  - Provision as needed, and scale on-demand
- Global, Available, and Scalable Capacity
  - Deploy globally, easily, cost-effectively, and quickly
- Higher-Level Managed Services
  - Lower operational cost by leveraging managed storage, database, analytics, application and deployment services
- Built-in Security
  - Leverage AWS' significant investment in security, simplify security testing, and use native security and encryption features
- Architecting for Cost
  - Fine grained billing, transparent costs, budgets and alerting tools
- Operating on AWS
  - Tooling, processes and best practices to support operational transitions

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Scaling Vertically



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Scaling Vertically

Examples of vertical scaling are:

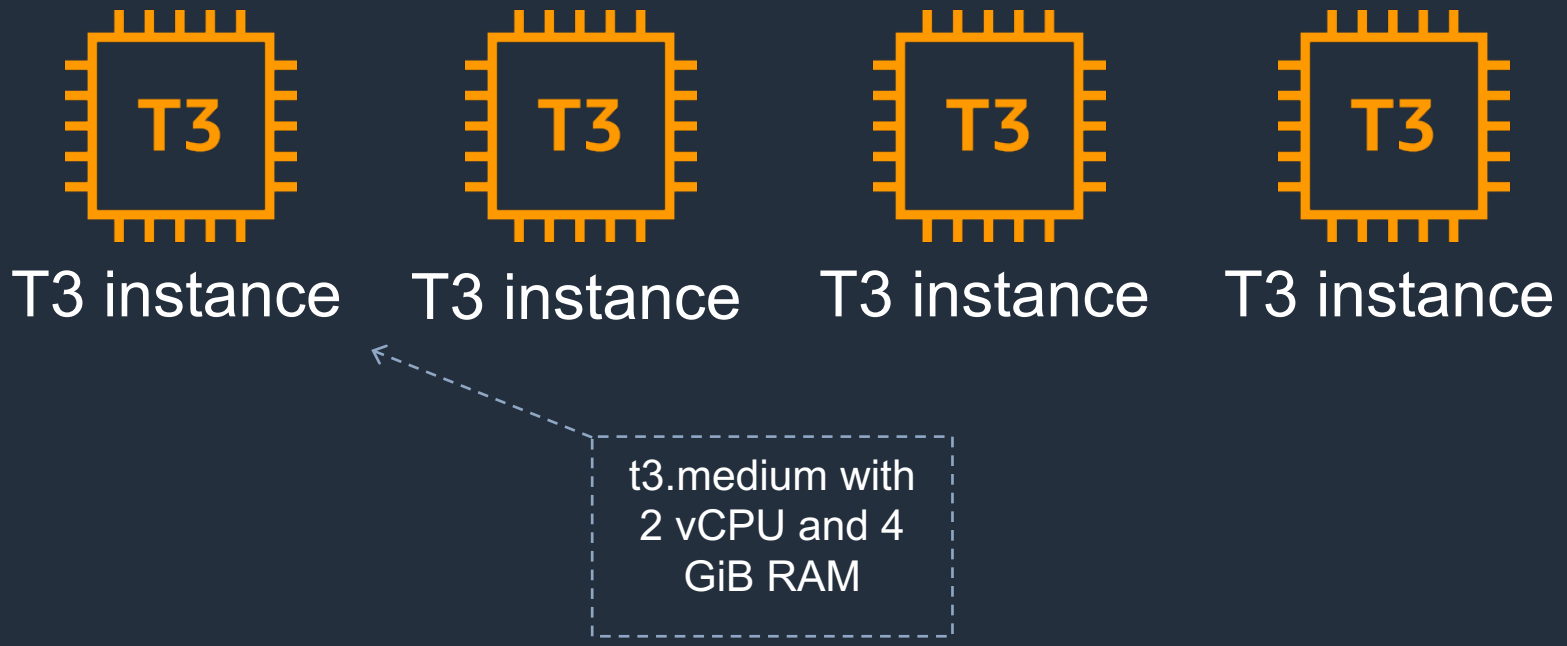
- Amazon EC2 instances
- Amazon RDS Database instances

Limitations of scaling vertically:

- Often requires manual intervention (though can be scripted/automated)
- Typically requires downtime
- Can reach a limit of scalability

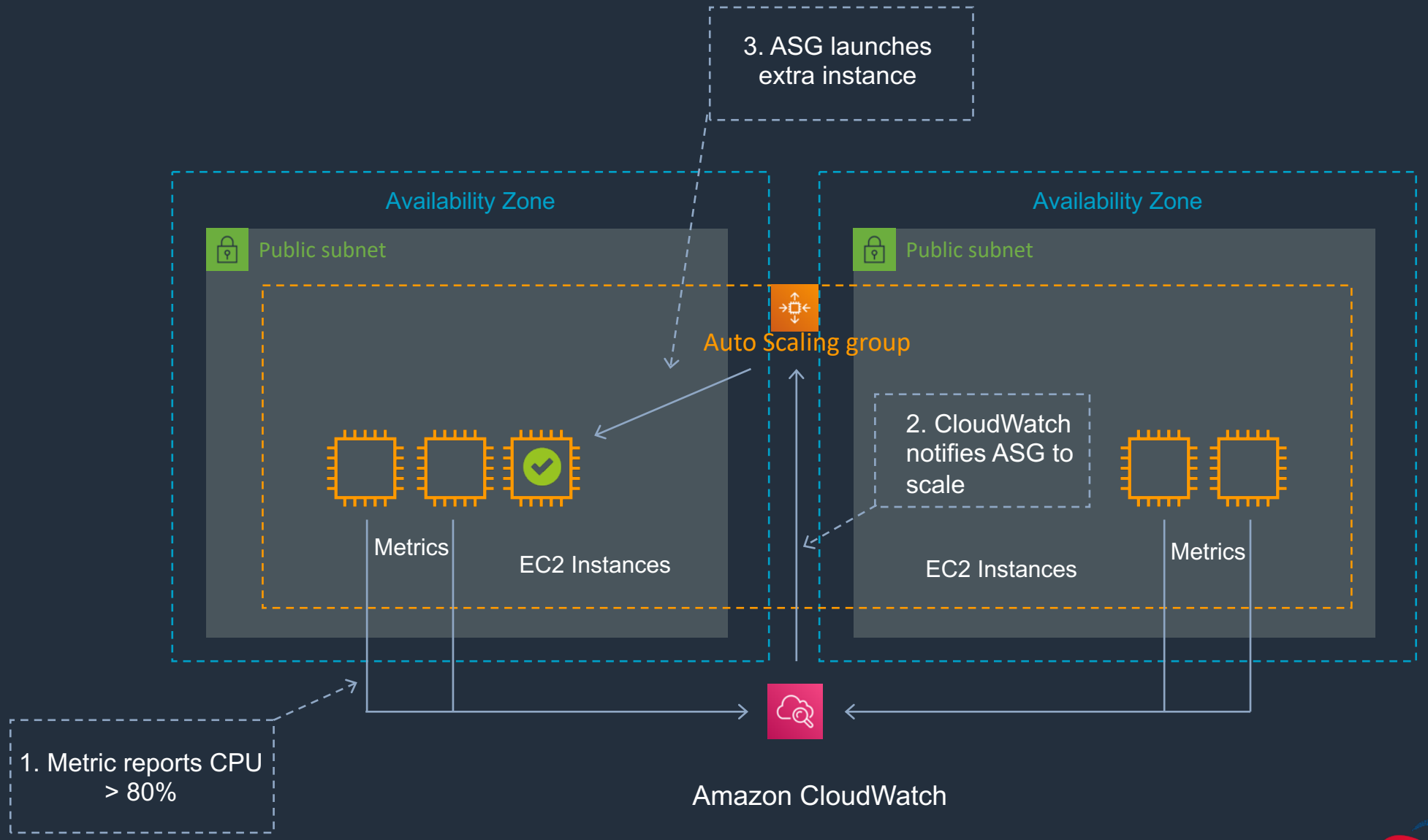
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Scaling Horizontally



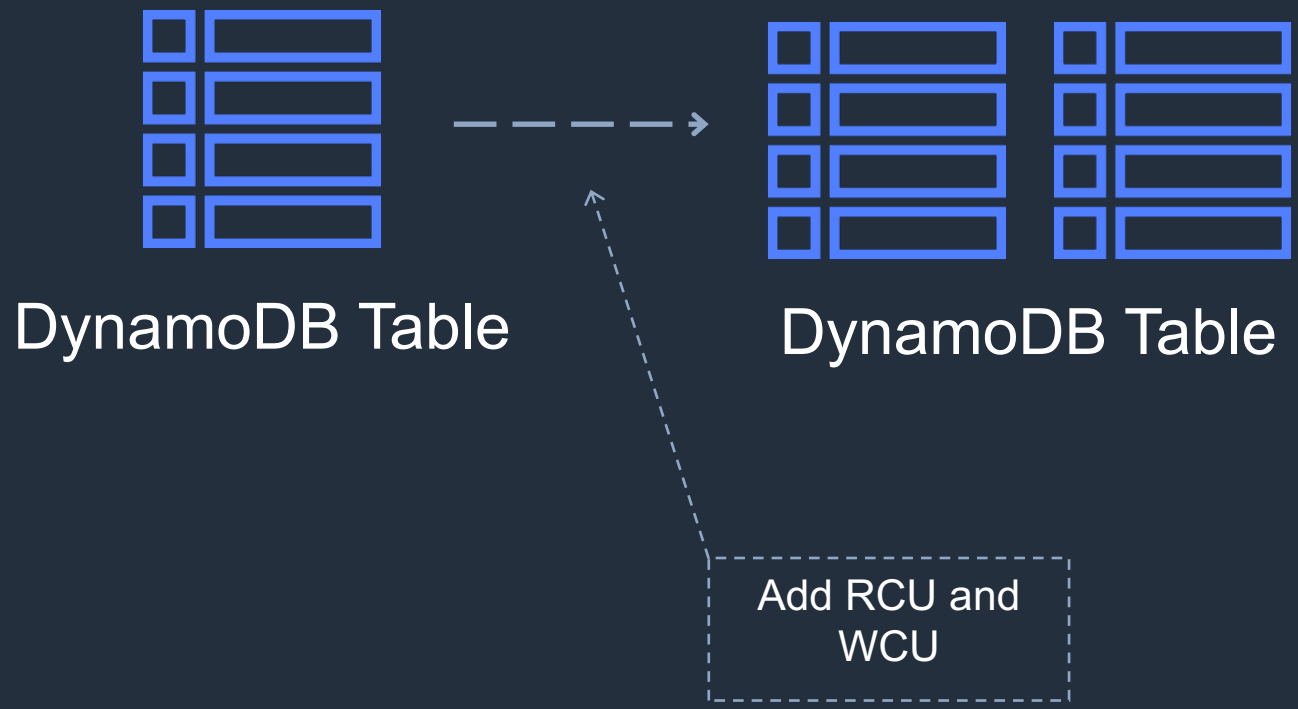
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Example of Horizontal Scaling with EC2



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Example of Horizontal Scaling with DynamoDB



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Scaling Horizontally

Examples of horizontal scaling are:

- Amazon EC2 Auto Scaling
- Amazon DynamoDB

Benefits of scaling horizontally:

- Seamless scaling, without downtime
- Can scale almost limitlessly (in some cases)

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Architecting for the Cloud: AWS Best Practices



https://d1.awsstatic.com/whitepapers/AWS_Cloud_Best_Practices.pdf

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Disposable resources instead of fixed servers

- With traditional infrastructure you work with fixed resources due to upfront costs and lead time
- With the cloud you can take advantage of the dynamically provisioned resources
- When designing for AWS, you can take advantage of the dynamically provisioned nature of cloud computing
- You can think of servers and other components as temporary resources
- You can launch as many as you need, and use them only for as long as you need them

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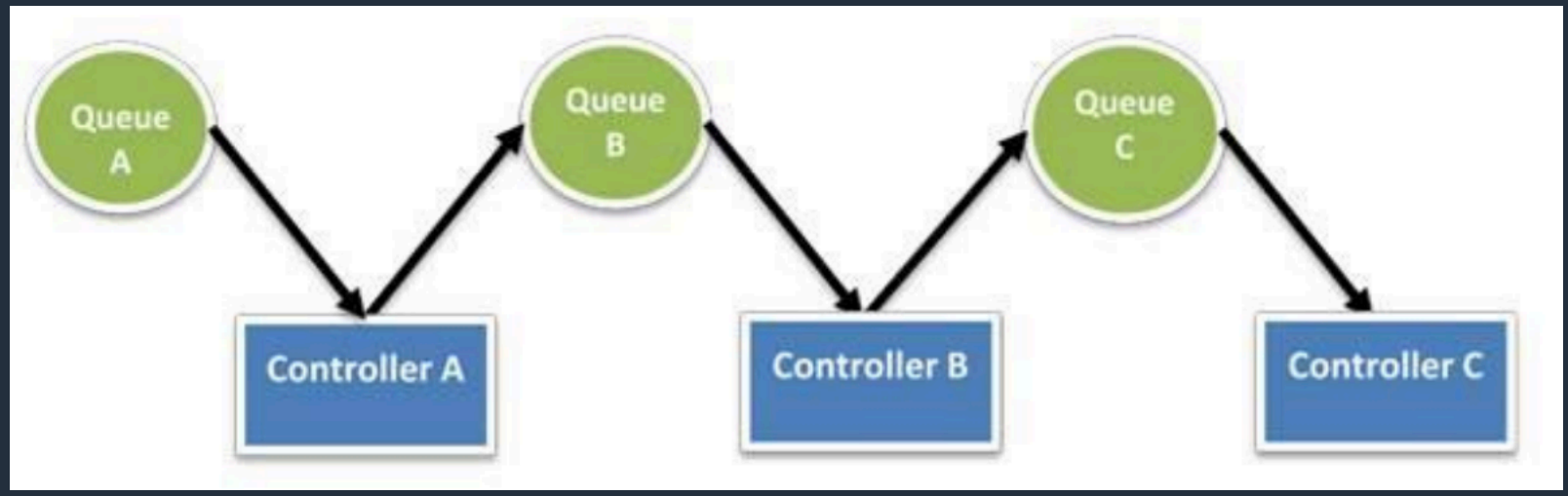
Automation

- Use automation tools on AWS to improve system stability and organizational efficiency
- Examples are:
  - Serverless Management and Deployment
  - Infrastructure Management and Deployment
  - Alarms and Events

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Loose Coupling

- Design IT systems to reduce interdependencies
- A change or a failure in one component should not cascade to other components
- Examples are:
  - Service Discovery – components should be able to “find” each other
  - Asynchronous Integration – introduce a layer between application components to store messages



Example of asynchronous integration with Amazon SQS

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Services, Not Servers

- This best practices advises customers to leverage more than just Amazon EC2
- Try to use the breadth of services available on AWS
- For example:
  - Amazon CloudFront for content delivery
  - ELB for load balancing
  - Amazon DynamoDB for NoSQL databases
  - Amazon CloudSearch for search workloads
  - Amazon Elastic Transcoder for video encoding

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Databases

- Use the right database technology for each workload
- Consider performance requirements, scalability, durability, functionality etc.
- Choose between Relational, NoSQL, Data Warehouse, Search, and Graph

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Removing Single Points of Failure

- Design for failure!
- Introduce redundancy – ensure if a component / resource fails, another can take over
- Detect failure – use health checks and alarms and try to automate detection and reaction
- Durable Data Storage – ensure your data aligns with your Recovery Point Objective (RPO) and Recovery Time Objective (RTO)
- Automated Multi-Data Center Resilience – be resilient in the face of a major disaster

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Optimize for Cost

- Right Sizing – use the best instance sizes and number of instances for cost efficiency
- Elasticity – horizontally scale as needed with changing demand
- Take Advantage of the Variety of Purchasing Options – use Reserved Instances and Spot Instances

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Caching

- Caching can be used to improve performance and cost efficiency
- Methods of caching include:
  - Application Data Caching – examples are Amazon ElastiCache and DynamoDB DAX
  - Edge Caching – key example is Amazon CloudFront

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Security

- Use AWS Features for Defense in Depth
- Share Security Responsibility with AWS
- Reduce Privileged Access
- Security as Code – for instance, use CloudFormation to repeatably build secure environments
- Real-Time Auditing – tools such as Trusted Advisor, AWS Config, and Amazon Inspector

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The Five Pillars of Operational Excellence

1) Operational Excellence
   - The operational excellence pillar includes the ability to run and monitor systems to deliver business value and to continually improve supporting processes and procedures
2) Security
   - The security pillar includes the ability to protect information, systems, and assets while delivering business value through risk assessments and mitigation strategies
3) Reliability
   - The reliability pillar includes the ability of a system to recover from infrastructure or service disruptions, dynamically acquire computing resources to meet demand, and mitigate disruptions such as misconfigurations or transient network issues

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The Five Pillars of Operational Excellence

4) Performance Efficiency

- The performance efficiency pillar includes the ability to use computing resources efficiently to meet system requirements and to maintain that efficiency as demand changes and technologies evolve

5) Cost Optimization

- The cost optimization pillar includes the ability to avoid or eliminate unneeded cost or suboptimal resources

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SECTION 22

Exam Preparation and Tips



Exam Preparation

1) Be an active learner





2) Use practice tests early





3) Dedicate regular time for learning

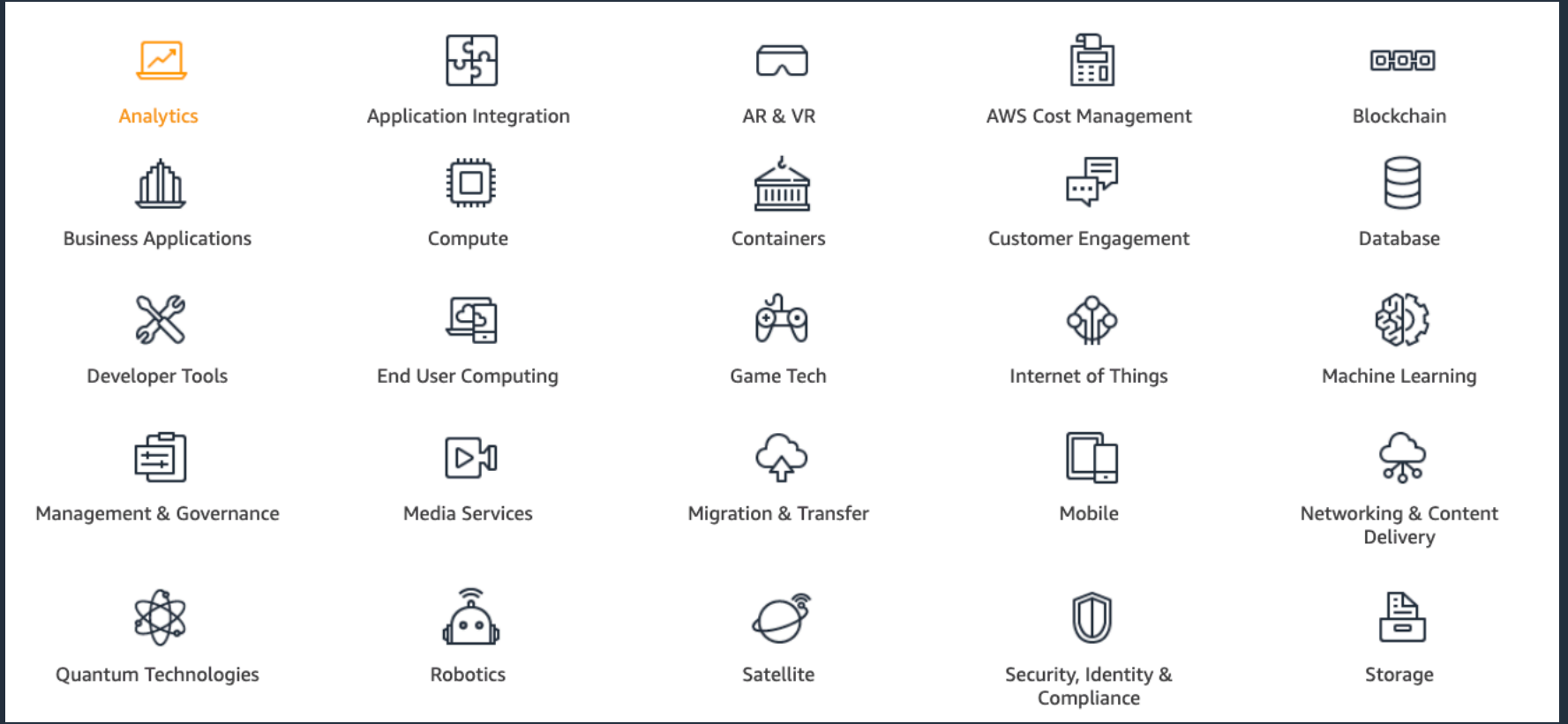


4) Don’t book until you’re ready



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Additional AWS Services



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Distractors

Which AWS services can be used to connect the AWS Cloud and on-premises resources? (Select TWO.)

1. AWS Managed VPN
2. Amazon Connect
3. Amazon CloudHSM
4. AWS Direct Connect
5. AWS Managed Services

Amazon Connect is an easy to use omnichannel cloud contact center that helps companies provide superior customer service at a lower cost

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Distractors seen recently

- Amazon Connect
- AWS Batch
- Amazon AppStream 2.0
- AWS X-Ray
- Amazon Lumberyard

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THE END

Hope you enjoyed the course!



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